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Probability and Stochastic Processes

A Volume in Honour of Rajeeva L. Karandikar

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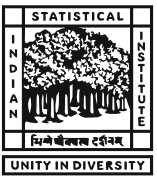
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Editors

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Preface

Every subject, including probability, is developing very rapidly in several directions. A beginner gets bewildered and lost. It is essential to bring out once in a while a brief status report to help the interested research workers. It would again be a huge task if we attempt to touch each direction. Even in one direction, if one tries to cover all the issues it becomes unmanageable for the author and confusing for the reader.

An online conference was held, during 29–31 March 2022, in honour of our colleague Prof. Rajeeva L. Karandikar, probabilist, statistician, psephologist, cryptologist, administrator and above all a kind and helpful human being.

On the morning of 29th March, there were two lectures: first by Sourav Chatterjee on "A new coefficient of correlation" and the second by M Vidyasagar on "Convergence of stochastic approximation algorithms via martingale and converse Lyapunov methods". In the evening, we had two presentations: Arup Bose on "Large dimensional random matrices and high dimensional statistics" and Kumarjit Saha on "Convergence to the Brownian web for a perturbed Howard model". On the second day morning, we had Riddhipratim Basu speaking on "First passage percolation on hyperbolic groups" and Shirshendu Ganguly speaking on "Fractal geometry in models of random growth". In the evening session, the lectures were delivered by Mathew Joseph on "Small-ball probabilities for the stochastic heat equation" and by Probal Chaudhuri on "A brief history and some observations on Mahalanobis' distance". On the final day, the morning session started with a lecture by Manjunath Krishnapur on "Random analytic functions from orthogonal polynomials on the unit circle followed" by Amarjit Budhiraja on "Domains of attraction of invariant distributions of the infinite Atlas model". The concluding session in the evening consisted of reminiscences by the participants after a brief presentation by Abhay G Bhat on "Glimpses of RLK's work over four decades".

There were about 180 participants who attended. After the sessions, it was felt by many participants that it would be nice to have the lectures brought out in book form so that researchers can consult these diverse topics at one place. It would also be helpful to the beginning researchers so that they can browse and get an idea of the background and tools to be picked up before they start their research career. The project of bringing out this book took shape. Unfortunately, some of the speakers,

due to several preoccupations and other constraints, could not write up. We provide a brief outline of each of the articles to indicate what the reader is in for to conclude the preface.

Even though random matrices first appeared in the context of multivariate statistical analysis by Wishart, the study of statistical properties of random matrices gained importance following Wigner's intuition, followed by confirmation, that the local statistical behaviours of energy levels of various nuclei are similar to the eigenvalues of certain random matrices. This is now a vast growing field, and the article by Monika Bhattacharya and Arup Bose provides a neat introduction to this and also helps in navigating through the literature.

One way to get interesting process is to start with a diffusion and constrain it to stay in an orthant—by reflecting when it is exiting the orthant. Such processes lead to several interesting questions about the behaviour of the process like existence of invariant/ergodic measures, their determination and properties. Amarjit Budhiraja and Sayan Banerjee discuss these issues with pointers to the literature.

For quite sometime one felt that the concept of correlation coefficient, though measures dependence to some extent, is not able to really measure the association between variables. The article of Sourav Chatterjee introduces a new measure and discusses its properties in a lucid manner.

Shirshendu Ganguly and Milind Hegde in a short survey outline the recent results on the fractal structure in the directed landscape. These are deemed to have a universal scaling limit within the KPZ universality class.

When you are studying a random process, you come across two interesting problems. First: if it has a limit, how fast is it approaching the limit? And the second: how long does the process stay close to its initial position? Mathew Joseph's article explains some recent results. Since this issue soon gets technical, he gives pointers to the literature.

The Burton–Keane result and proof for the uniqueness of infinite clusters are a significant milestone step in the area of percolation and stochastic geometry. Kumarjit Saha contributes a chapter in which he implements such an arguement to show that the two-dimensional directed spanning forest constructed on the Ginibre ensemble is connected almost surely.

Vidyasagar in a broad and detailed survey studies the global convergence properties of the stochastic approximation (SA) algorithm, first proposed by Robbins and Monro. The chapter includes a general theorem based on the existence of a Lyapunov function having suitable properties without any assumption on global exponential stability of the equilibrium. Alternate versions of SA introduced by Kiefer and Wolfowitz along with some extensions using the martingale approach are also included.

We thank all the speakers of the conference and also all of those who participated in the discussions. We thank the authors and reviewers for sparing their time.

Bangalore, India — Siva Athreya
Delhi, India — Abhay G. Bhatt
Chennai, India — B. V. Rao
October, 2023

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Abhay G. Bhatt is Professor, at the Theoretical Statistics and Mathematics Unit at the Indian Statistical Institute, Delhi Centre, India. His primary field of interest is in the area of probability theory and stochastic processes. He has worked extensively on problems in filtering theory, controlled Markov processes and Markov processes characterized via martingale problems.

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Rajeeva Laxman Karandikar: An Appreciation



Abhay Gopal Bhatt

1 Introduction

In an illustrious career spanning over four decades, Professor Rajeeva Laxman Karandikar (RLK) has made several seminal contributions to academics. He has been an outstanding researcher and has actively collaborated with several co-authors from around the world. His main research interests lie in the broad area of Probability theory, and specifically in the area of Stochastic processes. He has also worked with industry, wherein he has successfully made use of his theoretical knowledge to solve practical real-life problems. He is also an excellent administrator having served in various leadership positions both in the academic world and in the industry.

RLK has had his early training in Statistics and Probability. Both his parents were academicians, teaching in colleges in Indore. His mother was a Professor of Chemistry and his father of Mathematics and Statistics. Given this background at home, it was perhaps natural that he would be drawn to academics. His father was also an avid Bridge player. He had studied the combinatorial and statistical aspects of the tournament problem in masters pair bridge tournament. RLK inherited the interest in Bridge as well as in Combinatorics.

He completed his school and college education in Indore, graduating with a Bachelor's degree in Mathematics, Physics, and Statistics. Influenced by Professor M. G. Nadkarni, Karandikar applied to the *Indian Statistical Institute (ISI)* to continue his study towards a Master's degree in Statistics. At ISI, he could discuss Mathematics with an excellent array of teachers as well as with an amazing collection of sharp minds—his fellow students ranging from those pursuing an undergraduate degree to those working towards their doctoral degrees. Among his teachers were Professors J. K. Ghosh, A. Maitra, B. V. Rao, S. C. Bagchi, G. Jogesh Babu, and A. K. Roy who greatly influenced his academic career.

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After getting his M. Stat. degree he continued his studies in ISI for his doctoral degree, working under the supervision of B. V. Rao. The thesis included results on Pathwise formulae for stochastic differential equations and multiplicative stochastic integration with respect to semimartingales. He received his Ph.D. degree from ISI in 1981. He has continued to work in the general area of Stochastic calculus and his recent book *Introduction to Stochastic Calculus*, co-authored with B. V. Rao, approaches the subjects from a new angle.

During his Ph.D. years, he got introduced to finitely additive probabilities and went on to prove a meta theorem on limit theorems on finitely additive probability spaces.

Immediately after the successful completion of his Ph.D. he visited the *University of North Carolina at Chapel Hill (UNC)* and worked with Professor G. Kallianpur, starting a life-long collaborative partnership. During the 1980s, they embarked on a systematic study of Gaussian measures on the field of cylinder sets in a real separable Hilbert space, with the aim of applying the theory to Stochastic filtering in the set-up of finitely additive probability theory. A beautiful and lucid exposition of this theory is available in their jointly authored book *White noise theory of Prediction, Filtering and Smoothing*.

After a two-year stint at UNC, RLK returned to India and joined as an Associate Professor at the Delhi Centre of ISI in 1984. Apart from being a brilliant researcher, he was an excellent teacher. I joined ISI, Delhi as a Master's student in 1987. His enthusiasm in explaining a new concept and in discussing mathematics influenced all students in his class. I decided to continue my studies and enrolled for my Ph.D. degree under his supervision. This has been one of the best decisions of my life.

It was during this period that we looked at a result of Echeverria, characterizing stationary measures of Markov processes taking values in a locally compact, separable metric space and extended it to the case when the state space is a general complete, separable metric space. This extension was also critically used in proving the uniqueness of solutions of the measure valued equations of stochastic filtering theory, both in the finitely additive as well as the classical (countably additive) probability set-up.

In addition to the three main research themes mentioned above, RLK has worked in several diverse areas including mathematical finance, queueing theory, extreme value theory, game theory, option pricing, learning theory, gene prediction algorithms, cryptography, etc., to name just a few. His book *Introduction to option pricing theory*, written jointly with Kallianpur, gives a very good account of the classical Itô theory and its application to stochastic finance.

He has also actively engaged with institutions outside academics. He has imparted his expertise and advice to industry, the Indian defence establishment, the Central Bureau of Investigation, the Election Commission of India, etc.

RLK has also developed a statistical model for predicting seats based on opinion poll data in the complex scenario of Indian elections. This model, used for over two decades and during several state and general elections in India, has been fairly successful.

Several colleagues and students have benefitted by learning from him, discussing with him, and collaborating with him. He has an unending enthusiasm for the subject which rarely fails to be contagious.

Awards and Accolades

His work has been acknowledged over the decades by peers and the general science community as well as by the industry and the government.

He was awarded the *Young Scientist Medal* of the *Indian National Science Academy (INSA),* New Delhi, in 1985. He was elected *Fellow of the Indian Academy of Sciences (IAS)*, Bangalore, in 1994 and of INSA in 2005.

He was a recipient of the prestigious *S. S. Bhatnagar prize* in 1999 which is awarded by the Council for Scientific and Industrial Research (CSIR). He was the first recipient of the the *National Award in Statistics in honour of Professor C. R. Rao* instituted in 2000/01 by the Ministry of Statistics and Programme Implementation, Government of India (MOSPI). He was awarded the *P. C. Mahalanobis Gold Medal* by the *Indian Science Congress Association* (ISCA) in 2014.

During his sojourn in the industry, he served as the vice-president of *Cranes Software*, a Bangalore-based company. Earlier, he had also been directly involved in academic administration during his stint at ISI. Later RLK served as Director of *Chennai Mathematical Institute* for ten years and is currently the Chairman of the *National Statistical Commission* of India.

In the rest of this article, we give a brief overview of his research.

2 Stochastic Calculus

While Karandikar has worked in several different areas of probability, most of his research has concentrated around the theme of Stochastic Calculus.

Pathwise Formulae

His early work includes a pathwise formula for stochastic integrals and for solutions of stochastic differential equations driven by a semimartingale. Stochastic integral appears in several diverse applications. For example, in statistical inference involving continuous time processes, an estimator appears as a stochastic integral. Necessarily, it should be a function of observation paths. Similarly, trading strategies in mathematical finance can involve a stochastic integral and should be computable from the observed paths of the stock prices. However, the Itô Integral is defined only as a limit in probability and not as a function of the underlying paths.

RLK proved the existence of a universal map $\mathcal{I}$ with the integrand—a continuous adapted process and the integrator—a continuous semimartingale as arguments and whose output is a version of the corresponding stochastic integral (see (Karandikar, 1981)), i.e. he showed that there exists $\mathcal{I} : C([0, T], \mathbb{R}) \times C([0, T], \mathbb{R}) \to C([0, T], \mathbb{R})$ such that for continuous adapted process U and a continuous semimartingale V

$$\mathcal{I}_t(U, V) = \int_0^t U_s dV_s \quad \forall\, t \quad \text{a.s.}$$

In Karandikar (1981) he also extended these pathwise results to solutions of stochastic differential equations (SDE) driven by a continuous semimartingale. In particular, he proved the existence of a map $\mathcal{T}^{a,b} : \mathbb{R} \times C([0, T], \mathbb{R}) \to C([0, T], \mathbb{R})$ depending only the coefficients a and b such that for a continuous semimartingale V and a initial random variable X_0, $X_t = \mathcal{T}_t^{a,b}(X_0, V)$ is a solution of the SDE

$$dX_t = a(t, X_t)dt + b(t, X_t)dV_t.$$

In general, the stochastic integral is a measurable function of the integrand and the integrator, but it may not be amenable for use in applications. These pathwise formulae are thus very helpful for applications and have been well appreciated by several noted researchers.

These results were extended to r.c.l.l. semimartingales in Karandikar (1989, 1995).

Multiplicative Decomposition of Matrix-Valued Semimartingales

Around the same time RLK showed (see Karandikar, 1982) that a continuous non-singular matrix-valued semimartingale Y has a unique multiplicative decomposition $Y = NB$, where N is a matrix-valued local martingale and B is a matrix-valued process with paths of locally bounded variation.

These were extended to r.c.l.l. semimartingales in Karandikar (1991). In the spirit of the pathwise formulae, almost sure limit theorems for matrix-valued semimartingales were proved in Karandikar (1982). In fact starting from his Ph.D. thesis, RLK was interested in limit theorems. In a path-breaking paper (Karandikar, 1989), he introduced a notion of a dominating process for a semimartingale which was particularly useful in limit theorems. A notion of fast convergence for stochastic integrals in terms of the dominating processes was introduced. It is analogous to the relationship of almost sure convergence to the convergence in probability. Using this notion many of the standard theorems of measure theory, like the dominated convergence theorem or Fubini's theorem, found analogues for stochastic calculus.

A New Approach to Stochastic Calculus

In a recent jointly authored book with Rao (2018), stochastic calculus is developed in a novel way. A notion of *stochastic integrator* is developed and the corresponding stochastic integral is defined. It is shown that the class of stochastic integrators coincides with the class of semimartingales. This is different from the traditional approach of defining the integral via the Doob–Meyer decomposition.

3 White Noise Theory

During his student days, Karandikar was exposed to several different notions in mathematics, and in particular in probability. He got interested in studying probability on finitely additive spaces. In Karandikar (1982), he proved the following meta limit theorem. Considering the Dubins–Savage set-up of finitely additive probability, he showed that

> *Almost all limit theorems that are valid for a sequence of independent random variables in countably additive probability spaces are also valid – verbatim – in the finitely additive set-up.*

These include the laws of large numbers, central limit theorems, and laws of iterated logarithms. This was extended to the general finitely additive probability set-up in Karandikar (1988)

Nonlinear Filtering in the Finitely Additive Set-up

In the mid-1970s, G. Kallianpur (GK) had returned to India and was the Director of ISI, a post he continued to hold when RLK joined as a research scholar. However, GK returned to UNC soon after. After getting his Ph.D. degree, RLK visited the *Center for Stochastic Processes* at UNC, Chapel Hill, and during his two-year stint there, he started a defining collaboration with GK.

They embarked on a systematic and comprehensive study of finitely additive probability spaces and in particular of Gaussian measures on the field of cylinder sets in a real, separable Hilbert space. One of the aims of this study was to address the criticism of the classical theory of nonlinear filtering from some quarters of not being useful, since it was not implementable.

Quoting Kolmogorov (1950) in Kallianpur and Karandikar (1988) the authors pointed out that countably additive probability spaces *"occur only as idealized models of real random processes" ⋯ "(we) limit ourselves, arbitrarily to only those models which satisfy the axiom (of countable additivity)."* They studied the filtering model when the additive observation noise was assumed to be a white noise defined on a finitely additive probability space. In particular, they considered the observation model

$$O_t = h(X_t) + n_t$$

where n is the white noise, h is a known function of the signal process X, and O is the observation process. The signal X is assumed to be a Markov process (on a countably additive probability space).

In a series of papers, they studied the optimal filter π defined by

$$\langle \pi_t, f \rangle = E[f(X_t)|O_s : 0 \leq s \leq t],$$

deriving the equations of filtering, *viz.,* the Fujisaki–Kallianpur–Kunita and the Zakai equations, satisfied by the optimal filter, showing uniqueness as well as the Markov

property of the filter. Also shown was the continuous dependence of the filter on the signal process, i.e. robustness of the filter. In this set-up, they showed that when the signal process is finite dimensional, the equations of filtering are partial differential equations.

This collaboration continued for most of the decade. A nice summary can be found in their special invited paper in the Annals of Probability (Kallianpur and Karandikar, 1985). Their book Kallianpur and Karandikar (1988) gives a comprehensive treatment of the entire subject.

Karandikar continued to work on this theme with other co-authors. In (Bhatt et al., 2006) uniqueness of solution of the filtering equations under extremely general conditions—when the observation function h is allowed to be unbounded and discontinuous, for example—was proved.

4 Martingale Problems

I joined the Ph.D. programme at ISI in 1989. RLK had agreed to be my thesis advisor. He introduced me to the theory of martingale problems developed by Stroock and Varadhan (see Stroock & Varadhan, 1979). It is now widely used as a means of characterizing Markov processes.

Invariant Measures

Any Markov process is the unique solution of a martingale problem for a suitably chosen operator (see Ethier & Kurtz, 1986), which in general is a restriction of the generator of the Markov process. In particular, a process X is said to be a solution to the martingale problem for an operator A if

$$f(X_t) - \int_0^t Af(X_s)ds$$

is a martingale for all $f \in \text{domain}\ (A)$. If the solution is unique, then it is a Markov process.

Using this fact, and when the state space of the Markov process is a locally compact metric space, a sufficient condition for an invariant measure for the Markov process was proved in Echeverria (1982). This was extended to the situation when the state space is allowed to be a complete, separable metric space (see, Bhatt & Karandikar, 1993). It was shown that if a probability measure μ on the state space satisfies

$$\int Af d\mu = 0 \qquad \forall\ f \in \text{domain}(A)$$

then μ is a stationary distribution for the Markov process. One of the key steps in the proof was an *imbedding of the martingale problem into a separable compact*

space. The additional condition to be imposed was that the martingale problem is well-posed in the class of all measurable processes and not just in the class of all right continuous processes.

While the earlier results were applicable to operators A whose domain and range were subsets of bounded continuous functions, in later works (for example Bhatt et al., 2006) it was extended when the range was allowed to include discontinuous and unbounded functions. In particular, the results could be applied to diffusion processes with discontinuous and unbounded coefficients.

These results were further helpful in two specific directions.

Applications to Filtering in the Finitely Additive Case

It was shown that there is a one-to-one correspondence between solutions of the martingale problem for an operator A and solutions of the corresponding Fokker–Planck equations. This in turn extended the results of Kallianpur and Karandikar on the uniqueness of solution of measure valued equations of filtering (in the finitely additive set-up) when the state space is a complete, separable metric space and the range of the operator can include unbounded functions. (See Bhatt & Karandikar, 1995).

Applications to Classical Nonlinear Filtering

An interesting off-shoot of these circles of ideas was to use these results to show the uniqueness of solution to the measure valued equations in classical nonlinear filtering theory. Here the following observation model is considered.

$$Y_t = \int_0^t h(X_s)ds + W_t$$

where W is a Brownian motion. This can be thought of as the accumulated observation model unlike the *instantaneous* model in the finitely additive set-up. The equations of filtering in this classical set-up, namely, the Fujisaki–Kallianpur–Kunita and the Zakai equations were derived in the 1970s. In Bhatt et al. (1995), the uniqueness of solutions of these equations under fairly general conditions was proved.

Later these results were extended to include *the signal-noise correlated case* and also the *unbounded and discontinuous coefficients case*.

In a particular situation, when *the signal is non-Markovian but arises as a unique solution of a stochastic differential equation*, the uniqueness of the filtering equations was shown in Bhatt and Karandikar (1999). This was done by looking at the corresponding *historical process of the signal* which is a Markov process taking values in the path space, a complete, separable metric space case. Well-posedness of the *corresponding martingale operator* was shown so that the earlier results could be applied.

5 Psephology

During the late 1990s, Karandikar got interested in opinion polls and *election predictions*. Along with a team of a few young colleagues from the ISI and with collaborators from the *Center for Studies of Developing Societies* (CSDS), Delhi, he developed the *early seat prediction system* which was aired on national television during the live counting of votes of the Indian general elections of 1998. Since then for over 2 decades he had been a regular figure on Indian television channels at the time of state and general elections.

The Indian political system is quite complex in the sense that it boasts of hundreds of different political parties fighting the elections. The political equations also change from one state to another and sometimes from one region to another in the same state. Two parties may come together as one group in one state while they may be opponents in another or for the parliamentary elections. Moreover, these equations are quite fluid and keep changing from one election to another.

One of the important contributions of *psephologist* Karandikar was to develop a model which addresses these complexities and one which is implementable.

Votes to Seat Conversion

A natural facet of this complex political arena of Indian elections is that predicting seats for a political group becomes even more complicated. Based on voter preferences from a random sample of a reasonable size and using standard classical sampling theory, it would be relatively easy to predict the vote shares for a political party or group. However, the audience for the opinion polls are more interested in the prediction of seats rather than in the vote share. RLK's ''votes to seat" conversion formula, based on simple probabilistic and statistical methods, is unique and has been highly successful. A comprehensive report can be found in Karandikar (2020).

6 Other Excursions

While the preceding sections give a glimpse of the main research interest of Karandikar, it is to be noted that he has delved into many different research areas. It would be unwieldy to list all of them here.

RLK's interest in stochastic calculus has naturally led him to stochastic finance. His second book with Kallianpur (2000) works as a good text for someone who would like to work in the area.

Among other topics, he has worked on Quantum Itô Formula (with R. L. Hudson and K. R. Parthasarathy), Feynman Integrals & Cameron-Martin formula (with G. Kallianpur and D. Kannan), and on Boltzmann equations (with J. Horowitz). He has published on Queueing Theory (with V. G. Kulkarni) and on Extreme Value Theory (with L. de Haan).

He has collaborated with J. Horowitz and with M. Vidyasagar on Rates of convergence of empirical means and with N. R. Chaganty on Bootstrapped empirical probability measures. Recently, he has also looked at Learning Theory along with M. Vidyasagar.

He has published papers on Economic Game Theory (jointly with D. Mookherjee, D. Ray, and V-R Fernando) and on linear algebra with R. Bhatia. During the recent pandemic, he has collaborated with S. Mande and B. Chatterjee on some results of the COVID-19 pandemic.

In collaboration with S. Bhogle, he has developed a machine learning-based algorithm for predicting the win percentage of a team in limited overs cricket. This algorithm can also be tweaked to act as a replacement for the Duckworth–Lewis–Stern method for calculating target scores in rain-affected matches.

Apart from his work on elections, he has used his academic mastery to apply it to industry. He has developed a block cipher (Cryptology) for the Indian defence establishment and propounded a gene prediction algorithm (Bioinformatics). He has been consulted by the *Central Bureau of Investigation* of India in crime detection and has advised the *Election Commission of India* regarding sampling methodology for verification of the EVM-VVPAT systems.

This breadth of topics not only emphasizes his versatility in all topics of mathematics but also his clarity of thought and enthusiasm for the subject.

On a personal level, I have benefitted immensely by having him as a teacher, a guide, then a colleague and a friend.

Thank you RLK.

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Spectrum of High-Dimensional Sample Covariance and Related Matrices: A Selective Review



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1 Introduction

Suppose X is a $p \times n$ matrix of mean zero real-valued random variables, where the n columns of X are independent. The *unadjusted sample covariance matrix* $S = n^{-1}XX^*$ was introduced by Wishart (1928) and is one of the most important examples of a random matrix in statistics. In particular, if the n columns are Gaussian with a positive definite covariance matrix Σ_p, then S is called the *Wishart matrix* and the joint distribution of its eigenvalues can be written down explicitly. Many methods in multivariate statistics are based on this matrix (recall Hotelling's-T) and its eigenvalues and eigenvectors (recall principal component analysis).

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Rajeeva L. Karandikar has been a senior peer and a dear friend since 1976. It is a privilege and a pleasure to contribute this humble article in celebration of his distinguished career, and with fond remembrance of our frequent "ponderings" around the water bodies of the Indian Statistical Institute during late '70s and early '80's. It is then that I gained my first lucid exposure to many ideas from advanced probability theory.

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The behavior of these statistics change drastically if p is large. One of the reasons is that $n^{-1}XX^*$ is not a "good" estimate of Σ_p as p becomes large. Quite naturally, the problem of estimation of Σ_p as $p \to \infty$ has attracted several researchers. However this article does not address the issue of estimation of Σ_p, and one may refer to Bickel and Levina (2008a, b), Rothman et al. (2008, 2009), Cai et al. (2010, 2013b) and Cai and Liu (2011) for information on this issue.

Instead, we focus on the behavior of S through the properties of its eigenvalues, as $p, n \to \infty$. There are primarily two ways to study the behavior of random matrices whose dimensions grow. The first is to consider the *empirical spectral distribution* (ESD), which puts equal mass at all the eigenvalues, and study its weak limit (LSD). One may also study other features of the ESD, for example its "extreme values". In Sect. 2 we introduce the concepts of ESD and LSD. We state a general result which forms the basis of the trace-moment method that has been used in the literature to establish the convergence of the ESD for many symmetric random matrix sequences. The above result provides the readers with the general template for LSD proofs, even though in most cases the details are not easy. Incidentally, all proofs are omitted from this review. We also introduce an important summary of the ESD, namely the *linear spectral statistics*, a notion that has found frequent uses in high-dimensional statistics.

Since matrices are non-commutative objects, collections of random matrices cannot be treated as usual random variables. It is crucial to have a notion of *non-commutative probability*. In Sect. 3, the basic notions of this theory, including those of *moments*, *free cumulants* and *free independence*, as we need them, are introduced by an analogy with the concepts of moments, cumulants and independence in classical probability. We also define a notion of joint convergence, which is usually referred to as *algebraic convergence*. This theory offers a second way of studying random matrices, and the two approaches are related to each other.

In the context of the sample covariance matrix, the fundamental probability laws and non-commutative variables that arise are the *semi-circular law/variable*, the *Marčenko–Pastur law/variable* and the *Tracy–Widom law*. The Marčenko–Pastur law/variable is a special case of the compound free Poisson law/variable. Likewise, for the *cross-covariance matrices*, the *cross-covariance variable* and the *elliptic variable* become important. All these laws and variables that we shall need are introduced in Sect. 4. In Sect. 5 we collect together the high-dimensional matrices that we shall discuss. These include the *Wigner*, the S, the *Independent matrix*, the *separable, generalized, and cross-covariance matrices*, *Kendall's* τ and *Spearman's* ρ *matrices* and the *autocovariance matrices* in one and high dimensions.

In Sect. 6 we describe all the ESD and algebraic convergence results for the S and related matrices. Section 6.1 states one of the fundamental results in random matrix theory on the convergence of the ESD and the joint convergence of independent *Wigner matrices*. This serves as an important tool while analyzing other random matrices.

In Sect. 6.2, we describe results on the convergence of the ESD and the joint convergence of independent S matrices. If Σ_p is the identity matrix and $p \to \infty$, $p/n \to y \neq 0$, then the ESD of S converges (weakly) almost surely to the Marčenko–Pastur

law. On the other hand, if $p \to \infty$, $p/n \to 0$, then the ESD of $\sqrt{\frac{n}{p}}(S - I_p)$, where I_p is the identity matrix of order p, converges (weakly) almost surely to the semi-circular law. In the above two situations, if we have several independent S matrices, then they algebraically converge to freely independent Marčenko–Pastur variables or freely independent semi-circular variables according as $y \neq 0$ or $y = 0$. It then follows that the ESD of any symmetric matrix polynomial in these matrices also converges weakly, and the LSD is identified by the moments of the corresponding self-adjoint polynomial in these freely independent variables.

In S, the (i, j)th entry is the sample covariance between the ith and the jth coordinates of the n column vectors of X. Other measures of dependence could be used, especially when the observations are heavy tailed. Two common measures are Kendall's τ and Spearman's ρ. The ESD of the high-dimensional *Kendall's τ and Spearman's ρ matrices* also converge to suitably adjusted Marčenko–Pastur laws. This is discussed in Sect. 6.3.

One way to model dependence among rows and columns of X is to model it as $X = A^{1/2}ZB^{1/2}$ where A and B are both positive definite matrices of orders p and n respectively and Z is a $p \times n$ *Independent matrix* with entries that are all independent. The matrix $\tilde{S} = \frac{1}{n}XX^* = \frac{1}{n}A^{1/2}ZBZ^*A^{1/2}$ is called a *separable sample covariance matrix*. Properties of this matrix are discussed in Sect. 6.4.

Suppose we have two $p \times n$ matrices X and Y. Then $C = n^{-1}XY^*$ is called a *sample cross-covariance matrix*. These matrices are discussed in Sect. 6.5. If $Y = X$, then C reduces to an S matrix, but in general this is a non-symmetric matrix and has complex eigenvalues. Under a *pair-correlation* condition, here too the joint convergence holds in the two cases $y \neq 0$ and $y = 0$. The limit variables are free cross-covariance variables, and elliptic variables, respectively. Since C is a non-symmetric matrix, it is hard to establish its LSD. Akemann et al. (2021) showed that when the entries are Gaussian, the LSD is the uniform distribution on the interior of an ellipse, the length of whose axes depend on the value of the correlation.

Now suppose that $\{X_i\}$ is a real-valued mean zero stationary time series with the autocovariance sequence $\{\gamma(i)\}$. Then the covariance matrix of the vector $(X_0, \ldots X_{p-1})$ is given by $\Sigma_p = ((\gamma(i-j)_{1\leq i,j\leq p}))$. The *sample autocovariance matrix* $\hat{\Gamma}_p$ is obtained by replacing the entries of Σ_p by their moment estimators. In Sect. 6.6 we discuss the properties of $\hat{\Gamma}_p$ as $p \to \infty$. In particular, $\hat{\Gamma}_p$ is not "consistent" for Σ_p in the sense that their LSDs are not the same, and $\hat{\Gamma}_p$ can be modified to achieve consistency.

High-dimensional time series models, and the natural random matrices that arise in the study of these models, necessitate the extension of the joint convergence results for independent sample covariance matrices, especially by injecting *deterministic matrices* in the mix.

In Sect. 6.7 we deal with matrix polynomials whose basic building blocks are of the form $AZBZ^*A^*$ where A and B are suitable deterministic square matrices of orders p and n respectively, and Z is a $p \times n$ Independent matrix. We state

convergence results for these. We reap the benefits of these results when we consider p-dimensional time series models where p also grows with the sample size.

Let $X_{t,p}^{(n)}$ be a p dimensional mean zero stationary time series, $t = 1, \ldots, n$ where $n = n(p) \to \infty$ as $p \to \infty$. Such models are known as infinite-dimensional time series models. For simplicity we write $X_{t,p}$ for $X_{t,p}^{(n)}$. The uth order autocovariance is now the $p \times p$ matrix $\Gamma_{u,p} = \mathrm{E}(X_{t,p} X_{t,p}^*)$. Its moment estimate is the *sample autocovariance matrix* of order u, $\hat{\Gamma}_{u,p} = \frac{1}{n} \sum_{t=1}^{n-u} X_{t,p} X_{t+u,p}^*$. Except for the case $u = 0$, these matrices are non-symmetric. To study their asymptotic properties, suppose that $X_{t,p}$ is a stationary linear time series with an i.i.d. driving noise. Then, as in the case for $p = 1$, $\hat{\Gamma}_{0,p}$ is not consistent for $\Gamma_{0,p}$ but appropriately banded and tapered $\hat{\Gamma}_{0,p,n}$ are consistent. In Sect. 6.8 we show how the joint algebraic convergence of $\{\hat{\Gamma}_{u,p}\}$ can be established by using the results of Sect. 6.7. In particular, the ESD of any symmetric matrix polynomial in the sample autocovariance matrices, even those including more than one independent time series, converges. Incidentally, the convergence of the ESD for a non-symmetric polynomial has been shown only in a couple of very special cases in Bose and Hachem (2020, 2022). However, non-symmetric matrices are outside the purview of our discussion.

The *linear spectral statistics* (LSS) is an important summary of the ESD. Suppose f is a function. Then the corresponding LSS is nothing but the expectation of this function with respect to the ESD. Thus, if $\{\lambda_i\}$ are the eigenvalues of a $p \times p$ random matrix then the LSS equals $p^{-1} \sum f(\lambda_i)$. Testing and other types of inference in high-dimensional statistics is often performed with the help of suitable LSS. Note that the widely used *trace statistic* is a particular LSS. In Sect. 7 we state some joint asymptotic normality of the LSS over a suitable class of functions f, for S and some related matrices.

The study of the extreme eigenvalues of a high-dimensional random matrix is significantly harder than the study of its ESD and far fewer results are known. In Sect. 8 we state some results on the asymptotic distribution of the maximum eigenvalues of the S and some related matrices. The Tracy–Widom laws appear as the limit distribution.

A high-dimensional covariance matrix is said to be *spiked* if there are a few eigenvalues which are much larger than the rest of the eigenvalues. Such situations arise in signal processing and factor models. In Sect. 9, we discuss some results on the behavior of the eigenvalues of S in spiked models.

In Sect. 10 we discuss some methods of inference in high-dimensional models. These methods are all based on results for high-dimensional sample covariance matrices, including some that we have not discussed in the other sections.

Suppose the columns of the data matrix are independent and we wish to test for the independence across the rows. In Sect. 10.1 we discuss several tests. These include tests based on the maximum eigenvalue, the trace of second and fourth powers, log-determinant and the largest off-diagonal entries of S.

In Sect. 10.2 we discuss methods to estimate the number of spikes in a consistent manner. This is based on a confidence interval constructed using a Tracy–Widom

law for the largest non-spiked sample eigenvalue, and checking how many of the sample spiked eigenvalues are beyond this interval.

In Sect. 10.3 we consider two high-dimensional time series models and discuss testing the equality of some specific coefficient matrices. In Sect. 10.4 we present a simple graphical method which estimates the order of a high-dimensional time series.

2 Spectral Distribution and Linear Spectral Statistic

Suppose R_p is any $p \times p$ matrix. Its *empirical spectral measure* is the probability law which puts mass p^{-1} on the eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_p$. The corresponding distribution function is called the *empirical spectral distribution function*. We shall refer to either of them as ESD. For most of this article, the eigenvalues are all real, and in that case the ESD is given by

$$F_{R_p}(x) = p^{-1} \sum_{i=1}^{p} \mathbb{I}\{\lambda_i \le x\}, x \in \mathbb{R}.$$

If R_p is a random matrix, then the ESD is a random probability law. By taking an expectation, we can then define the *expected empirical spectral distribution function* (EESD) as

$$\mathrm{E}(F_{R_p}(x, y)) = \frac{1}{p} \sum_{i=1}^{p} \mathrm{P}\big[\mathcal{R}(\lambda_i) \le x,\ \mathcal{I}(\lambda_i) \le y\big]$$

where $\mathcal{R}(z)$ and $\mathcal{I}(z)$ represent the real and the imaginary part of a complex number z, respectively. We shall write $\mathrm{E}(F_{R_p})$ in brief. The corresponding probability law is called the *expected empirical spectral measure*.

Our interest is in the case where $p \to \infty$. The notions of convergence for the empirical distribution ESD/EESD in increasing order of strength are weak convergence of $\{\mathrm{E}(F_{R_p})\}$ and, weak convergence of $\{F_{R_p}\}$ in probability and almost surely. In all cases, we refer to the limit as the *limiting spectral distribution* (LSD) of $\{R_p\}$ and the corresponding probability law as the limiting spectral measure.

The study of the LSD of non-Hermitian matrices is extremely difficult. In contrast, real symmetric or Hermitian random matrices are relatively easier to handle, primarily due to the trace-moment formula and the lemma stated below. Note that in this case all the eigenvalues are real and the ESD and EESD are probability distributions on the real line. We will use the notation $m_h(\mu)$ and $m_h(F)$ to denote the hth moment of the probability measure μ and the distribution F, respectively. The trace and the average trace of a matrix A will be denoted respectively by $\mathrm{Tr}(A)$ and $\mathrm{tr}(A)$.

When R_p is real symmetric, the hth moment of F_{R_p} is given by

$$m_h(F_{R_p}) = \frac{1}{p}\sum_{i=1}^{p}\lambda_i^h = \frac{1}{p}\operatorname{Tr}(R_p^h) = \operatorname{tr}(R_p^h) = m_h(R_p), \quad \text{say.} \tag{2.1}$$

This is the *trace-moment* formula. It implies that the moments $m_h(\mathrm{E}(F_{R_p}))$ of the EESD are given by

$$m_h(\mathrm{E}(F_{R_p})) = \mathrm{E}\left[\frac{1}{p}\operatorname{Tr}(R_p^h)\right] = \mathrm{E}[m_h(R_p)].$$

Then the following lemma holds. For proof, see Bose (2018).

Lemma 2.1 *Suppose $\{R_p\}$ are real symmetric matrices. Consider the following conditions:*

(C1) For every $h \geq 1$, $\lim_{p\to\infty} E[m_h(R_p)] = m_h$ exists.

(C2) The moment sequence $\{m_h\}$ corresponds to a unique probability law, say F.

(C3) For every $h \geq 1$, $\lim_{p\to\infty} Var[m_h(R_p)] = 0$.

(C4) For every $h \geq 1$, $\sum_{p=1}^{\infty} E[m_h(R_p) - E(m_h(R_p))]^4 < \infty$.

Then we have the following:

(a) If (C1) and (C2) hold, then the EESD of R_p converges weakly to F.

(b) If in addition (C3) holds, then $\{F_{R_p}\}$ converges weakly in probability to F.

(c) If in addition (C4) holds, then $\{F_{R_p}\}$ converges weakly almost surely to F.

Suppose R_p is a (symmetric) random matrix. Now take a real-valued function f and consider its expectation with respect to the ESD, that is $p^{-1}\sum_{i=1}^{p} f(\lambda_i)$. This is called a *linear spectral statistics* (LSS) and is a popular class of summary statistics. For $f(x) = x^h$, the LSS is the hth moment of the ESD and, if we vary h, then these moments identify the ESD uniquely. Later we shall see several limit theorems for general spectral statistics in different models.

Another important way to conclude the convergence of the ESD/EESD for real symmetric matrices is to use the *Stieltjes transform*. For any real-valued random variable V, its Stieltjes transform is defined as

$$m(z) = \mathrm{E}\big(V - z\big)^{-1}, \quad \text{usually defined for all complex } z \text{ with } \mathcal{I}(z) > 0.$$

Note that the Stieltjes transform of the EESD of R_p is related to the resolvent and is given by

$$p^{-1}\mathrm{E}\left(\sum_{i=1}^{p}(\lambda_i - z)^{-1}\right) = \mathrm{E}\operatorname{tr}\big(A - zI_p\big)^{-1}.$$

A sequence of EESD converges if their Stieltjes transforms converge point-wise to a Stieltjes transform. Many, but not all, of the results discussed later can be proved by using this idea.

3 Non-commutative Probability

Matrices are non-commutative objects and so it is crucial to have a notion of non-commutative probability. We quickly review some basic notions from non-commutative probability theory. These will be vital in understanding the asymptotic behavior of high-dimensional random matrices. We develop these notions by analogy with classical probability. For further details, see Bose (2021), Mingo and Speicher (2017) and Nica and Speicher (2006).

In contrast to the basic notion of a probability space in classical probability, a primary notion in non-commutative probability is a $*$-*probability space*. This consists of a pair $(\mathcal{A}, \varphi)$, where $\mathcal{A}$ is a $*$-*algebra* with the identity $\mathbf{1}$, and φ (called a *state*) is a complex-valued linear functional on $\mathcal{A}$ such that $\varphi(\mathbf{1}) = 1$. The analogue of classical random variables are the *non-commutative variables/elements* of the algebra. The functional φ can be thought of as the analogue of the expectation operator in classical probability. We also demand that the φ is a positive state, that is, $\varphi(a^*a) \geq 0$ for all $a \in \mathcal{A}$.

In the context of random matrices, we shall encounter the following $*$-probability space.

Example 3.1 For any positive integer d, let

$$\mathcal{M}_d(\mathbb{C}) = \text{ set of all } \ d \times d \ \text{ matrices whose entries are complex-valued random variables,}$$

$$\text{tr}(a) = \frac{1}{d}\,\text{Tr}(a) = \frac{1}{d}\sum_{i=1}^{d}\alpha_{ii} \quad \text{for all } \ a = ((\alpha_{ij}))_{i,j=1}^{d} \in \mathcal{M}_d(\mathbb{C}),$$

$$\text{Etr}(a) = \frac{1}{d}\sum_{i=1}^{d}\text{E}(\alpha_{ii}) = d^{-1}\text{E}\,\text{Tr}(a).$$

Then $(\mathcal{M}_d(\mathbb{C}), \text{Etr})$ is a $*$-probability space. If we restrict the entries to be constants, then $(\mathcal{M}_d(\mathbb{C}), \text{tr})$ is also a $*$-probability space.

For a $*$-probability space $(\mathcal{A}, \varphi)$, the *moments* of a set of variables $\{a_i : i \in I\} \subset \mathcal{A}$ are the numbers $\{\varphi\left(\Pi(a_i, a_i^* : i \in I)\right) : \Pi \text{ is a finite degree monomial}\}$.

Suppose $a \in \mathcal{A}$ is a *self-adjoint* element, that is, $a^* = a$. Then μ_a is called the *probability law of* a if it is the *unique* law on $\mathbb{R}$ such that

$$\varphi(a^n) = \int_{\mathbb{R}} x^n \mu_a(dx), n = 1, 2, \ldots. \tag{3.1}$$

If a probability measure μ_a satisfies (3.1) and is compactly supported, then it is indeed unique. Most of the situations that we encounter later will fall under this category.

Recall the classical notions of moments and cumulants. When the *moment generating function* of a random vector exists in a neighborhood of zero, its logarithm is the *cumulant generating function* and the coefficients in its Taylor series expansion are the *cumulants*. Let $\mathcal{P}(n)$ be the set of all partitions of $\mathbb{Z}_n = \{1, 2, \ldots, n\}, n \geq 2$. Then $\mathcal{P}(n)$ is a finite *lattice* with the reverse refinement partial ordering and hence is equipped with a *Möbius function*, defined for every pair $(\pi,\ \sigma), \pi \leq \sigma, \pi, \sigma \in \mathcal{P}(n), n \geq 2$. The one–one relationship between moments and cumulants can be expressed through this Möbius function. See (3.5) and the subsequent discussion.

Since in a $*$-probability space, a moment depends on the order in which the variables appear in the monomial, the notion of cumulants is non-existent. This notion is replaced by the notion of *free cumulants*, which are connected in a one–one way to moments via the Möbius function on the set of *non-crossing partitions*. We describe this below briefly.

Any partition π of $\mathbb{Z}_n$ is called *crossing* if there exists $p_1 < q_1 < p_2 < q_2$ in $\mathbb{Z}_n$ so that the pairs $\{p_1, p_2\}$ and $\{q_1, q_2\}$ are in two different blocks. Otherwise, π is called *non-crossing*. For instance $\pi_1 = \{\{1, 4\}, \{2, 3\}\}, \{5, 6\}\}$ and $\pi_2 = \{\{1, 3, 5\}, \{2, 6\}, \{4\}, \}$ are respectively non-crossing and crossing partitions of $\mathbb{Z}_6$. Let $NC(n)$ be the set of all non-crossing partitions of $\mathbb{Z}_n$. It is a finite lattice and hence is equipped with a Möbius function, say μ.

Let $(\mathcal{A}, \varphi)$ be a $*$-probability space. Define *multi-linear functionals* $(\varphi_n)_{n\in\mathbb{N}}$ on $\mathcal{A}^n$ via

$$\varphi_n(a_1, a_2, \ldots, a_n) = \varphi(a_1 a_2 \cdots a_n). \tag{3.2}$$

The functions $\{\varphi_n, n \geq 1\}$ are *multiplicatively* extended to $\{\varphi_\pi, \pi \in NC(n), n \geq 1\}$ recursively as follows: if $\pi = \{V_1, V_2, \ldots, V_r\} \in NC(n)$ (V_i are the blocks of π) then

$$\varphi_\pi[a_1, a_2, \ldots, a_n] = \varphi(V_1)[a_1, a_2, \ldots, a_n] \cdots \varphi(V_r)[a_1, a_2, \ldots, a_n], \tag{3.3}$$

where $\varphi(V)[a_1, a_2, \ldots, a_n] = \varphi_s(a_{i_1}, a_{i_2}, \ldots, a_{i_s}) = \varphi(a_{i_1} a_{i_2} \cdots a_{i_s})$ if $V = \{i_1, i_2, \ldots, i_s\}$ with $i_1 < i_2 < \cdots < i_s$. Note that the order has been preserved and the two types of braces ( ) and [ ] in (3.2) and (3.3) have different uses. If $\mathbf{1}_n$ denotes the single-block partition of $\mathbb{Z}_n$, then

$$\varphi_{\mathbf{1}_n}[a_1, a_2, \ldots, a_n] = \varphi_n(a_1, a_2, \ldots, a_n) = \varphi(a_1 a_2 \cdots a_n). \tag{3.4}$$

The *joint free cumulant* of order n of $(a_1, a_2, \ldots, a_n)$ is defined as

$$\kappa_n(a_1, a_2, \ldots, a_n) = \sum_{\sigma \in NC(n)} \varphi_\sigma[a_1, a_2, \ldots, a_n] \mu(\sigma, \mathbf{1}_n). \tag{3.5}$$

It is called a *mixed free cumulant* if at least *one* pair a_i, a_j is different and $a_i \neq a_j^*$ for some $i \neq j$. For any $\epsilon_i \in \{1, *\}$, $1 \leq i \leq n$, $\kappa_n(a^{\epsilon_1}, a^{\epsilon_2}, \ldots, a^{\epsilon_n})$ is called a *marginal free cumulant* of order n of $\{a, a^*\}$. We point out that the relation between the classical cumulants and classical moments of random variables is also given by (3.5) once we replace $NC(n)$ by $\mathcal{P}_n$, take μ to be the Möbius function of $\mathcal{P}(n)$, $n \geq 1$, and the multiplicative extension is defined via all partitions.

The nth free cumulant of a self-adjoint element a is given by

$$\kappa_n(a) = \kappa_n(a, a, \ldots, a).$$

Let $\{\kappa_\pi : \pi \in NC(n), n \geq 1\}$ be the multiplicative extension of $\{\kappa_n, n \geq 1\}$. By using (3.5) and the Möbius function μ, the following relations can be shown:

$$\kappa_\pi[a_1, a_2, \ldots, a_n] = \sum_{\sigma \in NC(n):\ \sigma \leq \pi} \varphi_\sigma[a_1, a_2, \ldots, a_n]\mu(\sigma, \pi), \pi \in NC(n),\ n \geq 1 \quad (3.6)$$

$$\begin{aligned}
\kappa_{\mathbf{1}_n}[a_1, a_2, \ldots, a_n] &= \kappa_n(a_1, a_2, \ldots, a_n), \\
\kappa_2(a_1, a_2) &= \varphi(a_1 a_2) - \varphi(a_1)\varphi(a_2), \ \text{(called the covariance)} \\
\varphi(a_1 a_2 \cdots a_n) &= \sum_{\sigma \in NC(n):\ \sigma \leq \mathbf{1}_n} \kappa_\sigma[a_1, a_2, \ldots, a_n], \qquad (3.7) \\
\varphi_\pi[a_1, a_2, \ldots, a_n] &= \sum_{\sigma \in NC(n):\ \sigma \leq \pi} \kappa_\sigma[a_1, a_2, \ldots, a_n].
\end{aligned}$$

Eqs. (3.5) and (3.7) provide a one–one relation between free cumulants and moments and will be referred to as *moment free cumulant relations*. Again, the above relations also hold for classical moments and cumulants once we replace $NC(n)$ by $\mathcal{P}_n$, take μ to be the Möbius function of $\mathcal{P}(n)$, $n \geq 1$, and the multiplicative extension is defined by using all partitions.

In classical probability, when the moment generating function exists in a neighborhood of zero, independence is equivalent to vanishing of all the mixed cumulants. By analogy, in non-commutative probability, we can define variables $\{a_i : i \in I\}$ to be *freely independent* (or free in short) if all their (adjoints included) mixed free cumulants vanish. Among several notions of independence available in the non-commutative set up, free independence turns out to be the most useful one, in particular in the context of high-dimensional random matrices. Analogous to the classical case, the existence of free variables is guaranteed by a free product construction of non-commutative probability spaces.

Suppose a and b are two free self-adjoint variables with probability laws μ_a and μ_b and suppose $a = x^2$ for some variable x. Then the *free product convolution* of μ_a and μ_b is defined as the probability law of xbx.

Finally, we need a notion of *convergence*. It is now most natural to define it via convergence of moments. Let $(\mathcal{A}_n, \varphi_n)$, $n \geq 1$ and $(\mathcal{A}, \varphi)$ be $*$-probability spaces. We say that $a^{(n)} \in \mathcal{A}_n$ *converges in $*$-distribution* (write $a^{(n)} \to a$) to $a \in \mathcal{A}$ if

$$\lim \varphi_n(\Pi(a^{(n)}, a^{(n)*})) = \varphi(\Pi(a, a^*)) \ \text{ for all polynomials } \ \Pi. \quad (3.8)$$

This can be extended to joint convergence in a natural way. For an index set I, $\{a_i^{(n)} : i \in I\}$ from $\mathcal{A}_n$ are said to *converge (jointly)* to $\{a_i : i \in I\}$ from $\mathcal{A}$ (we write $\{a_i^{(n)} : i \in I\} \to \{a_i : i \in I\}$) if

$$\Pi(\{a_i^{(n)}, a_i^{*(n)} : i \in I\}) \xrightarrow{*} \Pi(\{a_i, a_i^* : i \in I\}) \text{ for all } \Pi. \tag{3.9}$$

If the limit variables $\{a_i, i \in I\}$ are free then we say that $\{a_i^{(n)}, i \in I\}$ are free in the limit, or are *asymptotically free*. Later we shall encounter plenty of examples of asymptotically free variables that arise from the consideration of high-dimensional random matrices.

4 Useful Probability Laws and Non-commutative Variables

We introduce a few probability laws and non-commutative variables that will be crucial in the study of large dimensional sample covariance matrices.

A self-adjoint variable v_0 is a *Bernoulli variable* with parameter α if $\varphi(v_0^r) = \alpha$ for all positive integers r. Note that then $v_1 = \mathbf{1} - v_0$ is Bernoulli with parameter $1 - \alpha$ and $v_0 v_1 = 0$.

The *standard semi-circular probability law* has probability density

$$f(x) = \begin{cases} \frac{1}{2\pi}\sqrt{4 - x^2} & \text{if } -2 < x < 2, \\ 0 & \text{otherwise.} \end{cases} \tag{4.1}$$

If a random variable X obeys the (standard) semi-circular law, then, for any $\sigma > 0$, the law of σX is also called the semi-circular law, with variance σ^2. This law occupies a central position in non-commutative probability and is also known as the *free Gaussian law*. The moment sequence of the standard semi-circular law is given by

$$m_h = \begin{cases} \frac{1}{n+1}\binom{2n}{n} & \text{if } h = 2n, \\ 0 & \text{if } h \text{ is odd.} \end{cases} \tag{4.2}$$

A self-adjoint variable s in a $*$-probability space is a (standard) *semi-circular variable* if its moment sequence is as in (4.2). This s need not be a classical random variable. Its free cumulants have the same pattern as the classical cumulants of a standard Gaussian variable:

$$\kappa_h(s) = \begin{cases} 1 & \text{if } h = 2, \\ 0 & \text{otherwise.} \end{cases} \tag{4.3}$$

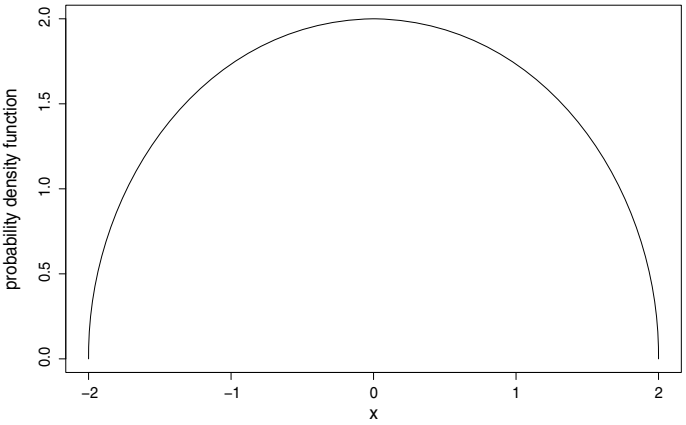


Fig. 1 Density of the standard semi-circular law

The number $C_n = \frac{1}{n+1}\binom{2n}{n}$ is the nth Catalan number. It also equals the cardinalities of $NC(n)$, and of the set of non-crossing pair-partitions of $\mathbb{Z}_{2n}$ (Fig. 1).

The analogue of the multivariate Gaussian family in non-commutative probability is the *semi-circular family*. Self-adjoint variables $\{s_i, 1 \leq i \leq n\}$ in an NCP $(\mathcal{A}, \varphi)$ are said to form a *semi-circular family* if every s_i is a semi-circular variable (perhaps with different variances) and all mixed free cumulants of order greater than two vanish. The matrix $((\kappa_2(s_i, s_j)))$ is positive semi-definite. If in addition this matrix is diagonal, then $\{s_i\}$ are free.

Example 4.1 It is not always easy to calculate any joint moment, even when the variables are free. However, since third and higher order free cumulants of a semi-circular variable vanish, calculation of moments are relatively easier when semi-circular variables are involved. Here is one example which will be useful to us later.

Suppose $s_\tau, 1 \leq \tau \leq m$, are free semi-circular variables. Suppose v_0 is a Bernoulli variable with parameter α, and which is free of $\{s_\tau\}$. Let $v_1 = 1 - v_0$. Then it can be shown that

$$\varphi\left(\prod_{l=1}^{k} v_0 s_{\tau_l} v_1 s_{\tau_l} v_0\right) = \sum_{\pi \in B_k} \alpha^{k+1-|\pi|}(1-\alpha)^{|\pi|},$$

where B_k is the set of all non-crossing partitions $\pi = \cup_{i=1}^{m} \pi_i$ of $\{1, 2, \ldots, k\}$ such that each π_i is a non-crossing partition of $\{l \in \{1, 2, \ldots, k\} : \ \tau_l = i\}$ and $|\pi|$ is the number of blocks in π. In particular, if v_0 is free of a single semi-circular s, then

$$\varphi\left((v_0 s v_1 s v_0)^k\right) = \sum_{\pi \in NC(k)} \alpha^{k+1-|\pi|}(1-\alpha)^{|\pi|}. \tag{4.4}$$

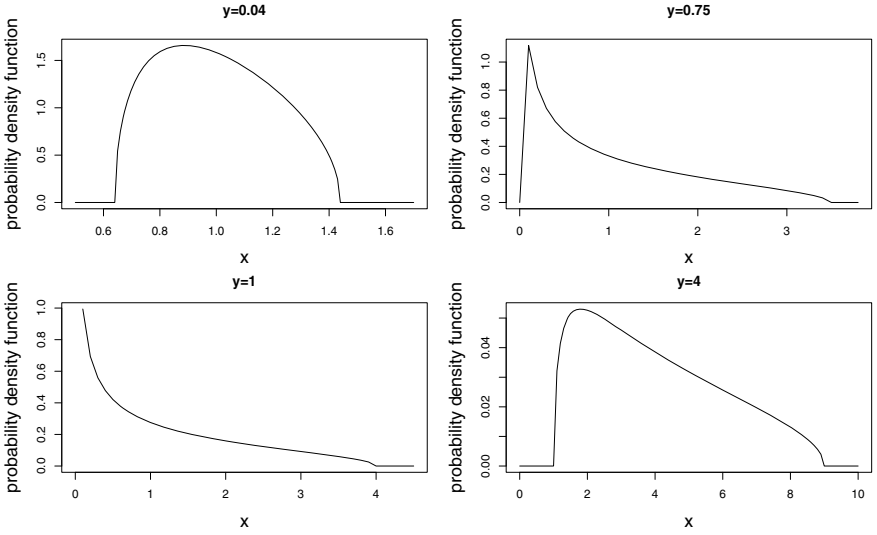


Fig. 2 Density of the Marčenko–Pastur law. The mass at 0 for $y > 1$ is not shown

A self-adjoint variable a is said to be *compound free Poisson* with rate λ and *jump distribution* F, if its free cumulants are given by

$$\kappa_n(a) = \lambda \mathrm{E}(Y^n), \quad \text{for all } \ n \geq 1, \tag{4.5}$$

where Y is a random variable with distribution F. Incidentally, if X is a compound Poisson random variable, then its *cumulants* are as given in the right side of (4.5).

The *Marčenko–Pastur* law is parametrized by $y > 0$ and arises as the LSD of S. It has a positive mass $1 - \frac{1}{y}$ at 0 if $y > 1$. Elsewhere it has the density

$$f_y(x) = \begin{cases} \frac{1}{2\pi xy}\sqrt{(b-x)(x-a)} & \text{if } a \leq x \leq b, \\ 0 & \text{otherwise,} \end{cases} \tag{4.6}$$

where

$$a = a(y) = (1 - \sqrt{y})^2 \ \text{ and } \ b = b(y) = (1 + \sqrt{y})^2.$$

We shall denote this probability law by MP_y (Fig. 2).

Since the law is compactly supported, its moments determine the law uniquely. It is easily checked that these are given by

$$m_k(\mathrm{MP}_y) = \sum_{t=0}^{k-1} \frac{1}{t+1} \binom{k}{t} \binom{k-1}{t} y^t, \ k \geq 1. \tag{4.7}$$

A self-adjoint variable mp_y in a $*$-probability space will be called a *Marčenko–Pastur variable*, if its moments are as in (4.7). These moments yield the free cumulants as

$$\kappa_n(\mathrm{mp}_y) = y^{n-1}. \tag{4.8}$$

Thus, mp_y is a compound free Poisson with rate y^{-1} and jump distribution F with full mass at y. By abuse of notation, we shall use mp_y later also to denote a random variable with the MP_y distribution.

The semi-circular and the Marčenko–Pastur laws/variables are related. Suppose s is a semi-circular variable. Then s^2 is a free Poisson variable with rate 1, that is, it is an mp_1 variable. In general, suppose s is a semi-circular variable, v_0 is a Bernoulli variable with parameter $y/(1+y)$ and $v_1 = 1 - v_0$. Then using (4.4), (4.7) and (4.8),

$$m_k(\mathrm{MP}_y) = \sum_{\pi \in NC(k)} y^{k-|\pi|} = \frac{(1+y)^{k+1}}{y} \varphi\Big((v_0 s v_1 s v_0)^k \Big).$$

We shall also need the concept of an *elliptic variable* and a *cross-covariance* variable. These will appear in the convergence of the so-called cross-covariance matrices (defined in the next section). An element e is said to be a (standard) *elliptic* variable with parameter ρ, $-1 \le \rho \le 1$, if its free cumulants of order one and of order greater than 2 are zero, and its second order free cumulants are given by $\kappa_2(e,e) = \kappa_2(e^*,e^*) = \rho$ and $\kappa_2(e,e^*) = \kappa_2(e^*,e) = 1$. An elliptic variable has the following representation. Suppose s_1 and s_2 are two free standard semi-circular variables. Then the variable $e = \sqrt{\frac{1+\rho}{2}} s_1 + \sqrt{-1}\sqrt{\frac{1-\rho}{2}} s_2$ is an elliptic variable with parameter ρ. If $\rho = 1$ then e is self-adjoint and is semi-circular.

The next variable has a more complicated free cumulant structure. An element c is a *cross-covariance variable* with parameters ρ and y, $0 < y < \infty$, if its free cumulants are given by

$$\kappa_k(c^{\eta_1}, c^{\eta_2}, \ldots, c^{\eta_k}) = \begin{cases} y^{k-1} \rho^{S(\eta_1,\ldots,\eta_k)} & \text{if } \ \rho \neq 0, \\ y^{k-1} \mathbb{I}\{S(\eta_1,\ldots,\eta_k) = 0\} & \text{if } \ \rho = 0, \end{cases}$$

where

$$S(\eta_1,\ldots,\eta_k) = \sum_{u=1}^{k-1} \mathbb{I}\{\eta_u = \eta_{u+1}\} + \mathbb{I}\{\eta_k = \eta_1\} \text{ and } \eta_1, \eta_2, \ldots, \eta_k \in \{1, *\}. \tag{4.9}$$

When $\rho = 1$, c is self-adjoint with free cumulants $\kappa_k(c) = y^{k-1}$. So, it is an mp_y variable.

The *Tracy–Widom probability law* is defined via a Painlevé function $q(\cdot)$ of Type II. Let $q(t)$ be the solution of the differential equation

$$\frac{d^2}{dt^2}q(t) = tq(t) + 2(q(t))^3$$

that satisfies the boundary condition $\lim_{t\to\infty}(\mathrm{Ai}(t))^{-1}q(t) = 1$ where $\mathrm{Ai}(x) = \frac{1}{\pi}\int_0^\infty \cos\left(\frac{t^3}{3} + xt\right)dt$ is the Airy function. Then the Tracy–Widom probability law is defined for three values of a parameter $\beta = 1, 2, 4$ via $q(\cdot)$ as follows. Its distribution function F_β is given by

$$F_\beta(x) = \begin{cases} e^{-\int_x^\infty (t-x)q^2(t)dt}, & \text{if } \beta = 2, \\ \\ e^{-\frac{1}{2}\int_x^\infty q(t)dt}(F_2(x))^{1/2}, & \text{if } \beta = 1 \end{cases}$$

and

$$F_4(2^{-1/2}x) = \cosh\left(-\frac{1}{2}\int_x^\infty q(t)dt\right)(F_2(x))^{1/2}.$$

These distributions for $\beta = 1$ and 2 arise in a natural way in random matrix theory as the asymptotic distribution of the largest eigenvalue of the Wigner and the sample variance–covariance matrices. See Tracy and Widom (1994), Johnstone (2001) and Johansson (2000). The case $\beta = 4$ arises in the asymptotic distribution of the largest eigenvalue of the Wigner matrix with quaternion random entries, and is outside the purview of this article (Fig. 3).

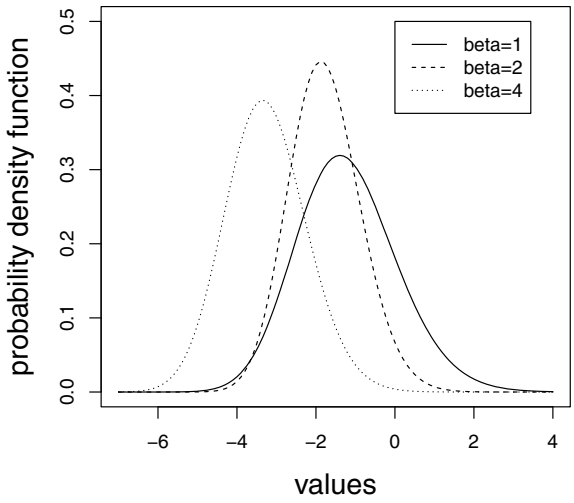


Fig. 3 Density of the Tracy–Widom law for $\beta = 1, 2, 4$

5 Matrices of Our Interest

We now give a quick description of the matrices that we shall encounter in this article. *Wigner matrix*. Wigner (1955) introduced a symmetric random matrix while discussing statistical models for heavy nuclei atoms in physics. Voiculescu (1991) discovered a deep connection of this matrix with free independence. The symmetric (unscaled) Wigner matrix W_n is defined as

$$W_n := \begin{bmatrix} x_{1,1,n} & x_{1,2,n} & x_{1,3,n} & \dots & x_{1,n-1,n} & x_{1,n,n} \\ x_{1,2,n} & x_{2,2,n} & x_{2,3,n} & \dots & x_{2,n-1,n} & x_{2,n,n} \\ & & & \vdots & & \\ x_{1,n,n} & x_{2,n,n} & x_{3,n,n} & \dots & x_{n-1,n,n} & x_{n,n,n} \end{bmatrix}$$

where $\{x_{i,j,n} : 1 \le i \le j \le n\}$ is a collection of independent random variables with mean 0 and variance 1.

Independent matrix. If we drop the symmetry in the Wigner matrix and consider the $p \times n$ matrix all whose entries are independent with mean 0 and variance 1, then we call it the Independent matrix.

Sample covariance matrix. Let $X = ((x_{i,j,p}))_{1\le i\le p, 1\le j\le n}$ be a $p \times n$ random matrix. Then $S = n^{-1}XX^*$ is the *unadjusted sample covariance matrix* (in short, the S matrix) and X is called the data matrix of S.

Kendall's τ and Spearman's ρ matrices. Asymptotic properties of S matrices are usually derived under finiteness of some or all moments. Most of these results cannot be extended to heavy tailed cases. See Heiny and Yao (2020) for a detailed discussion on this. Note that for S, the (i, j)th entry is the sample covariance between the ith and the jth coordinates of the n column vectors of X. Two common measures that could be used instead are Kendall's τ and Spearman's ρ. Kendall's τ for bivariate data $\{(x_i, y_i) : 1 \le i \le n\}$ is a U-statistics of order 2 defined as

$$\tau = \frac{1}{\binom{n}{2}} \sum_{1\le i<j\le n} \text{Sign}(x_i - x_j)\text{Sign}(y_i - y_j) \quad \text{where} \quad \text{Sign}(a) = (|a| + \mathbb{I}(a = 0))^{-1}a, \tag{5.1}$$

$\mathbb{I}(A)$ is 1 if A occurs and 0 otherwise. The value of τ lies in the interval $[-1, 1]$ where $\tau \pm 1$ indicates perfect agreement/disagreement.

Suppose $X = ((x_{i,j}))$ is a $p \times n$ random matrix. Then the (k, l)th entry of Kendall's τ matrix T_p is the value of τ calculated for the n pairs $\{(x_{k,i}, x_{l,i}) : 1 \le i \le n\}$. Clearly, the diagonal entries of T_p are all 1.

Spearman's ρ for bivariate data $\{(x_i, y_i) : 1 \le i \le n\}$ is asymptotically equivalent to the following U-statistic:

$$\rho = \frac{1}{\binom{n}{3}} \sum_{1 \le i < j < k \le n} \Big[\text{Sign}(x_i - x_j)\text{Sign}(y_i - y_k) + \text{Sign}(x_j - x_i)\text{Sign}(y_j - y_k) + \text{Sign}(x_k - x_i)\text{Sign}(y_k - y_j) \Big].$$

Now suppose we have the data matrix $X = ((x_{i,j}))$. Similar to T_p, Spearman's ρ matrix R_p is defined with its (k,l)th entry as Spearman's ρ for $\{(x_{k,i}, x_{l,i}) : 1 \le i \le n\}$. We refer to El Maache and LePage (2003) for more details.

Separable sample covariance matrices. Often the data matrix X in $S = n^{-1}XX^*$ is taken to be an Independent matrix. Then, in particular, the covariance matrix of each column vector of X is the identity matrix I_p and that of each row vector is I_n. One way to model dependence among rows and columns is to model the data matrix X as $X = A^{1/2}ZB^{1/2}$ where A and B are both positive definite matrices of orders p and n respectively and Z is a $p \times n$ Independent matrix. This model is called *separable* since the variance–covariance matrix of $vec(X)$ is equal to $A \otimes B$ where $\otimes$ denotes the Kronecker product. The matrix $\tilde{S} = \frac{1}{n}XX^* = \frac{1}{n}A^{1/2}ZBZ^*A^{1/2}$ is called a *separable sample covariance matrix*.

Cross-covariance matrix. Let $X = ((x_{i,j,p}))$ and $Y = ((y_{i,j,p}))$ be two $p \times n$ Independent matrices (not necessarily independent of each other). Then the matrix $C = n^{-1}XY^*$ is a *sample cross-covariance matrix*. When $X = Y$, C reduces to an S matrix. However, in general C is not a symmetric matrix and hence will have complex eigenvalues.

Autocovariance matrix of a real time series. Suppose $X = \{X_t\}$ is a mean zero *stationary* process and $\text{E}(X_t^2) < \infty$. Then its *autocovariance function* is defined as $\gamma_X(k) = cov(X_0, X_k)$, $k = 0, 1, \ldots$. This function is of fundamental importance in time series analysis. Brockwell and Davis (1991) is a good source of detailed information. Observe that the covariance matrix of $(X_0, \ldots, X_{n-1})$ is given by $\Sigma_n(X) = ((\gamma_X(i-j)))_{1 \le i,j \le n}$. Based on the sample $\{X_0, \ldots, X_n\}$, we may estimate the autocovariances by $\hat{\gamma}_X(k) = n^{-1} \sum_{i=1}^{n-|k|} X_i X_{i+|k|}$. These estimates lead to the *sample autocovariance matrix* $\hat{\Gamma}_n(X) = ((\hat{\gamma}_X(i-j)))_{1 \le i,j \le n}$ which is an estimate of $\Sigma_n(X)$. The common divisor n is used for all $\hat{\gamma}_k$ to ensure that $\hat{\Gamma}_n(X)$ is non-negative definite.

Generalized covariance matrix. Suppose $\{Z_\tau : 1 \le \tau \le m\}$ is a collection of $p \times n$ Independent matrices with independence across τ as well. For $1 \le \tau \le m$, let $\{A_{1\tau}, A_{2\tau}\}$ and $\{B_\tau\}$ be collections of order $p \times p$ and $n \times n$ matrices, respectively. Taking a cue from the separable model, let $G_\tau = A_{1\tau}Z_\tau B_\tau Z_\tau^* A_{2\tau}^*$, $1 \le \tau \le m$. Then any finite degree matrix polynomial in $\{G_\tau : 1 \le \tau \le m\}$ is known as a *generalized covariance matrix*. These types of matrices arise in a natural manner in the context of high-dimensional linear time series models.

Autocovariance matrices for high-dimensional linear time series. Let $X_{t,p}^{(n)}$ be a p dimensional linear time series given by

$$X_{t,p}^{(n)} = \sum_{j=0}^{q} \psi_{j,p} \varepsilon_{t-j}, \quad t, n \geq 1 \text{ (almost surely)}, \tag{5.2}$$

where $\{\varepsilon_t\}$ are p dimensional random vectors, whose components are independent, and the vectors are also independent across t, with mean 0 and variance–covariance matrix I_p, and $\{\psi_{j,p}\}$ are $p \times p$ *parameter matrices*. Note that $X_{t,p}^{(n)}$ is a triangular sequence, but for convenience, we shall write $X_{t,p}$ for $X_{t,p}^{(n)}$. When $q = \infty$, additional conditions on $\{\psi_{j,p}\}$ are imposed so that the above infinite series is summable. Then $\{X_{t,p}\}$ is a stationary time series and the *population autocovariance matrix* of order u is given by

$$\mathrm{E} X_{t,p} X_{t,p+u}^* = \Gamma_{u,p} = \sum_{j=0}^{q-u} \psi_{j,p} \psi_{j+u,p}^*, \text{ for all } u \geq 0. \tag{5.3}$$

This high-dimensional linear model includes the high-dimensional MA, AR and ARMA models.

A method of moment estimator of $\Gamma_{u,p}$ is given by the $p \times p$ matrix

$$\hat{\Gamma}_{u,p,n} = \frac{1}{n} \sum_{t=1}^{n-u} X_{t,p} X_{t+u,p}^*. \tag{5.4}$$

$\hat{\Gamma}_{u,p,n}$ is called the *sample autocovariance matrix*. We shall see later the connection between these matrices and the generalized covariance matrices.

6 LSD and Joint Algebraic Convergence Results

6.1 Wigner Matrix

We begin with one of the fundamental results in random matrix theory which has wide implications. First, it says that the semi-circular law arises as the LSD of the Wigner matrix. Second, it gives the asymptotic freeness of independent Wigner matrices, as well as the asymptotic freeness of Wigner matrices with suitable deterministic matrices. Third, the S matrices can be connected to Wigner matrices via embedding, and that will help us to conclude several asymptotic properties of S. We need the following assumption.

Assumption I For every n, $\{x_{i,j,p}\}$ are independent with mean zero and variance 1, and

$$\sup_{i,j,p} \mathrm{E}(|x_{i,j,p}|^k) \leq B_k < \infty \text{ for all } k \geq 1.$$

The proof of the following theorem is essentially based on the moment method, but involves intricate calculations. For details, we refer to the proofs of Theorems 8.1.2(b), 9.2.1 and 11.3.1 given in Bose (2021).

Theorem 6.1 *(a) Let W be a $p \times p$ Wigner matrix with the independent entries $\{x_{i,j,p} : 1 \leq i \leq j \leq p, p \geq 1\}$ which satisfy Assumption I. Then, as $p \to \infty$, the ESD of $\{p^{-1/2}W\}$ converges to the semi-circular law almost surely.*

(b) Independent $p \times p$ Wigner matrices $p^{-1/2}W_\tau$, $1 \leq \tau \leq m$ whose entries satisfy Assumption I, when considered as elements of the $*$-probability space $(\mathcal{M}_p(\mathbb{C}), p^{-1}E\,\mathrm{Tr})$, converge jointly to free (standard) semi-circular variables.

(c) Suppose $p^{-1/2}W_\tau$, $1 \leq \tau \leq m$ are independent Wigner matrices whose entries satisfy Assumption I. Suppose $\{D_r, 1 \leq r \leq t\}$ are $p \times p$ deterministic matrices which converge jointly with respect to the state tr. Further, the entries $\{d_{i,j,r}\}$ satisfy the condition

$$\sup_{k \geq 1} \sup_{p \geq 1} \max_{1 \leq r \leq t} \left[\frac{1}{p} \sum_{1 \leq i_j \leq p, j = 1, \ldots, k} |d_{i_1,i_2,r} d_{i_2,i_3,r} \cdots d_{i_k,i_1,r}| \right]^{\frac{1}{k}} < \infty.$$

Then $\{p^{-1/2}W_\tau, 1 \leq \tau \leq m, D_r, 1 \leq r \leq t\}$ converge jointly. Moreover, the Wigner matrices are asymptotically free of the deterministic matrices.

6.2 Sample Covariance Matrix

Recall the unadjusted sample covariance matrix $S = n^{-1}XX^*$ defined earlier, where $X = ((x_{i,j,p}))_{1 \leq i \leq p, 1 \leq j \leq n}$. Here $p \to \infty$ and n is taken to be a function of p. Just like the Wigner matrices, the LSD exists. Moreover, these matrices converge jointly and enjoy asymptotic freeness. There are several proofs known for these results. In particular the convergence of the ESD can be deduced either by the moment method or through the convergence of the *Stieltjes transform*. However, we shall discuss briefly a proof that uses tools from non-commutative probability theory, using Theorem 6.1(c) for Wigner matrices. This has the added advantage in that it establishes the joint convergence of several independent S matrices as well as identifies the limits in terms of free variables. Since we work with moments, we shall require all the moments to be finite. However, see the discussion before the statement of Theorem 6.3. We first assume that $p/n \to y \neq 0$.

Theorem 6.2 *Let S_τ, $1 \le \tau \le m$, be independent S matrices with entries $\{x_{i,j,p}^{(\tau)}\}$ that satisfy Assumption I. Suppose $\frac{p}{n} \to y > 0$ as $p \to \infty$. Then the following hold:*

(a) The matrices $(S_1, \ldots, S_m)$ as elements of $(\mathcal{M}_p(\mathbb{C}), p^{-1}E\,\mathrm{Tr})$ converge jointly to $(a_1, \ldots, a_m)$ (say) where $\{a_i\}$ are free mp_y variables.

(b) The ESD of any S_τ converges weakly almost surely to the MP_y law with parameter y.

(c) The ESD of any symmetric matrix polynomial $\Pi\big(\{S_\tau\}\big)$ converges weakly almost surely to the law of $\Pi\big(\{a_\tau\}\big)$.

The above result holds mutatis mutandis if the jth matrix is based on $n = n_j$ observations and $p/n_j \to y_j$ which are not necessarily equal. Part (a) is due to Capitaine and Casalis (2004) and a proof is available in Bose (2021). The original version of Part (b) is due to Marčenko and Pastur (1967). The proofs of Parts (b) and (c) can be found in Bose and Bhattacharjee (2018).

We sketch a proof of Theorem 6.2 that uses non-commutative probability and invokes Theorem 6.1(c). For this purpose, we use an *embedding*. Any $p \times p$ matrix A and any $n \times n$ matrix B, are embedded in matrices $\bar{A}$ and $\underline{B}$ of order $p + n$ as below

$$\bar{A} = \begin{pmatrix} A & O_{p\times n} \\ O_{n\times p} & O_{n\times n} \end{pmatrix} \quad \text{and} \quad \underline{B} = \begin{pmatrix} O_{p\times p} & O_{p\times n} \\ O_{n\times p} & B \end{pmatrix}.$$

For the collection of matrices $\{A_\tau : 1 \le \tau \le m\}$ and $\{B_\tau : 1 \le \tau \le m\}$ of order p and n respectively, and for any polynomial Π, we have

$$\begin{aligned}
\overline{\Pi(A_\tau : 1 \le \tau \le m)} &= \Pi(\bar{A}_\tau : 1 \le \tau \le m), \\
\underline{\Pi(B_\tau : 1 \le \tau \le m)} &= \Pi(\underline{B}_\tau : 1 \le \tau \le m), \\
\mathrm{Tr}(\Pi(A_\tau : 1 \le \tau \le m)) &= \mathrm{Tr}(\Pi(\bar{A}_\tau : 1 \le \tau \le m)), \\
\mathrm{Tr}(\Pi(B_\tau : 1 \le \tau \le m)) &= \mathrm{Tr}(\Pi(\underline{B}_\tau : 1 \le \tau \le m)).
\end{aligned}$$

For each $1 \le \tau \le m$, let W_τ be a Wigner matrix of order $p + n$ whose sub-matrix constructed by the first p rows and the last n columns of W_τ is $X_\tau = ((x_{i,j,p}^{(\tau)}))$. Note that

$$\bar{S}_\tau = \frac{p+n}{n}\bar{I}\frac{W_\tau}{\sqrt{p+n}}\underline{I}\frac{W_\tau}{\sqrt{p+n}}\bar{I}, \tag{6.1}$$

and therefore for any polynomial Π, we have

$$\begin{aligned}
\lim \frac{1}{p}\mathrm{E}\,\mathrm{Tr}\left(\Pi(\{S_\tau\})\right) &= \lim \frac{p+n}{p}\frac{1}{p+n}\mathrm{E}\,\mathrm{Tr}\left(\Pi(\{n^{-1}(p+n)\bar{S}_\tau\})\right) \\
&= \frac{1+y}{y}\lim \frac{1}{p+n}\mathrm{E}\,\mathrm{Tr}\left(\Pi(\{n^{-1}(p+n)\bar{S}_\tau\})\right).
\end{aligned}$$

Clearly, $\bar{I}$ and $\underline{I}$ as elements of $(\mathcal{M}_{p+n}(\mathbb{C}), (p+n)^{-1}\,\mathrm{Tr})$ converge respectively to Bernoulli variables, say v_0, and v_1, in $(\mathcal{A}, \varphi)$ with

$$\varphi(v_0^r) = y(1+y)^{-1} \text{ and } \varphi(v_1^r) = (1+y)^{-1}, \text{ for all } r \geq 1.$$

Also, by Theorem 6.1(b), the matrices $\{(p+n)^{-1/2}W_\tau\}$ in $(\mathcal{M}_{p+n}(\mathbb{C}), (p+n)^{-1}\mathrm{E}\,\mathrm{Tr})$ converge jointly to free semi circular variables $\{s_\tau\}$ in $(\mathcal{A}, \varphi)$. Moreover, by Theorem 6.1(c), $\{v_0, v_1\}$ and $\{s_\tau\}$ are free. Note that since $\bar{I} + \underline{I} = I_{n+p}$, we get $v_0 + v_1 = \mathbf{1}$.

However, there is a subtle point which has its roots in the role of non-crossing partitions in moments, and the reader has to accept it on faith. For all the polynomials we are dealing with, there is always a Wigner matrix between $\bar{I}$ and $\underline{I}$. This has the consequence crucial effect that when we compute the limiting moments of the given polynomials, v_0 and v_1 can be treated as free. This happens only for the polynomials in the matrices that we are working with. Hence, we treat v_0 and v_1 as free and proceed. We have

$$\lim \frac{1}{p}\mathrm{E}\,\mathrm{Tr}\left(\Pi(\{S_\tau\})\right) = \frac{1+y}{y}\varphi\big(\Pi(\{(1+y)v_0 s_\tau v_1 s_\tau v_0\})\big). \tag{6.2}$$

By freeness of $\{v_0, v_1\}$ and $\{s_\tau\}$, and using Example 4.1, we have

$$\varphi\left(\prod_{l=1}^{k} v_0 s_{\tau_l} v_1 s_{\tau_l} v_0\right) = \frac{y}{(1+y)^{k+1}} \sum_{\pi \in B_k} y^{k-|\pi|}.$$

Thus, by (6.2), we have

$$\lim \frac{1}{p}\mathrm{E}\,\mathrm{Tr}\left(\prod_{l=1}^{k} S_{\tau_l}\right) = \sum_{\pi \in B_k} y^{k-|\pi|},$$

which establishes Theorem 6.2(a), i.e. asymptotic freeness of $\{S_\tau\}$, by invoking (4.8). This joint convergence guarantees the condition (C1) for any polynomial Π. Additional technicalities are required to establish (C2) and (C4) in order to prove Theorem 6.2(b) and (c). We refer to Bhattacharjee and Bose (2017) for the details.

Remark 6.1 (a) Full force of Assumption I is not needed to ensure the almost sure weak convergence of the ESD for particular polynomials. Rather, the uniform boundedness of only a few moments, depending on the nature of the polynomial, suffices. This is achieved by truncation arguments. The joint convergence in Part (a) allows us to express the limit distribution for any polynomial in Part (c) in terms of the free variables a_τ, $1 \leq \tau \leq m$ given in Part (a). For example the LSD of $S_2S_1S_2$ is identified by the moments of the self-adjoint variable $a_2a_1a_2$.

(b) In this paper, we mainly focus on the S matrices driven by either i.i.d. or independent observations. Other variations include observations which are heavy tailed, or

Table 1 Simulated and actual values for the pair $(p^{-1}\mathrm{E}\,\mathrm{Tr}(S_2S_1S_2)^r, \varphi(a_2a_1a_2)^r)$ with $n = 500$, $p = yn$

	$y = 0.5$	$y = 1$	$y = 1.5$	$y = 2$
$r = 1$	(1.49, 1.5)	(1.98, 2)	(2.49, 2.5)	(2.99, 3)
$r = 2$	(6.68, 6.69)	(17.59, 18)	(37.23, 37.93)	(70.4, 71)
$r = 3$	(41.06, 41.47)	(222.78, 224)	(814.32, 815.96)	(2828.2, 2829)

that have different distributions and unequal variances. See Belinschi et al. (2009), Bose et al. (2010), Benaych-Georges and Cabanal-Duvillard (2012) and Noiry (2018) for some of these results. See also Bose and Sen (2022) for a unified way of looking at all these matrices and also for a comprehensive list of articles on such variations of the S matrix.

Table 1 shows closeness between the simulated and actual values for the pair $(p^{-1}\mathrm{E}\,\mathrm{Tr}(S_2S_1S_2)^r, \varphi(a_2a_1a_2)^r)$ with $n = 500$, $p = yn$ for Gaussian observations. Throughout this paper, we use Gaussian observations for all the tables and figures since the stated results do not depend on the distribution of the observations.

We now discuss the case $\frac{p}{n} \to 0$ as $p \to \infty$.

Theorem 6.3 *Let $\{S_\tau\}$ be independent S matrices with entries $\{x_{i,j,n}^{(\tau)}\}$ that satisfy Assumption I and let $\{H_\tau = \sqrt{np^{-1}}(S_\tau - I_p)\}$. Suppose $\frac{p}{n} \to 0$ as $p \to \infty$. Then the following hold:*

(a) The matrices $\{H_\tau\}$ as elements of $(\mathcal{M}_p, p^{-1}E\,\mathrm{Tr})$ converge jointly to free standard semi-circular variables $\{s_\tau\}$.

(b) The ESD of any H_τ converges weakly almost surely to the standard semi-circular law.

(c) The ESD of any symmetric matrix polynomial $\Pi(\{H_\tau\})$ converges weakly almost surely to the law of $\Pi(\{s_\tau\})$.

As before, the result above continues to hold if we have sample sizes n_τ for S_τ, provided $p/n_\tau \to 0$ for all τ. For a proof of the above theorem, see Bose (2021). The embedding technique used for the case $p/n \to y \in (0, \infty)$ does not work when $p/n \to 0$ and one needs to use the moment method. This comment holds for all the random matrices discussed in this article (Table 2).

Table 2 Simulated and actual values for the pairs $(\frac{1}{p}\mathrm{E}\,\mathrm{Tr}(H_2H_1H_2)^r, \varphi(s_2s_1s_2)^r)$ with $p = 10^3$, $n = p^{1.5}$

$r = 1$	$r = 2$	$r = 3$	$r = 4$
(0, 0)	(0.98, 1)	(0, 0)	(3.94, 4)

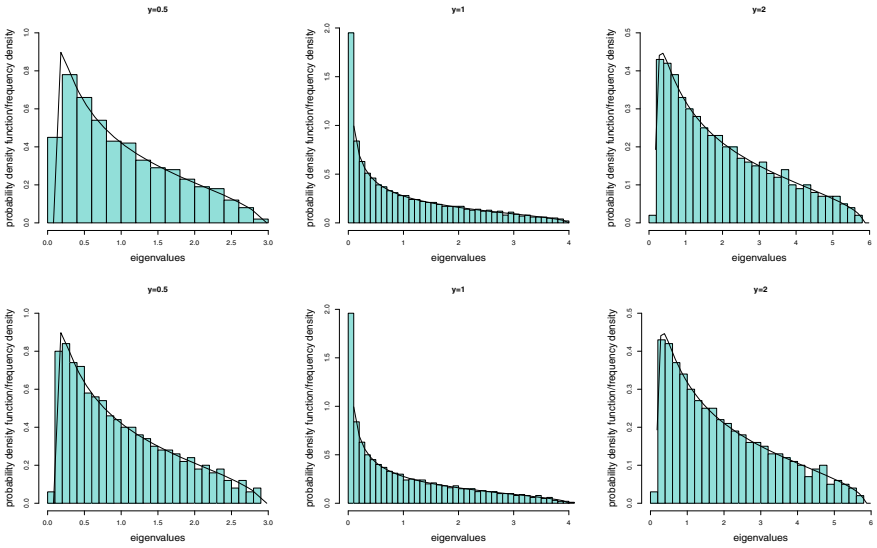


Fig. 4 Histogram of eigenvalues of $\frac{(3T_p - I_p)}{2}$ (first row) and R_p (second row) for $y = 0.5, 1, 2$, $p = 1000$ and $n = p/y$ along with the density function of MP_y

6.3 *Kendall's T_p and Spearman's R_p Matrices*

Recall the definitions of T_p and R_p from Sect. 5. The entries of these matrices are formed by using the variables $x_{i,j,n}$ of the data matrix $X = ((x_{i,j,n}))$ where we have agreed to suppress the index n. Consider the following assumptions.

(RC1) $\{x_{i,j,n}\}$ is a collection of independent random variables.

(RC2) For each n, $p \geq 1$ and $1 \leq i \leq p$, $\{x_{i,j,n} : 1 \leq j \leq n\}$ are identically distributed.

(RC3) Distributions of $\{x_{i,j,n}\}$ are absolutely continuous with respect to the Lebesgue measure.

We then have the following theorem. Parts (a) and (b) were established in Bai and Zhou (2008) and Bandeira et al. (2017), respectively.

Theorem 6.4 *Suppose the assumptions (RC1)–(RC3) hold and $p/n \to y \in (0, \infty)$. Then*

(a) The ESD of R_p converges weakly almost surely to MP_y.

(b) The ESD of $\frac{(3T_p - I_p)}{2}$ *converges weakly in probability to* MP_y *(Fig. 4).*

6.4 Separable Sample Covariance Matrix

Suppose $\tilde{S}_\tau = p^{-1}YY^*$, $1 \le \tau \le m$, are separable covariance matrices where $Y_\tau = A_\tau^{1/2} X_\tau B_\tau^{1/2}$. Here $\{A_\tau\}$ and $\{B_\tau\}$, $1 \le \tau \le m\}$ are collections of positive definite matrices of order p and n, respectively. We then have the following result. We state only the joint algebraic convergence result. The convergence of the ESD of any symmetric matrix polynomial follows along the usual lines. The operator norm of a matrix A is defined as

$$||A||_2 = \sqrt{\lambda_{\max}(A^*A)}$$

where $\lambda_{\max}$ denotes the largest eigenvalue.

Theorem 6.5 *Suppose $p/n \to y$, $0 < y < \infty$. Suppose that the entries $\{x_{i,j,p}^{(\tau)}\}$ of X_τ satisfy Assumption I. Assume that $\sup_p ||A_\tau||_2 + \sup_n ||B_\tau||_2 < \infty$. Further, the collections $(A_1, A_2, \ldots, A_m)$ and $(B_1, B_2, \ldots, B_m)$ as elements of $(\mathcal{M}_p(\mathbb{C}), p^{-1}\,\mathrm{Tr})$ and $(\mathcal{M}_n(\mathbb{C}), n^{-1}\,\mathrm{Tr})$ each converge jointly. Then $(\tilde{S}_1, \tilde{S}_2, \ldots, \tilde{S}_m)$ as elements of $(\mathcal{M}_p(\mathbb{C}), p^{-1}E\,\mathrm{Tr})$ converge jointly to some $(\tilde{s}_1, \tilde{s}_2, \ldots, \tilde{s}_m)$.*

This can be established by using the embedding argument given earlier. As a byproduct of that argument, the limit moments of any polynomial in $\tilde{S}_\tau$, $1 \le \tau \le m$, can be expressed in terms of the following variables. Note that $\{A_\tau\}$ and $\{B_\tau\}$ are elements of different spaces but they can be brought on one common space by embedding. So, consider three collections of variables $\{s_\tau\}$, $\{a^{(\tau)}\}$, $\{b^{(\tau)}\}$, $1 \le \tau \le m$ in some $*$-probability space such that $\{s_\tau\}$ are free semi-circular variables, and these three collections are free of each other. Further, for any polynomial Π,

$$\varphi\big(\Pi(\{a^{(\tau)}\})\big) = \frac{y}{1+y} \lim_{p\to\infty} \frac{1}{p} \mathrm{Tr}\left(\Pi(\{A_\tau^{1/2}\})\right)$$
$$\varphi\big(\Pi(\{b^{(\tau)}\})\big) = \frac{1}{1+y} \lim_{n\to\infty} \frac{1}{n} \mathrm{Tr}\left(\Pi(\{B_\tau\})\right).$$

Then, the limiting moments for any polynomial Π can be written as

$$\lim_{p\to\infty} \frac{1}{p} \mathrm{Tr}\left(\Pi(\{\tilde{S}_\tau\}\right) = \frac{1+y}{y} \varphi\big(\Pi(\{(1+y)a^{(\tau)} s_\tau b^{(\tau)} s_\tau a^{(\tau)}\})\big).$$

Results similar to those given for the ESD of S matrices can then be easily established. The ESD of any symmetric matrix polynomial $\Pi(\tilde{S}_\tau : 1 \le \tau \le m)$ converges weakly almost surely. The corresponding LSD is uniquely determined by the above moments. More details on these results are available in Bhattacharjee and Bose (2017). The LSD result for a single separable covariance matrix is available also in Zhang (2006).

Next suppose that $p \to \infty$, $p/n \to 0$. In this case consider the centered and scaled matrices,

$$F_\tau = \sqrt{n/p}(\tilde{S}_\tau - (n^{-1}\,\mathrm{Tr}(B_\tau))A_\tau),\ 1 \le \tau \le m.$$

Table 3 First three columns give the simulated (for $n = 1000$, $p = yn$) and exact values of $(p^{-1}\mathrm{E}\,\mathrm{Tr}(\tilde{S}_1\tilde{S}_2 + \tilde{S}_2^*\tilde{S}_1^*)^r, \varphi(\tilde{s}_1\tilde{s}_2 + \tilde{s}_2^*\tilde{s}_1^*)^r)$ when $A_1^{1/2} = P_1 + P_{-1}$, $A_2^{1/2} = P_3 + P_{-3}$, $B_1 = P_2 + P_{-2} + P_0$, $B_2 = P_4 + P_{-4} + P_0$, where P_j is the matrix with 1 on the jth main diagonal. The last column gives the simulated ($p = 1000, n = p^{1.5}$) and exact values of $(p^{-1}\mathrm{E}\,\mathrm{Tr}(F_1F_2 + F_2^*F_1^*)^r, \varphi(f_1f_2 + f_2^*f_1^*)^r)$

	$y = 0.5$	$y = 1$	$y = 2$	$y = 0$
$r = 1$	(11.96, 12)	(11.98, 12)	(11.99, 12)	(−0.005, 0)
$r = 2$	(841.98, 844)	(1622.78, 1624)	(3839.98, 3832)	(430.6, 432)

To establish the joint convergence, curiously, we do not need the joint convergence of $\{B_\tau\}$. Instead, it is enough to assume that

$$\lim_{n\to\infty} n^{-1}\,\mathrm{Tr}(B_{\tau_1}) = \lambda_{\tau_1} \quad \text{and} \quad \lim_{n\to\infty} n^{-1}\,\mathrm{Tr}(B_{\tau_1}B_{\tau_2}) = \sigma_{\tau_1,\tau_2},\ 1 \le \tau_1, \tau_2 \le m.$$

Then $(F_1, F_2, \ldots, F_m)$ as elements of $(\mathcal{M}_p(\mathbb{C}), p^{-1}\mathrm{E}\,\mathrm{Tr})$ converge jointly to $(f_1, f_2, \ldots, f_m)$. The ESD of any symmetric polynomial in $\{F_\tau, F_\tau^*\}$ converges weakly almost surely. These limits can be expressed as follows.

Let $\{a_\tau^{(0)} : 1 \le \tau \le m\}$ and the semi-circular family $\{s_\tau : 1 \le \tau \le m\}$ be defined in $(\mathcal{A}, \varphi)$ such that, for any polynomial Π,

$$\varphi\big(\Pi(\{a_\tau^{(0)}\})\big) = \lim \frac{1}{p}\,\mathrm{Tr}\left(\Pi(\{A_\tau\})\right) \quad \text{and}$$
$$\kappa_2(s_{\tau_1}, s_{\tau_2}) = \sigma_{\tau_1,\tau_2}, \quad \text{for all } 1 \le \tau_1, \tau_2 \le m$$

and the collections $\{s_\tau\}$ and $\{a_\tau^{(0)}\}$ are free. Then the LSD of F_τ is identical to the law of $\sqrt{a_\tau^{(0)}}s_\tau\sqrt{a_\tau^{(0)}}$. Also $\sum_{k=1}^K \prod_{l=1}^{r_k} F_{\tau_{kl}}$ from $(\mathcal{M}_p(\mathbb{C}), p^{-1}\mathrm{E}\,\mathrm{Tr})$ converges to $\sum_{k=1}^K \prod_{l=1}^{r_k} \sqrt{a_{\tau_{kl}}^{(0)}}s_{\tau_{kl}}\sqrt{a_{\tau_{kl}}^{(0)}}$ in $(\mathcal{A}, \varphi)$ for all $\tau_{kj} \in \{1, 2, \ldots, m\}$, $1 \le j \le r_k$, $1 \le k \le K$. More details on the above results are available in Bhattacharjee and Bose (2016a). The LSD result for a single separable covariance matrix is also available in Wang and Paul (2014) (Table 3).

6.5 Cross-Covariance Matrix

Recall the cross-covariance matrix from Sect. 5.

Assumption II The bivariate vectors $\{(x_{i,j,p}, y_{i,j,p})\}$ are independent with zero mean, unit variances, and correlation between $x_{i,j,p}$ and $y_{i,j,p}$ is ρ. Moreover,

$$\sup_{i,j,p} \mathrm{E}\big(|X_{i,j,p}|^k + |Y_{i,j,p}|^k\big) < B_k < \infty.$$

Table 4 Closeness of $p^{-1}\mathrm{E}\,\mathrm{Tr}(C_1C_2 + C_2^*C_1^*)^r$ and $\varphi(c_1c_2 + c_2^*c_1^*)^r$, $n = 1000$, $p = yn$

	$y = 0.5, \rho = -0.5$	$y = 1, \rho = -0.5$	$y = 2, \rho = -0.5$
$r = 1$	(0.49, 0.5)	(0.5, 0.5)	(0.49, 0.5)
$r = 2$	(1.47, 1.5)	(3.55, 3.62)	(10.77, 10.87)
$r = 3$	(3.9, 4.09)	(11.62, 11, 84)	(43.8, 44.47)
	$y = 0.5, \rho = 0$	$y = 1, \rho = 0$	$y = 2, \rho = 0$
$r = 1$	(0.002, 0)	(0.002, 0)	(0.003, 0)
$r = 2$	(0.47, 0.5)	(1.98, 2)	(7.97, 8)
$r = 3$	(0.02, 0)	(−0.02, 0)	(0.01, 0)
	$y = 0.5, \rho = 0.5$	$y = 1, \rho = 0.5$	$y = 2, \rho = 0.5$
$r = 1$	(0.5, 0.5)	(0.5, 0.5)	(0.49, 0.5)
$r = 2$	(1.43, 1.5)	(3.48, 3.62)	(10.8, 10.87)
$r = 3$	(3.93, 4.09)	(11.59, 11.84)	(43.77, 44.47)

Apparently, there are only a few articles on the behavior of general cross-covariance matrices. Akemann et al. (2008) and Vinayak and Benet (2014) respectively analyzed its characteristic polynomial and its spectral domain. Akemann et al. (2021) established the weak convergence of the ESD when the entries are complex Gaussian, $0 < y < \infty$, and $\rho \in [-1, 1]$.

The following theorem states the joint convergence of independent cross-covariance matrices to freely independent cross-covariance variables when $p/n \to y \neq 0$. Theorem 6.7 treats the case $y = 0$. These results were established by Bhattacharjee et al. (2022).

Theorem 6.6 *Suppose $\{C_\tau\}$ are independent $p \times p$ cross-covariance matrices based on $n = n_\tau$ observations, respectively. Suppose the entries of each C_τ satisfy Assumption II and with correlation ρ_τ, $1 \le \tau \le m$. Suppose $p \to \infty$, $p/n_\tau \to y_\tau$, $0 < y_\tau < \infty$ for all τ. Then the following statements hold.*

(a) As elements of $(\mathcal{M}_p(\mathbb{C}), p^{-1}E\,\mathrm{Tr})$, $\{C_\tau\}$ converge jointly to free variables $\{c_\tau\}$ where each c_τ is a cross-covariance variable with parameters (ρ_τ, y_τ). The convergence also holds with respect to the state tr almost surely.

(b) Let $\Pi = \Pi(\{C_\tau\})$ be any finite degree real symmetric matrix polynomial in $\{C_\tau, C_\tau^*\}$. Then the ESD of Π converges weakly almost surely to the compactly supported probability law of the self-adjoint variable $\Pi(\{c_\tau\})$ (Table 4).

Theorem 6.7 *Suppose all conditions in the above theorem hold, except that $p/n_\tau \to y_\tau = 0$ for all τ. Consider the matrices $E_\tau = \sqrt{n_\tau p^{-1}}\big(C_\tau - \rho_\tau I_p\big)$, $1 \le \tau \le m$. Then the following hold:*

(a) As elements of $(\mathcal{M}_p, p^{-1}E\,\mathrm{Tr})$, $\{E_\tau\}$ converge jointly to free elliptic variables $\{e_\tau\}$, with parameters $\{\rho_\tau^2\}$. The convergence also holds with respect to the state tr almost surely.

Table 5 Closeness of $p^{-1}\mathrm{E}\,\mathrm{Tr}(E_1E_2 + E_2^*E_1^*)^r$ and $\varphi(e_1e_2 + e_2^*e_1^*)^r$ for $p = 1000$, $n = p^{1.5}$

	$\rho = -0.5$	$\rho = 0$	$\rho = 0.5$
$r = 1$	(0.002, 0)	(−0.002, 0)	(0.001, 0)
$r = 2$	(1.98, 2)	(1.974, 2)	(1.998, 2)
$r = 3$	(−0.02, 0)	(0.03, 0)	(0.02, 0)

(b) The ESD of any finite degree real symmetric matrix polynomial $\Pi = \Pi\big(\{E_\tau, E_\tau^*\}\big)$ *converges weakly a.s. to the compactly supported probability law of the self-adjoint variable* $\Pi\big(\{e_\tau, e_\tau^*\}\big)$.

Note that the above results are silent about the convergence of ESD of C_τ. As pointed out earlier, since C_τ is a non-symmetric matrix, the convergence of its ESD cannot be approached by the moment method. The work of Akemann et al. (2021) mentioned above seems to be the only one in this direction (Table 5).

6.6 Autocovariance Matrix of a Real Time Series

Suppose $X = \{X_t\}$ is a mean zero *stationary* process and $\mathrm{E}(X_t^2) < \infty$ with the *autocovariance function* $\gamma_X(k) = cov(X_0, X_k)$, $k \geq 0$. The covariance matrix of $(X_0, \ldots X_{n-1})$ is given by

$$\Sigma_n(X) = ((\gamma_X(i-j)))_{1\leq i,j\leq n}.$$

Based on the sample $\{X_0, \ldots, X_n\}$, the sample autocovariances are given by

$$\hat{\gamma}_X(k) = n^{-1}\sum_{i=1}^{n-|k|} X_i X_{i+|k|}$$

which give the *sample autocovariance matrix*

$$\hat{\Gamma}_n(X) = ((\hat{\gamma}_X(i-j)))_{1\leq i,j\leq n}.$$

Before we give results for $\hat{\Gamma}_n(X)$, let us recall some results for $\Sigma_n(X)$ which is deterministic. To every sequence $\{\gamma_X(i)\}$, there is a unique *spectral distribution*, say $F_X(\cdot)$, which satisfies the equation

$$\gamma_X(h) = \int_{(0,\ 1]} \exp(2\pi i h x) dF_X(x) \ \text{ for all } \ h = 0, 1, 2, \ldots. \tag{6.3}$$

Let us assume that

$$\sum_{k=1}^{\infty} |\gamma_X(k)| < \infty. \tag{6.4}$$

Then $F_X(\cdot)$ has a density (known as the *spectral density*),

$$f_X(t) = \sum_{k=-\infty}^{\infty} \exp(-2\pi i t k)\gamma_X(k), \ t \in (0, \ 1]. \tag{6.5}$$

Note that $\Sigma_n(X)$ is a *symmetric Toeplitz matrix* (each diagonal has the same entries). From Szegö's theory for Toeplitz operators (see Böttcher & Silbermann, 1999), if (6.4) holds, then the LSD of $\Sigma_n(X)$ equals (the distribution of) $f_X(U)$ where U is uniformly distributed on $(0, \ 1]$.

To study $\hat{\Gamma}_n(X)$, we shall assume that $X = \{X_t\}_{t\in\mathbb{Z}}$ is a *linear process*

$$X_t = \sum_{k=0}^{\infty} \theta_k \varepsilon_{t-k}. \tag{6.6}$$

The following results were established by Basak et al. (2014). Proofs of these results as well as others on related matrices can be found in Bose (2018).

Assumption III. (a) For all t, $\mathrm{E}[\varepsilon_t] = 0$ and $\mathrm{E}[\varepsilon_t^2] = 1$.

(b) Either $\{\varepsilon_t\}$ are i.i.d. or they are independent and uniformly bounded.

(c) $\sum_{j=0}^{\infty} |\theta_j| < \infty$.

Theorem 6.1 *Suppose Assumption III holds. Then the ESD of* $\hat{\Gamma}_n(X)$ *converges weakly almost surely. The LSD does not depend on the distribution of* $\{\varepsilon_t\}$.

Unfortunately, no easy description of the above LSD or its moments are available. The LSD is known to have unbounded support, at least when $\{\theta_i\}$ is a non-negative sequence. In contrast, the LSD of $\Sigma_n(X)$ always has bounded support. Hence this shows the "inconsistency" of the sample autocovariance matrix. See Figs. 5 and 6.

We can achieve consistency by modifying $\hat{\Gamma}_n(X)$. This involves borrowing the well-known techniques of *tapering* and *banding* used in the estimation of the spectral density. Let $\{m = m_n\} \to \infty$ be a sequence of integers and let $K(\cdot)$ be a non-negative *kernel function*. Let

$$\hat{f}_X(t) = \sum_{k=-m}^{m} K(k/m) \exp(-2\pi i t k)\hat{\gamma}_X(k), \ t \in (0, \ 1] \tag{6.7}$$

be a *kernel density estimate* of $f_X(\cdot)$. It is a spectral density in its own right with the autocovariance sequence as

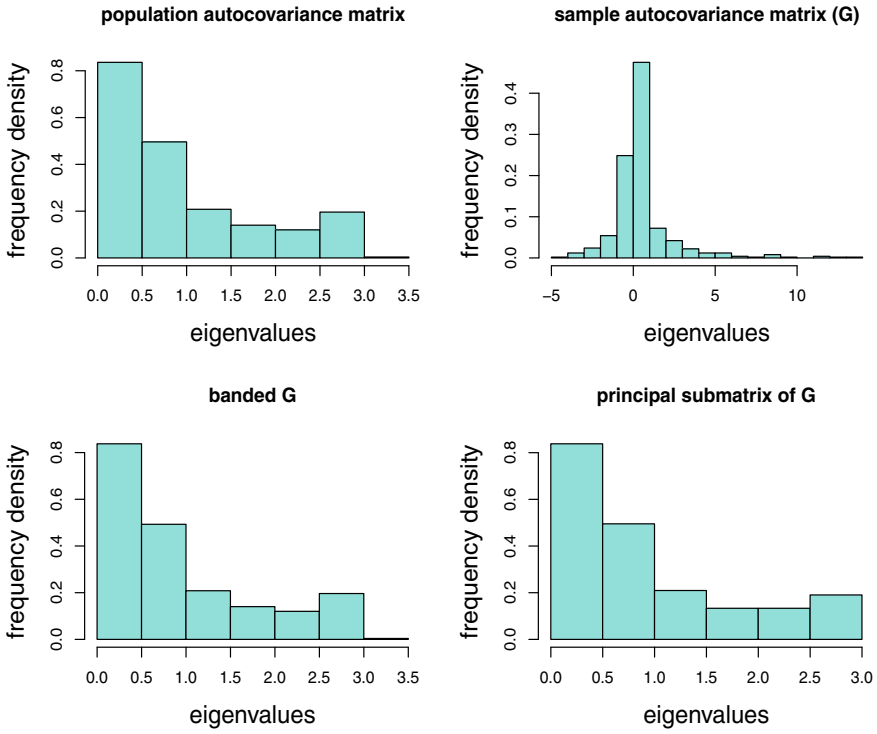


Fig. 5 Histogram of the ESD of $\Sigma_n(X)$, $\hat{\Gamma}_n(X)$ (labeled as G), $\hat{\Gamma}_{n,K}(X)$ and $\hat{\Gamma}_{n,B}(X)$ with $m_n = n^{3/4}$, $n = 10^5$ and AR(1) process with AR parameter 0.5

$$\begin{aligned}\gamma_K(h) &= \int_{(0,\ 1]} \exp(2\pi i h x)\, \hat{f}_X(x) dx \\ &= \sum_{k=-m}^{m} K(k/m) \int_{(0,\ 1]} \exp\{2\pi i h x - 2\pi i x k\} \hat{\gamma}_X(k) dx \\ &= K(h/m)\hat{\gamma}_X(h), \text{ for } -m \le h \le m.\end{aligned}$$

The other autocovariances are zero. Consider the *tapered* sample autocovariance matrix

$$\hat{\Gamma}_{n,K}(X) = ((K((i-j)/m)\hat{\gamma}_X(i-j)))_{1\le i,j\le n}. \tag{6.8}$$

The choice of $K(x) = I_{\{|x|\le 1\}}$ yields the *banded* version of $\hat{\Gamma}_n(X)$, considered in McMurry and Politis (2010). Another type of banding considered in McMurry and Politis (2010) is to use the $m_n \times m_n$ principal sub-matrix of $\hat{\Gamma}_n(X)$. We call this matrix $\hat{\Gamma}_{n,B}(X)$.

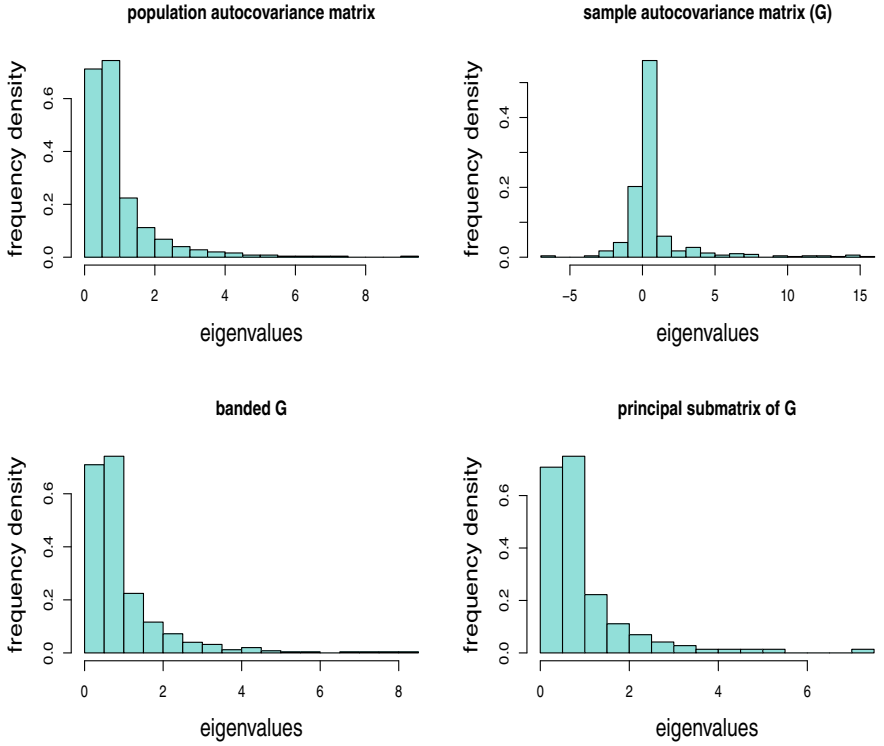


Fig. 6 Histogram of the ESD of $\Sigma_n(X)$, $\hat{\Gamma}_n(X)$ (labeled as G), $\hat{\Gamma}_{n,K}(X)$, and $\hat{\Gamma}_{n,B}(X)$ with $m_n = n^{3/4}$, $n = 10^5$ for the MA(1000) process with MA parameters $(-1)^u u^{-1}$ at lag u

We have the following theorem on these tapered and banded estimates.

Theorem 6.2 *Suppose Assumption III holds and $m_n \to \infty$ such that $m_n/n \to 0$.*

(a) Let $K(x)$, $|x| \leq 1$ be a bounded, symmetric, non-negative definite kernel which is continuous at 0 and $K(0) = 1$. Then the ESD of $\hat{\Gamma}_{n,K}(X)$ converges weakly almost surely to $f_X(U)$.

(b) Suppose $m_n \to \infty$ such that $m_n/n \to 0$. Then the ESD of $\Gamma_{n,B}(X)$ converges weakly almost surely to $f_X(U)$.

6.7 Generalized Covariance Matrices

Recall the generalized covariance matrices $\{G_\tau\}$ from Sect. 5. Their joint convergence can be addressed by the embedding trick discussed in Sect. 6.2. For details of these results we refer to Bhattacharjee and Bose (2017). Here we give a brief outline.

Suppose $\{A_{j\tau} : \ 1 \le \tau \le m, \ \ j = 1, 2\}$ converge in the algebraic sense. Further suppose that $\sup_p ||A_\tau||_2 + \sup_n ||B_\tau||_2 < \infty$, and the entries of X_τ satisfy Assumption I.

First consider the case $p/n \to y \in (0, \infty)$. Suppose further that $\{B_\tau : \ 1 \le \tau \le m\}$ converge. Then $\{G_\tau : \ 1 \le \tau \le m\}$ converge jointly under the state Etr. The limiting moments of any polynomial in $\{G_\tau, G_\tau^*\}$ can be expressed in terms of the following variables. Consider a $*$-probability space $(\mathcal{A}, \varphi)$ where we have three collections of variables as described below

(i) $(s_\tau : \ 1 \le \tau \le m)$ are free standard semi-circular variables.

(ii) $\{a_j^{(\tau)} : \ 1 \le \tau \le m, \ \ j = 1, 2\}$ and $(b^{(1)}, b^{(2)}, \ldots, b^{(m)})$ are the other two collections, essentially arising from the limits of the matrices $\{A_{j\tau}\}$ and $\{B_\tau\}$, except that we need to compensate for the embedding. Hence, for any polynomial Π, they are required to satisfy

$$\varphi\big(\Pi(\{a_j^{(\tau)}, a_j^{(\tau)*}\})\big) = \frac{y}{1+y} \lim_{p\to\infty} \frac{1}{p} \operatorname{Tr}\left(\Pi(\{A_{j\tau}, A_{j\tau}^*\})\right),$$
$$\varphi\big(\Pi(\{b^{(\tau)}\})\big) = \frac{1}{1+y} \lim_{n\to\infty} \frac{1}{n} \operatorname{Tr}\left(\{\Pi(B_\tau\})\right).$$

(iii) The three collections are free of each other.

Then, for any polynomial Π, we have

$$\lim_{p\to\infty} \frac{1}{p} \operatorname{Tr}\left(\Pi(\{G_\tau, G_\tau^*\})\right) = \frac{1+y}{y} \varphi\big(\Pi(\{(1+y)a_1^{(\tau)} s_\tau b^{(\tau)} s_\tau a_2^{(\tau)*}, (1+y) a_2^{(\tau)} s_\tau b^{(\tau)*} s_\tau a_1^{(\tau)*}\})\big).$$

Moreover, the ESD of any real symmetric matrix polynomial $\Pi(\{G_\tau, G_\tau^*\})$ converges weakly almost surely, and the LSD is uniquely determined by the moments appearing on the right side of the above equation.

Now consider the case $p/n \to 0$. As in the case for the S matrix (see Theorem 6.3), now we need a different scaling and an additional centering. We continue to assume that $\{A_{j\tau}\}$ converge jointly. However, for $\{B_\tau\}$, it now suffices to assume only that their first two moments converge:.

$$\lim_{n\to\infty} n^{-1} \operatorname{Tr}(B_{\tau_1}^{\epsilon_1}) = \lambda_{\tau_1}^{\epsilon_1} \ \text{ and } \ \lim_{n\to\infty} n^{-1} \operatorname{Tr}(B_{\tau_1}^{\epsilon_1} B_{\tau_2}^{\epsilon_2}) = \sigma_{\tau_1,\tau_2}^{(\epsilon_1,\epsilon_2)}, \ 1 \le \tau_1, \tau_2 \le m, \ \epsilon_1, \epsilon_2 = 1, *.$$

Instead of writing the general joint convergence result, we provide a flavor by considering a single polynomial matrix $\tilde{G} = \sum_{k=1}^K \prod_{l=1}^{r_k} G_{\tau_{kl}}$. Recall that for the S matrix, we used the centering I_p. The appropriate centering for $\tilde{G}$ turns out to be $C = \sum_{k=1}^K \left(\prod_{l=1}^{r_k} \frac{1}{n} \operatorname{Tr}(B_{\tau_{kl}})\right) \left(\prod_{l=1}^{r_k} A_{l\tau_{kl}} A_{2\tau_{kl}}\right)$. Then $\sqrt{n/p}(\tilde{G} - C)$ in $(\mathcal{M}_p(\mathbb{C}), p^{-1}\mathrm{E}\operatorname{Tr})$ converges.

The limit variable can be described in a manner similar to that given in the $y \neq 0$ case. Let $(\mathcal{A}, \varphi)$ be a $*$-probability space where we have *two* collections, $\{a_{j\tau}^{(0)}\}$ and $\{w_\tau\}$ which are free of each other and their moments/free cumulants satisfy the following: for any polynomial Π,

$$\varphi\big(\Pi(\{a_{j\tau}^{(0)}\})\big) = \lim \frac{1}{p} \operatorname{Tr}\left(\Pi(\{A_{j\tau}, A_{j\tau}^*\})\right) \quad \text{and}$$

$$\kappa_r(w_{\tau_1}^{\epsilon_1}, w_{\tau_2}^{\epsilon_2}, \ldots, w_{\tau_r}^{\epsilon_r}) = \begin{cases} \sigma_{\tau_1,\tau_2}^{(\epsilon_1,\epsilon_2)} & \text{for } r = 2 \text{ and all } 1 \le \tau_1, \tau_2 \le m, \epsilon_1, \epsilon_2 = 1, *, \\ 0 & \text{otherwise.} \end{cases}$$

Then $\sqrt{n/p}(\tilde{G} - C)$ as an element of $(\mathcal{M}_p(\mathbb{C}), p^{-1}\mathrm{E}\,\mathrm{Tr})$ converges to

$$\sum_{k=1}^{K} \sum_{l=1}^{r_k} \left(\prod_{u=1}^{l-1} a_{1\tau_{ku}}^{(0)} a_{2\tau_{ku}}^{(0)*} \lambda_{\tau_{ku}}^{(1)} \right) \left(a_{1\tau_{kl}}^{0} w_{\tau_{kl}} a_{2\tau_{kl}}^{(0)*}\right) \left(\prod_{u=l+1}^{r_k} a_{1\tau_{ku}}^{(0)} a_{2\tau_{ku}}^{(0)*} \lambda_{\tau_{ku}}^{(1)} \right).$$

In general, joint convergence of the centered and scaled $\{G_\tau\}$ matrices holds. With some additional arguments, as a consequence, ESD of any real symmetric polynomial in these matrices converges weakly almost surely. For more details see Bhattacharjee and Bose (2016a).

6.8 *High-Dimensional Autocovariance Matrices*

Recall the sample autocovariance matrix $\{\hat{\Gamma}_{u,p,n}\}$ from Sect. 5. We now connect these to the generalized covariance matrices of the previous section. Let Z be a $p \times n$ Independent matrix whose (i, j)th entry is the ith component of ε_j. Also let P_u be a square matrix of order n whose (i, j)th entry is 1 only if $j - i = u$ and the other entries are 0. Note that $P_0 = I_n$ and P_u is not symmetric for $u \neq 0$. Moreover, $P_u = P_{-u}^*$ and $P_u P_{-u} - I_n$ is a diagonal matrix with rank at most u. It is easy to see that, for finite q, we have

$$\hat{\Gamma}_{u,p,n} = \sum_{j,j'=0}^{q} \psi_{j,p} Z P_{j'-j+u} Z^* \psi_{j',p}^* + \text{a matrix with rank at most } 2q. \quad (6.9)$$

Thus the sample autocovariance matrices are low rank perturbation of the generalized covariance matrices $\{\psi_{j,p} Z P_{j'-j+u} Z^* \psi_{j',p}^*\}$.

Hence if the entries of Z in (6.9) satisfy Assumption I, and $\{\psi_{j,p}\}$, satisfy the assumptions given for $\{A_{j\tau}\}$ in Sect. 6.7, then $\{\hat{\Gamma}_{u,p,n}\}$ also converge jointly, when q is finite. Under additional assumptions on coefficient matrices $\{\psi_{j,p}\}$, even when $q = \infty$, the existence of the LSD of symmetric polynomials is also guaranteed. See Bhattacharjee and Bose (2016b, 2019) for more details. This LSD result will be useful to us in applications later. The LSD of specific symmetric polynomials in sample autocovariance matrices are also discussed in Wang et al. (2011, 2017), Pfaffel and Schlemm (2011), Yao (2012) and Liu et al. (2015) under some stronger assumptions such as simultaneous diagonalizability of $\{\psi_{j,p}\}$.

7 Linear Spectral Statistics: Gaussian Limit

A key statistic in high-dimensional statistical inference is the LSS, and the more specific case of the trace statistic is also commonly used. For example, Schott (2001) and Ledoit and Wolf (2002) considered statistically testing certain hypothesis regarding two high-dimensional covariance matrices, and the test statistics are all based on LSS of two independent sample variance–covariance matrices. Another example is the following: the performance of the likelihood ratio test statistic in high dimensions is significantly affected by the largeness of the dimension, and Bai et al. (2009) gave several modified statistics using the LSS and thereby achieved improved performance.

Jonsson (1982) obtained the asymptotic normality for LSS of $S = n^{-1}XX^*$ by assuming the Gaussianity of X. Bai and Silverstein (2008) established the asymptotic normality of LSS of $S = AZZ^*A$, assuming that A is a $p \times p$ symmetric matrix with full rank and $\sup_p ||A||_2 < \infty$ and the ESD of A^2 converges weakly to the non-random probability distribution H and the entries of Z are all i.i.d. with mean 0, variance 1 and fourth moment 3. Later Pan and Zhou (2008) improved this result by assuming only the finiteness of the fourth order moment. We give this result here. Let F_S and F_{A^2} denote the distribution functions of the ESDs of $S = n^{-1}AXX^*A$, and A^2, respectively. Consider the free product of the two laws $\mathrm{MP}_{p/n}$ and F_{A^2} and let $F_{p/n,A^2}$ denote the distribution function of this law. For any function f, let

$$X_n(f) = \int f(x)pd(F_S(x) - F_{p/n,A^2}(x)).$$

Theorem 7.1 *Suppose the entries of Z are all i.i.d. with mean 0, variance 1 and finite fourth moment. Suppose A is $p \times p$ symmetric matrix, with $\sup_p ||A||_2 < \infty$ and the ESD of A^2 converges weakly to a non-random probability distribution H. Let $f_1, f_2, \ldots, f_k$ be functions analytic on an open region contained in the support of $F_{y,H}$. Then $(X_n(f_1), X_n(f_2), \ldots, X_n(f_k))$ converges weakly to a Gaussian vector $(X_{f_1}, X_{f_2}, \ldots, X_{f_k})$.*

If the fourth moment of each entry of Z equals 3*, then the expectation and covariance function of $(X_{f_1}, X_{f_2}, \ldots, X_{f_k})$ are as follows:*

$$EX_f = -\frac{1}{2\pi i}\oint_C f(z)\frac{y\int \frac{(s(z))^3t^2dH(t)}{(1+ts(z))^2}}{\left(1 - y\int \frac{(s(z))^2t^2dH(t)}{(1+ts(z))^2}\right)^2}dz,$$

$$\mathbb{C}\mathrm{ov}(X_{f_1}, X_{f_2}) = -\frac{1}{2\pi^2}\oint_{C_1}\oint_{C_2}\frac{f_1(z_1)f_2(z_2)}{(s(z_1) - s(z_2))^2}s'(z_1)s'(z_2)dz_1dz_2 \quad \textit{where}$$

$$s(z) = -\left(z - y\int \frac{t}{1+ts(z)}dH(t)\right)^{-1}$$

the integrals are along contours (non-overlapping for the covariance function) which are closed and positively oriented and enclose the support of $F_{y,H}$.

Remark 7.1 The trace is a special case of the LSS, and is relatively easier to handle. Asymptotic normality results are available for the trace of polynomials in sample autocovariance matrices of one or more independent high-dimensional linear time series models. In particular, they apply to the LSS of polynomials in independent separable sample covariance matrices. See Bhattacharjee and Bose (2016a) and Bhattacharjee and Bose (2019) for these result in the two cases $p/n \to y > 0$ and $p/n \to 0$, respectively.

Now we consider the LSS for Kendall's T_p and Spearman's R_p matrices. Let F_{T_p} be the empirical distribution function of T_p, and $\tilde{F}^p$ be the distribution function of $\frac{2}{3}\text{mp}_{p/n} + \frac{1}{3}$. Let $m(z)$ be the Stieltjes transform of $\frac{2}{3}\text{mp}_y + \frac{1}{3}$ and $m'(z)$ be the derivative of $m(z)$ with respect to z. Define $G_p(x) = p(F_{T_p}(x) - \tilde{F}^p(x))$ and $D_y(z) = 1 + \frac{2}{3}ym(z)$. The following result on asymptotic normality of LSS of T_p is established in Li et al. (2021).

Theorem 7.2 *Suppose Assumptions (RC1)–(RC3) hold and* $p/n \to y \in (0, \infty)$. *Let* $f_1, f_2, \ldots, f_k$ *be functions on* $\mathbb{R}$ *and analytic on an open interval containing the support of the law of* $\frac{2}{3}\text{mp}_y + \frac{1}{3}$. *Then the random vector*

$$\left(\int f_1(x) dG_p(x), \int f_2(x) dG_p(x), \ldots, \int f_k(x) dG_p(x) \right)$$

converges weakly to a Gaussian random vector $(X_{f_1}, X_{f_2}, \ldots, X_{f_k})$ *with mean*

$$EX_f = -\frac{1}{2\pi i} \oint_\gamma f(z) \left[\frac{36ym^3(z)D_y(z)}{(4ym^2(z) - 9D_y^2(z))62} - \frac{2y^2m^3(z)(D_y^2(z) + 2D_y(z) + 4)}{4ym^2(z) - 9D_y^2(z)} \right] dz$$
$$-\frac{1}{2\pi i} \oint_\gamma f(z) \frac{8cm^3(z)}{D_y(z)(4cm^2(z) - 9D_y^2(z))} dz, \quad (7.1)$$

and covariances

$$\mathbb{C}\text{ov}(X_{f_i}, X_{f_j}) = -\frac{1}{2\pi^2} \oint f_i(z_i) f_j(z_j) \frac{m'(z_i)m'(z_j)}{(m(z_i) - m(z_j))^2} dz_i dz_j$$
$$+\frac{1}{\pi^2} \oint f_i(z_i) f_j(z_j) \frac{2ym'(z_i)m'(z_j)}{9D_y^2(z_i)D_y^2(z_j)} dz_i dz_j \quad (7.2)$$

where the contours in (7.1) and (7.2) (two contours in (7.2) are assumed to be non-overlapping) are closed, and taken in the positive direction in the complex plane, each enclosing the support of the law of $\frac{2}{3}\text{mp}_y + \frac{1}{3}$.

Following result on asymptotic normality of trace of different powers of R_p is established in Bao et al. (2015).

Theorem 7.3 *Suppose (RC1)–(RC3) hold and $p/n \to y \in (0, \infty)$. Then the following statements hold.*

(a) For all $r \geq 2$, we have the following expansion for the expectation function:

$$E(\mathrm{Tr}(R_p^r)) = \frac{n^r}{(n-1)^{r-1}} \sum_{j=0}^{r-1} \frac{1}{j+1} \binom{r}{j} \binom{r-1}{j} \left(\frac{n-1}{p}\right)^j - \frac{1}{2} \sum_{j=0}^{r} \binom{r}{j}^2 y^j$$
$$+2y^{1+r} \sum_{j=0}^{r} \binom{r}{j} \left(\frac{1-y}{y}\right)^j \left[\binom{2r-j}{r-1} - \binom{2r+1-j}{r-1}\right]$$
$$+\frac{1}{4}((1-\sqrt{y})^{2r} + (1+\sqrt{y})^{2r}) + o(1).$$

(b) The random vector $(\mathrm{Tr}(R_p^{r_u}) - E(\mathrm{Tr}(R_p^{r_u})) : 1 \leq u \leq k)$, for all $r_u \geq 2$, $1 \leq u \leq k$, $k \geq 1$, converges weakly to a k dimensional Gaussian random vector with mean 0 and the (α, β)th entry of its covariance matrix is given by (Fig. 7)

$$2y^{r_\alpha + r_\beta} \sum_{j_1=0}^{r_\alpha - 1} \sum_{j_2=0}^{r_\beta} \binom{r_\alpha}{j_1} \binom{r_\beta}{j_2} \left(\frac{1-y}{y}\right)^{j_1+j_2} \left[\sum_{l=1}^{r_\alpha - j_1} l \binom{2r_\alpha - 1 - j_1 - l}{r_\alpha - 1} \binom{2r_\beta - 1 - j_2 + l}{r_\beta - 1}\right]$$
$$-2y^{r_\alpha + r_\beta + 1} \sum_{j_1=0}^{r_\alpha} \sum_{j_2=0}^{r_\beta} \binom{r_\alpha}{j_1} \binom{r_\beta}{j_2} \left(\frac{1-y}{y}\right)^{j_1+j_2} \binom{2r_\alpha - j_1}{r_\alpha - 1} \binom{2r_\beta - j_2}{r_\beta - 1}.$$

8 Extreme Eigenvalues: Tracy–Widom Laws

Let $X_{(1)} = ((x_{(1),i,j,n}))$ and $X_{(2)} = ((x_{(2),i,j,n}))$ be $p \times n$ matrices where the entries $x_{(1),i,j,n}$'s and $x_{(2),i,j,n}$ are all *i.i.d.* with real and *complex* entries, respectively. For $\beta = 1, 2$, define $S_{(\beta)} = \frac{1}{n} X_{(\beta)} X_{(\beta)}^*$. For any square symmetric matrix A, let $\lambda_{\max}(A)$ denote the largest eigenvalue of A. Consider the following assumptions:

(TW1) $\mathrm{E}(x_{(1),i,j,n}) = 0, \mathrm{E}|x_{(1),i,j,n}|^2 = 1.$

(TW2) $\mathrm{E}(x_{(2),i,j,n}) = 0, \mathrm{E}(x_{(2),i,j,n}^2) = 0, \mathrm{E}|x_{(2),i,j,n}|^2 = 1.$

The following results have been established in Johnstone (2001); Johansson (2000) respectively for $\beta = 1$ and 2.

Theorem 8.1 *Suppose $p/n \to y \in (0, \infty)$, (TW1) and (TW2) hold where the entries of $X_{(1)}$ and $X_{(2)}$ are Gaussian. Then, for $\beta = 1, 2$, we have*

$$\mathbb{P}\left(\frac{\lambda_{\max}(S_{(\beta)}) - \mu_{n,p}}{\sigma_{n,p}} \leq t\right) \to F_\beta(t) \quad \forall\, t \in \mathbb{R} \ \textit{where}$$
$$\mu_{n,p} = (1 + \sqrt{p/n})^2 \quad \textit{and} \quad \sigma_{n,p} = n^{-2/3} (n/p)^{1/6} (1 + \sqrt{p/n})^{4/3}.$$

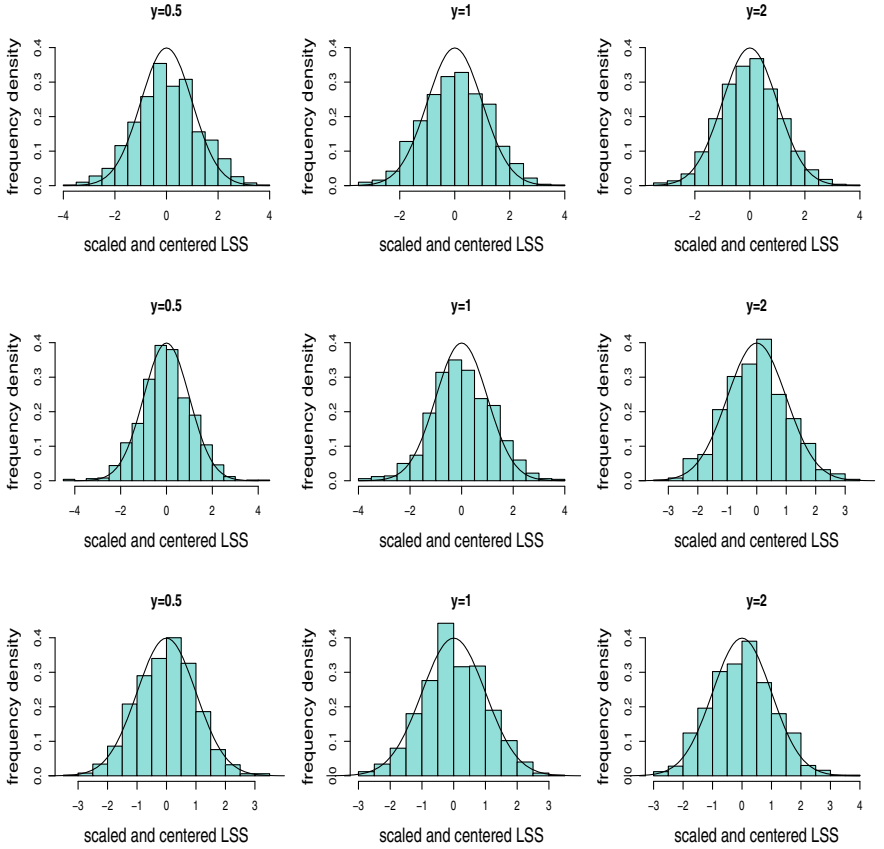


Fig. 7 Histogram (based on 1000 replications) of the centered and scaled $\mathrm{Tr}(AXX^*A)$ (first row), $\mathrm{Tr}\, T_p$ (second row) and $\mathrm{Tr}\, R_p$ (third row) with $A = P_1 + P_{-1} + P_0$, $y = 0.5, 1, 2$, $n = 2000$ and $p = ny$. Observations are standard Gaussian

Pillai and Yin (2014) relaxed the Gaussian assumption and instead assumed the following:

(TW3) Each entry has a sub-exponential decaying tail, that is, there exists $C, \nu > 0$ such that for all sufficiently large u,

$$\mathbb{P}(|x_{(\beta),i,j,n}| > u) \leq Ce^{-u^{\nu}}, \quad \forall\, i, j, n \text{ and } \beta.$$

This has been further generalized as follows. Let $\tilde{S}_{(\beta)} = \frac{1}{n}\Sigma_p^{1/2} X_{(\beta)} X_{(\beta)}^* \Sigma_p^{1/2}$ for $\beta = 1, 2$ where Σ_p is a positive definite matrix of order p. Let $\sigma_{1,p} \geq \sigma_{2,p} \geq \cdots \geq \sigma_{p,p}$ be the eigenvalues of Σ_p and $\hat{\rho}(\cdot)$ be the distribution function of the ESD of Σ_p. We need the following assumption on Σ_p.

(TW4) $\liminf_{p\to\infty} \sigma_{p,p} > 0$, $\limsup_{p\to\infty} \sigma_{1,p} < \infty$ and $\limsup_{n,p\to\infty} \sigma_{1,p}\xi_{n,p} < 1$ where $\xi_{n,p}$ is the unique solution of $\int_{\mathbb{R}} \left(\frac{t\xi_{n,p}}{1-t\xi_{n,p}}\right)^2 d\hat{\rho}(t) = \frac{n}{p}$ in the interval $[0, \sigma_{1,p}^{-1}]$.

Following theorem, due to Lee and Schnelli (2016), states the asymptotic distribution of the maximum eigenvalue of $\tilde{S}_{(\beta)}$.

Theorem 8.2 *Suppose Assumptions (TW1)–(TW4) hold and* $p/n \to y \in (0, \infty)$. *Then for all* $\beta = 1, 2$, *we have*

$$\mathbb{P}\left(\frac{\lambda_{\max}(\tilde{S}_{(\beta)}) - \tilde{\mu}_{n,p}}{n^{-2/3}\tilde{\sigma}_{n,p}^{-1}} \le t\right) \to F_\beta(t) \ \ \forall\, t \in \mathbb{R} \ \textit{where}$$

$$\tilde{\mu}_{n,p} = \frac{1}{\xi_{n,p}}\left(1 + \frac{p}{n}\int_{\mathbb{R}}\left(\frac{t\xi_{n,p}}{1-t\xi_{n,p}}\right) d\hat{\rho}(t)\right) \ \textit{and}$$

$$\tilde{\sigma}_{n,p}^{-3} = \frac{p}{n}\int_{\mathbb{R}}\left(\frac{t\xi_{n,p}}{1-t\xi_{n,p}}\right)^3 d\hat{\rho}(t) + \xi_{n,p}^{-3}.$$

Remark 8.1 If $\Sigma_p = I_p$, then the distribution function $\hat{\rho}(\cdot)$ of the ESD of Σ_p is degenerate at 1. Therefore, the equation $\int_{\mathbb{R}} \left(\frac{t\xi_{n,p}}{1-t\xi_{n,p}}\right)^2 d\hat{\rho}(t) = \left(\frac{\xi_{n,p}}{1-\xi_{n,p}}\right)^2 = \frac{n}{p}$ has only one solution $\xi_{n,p} = \frac{1}{1+\sqrt{p/n}}$ in $[0, 1]$ and consequently $\tilde{\mu}_{n,p} = \mu_{n,p}$ and $\tilde{\sigma}_{n,p}^{-1}\sigma_{n,p} \to 1$. This shows that Theorem 8.1 is a special case of Theorem 8.2 (Fig. 8).

For the largest eigenvalues of R_p and T_p, the following results have been established in Bao (2019b, a). Recall $\mu_{n,p}$ from Theorem 8.1.

Theorem 8.3 *Suppose (RC1)–(RC3) hold and* $p/n \to y \in (0, \infty)$. *Then for all* $t \in \mathbb{R}$,

$$\mathbb{P}\left[n^{2/3}\left(\frac{p}{n}\right)^{1/6}\mu_{n,p}^{-2/3}\left(\lambda_{\max}(R_p) - \mu_{n,p}\right) \le t\right] \to F_1(t) \ \textit{and},$$

$$\mathbb{P}\left[\frac{3}{2}n^{2/3}\left(\frac{p}{n}\right)^{1/6}\mu_{n,p}^{-2/3}\left(\lambda_{\max}(T_p) - \frac{2}{3}\mu_{n,p} - \frac{1}{3}\right) \le t\right] \to F_1(t).$$

9 Spiked Covariance Matrices

Consider the data matrix $Y = AX$ where X is an Independent matrix of order $(p + l) \times n$ and A is a deterministic matrix of order $p \times (p + l)$. Clearly, the variance–covariance matrix of each column of Y is

$$\Sigma = AA^*.$$

Let $\mu_1 \ge \mu_2 \ge \cdots \ge \mu_p$ be the eigenvalues of Σ. The sample covariance matrix $S = n^{-1}YY^* = n^{-1}AXX^*A^*$ has spectral decomposition $S = \hat{V}\hat{\Lambda}\hat{V}^*$ where

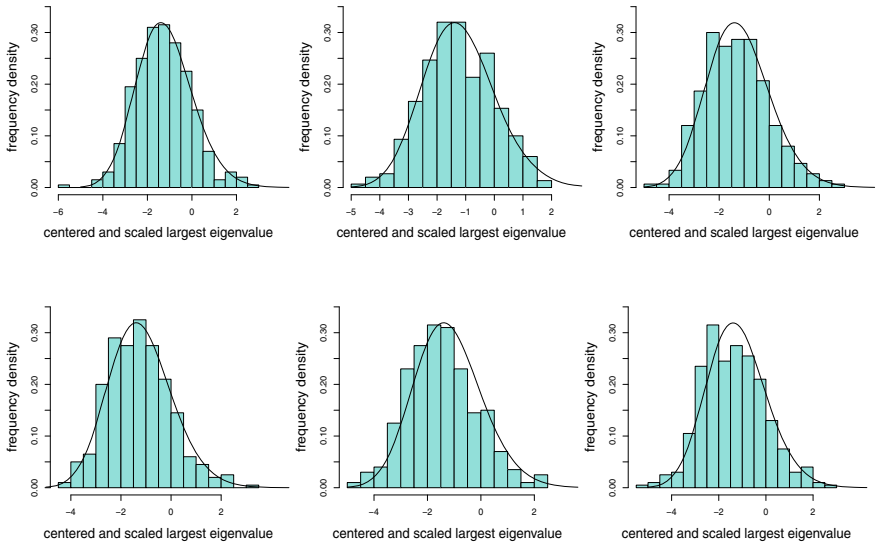


Fig. 8 Histogram of appropriately centered and scaled largest eigenvalues of $S_{(1)}$ based on 500 iterations: first row $\Sigma_p = I_p$; second row Σ_p is diagonal with odd and even diagonal elements as 1 and 2. Column one $n = 200$, $p = 200$; column two, $n = 200$, $p = 400$; column three, $n = 400$, $p = 200$. Density of F_1 is overlaid

$\hat{\Lambda} = \text{diag}(\hat{\mu}_1, \hat{\mu}_2, \ldots, \hat{\mu}_p)$ and $\hat{V}$ is an orthogonal matrix. The columns $\hat{v}_1, \hat{v}_2, \ldots, \hat{v}_p$ of $\hat{V}$ are eigenvectors of S and $\hat{\mu}_1 \geq \hat{\mu}_2 \geq \cdots \geq \hat{\mu}_p$ are eigenvalues of S. In the spiked setting, we have K many spiked eigenvalues which are separated from the rest. More specifically, $\mu_1 \geq \mu_2 \geq \cdots \geq \mu_K$ tend to infinity as $p \to \infty$ while $\mu_{K+1} \geq \mu_{K+2} \geq \cdots \geq \mu_p$ remain bounded.

An example of a spiked model is the factor model where

$$Y = T_1 F + T_2 Z = (T_1 \quad T_2) \begin{pmatrix} F \\ Z \end{pmatrix}$$

where T_1 is a $p \times K$ deterministic matrix of factor loading, F is a $K \times n$ matrix of factors, T_2 is a $p \times p$ deterministic matrix and Z is a $p \times n$ Independent matrix. A common assumption in the factor model is that the singular values of the factor loading matrix T_1 are significantly larger than those of T_2 associated with the noise component, otherwise the signals will be overwhelmed by noise. Note that in this example, $A = (T_1 \quad T_2)$ is not a square matrix, hence it will be necessary to take A as a rectangular matrix.

We shall need some notation. Write the singular value decomposition of A as $A = V\Lambda^{1/2}U$ so that V and U are matrices of order $p \times p$ and $p \times (p + l)$ respectively satisfying $UU^* = VV^* = I_p$ and $\Lambda = \text{Diag}(\mu_1, \mu_2, \ldots, \mu_p)$ is a diagonal matrix. Let u_j be the jth column of U and u_{ij} be the ith entry of u_j. Also x_j denotes the

jth column of X. Partition U as $U^* = (U_1^* \quad U_2^*)$ where U_1 and U_2 are of order $K \times (p+l)$ and $(p-K) \times (p+l)$, respectively. Also write

$$\Lambda = \begin{pmatrix} \Lambda_S & O_{K\times(p-K)} \\ O_{(p-K)\times K} & \Lambda_P \end{pmatrix}, \quad \Lambda_S = \text{diag}(\mu_1, \mu_2, \ldots, \mu_K), \quad \Lambda_P = \text{diag}(\mu_{K+1}, \mu_{K+2}, \ldots, \mu_p).$$

Define $\Sigma_1 = U_2^* \Lambda_P U_2$.

Analysis of spiked covariance matrix was pioneered by the work of Johnstone (2001) and it has been further extended in various directions by Baik et al. (2005), Baik and Silverstein (2006), Bai and Yao (2008), Johnstone and Lu (2009), Jung and Marron (2009), Cai et al. (2013a, 2015) and Wang and Fan (2017).

Paul (2007) established that if $p/n \to y \in (0, \infty)$ and $\mu_1 \le (1 + \sqrt{y})$, then $\hat{v}_1$ is asymptotically almost surely orthogonal to v_1. Thus $\hat{v}_1$ is not a useful estimate of v_1. Even when $\mu_1 > (1 + \sqrt{y})$, the angle between $\hat{v}_1$ and v_1 does not converge to 0 unless $\mu_1 \to \infty$.

Motivated by this, Cai et al. (2020) only considered the case where $\mu_K \to 0$ and established Theorems 9.1–9.4 stated below. Similar results for complex-valued data are also established in Onatski (2009).

Consider the following assumptions.

(S1) Columns of X are i.i.d. and entries of X are independently distributed with mean 0, variance 1 and uniformly bounded fourth order moment.

(S2) $n^{-1}p > C_1 > 0$, $d_i := \frac{p}{n\mu_i} \to 0$ for all $i = 1, 2, \ldots, K$, $\sup_{p \ge 1} \max_{k+1 \le i \le p} |\mu_i| < C_2$ for some $C_2 > 0$, $Kn^{-1/6} \to 0$ and $K^2 d_K \to 0$.

(S2-1) $n^{-1}p \to 0$, $\mu_i \to \infty$ for all $1 \le i \le K$, $p^{-1}K \to 0$, $Kn^{-1/6} \to 0$.

Assumption (S2) or (S2-1) ensures that the spiked eigenvalues of Σ diverges to ∞ and restricts appropriately the growth of p, n, K and the largest non-spiked eigenvalue μ_{K+1} of Σ.

The following theorem states that $\mu_i^{-1}\hat{\mu}_i \to 1$ in probability and provides a uniform convergence rate for all $1 \le i \le K$.

Theorem 9.1 *Suppose (S1), and either of (S2) or (S2-1) hold. Then, uniformly for all* $1 \le i \le K$, *we have*

$$\frac{\hat{\mu}_i}{\mu_i} - 1 = O_{\mathbb{P}}(d_i + n^{-1}K^4 + \mu_i^{-1}).$$

Note that $\mu_i^{-1}\hat{\mu}_i \to 1$ in probability does not imply that $\hat{\mu}_i$ is a good estimator of μ_i due to the fact that μ_i itself tends to ∞. Also Theorem 9.1 does not provide any insight about how the leading eigenvalues of sample covariance matrices are affected by the non-spiked eigenvalues of Σ. This insight will be provided in Theorems 9.2 and 9.3. Consider the following assumptions. Theorem 9.2 considers the case where all the spiked eigenvalues of Σ have multiplicity exactly 1. Theorem 9.3 allows the multiplicity to be more than 1.

(S3) There exists a positive constant C not depending on n and p such that $\mu_{i-1}\mu_i^{-1} \geq C > 1$ for all $i = 2, 3, \ldots, K$.

(S4) Let $\gamma_{4i} = \mathrm{E}(x_{i,j,p})^4$. The following limits exist:

$$\tilde{\sigma}_{ii} = \lim_{p\to\infty}\Big[\sum_{s=1}^{p+l}(\gamma_{4s}-3)u_{is}^4 + 2\Big] \quad \text{and} \quad \tilde{\sigma}_{ij} = \lim_{p\to\infty}\sum_{s=1}^{p+l}(\gamma_{4s}-3)u_{is}^2u_{js}^2 \ \text{ for } i \neq j.$$

Assumption (S3) ensures that the spiked eigenvalues of Σ are separated enough from each other. Limits in (S4) appear as the asymptotic variances and covariances of $\hat{\mu}_i$ for $1 \leq i \leq K$.

For any $\theta \neq 0$, let $\tilde{m}_\theta(z)$ be the unique solution of the following equation:

$$\tilde{m}_\theta(z) = -\left(z - n^{-1}\,\mathrm{Tr}\left(I + \tilde{m}_\theta(z)\frac{\Sigma_1}{\theta}\right)^{-1}\frac{\Sigma_1}{\theta}\right)^{-1}, \quad z \in \mathbb{C}^+, \tag{9.1}$$

$$\tilde{m}_\theta(1) = \lim_{z\in\mathbb{C}^+, z\to 1}\tilde{m}_\theta(z). \tag{9.2}$$

Let θ_i be the solution of $\tilde{m}_{\theta_i}(1) + \mu_i^{-1}\theta_i = 0$.

Theorem 9.2 *Suppose (S1), (S3), (S4) and either of (S2) or (S2-1) hold. Then*

$$\sqrt{n}\left(\frac{\hat{\mu}_1-\theta_1}{\theta_1}, \frac{\hat{\mu}_2-\theta_2}{\theta_2}, \ldots, \frac{\hat{\mu}_K-\theta_K}{\theta_K}\right) \xrightarrow{\mathcal{D}} \mathcal{N}(0, \tilde{\Sigma})$$

where the (i, j)th element of $\tilde{\Sigma}$ is $\tilde{\sigma}_{ij}$. Moreover,

$$|v_i^*\hat{v}_j| \xrightarrow{\mathbb{P}} \mathbb{I}(i = j) \ \textit{for all}\ 1 \leq i, j \leq K. \tag{9.3}$$

Equation (9.3) shows that the sample eigenvector $\hat{v}_i$ is a good estimator of v_i up to a sign difference for $1 \leq i \leq K$.

Let the distinct values in $\{\mu_1, \mu_2, \ldots, \mu_K\}$ in descending order be $\alpha_1, \alpha_2, \ldots, \alpha_L$. Let the multiplicity of α_i be n_i. Each n_i is fixed and does not vary with n.

The following assumptions are required to state the next result.

(S5) There exists a positive constant C not depending on n and p such that $\alpha_{i-1}\alpha_i^{-1} \geq C > 1$ for all $i = 2, 3, \ldots, L$. Also $Kn^{-1/6} \to 0$.

(S6) The following limit exists:

$$G(r_i, k_1, k_2, l_1, l_2) = \lim_{n\to\infty} n^2 \times \mathbb{C}\mathrm{ov}(u'_{r_i+k_1}x_1u'_{r_i+l_1}x_1, u'_{r_i+k_2}x_1u'_{r_i+l_2}x_1),$$

$$\text{where} \quad r_i = \sum_{j=0}^{i-1} n_j, \quad n_0 = 0, \quad 1 \leq k_1, k_2, l_1, l_2 \leq n_i, \quad 1 \leq i \leq L.$$

Assumption (S5) guarantees that distinct spiked eigenvalues of Σ are far enough from each other and their multiplicities are finite. The limits in (S6) appear as asymptotic variance and covariances in the following theorem.

Theorem 9.3 *Suppose (S1), (S5), (S6) and either of (S2) or (S2-1) hold. Then*

$$\sqrt{n}\tilde{\theta}_i^{-1}(\hat{\mu}_{r_i+1}-\tilde{\theta}_i,\hat{\mu}_{r_i+2}-\tilde{\theta}_i,\ldots,\lambda_{r_i+n_i}-\tilde{\theta}_i)\xrightarrow{\mathcal{D}}\mathcal{R}_i \quad \textit{where} \quad \tilde{\theta}_i=\alpha_i\left(1+\frac{p-K}{n}\int\frac{t\,dF_{\Lambda_p}(t)}{\alpha_i-t}\right),$$

where F_{Λ_p} is the ESD of Λ_p, $\mathcal{R}_i$ are the eigenvalues of an $n_i\times n_i$ Gaussian matrix $\mathcal{G}_i$ whose entries have mean 0 and covariances between (k_1,l_1)th and (k_2,l_2)th entries of $\mathcal{G}$ is $G(r_i,k_1,k_2,l_1,l_2)$.

Moreover, for all $1\le i\le K-1$, we have

$$\sum_{j=r_i+1}^{r_i+n_i}(v_i^*\hat{v}_j)^2\xrightarrow{\mathbb{P}}1,\quad\forall\ i\in\{r_i+1,r_i+2,\ldots,r_i+n_i\},$$
$$\sum_{j=r_i+1}^{r_i+n_i}(v_i^*\hat{v}_j)^2\xrightarrow{\mathbb{P}}0,\quad\forall\ 1\le i\le r_i,\ r_{i+1}+1\le i\le K. \tag{9.4}$$

Equation (9.4) shows that the sample eigenvectors $\{\hat{v}_j:\ j=r_j+1,r_j+2,\ldots,r_j+n_j\}$ are close to the space spanned by $\{v_j:\ j=r_j+1,r_j+2,\ldots,r_j+n_j\}$.

The following theorem states that the appropriately centered and scaled largest non-spiked eigenvalue of S converges weakly to the Tracy–Widom law F_1. Let $\mathbb{C}^+=\{a+ib:\ a\in\mathbb{R},b>0\}$.

(S7) Let $m_{\Sigma_1}(z)$ be the Stieltjes transform of the LSD of $X^*\Sigma_1X$ and let γ_+ be the right most end point of the LSD. Suppose that $\limsup_{n,p}d\mu_{K+1}<1$ where $d=-\lim_{z\in\mathbb{C}^+\to\gamma_+}m_{\Sigma_1}(z)$.

Theorem 9.4 *Suppose Assumptions I, (S2), (S7) and either of (S3) or (S5) hold and $ln^{-1/6}\to 0$, $np^{-1}>C$ for some $C>0$. Then (Table 6)*

$$\mathbb{P}(n^{2/3}(\varrho_n)^{-1}(\hat{\mu}_{K+1}-\gamma_+)\le t)\to F_1(t),\quad\forall\ t\in\mathbb{R}\ \textit{where}$$
$$\varrho_n^3=d^{-3}\left(1+\frac{p-K}{n}\int\left(\frac{td}{1-td}\right)^3dF_{\Lambda_p}(t)\right).$$

Table 6 Values of the (mean, variance) based on 1000 iterations of $|v_i^* \hat{v}_j|$ for $1 \le i, j \le 3$ in the same model as in Fig. 9

	$\lvert\hat{v}_1^* v_1\rvert$	$\lvert\hat{v}_1^* v_2\rvert$	$\lvert\hat{v}_1^* v_3\rvert$
$y = 0.5$	(0.98, 0.004)	(0.0003, 0.003)	(0.0004, 0.005)
$y = 1$	(0.99, 0.006)	(0.0003, 0.002)	(0.0005, 0.008)
$y = 2$	(0.98, 0.002)	(0.0002, 0.003)	(0.0002, 0.004)
	$\lvert\hat{v}_2^* v_1\rvert$	$\lvert\hat{v}_2^* v_2\rvert$	$\lvert\hat{v}_2^* v_3\rvert$
$y = 0.5$	(0.0002, 0.008)	(0.99, 0.003)	(0.0002, 0.003)
$y = 1$	(0.0004, 0.003)	(0.99, 0.005)	(0.0006, 0.02)
$y = 2$	(0.0005, 0.003)	(0.99, 0.007)	(0.0005, 0.002)
	$\lvert\hat{v}_3^* v_1\rvert$	$\lvert\hat{v}_3^* v_2\rvert$	$\lvert\hat{v}_3^* v_3\rvert$
$y = 0.5$	(0.0004, 0.007)	(0.0003, 0.008)	(0.99, 0.004)
$y = 1$	(0.0004, 0.005)	(0.0003, 0.005)	(0.98, 0.004)
$y = 2$	(0.0007, 0.003)	(0.0004, 0.002)	(0.99, 0.002)

10 Applications

In this section we discuss some statistical applications.

10.1 Test of Independence

Let $\{X_i = (x_{1i}, x_{2i}, \ldots x_{pi})' : \ 1 \le i \le n\}$ be a collection of i.i.d. random vectors where $\{x_{ij}\}$ have mean 0 and variance 1. Consider the hypothesis

$$H_0 : \quad \text{the collections } \{(x_{ij} : \ 1 \le i \le n\} \text{ are independent across } 1 \le j \le p.$$

Several test statistics have been used to test the above hypothesis. These are based on largest eigenvalue, Frobenius norm, logarithm of determinant and largest off-diagonal entries of sample covariance matrix S, Spearman's ρ matrix R and Kendall's τ matrix T computed for the observed data matrix $X = ((x_{ji}))_{p\times n}$. We list below some of them:

$$\mathbb{T}_{11} = \frac{n\lambda_{\max}(S) - (\sqrt{p} + \sqrt{n})^2}{(\sqrt{n} + \sqrt{p})(p^{-1/2} + n^{-1/2})^{1/3}}, \quad \mathbb{T}_{12} = \frac{\mathrm{Tr}(S^2) - \mathrm{E}Tr(S^2)}{\sqrt{\mathrm{Var}(\mathrm{Tr}(S^2))}},$$

$$\mathbb{T}_{13} = \frac{\mathrm{Tr}(S^4) - \mathrm{E}(\mathrm{Tr}(S^4))}{\sqrt{\mathrm{Var}(\mathrm{Tr}(S^4))}}, \quad \mathbb{T}_{14} = \frac{\log(\mathrm{Det}(S)) - \mathrm{E}(\log(\mathrm{Det}(S)))}{\sqrt{\mathrm{Var}(\log(\mathrm{Det}(S)))}},$$

$$\mathbb{T}_{15} = n\left(\max_{1\le i<j\le p} |s_{ij}|\right)^2 - 4\log p + \log\log p, \quad \mathbb{T}_{16} = \mathbb{T}_{12} + n^{-1/2}\mathbb{T}_{15},$$

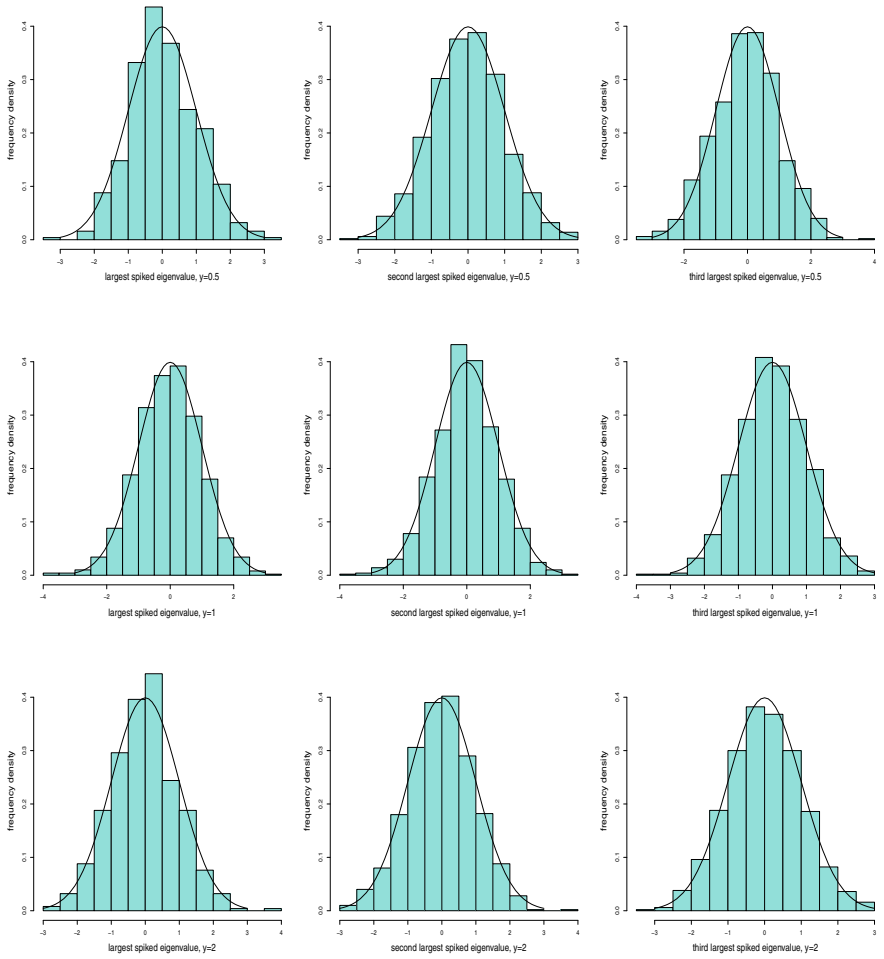


Fig. 9 Histograms (based on 1000 replications) of $\sqrt{n}(\sqrt{2}\theta_i)^{-1}(\hat{\mu}_i - \theta_i)$ where $\theta_i = 3^{3-i}\sqrt{n}\left(1 + n^{-1}(p-3)(3^{3-i}\sqrt{n}-1)^{-1}\right)$, $i = 1, 2, 3$ with the choice $K = 3$, $\Lambda_S = n^{1/2}\text{diag}(9, 3, 1)$, $\Lambda_P = I_{p-3}$, $\Sigma = V\Lambda V^*$ where V is an orthogonal matrix which diagonalizes $\Omega = ((0.5^{|i-j|}))$. The entries of X are i.i.d. standard Gaussian, $n = 1000$, $y = 0.5, 1, 2$, $p = ny$. Standard normal density function is also added with each histogram for showing their closeness

$$\mathbb{T}_{21} = \frac{n\lambda_{\max}(R) - (\sqrt{p}+\sqrt{n})^2}{(\sqrt{n}+\sqrt{p})(p^{-1/2}+n^{-1/2})^{1/3}}, \quad \mathbb{T}_{22} = \frac{\text{Tr}(R^2) - \text{E}(\text{Tr}(R^2))}{\sqrt{\text{Var}(\text{Trace}(R^2))}},$$
$$\mathbb{T}_{23} = \frac{\text{Tr}(R^4) - \text{E}(\text{Tr}(R^4))}{\sqrt{\text{Var}(\text{Tr}(R^4))}}, \quad \mathbb{T}_{24} = \frac{\log(\text{Det}(R)) - \text{E}(\log(\text{Det}(R)))}{\sqrt{\text{Var}(\log(\text{Det}(R)))}},$$
$$\mathbb{T}_{25} = n\left(\max_{1\le i<j\le p}|r_{ij}|\right)^2 - 4\log p + \log\log p, \quad \mathbb{T}_{26} = \mathbb{T}_{24} + n^{-1/2}\mathbb{T}_{25},$$

$$\mathbb{T}_{31} = \frac{(n/2)(3\lambda_{\max}(T) - 1) - (\sqrt{p} + \sqrt{n})^2}{(\sqrt{n} + \sqrt{p})(p^{-1/2} + n^{-1/2})^{1/3}}, \quad \mathbb{T}_{32} = \frac{\mathrm{Tr}(T^2) - \mathrm{E}(\mathrm{Tr}(T^2))}{\sqrt{\mathrm{Var}(\mathrm{Tr}(T^2))}},$$
$$\mathbb{T}_{33} = \frac{\mathrm{Tr}(T^4) - \mathrm{E}(\mathrm{Tr}(T^4))}{\sqrt{\mathrm{Var}(\mathrm{Tr}(T^4))}}, \quad \mathbb{T}_{34} = \frac{\log(\mathrm{Det}(T)) - \mathrm{E}(\log(\mathrm{Det}(T)))}{\sqrt{\mathrm{Var}(\log(\mathrm{Det}(T)))}},$$
$$\mathbb{T}_{35} = n\left(\max_{1\le i<j\le p} |t_{ij}|\right)^2 - 4\log p + \log\log p$$

where s_{ij}, r_{ij} and t_{ij} are the (i, j)-th entry of S, R and T, respectively. The following limit results are known for the above statistics under H_0. We omit the details of the assumptions under which these results hold.

(i) The test statistics $\mathbb{T}_{i1}$, $1 \le i \le 3$, converge weakly to the Tracy–Widom law with $\beta = 1$. This follows by Theorems 8.1 and 8.3.

(ii) The test statistics $\mathbb{T}_{ij}$, $1 \le i \le 3$, $j = 2, 3, 4$, converge weakly to the standard normal distribution by Theorems 7.1, 7.2 and 7.3. Simplified expressions for some of the centering and scaling in $\{\mathbb{T}_{ij} : 1 \le i \le 3, \ j = 2, 3, 4\}$ can be found in Pan and Zhou (2008), Bao et al. (2015) and Li et al. (2021). Otherwise, their suitable bootstrap estimates by simulating several independent S, R and T matrices under H_0 can be used instead of original centering and scaling.

(iii) Asymptotic distribution of $\mathbb{T}_{i5}$ is known to be the Gumbel distribution and that of $\mathbb{T}_{i6}$ to be the standard normal distribution. These follow from the works of Jiang (2004) and Zhou (2007) for $i = 1, 2$, respectively.

(iv) There does not appear to be any result known for the weak convergence of $\mathbb{T}_{35}$. However, a critical region based on $\mathbb{T}_{35}$ has been proposed using the multiplier bootstrap technique by Chen (2018).

(v) The test statistic analogous to $\mathbb{T}_{16}$ and $\mathbb{T}_{26}$ for the matrix T does not appear to have been studied in the literature.

(vi) The statistics defined above for the S matrix have obvious analogues for the sample correlation matrix. The asymptotic results for these statistics are very similar to those for the S matrix. See Jiang (2004), Schott (2005), Zhou (2007), Bao et al. (2012), Jiang and Yang (2013) and Gao et al. (2017) for the details.

It follows from the discussions above and those in Sects. 7 and 8 that the test statistics $\mathbb{T}_{ij}$ are expected to be larger in their absolute values under the alternative hypotheses. Thus we may reject H_0 for large values of $|\mathbb{T}_{ij}|$.

Simulations. Let J_p denote the $p \times p$ matrix whose all elements are 1. Consider a $p \times n$ matrix $Y = ((y_{ij}))$ with i.i.d. mean 0 and variance 1 entries. Define $X = \Sigma^{1/2} Y$ where Σ is a $p \times p$ positive definite matrix. Consider the following specifications for Σ and the distributions of y_{ij}. Below we choose different values of $c \ge 0$.

Model 1-c-1: $y_{ij} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1)$ and $\Sigma = (1 - p^{-1}4c)I_p + p^{-1}4cJ_p$.

Model 2-c-1: $y_{ij} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1)$ and $\Sigma = I_p + c((n^{1/4}\mathbb{I}(\max(i,j) = 2, \min(i,j) = 1)))$.

Model 1-c-2: $y_{ij} \overset{\text{i.i.d.}}{\sim} \sqrt{\frac{31}{33}}t_{33}$ and $\Sigma = (1 - p^{-1}4c)I_p + p^{-1}4cJ_p$.

Model 2-c-2: $y_{ij} \overset{\text{i.i.d.}}{\sim} \sqrt{\frac{31}{33}}t_{33}$ and $\Sigma = I_p + c((n^{1/4}\mathbb{I}(\max(i,j) = 2, \min(i,j) = 1)))$.

Model 1-c-3: $y_{ij} \overset{\text{i.i.d.}}{\sim} \sqrt{\frac{15}{17}}t_{17}$ and $\Sigma = (1 - p^{-1}4c)I_p + p^{-1}4cJ_p$.

Model 2-c-3: $y_{ij} \overset{\text{i.i.d.}}{\sim} \sqrt{\frac{15}{17}}t_{17}$ and $\Sigma = I_p + c((n^{1/4}\mathbb{I}(\max(i,j) = 2, \min(i,j) = 1)))$.

Model 1-c-4: $y_{ij} \overset{\text{i.i.d.}}{\sim} \sqrt{\frac{1}{2}}t_4$ and $\Sigma = (1 - p^{-1}4c)I_p + p^{-1}4cJ_p$.

Model 2-c-4: $y_{ij} \overset{\text{i.i.d.}}{\sim} \sqrt{\frac{1}{2}}t_4$ and $\Sigma = I_p + c((n^{1/4}\mathbb{I}(\max(i,j) = 2, \min(i,j) = 1)))$.

In the above, t_i denotes a t distributed random variable with i degrees of freedom. The various choices of i are based on the asymptotic distribution results. Note that Models 1-0-i and 2-0-i are identical for all $i = 1, 2, 3, 4$. Moreover, $c = 0$ corresponds to the null hypothesis.

Based on the numerical results displayed in Tables 7, 8 and 9, we draw the following conclusions.

Table 7 Empirical size (1000 iterations) for Models 1-0-i with $i = 1, 2, 3$ and empirical power (1000 iterations) for Models j-1-i with $j = 1, 2, i = 1, 2, 3$ based on the test statistics $\mathbb{T}_{1i}, 1 \leq i \leq 6$, when $p = n = 1000$

	1-0-1	1-0-2	1-0-3	1-0-4		1-1-1	1-1-2	1-1-3	1-1-4
$\mathbb{T}_{11}$	0.042	0.022	0.015	0.017	$\mathbb{T}_{11}$	0.93	0.48	0.39	0.33
$\mathbb{T}_{12}$	0.049	0.047	0.046	0.017	$\mathbb{T}_{12}$	0.98	0.97	0.98	0.45
$\mathbb{T}_{13}$	0.045	0.043	0.017	0.02	$\mathbb{T}_{13}$	0.94	0.97	0.37	0.34
$\mathbb{T}_{14}$	0.044	0.044	0.012	0.014	$\mathbb{T}_{14}$	0.94	0.48	0.33	0.39
$\mathbb{T}_{15}$	0.044	0.016	0.023	0.019	$\mathbb{T}_{15}$	0.85	0.42	0.45	0.39
$\mathbb{T}_{16}$	0.049	0.022	0.017	0.015	$\mathbb{T}_{16}$	0.98	0.44	0.35	0.48
	2-1-1	2-1-2	2-1-3	2-1-4					
$\mathbb{T}_{11}$	0.9	0.42	0.44	0.39					
$\mathbb{T}_{12}$	0.88	0.85	0.86	0.52					
$\mathbb{T}_{13}$	0.85	0.84	0.46	0.48					
$\mathbb{T}_{14}$	0.82	0.48	0.42	0.34					
$\mathbb{T}_{15}$	0.98	0.45	0.35	0.39					
$\mathbb{T}_{16}$	0.99	0.52	0.47	0.42					

Table 8 Empirical size (1000 iterations) for Models 1-0-i with $i = 1, 2, 3$ and empirical power (1000 iterations) for Models j-1-i with $j = 1, 2, i = 1, 2, 3$ based on the test statistics $\mathbb{T}_{2i}, 1 \leq i \leq 6$, when $p = n = 1000$

	1-0-1	1-0-2	1-0-3	1-0-4		1-1-1	1-1-2	1-1-3	1-1-4
$\mathbb{T}_{21}$	0.045	0.045	0.054	0.044	$\mathbb{T}_{21}$	0.92	0.94	0.94	0.93
$\mathbb{T}_{22}$	0.049	0.048	0.048	0.048	$\mathbb{T}_{22}$	0.98	0.98	0.99	0.98
$\mathbb{T}_{23}$	0.047	0.047	0.053	0.047	$\mathbb{T}_{23}$	0.98	0.97	0.98	0.97
$\mathbb{T}_{24}$	0.045	0.046	0.046	0.045	$\mathbb{T}_{24}$	0.98	0.97	0.97	0.97
$\mathbb{T}_{25}$	0.049	0.051	0.048	0.051	$\mathbb{T}_{25}$	0.86	0.83	0.87	0.87
$\mathbb{T}_{26}$	0.049	0.052	0.049	0.052	$\mathbb{T}_{26}$	0.98	0.97	0.98	0.99
	2-1-1	2-1-2	2-1-3	2-1-4					
$\mathbb{T}_{21}$	0.89	0.87	0.9	0.88					
$\mathbb{T}_{22}$	0.87	0.85	0.88	0.87					
$\mathbb{T}_{23}$	0.88	0.89	0.87	0.88					
$\mathbb{T}_{24}$	0.9	0.85	0.86	0.85					
$\mathbb{T}_{25}$	0.97	0.98	0.98	0.97					
$\mathbb{T}_{26}$	0.98	0.98	0.98	0.98					

Table 9 Empirical size (1000 iterations) for Models 1-0-i with $i = 1, 2, 3$ and empirical power (1000 iterations) for Models j-1-i with $j = 1, 2, i = 1, 2, 3$ based on the test statistics $\mathbb{T}_{3i}, 1 \leq i \leq 5$, when $p = n = 1000$

	1-0-1	1-0-2	1-0-3	1-0-4		1-1-1	1-1-2	1-1-3	1-1-4
$\mathbb{T}_{31}$	0.047	0.046	0.053	0.054	$\mathbb{T}_{31}$	0.94	0.95	0.94	0.94
$\mathbb{T}_{32}$	0.049	0.05	0.05	0.049	$\mathbb{T}_{32}$	0.99	0.99	0.99	0.99
$\mathbb{T}_{33}$	0.054	0.047	0.046	0.053	$\mathbb{T}_{33}$	0.99	0.98	0.98	0.98
$\mathbb{T}_{34}$	0.045	0.054	0.053	0.047	$\mathbb{T}_{34}$	0.98	0.98	0.97	0.97
$\mathbb{T}_{35}$	0.051	0.052	0.048	0.049	$\mathbb{T}_{35}$	0.87	0.85	0.85	0.88
	2-1-1	2-1-2	2-1-3	2-1-4					
$\mathbb{T}_{31}$	0.85	0.88	0.86	0.88					
$\mathbb{T}_{32}$	0.88	0.84	0.87	0.85					
$\mathbb{T}_{33}$	0.88	0.85	0.88	0.85					
$\mathbb{T}_{34}$	0.88	0.87	0.88	0.85					
$\mathbb{T}_{35}$	0.99	0.98	0.99	0.98					

1. All the test statistics perform much better when the observations are Gaussian compared to when they are t-distributed.
2. Performance of the test statistics, based on the S matrix, seems to be poorer for lesser degrees of freedom for the t-distributions.
3. The test statistics, based on Spearman's ρ matrix and Kendall's τ matrix, turn out to be robust to the behavior of the tail of the distribution.

4. Performance of the test statistics, based on Spearman's ρ matrix and Kendall's τ matrix, are equivalent with a slight edge for Kendall's τ matrix.
5. Model 1-1-j, for $1 \leq j \leq 4$, corresponds to an alternative hypothesis and contains many small disturbances which are responsible for the deviation from the null hypothesis. In this case, for each $1 \leq i \leq 3$, the test statistics $\{\mathbb{T}_{ij} : \ j = 2, 3, 4\}$ perform better than $\mathbb{T}_{i5}$. Also $\mathbb{T}_{i2}$ outperforms $\mathbb{T}_{i3}$ and $\mathbb{T}_{i4}$ for all $i = 1, 2, 3$.
6. Model 2-1-j, for $1 \leq j \leq 4$, corresponds to an alternative hypothesis and contains only a single large disturbance which is responsible for the deviation from the null hypothesis. In this case, for each $1 \leq i \leq 3$, the test statistics $\{\mathbb{T}_{ij} : \ j = 2, 3, 4\}$ perform worse than $\mathbb{T}_{i5}$.
7. For $i = 1, 2$, the test statistic $\mathbb{T}_{i6}$ is a combination of $\mathbb{T}_{i2}$ and $\mathbb{T}_{i5}$. Its performance is equivalent to $\mathbb{T}_{i2}$ for Models 1-1-j and equivalent to $\mathbb{T}_{i5}$ for Models 2-1-j, $i = 1, 2$, $j = 1, 2, 3, 4$. In the absence of any prior knowledge about the disturbances, these are recommended over the others.

10.2 Estimating the Number of Spikes

Identifying the number of spikes is an important problem for a range of statistical applications, including determining the number of significant principal components and the number of factors in a factor model. As an application of the results discussed in Sect. 9, a procedure to identify the number of spiked eigenvalues is proposed in Cai et al. (2020).

We shall estimate the number of spikes K through an iterative procedure, based on Theorem 9.4. From that theorem, one can observe that $[\hat{\mu}_{K+1} - \omega^* \varrho_n n^{-2/3}, \hat{\mu}_{K+1} + \omega^* \varrho_n n^{-2/3}]$ is a confidence interval for γ_+. Here ω^* is a suitable critical value determined from the Tracy–Widom law with $\beta = 1$. This, together with Theorem 9.2, implies that the number of eigenvalues of S which lie beyond $(\gamma_+ + \omega^* \varrho_n n^{-2/3} \log n)$ is an estimate of the number of spikes K. Incidentally, the quantity $\log n$ can be replaced by any sequence tending to ∞. In practice γ_+ and ϱ_n are unknown and need to be estimated.

It can be argued (details are omitted) that an estimator of ϱ_n is given by

$$\hat{\varrho}_n = \left(- \frac{\frac{1}{n-[n^{1/6}]} \sum_{i=[n^{1/6}]+1}^{n} \frac{1}{(\hat{\mu}_i - z_0)^3}}{\left(\frac{1}{n-[n^{1/6}]} \sum_{i=[n^{1/6}]+1}^{n} \frac{1}{(\hat{\mu}_i - z_0)^3} \right)^3} \right)^{1/3} \quad \text{where} \quad z_0 = \hat{\mu}_{[n^{1/6}]} + n^{-4/9}.$$

An initial estimate for γ_+ is then given by

$$\hat{\gamma}_{+0} = \hat{\mu}_{[n^{1/6}]} + n^{-5/9} \log n. \tag{10.1}$$

Table 10 Performance of $\hat{K}$ is recorded. Each quartile is computed based on 1000 iterations and their mean and variances are computed based on 500 observations. The model and the parameters are as in Fig. 9

	$y = 0.5$	$y = 1$	$y = 2$
(Mean of median, variance of median)	(3, 0.003)	(3, 0.002)	(3, 0.005)
(Mean of 1st quartile, variance of 1st quartile)	(2.996, 0.003)	(2.998, 0.005)	(2.994, 0.006)
(Mean of 3rd quartile, variance of 3rd quartile)	(3.003, 0.004)	(3.004, 0.003)	(3.004, 0.008)

By Theorem 9.4, there is at least one eigenvalue in the interval $[\hat{\gamma}_{+0}, \hat{\gamma}_{+0} + \omega^* m_n \varrho_n n^{-2/3}]$, where $m_n \to \infty$. We can now describe the iterative steps to obtain an estimate of K.

Step 1: Define the initial value $\hat{\gamma}_{+0}$ as in (10.1).

Step 2: Suppose that we have computed $\hat{\gamma}_{+(m-1)}$. If there is at least one eigenvalue of S in the interval $(\hat{\gamma}_{+(m-1)}, \hat{\gamma}_{+(m-1)} + 2.02(\log n)\varrho_n n^{-2/3})$, where 2.02 is the 99% quantile of Tracy–Widom law F_1, we compute $\hat{\gamma}_{+m} = \hat{\gamma}_{+(m-1)} + 2.02 \log n \varrho_n n^{-2/3}$. Here $\log n$ can be also replaced by the other sequence tending to infinity too. Otherwise the iteration stops.

Step 3: After getting $\hat{\gamma}_{+m}$, we return to Step 2 until the iteration stops.

Step 4: Denote the final value of the above iteration by $\hat{\gamma}_{+\text{end}}$. We define $\hat{K}$ to be the number of eigenvalues larger than $\hat{\gamma}_{+\text{end}}$.

Cai et al. (2020) established that, under the conditions of Theorems 9.2 and 9.4, we have $\hat{K} = K$ with high probability (Table 10).

10.3 Testing for High-Dimensional Time Series

Asymptotic normality of trace of polynomials in sample autocovariance matrices helps to test some hypotheses in linear time series models. Throughout this section, we assume that $p/n \to y \in (0, \infty)$. Similar result for $p/n \to 0$ can be found in Bose and Bhattacharjee (2018) and Bhattacharjee and Bose (2019). Weak convergence of the following test statistics $\mathbb{T}_i, i = 4, 5, 6, 7$, under both null and alternative hypotheses, follow from the asymptotic normality of trace of polynomials derived in Bhattacharjee and Bose (2016b, 2017).

One sample tests: Consider the model (5.2) with $\psi_{0,p} = I_p$. We write ψ_j for $\psi_{j,p}$. Let ε_{it} be the ith component of ε_t. Suppose $\{\varepsilon_{it}\}$ are independent across both t and i

Table 11 Empirical size for $c = 0$ and empirical power for $c = 1$ based on 1000 iterations for $y = 0.5, 1, 2$, $p = ny$, $n = 500, 1000, 2000$. The model is (5.2) where $\{\varepsilon_{ij}\}$ are standard normal, $q = 3$ and $\psi_k = ((0.5^{|i-j|}\mathbb{I}(|i-j| = 0, k))$. We consider $A = \psi_3 + c\psi_2$. Note that $c = 0$ corresponds to the null model

$c = 0$	$y = 0.5$	$y = 1$	$y = 2$	$c = 1$	$y = 0.5$	$y = 1$	$y = 2$
$n = 500$	0.042	0.055	0.045	$n = 500$	0.95	0.97	0.96
$n = 1000$	0.053	0.047	0.047	$n = 1000$	0.97	0.98	0.98
$n = 2000$	0.048	0.049	0.051	$n = 2000$	1	1	1

and $\sup_{n,p} \max_{i,t} \mathrm{E}|\varepsilon_{it}|^k \leq C^k$ for some $C > 0$ and all $k \geq 1$. Also assume that $\{\psi_j\}$ are norm-bounded and converge jointly.

Example 10.1 Let A be a $p \times p$ matrix with a non-degenerate LSD. Suppose we wish to test

$$H_0 : \psi_q = A \text{ against } H_1 : \psi_q \neq A.$$

As $\Gamma_q = \psi_q^*$, this is equivalent to testing

$$H_0 : \Gamma_q = A^* \text{ against } H_1 : \Gamma_q \neq A^*.$$

One can use the test statistic

$$\mathbb{T}_4 = \big(\mathrm{Tr}(\hat{\Gamma}_q) - \mathrm{Tr}(A)\big)/\hat{\lambda}_{1,n} \quad \text{where } \hat{\lambda}_{1,n}^2 \text{ is consistent for } \mathrm{Var}(\mathrm{Tr}(\hat{\Gamma}_q)).$$

Then $\mathbb{T}_4$ converges weakly to $\mathcal{N}(0, 1)$ under H_0. Under H_1, the statistic $\mathbb{T}_4$ with centering replaced by $\mathrm{Tr}(\Gamma_q)$ converges weakly to the standard normal distribution. Therefore, we can reject H_0 for large values of $|\mathbb{T}_4|$ (Table 11).

Example 10.2 Let A_j, $1 \leq j \leq q$, be $p \times p$ matrices with non-degenerate LSDs. Suppose we wish to test

$$H_0 : \psi_j = A_j \ \forall\ 1 \leq j \leq q \quad \text{against } H_1 : \psi_j \neq A_j \text{ for at least one } 1 \leq j \leq q.$$

Let $\hat{G}_q = \sum_{j=1}^q \hat{\Gamma}_j$, $G_q = \sum_{j=1}^q \Gamma_j$ and $\hat{\lambda}_{2,n}^2$ be consistent for the variance of trace of $\hat{G}_q$. Consider the test statistic

$$\mathbb{T}_5 = \big(\mathrm{Tr}(\hat{G}_q) - \mathrm{Tr}(G_q)\big)/\hat{\lambda}_{2,n}.$$

Then $\mathbb{T}_5$ converges weakly to $\mathcal{N}(0, 1)$ under H_0. Under H_1, the statistic $\mathbb{T}_5$ with centering replaced by $\mathrm{Tr}\, G_q$ converges weakly to the standard normal distribution. Therefore, we can reject H_0 for large values of $|\mathbb{T}_5|$ (Table 12).

Table 12 We consider the model (5.2) with standard normal $\{\varepsilon_{ij}\}, q = 3$ and $\psi_k = ((0.5^{|i-j|}\mathbb{I}(|i - j| = 0, k))$. We consider $A_k = \psi_k + c\psi_{k-1}, k = 1, 2, 3$. Note that $c = 0$ corresponds to null model and otherwise it is an alternative model. We record empirical size for $c = 0$ and empirical power for $c = 1$ based on 1000 iterations for $y = 0.5, 1, 2$, $p = ny, n = 500, 1000, 2000$

$c = 0$	$y = 0.5$	$y = 1$	$y = 2$	$c = 1$	$y = 0.5$	$y = 1$	$y = 2$
$n = 500$	0.046	0.045	0.054	$n = 500$	0.96	0.95	0.96
$n = 1000$	0.053	0.052	0.047	$n = 1000$	0.99	0.98	0.98
$n = 2000$	0.05	0.049	0.049	$n = 2000$	1	1	1

Two sample tests: Consider the model (5.2) with $\psi_{0,p} = I_p$ and

$$Y_{t,p}^{(n)} = \eta_t + \sum_{j=1}^{q} \Psi_{j,p}\eta_{t-j}, \quad n, p \geq 1. \tag{10.2}$$

Let η_{it} be the ith component of η_t. Suppose $\{\eta_{it}\}$ are independent across both t and i and $\sup_{n,p} \max_{i,t} \mathrm{E}|\eta_{it}|^k \leq C^k$ for some $C > 0$ and all $k \geq 1$. Also assume $\{\Psi_j\}$ are norm-bounded and converge jointly. Moreover, $\{\varepsilon_t\}$ and $\{\eta_t\}$ are independent. We shall write Ψ_j for $\Psi_{j,p}$ and Y_t for $Y_{t,p}^{(n)}$. Let us denote the population and sample autocovariance matrices of order u of $\{X_t\}$ as $\Gamma_{u,X}$ and $\hat{\Gamma}_{u,X}$, respectively. Also denote the population and sample autocovariance matrices of order u of $\{Y_t\}$ as $\Gamma_{u,Y}$ and $\hat{\Gamma}_{u,Y}$, respectively.

Example 10.3 Suppose we wish to test

$$H_0 : \psi_q = \Psi_q \text{ against } H_1 : \psi_q \neq \Psi_q.$$

As $\Gamma_{q,X} = \psi_q^*$ and $\Gamma_{q,Y} = \Psi_q^*$, this is equivalent to testing

$$H_0 : \Gamma_{q,X} = \Gamma_{q,Y} \text{ against } H_1 : \Gamma_{q,X} \neq \Gamma_{q,Y}.$$

One can use the test statistic

$$\mathbb{T}_6 = \left(\mathrm{Tr}(\hat{\Gamma}_{q,X}) - \mathrm{Tr}(\hat{\Gamma}_{q,Y})\right)/\hat{\lambda}_{3,n}$$

where $\hat{\lambda}_{3,n}^2$ is consistent for $\mathrm{Var}(\mathrm{Tr}\,\hat{\Gamma}_{q,X}) + \mathrm{Var}(\mathrm{Tr}\,\hat{\Gamma}_{q,X})$. The statistic $\mathbb{T}_6$ converges weakly to $\mathcal{N}(0, 1)$ under H_0. Under H_1, the statistic $\mathbb{T}_6$ with additional centering $\mathrm{Tr}(\Gamma_{q,X} - \Gamma_{q,Y})$ converges weakly to the standard normal variable. Therefore, we can reject H_0 for large values of $|\mathbb{T}_6|$ (Table 13).

Table 13 We consider the models (5.2) and (10.2) with standard normal $\{\varepsilon_{ij}\}$ and $\{\eta_{ij}\}$, $q = 3$ and $\psi_k = ((0.5^{|i-j|}\mathbb{I}(|i-j| = 0, k))$. We consider $\Psi_k = \psi_k + \psi_{k-1}$, $k = 1, 2$ and $\Psi_3 = \psi_3 + c\psi_3$. Note that $c = 0$ corresponds to null model and otherwise it is an alternative model. We record empirical size for $c = 0$ and empirical power for $c = 1$ based on 1000 iterations for $y = 0.5, 1, 2$, $p = ny$, $n = 500, 1000, 2000$

$c = 0$	$y = 0.5$	$y = 1$	$y = 2$	$c = 1$	$y = 0.5$	$y = 1$	$y = 2$
$n = 500$	0.053	0.055	0.046	$n = 500$	0.97	0.97	0.96
$n = 1000$	0.047	0.048	0.047	$n = 1000$	0.99	0.99	0.99
$n = 2000$	0.05	0.049	0.05	$n = 2000$	1	1	1

Table 14 We consider the models (5.2) and (10.2) with standard normal $\{\varepsilon_{ij}\}$ and $\{\eta_{ij}\}$, $q = 3$ and $\psi_k = ((0.5^{|i-j|}\mathbb{I}(|i-j| = 0, k))$ and $\Psi_k = \psi_k + c\psi_{k-1}$, $k = 1, 2, 3$. Note that $c = 0$ corresponds to null model and otherwise it is an alternative model. We record empirical size for $c = 0$ and empirical power for $c = 1$ based on 1000 iterations for $y = 0.5, 1, 2$, $p = ny$, $n = 500, 1000, 2000$

$c = 0$	$y = 0.5$	$y = 1$	$y = 2$	$c = 1$	$y = 0.5$	$y = 1$	$y = 2$
$n = 500$	0.045	0.054	0.047	$n = 500$	0.97	0.97	0.95
$n = 1000$	0.052	0.053	0.048	$n = 1000$	0.99	0.98	0.99
$n = 2000$	0.05	0.05	0.051	$n = 2000$	1	1	1

Example 10.4 Suppose we wish to test

$$H_0 : \psi_j = \Psi_j \; \forall \; 1 \le j \le q \quad \text{against} \quad H_1 : \psi_j \ne \Psi_j \quad \text{for atleast one } 1 \le j \le q.$$

Let $\hat{G}_{q,X} = \sum_{j=1}^{q} \hat{\Gamma}_{j,X}$, $G_{q,X} = \sum_{j=1}^{q} \Gamma_{j,X}$, $\hat{G}_{q,Y} = \sum_{j=1}^{q} \hat{\Gamma}_{j,Y}$, $G_{q,Y} = \sum_{j=1}^{q} \Gamma_{j,Y}$ and $\hat{\lambda}_{4,n}^2$ be consistent for the variance of trace of $(\hat{G}_{q,X} - \hat{\Gamma}_{q,Y})$. Consider the test statistic

$$\mathbb{T}_7 = \left(\text{Tr}\, \hat{G}_{q,X} - \text{Tr}\, \hat{G}_{q,Y}\right)/\hat{\lambda}_{4,n}.$$

Then $\mathbb{T}_7$ converges weakly to $\mathcal{N}(0, 1)$ under H_0. Under H_1, the statistic $\mathbb{T}_7$ with additional centering $\text{Tr}\left(G_{q,X} - G_{q,Y}\right)$ converges weakly to the standard normal variable. Therefore, we can reject H_0 for the large values of $|\mathbb{T}_7|$ (Table 14).

10.4 MA Order Determination

In this section, we shall discuss a graphical method of determining the unknown order of a high-dimensional moving average process. The material of this section is taken from Bose and Bhattacharjee (2018) where a similar method for a high-dimensional autoregressive process is also discussed.

Suppose we wish to determine the order q of a univariate MA(q). For this process, the population autocovariances of order greater than q are all zero. On the other hand, under reasonable conditions, sample autocovariances are consistent estimators for the population autocovariances. Hence, a plot of the sequence of sample autocovariances provides a graphical method to identify q; if the plot falls below a small threshold for $u > \hat{q}$, then $\hat{q}$ is an estimate of q.

Now consider the infinite-dimensional MA(q) process defined in (5.2) with unknown parameter matrices $\{\psi_j\}$ and we wish to determine q. Then we know that Γ_u is the null matrix for all $u > q$. Suppose $\{X_{t,p} : 1 \le t \le n\}$ is a sample of size n from the process (5.2). Unlike the finite-dimensional case, the sample autocovariance matrices $\{\hat{\Gamma}_u\}$ are not consistent for the population autocovariance matrices $\{\Gamma_u\}$. Hence, the idea described for univariate MA processes above cannot be extended naively.

The LSD results discussed in Sect. 6 for $\{\hat{\Gamma}_u\}$ can be used efficiently for our purpose. Consider the following symmetric polynomials,

$$\begin{aligned}
\hat{\Pi}_{1u} &= \hat{\Gamma}_u \hat{\Gamma}_u^*, \quad \Pi_{1u} = \Gamma_u \Gamma_u^*, \\
\hat{\Pi}_{2u} &= \hat{\Gamma}_u \hat{\Gamma}_u^* + \hat{\Gamma}_{u+1} \hat{\Gamma}_{u+1}^*, \quad \Pi_{2u} = \Gamma_u \Gamma_u^* + \Gamma_{u+1} \Gamma_{u+1}^*, \\
\hat{\Pi}_{3u} &= \hat{\Gamma}_u + \hat{\Gamma}_u^*, \quad \Pi_{3u} = \Gamma_u + \Gamma_u^*.
\end{aligned} \tag{10.3}$$

If the model (5.2) satisfies the assumptions stated in Sect. 10.3, then the LSD of $\hat{\Pi}_{1u}$ (or of $\hat{\Pi}_{2u}$, $\hat{\Pi}_{3u}$) are identical for $u > q$ and are different for $0 \le u \le q$ when $p/n \to y \in (0, \infty)$. Moreover, when $p/n \to 0$, the LSD of $\hat{\Pi}_{1u}$, $\hat{\Pi}_{2u}$ and $\hat{\Pi}_{3u}$ are respectively identical with the LSD of Π_{1u}, Π_{2u} and Π_{3u} and these LSD are degenerate at 0 for $u > q$.

Suppose we know that the observations are from the time series (5.2) where the unknown q is the parameter of interest and $\{\psi_j\}$ are the nuisance parameters. The above result can be used to identify graphically the value of q as follows. We plot the CDF of the ESD (call it ECDF) of $\hat{\Pi}_{1u}$ (or of $\hat{\Pi}_{2u}$, $\hat{\Pi}_{3u}$) for first few sample autocovariance matrices. When p is moderate compared to n, we say that $\hat{q}$ is an estimate of q if the ECDF of $\hat{\Pi}_{1u}$ (or $\hat{\Pi}_{2u}$, $\hat{\Pi}_{3u}$) with order $u > \hat{q}$ empirically coincide with each other. Similarly, when p is small compared to n, $\hat{q}$ is an estimate of q, if the ECDF of $\hat{\Pi}_{1u}$ (or $\hat{\Pi}_{1u}$, $\hat{\Pi}_{3u}$) with $u > \hat{q}$ is degenerate at 0. Note that the estimation of the parameter matrices is not required to implement this method.

Simulations. Let $\varepsilon_t \sim \mathcal{N}_p(0, I_p)$ $\forall t$. Suppose $A_p = 0.5 I_p$, $B_p = 0.5(I_p + J_p)$, $C_p = ((I(1 \le i = j \le [p/2]) - I([p/2] < i = j \le p)))$ and $D_p = ((I(i + j = p + 1)))$. Consider the following models.

Model 1 $X_t = \varepsilon_t$.

Model 2 $X_t = \varepsilon_t + A_p \varepsilon_{t-1}$.

Model 3 $X_t = \varepsilon_t + B_p \varepsilon_{t-1}$.

Model 4 $X_t = \varepsilon_t + C_p \varepsilon_{t-1} + D_p \varepsilon_{t-2}$.

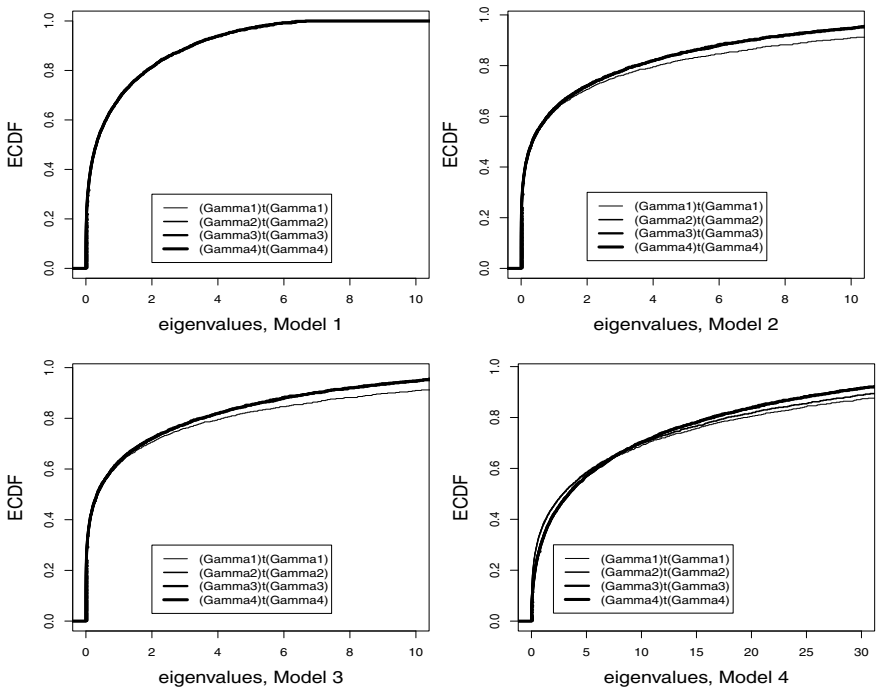


Fig. 10 ECDF of $\hat{\Pi}_{1u}, 1 \leq u \leq 4, n = p = 500$. In the graphs, Gamma u $= \hat{\Gamma}_u$ and t(Gamma u) $= \hat{\Gamma}_u^*$

Note that the LSD of A_p, B_p, C_p and D_p exist and $\{C_p, D_p\}$ converge jointly. Hence, the above discussion on the LSD of symmetric polynomials in sample autocovariance matrices holds true. Suppose we have a time series sample of size n, say S_i, from Model i, $1 \leq i \leq 4$. Note that no replication is involved.

Case a: $p/n \to y > 0$. We take $p = n = 500$. Pretending that we do not know from which model our sample is, we wish to determine q. For each $1 \leq i \leq 4$, we plot the ECDF of $\hat{\Pi}_{1u}$ for $1 \leq u \leq 4$, based on the sample S_i, in the same graph. See Fig. 10.

Note that the ECDF of $\hat{\Pi}_{1u}$ coincide in the following cases: (a) for all $u > 0$ in Model 1, (b) for $u > 1$ in Models 2 and 3 and (c) for $u > 2$ in Model 4. Hence, $\hat{q}$ is 0, 1, 1 and 2 respectively for Models 1–4. Thus, the above method identifies q precisely. Incidentally, since LSD of A_p and B_p are identical, the ECDF for Models 2 and 3 are almost identical. Similar behavior was observed for the polynomial $\hat{\Pi}_{2u}$ (see Fig. 11).

Case b: $p/n \to 0$. For Models 1–4, we draw *one* sample with $n = 500$, $p = n^{0.9} \sim 269$. We plot the ECDF of $\hat{\Pi}_{3u}$ for $1 \leq u \leq 4$ in Fig. 12. We have the same conclusions as in Case a.

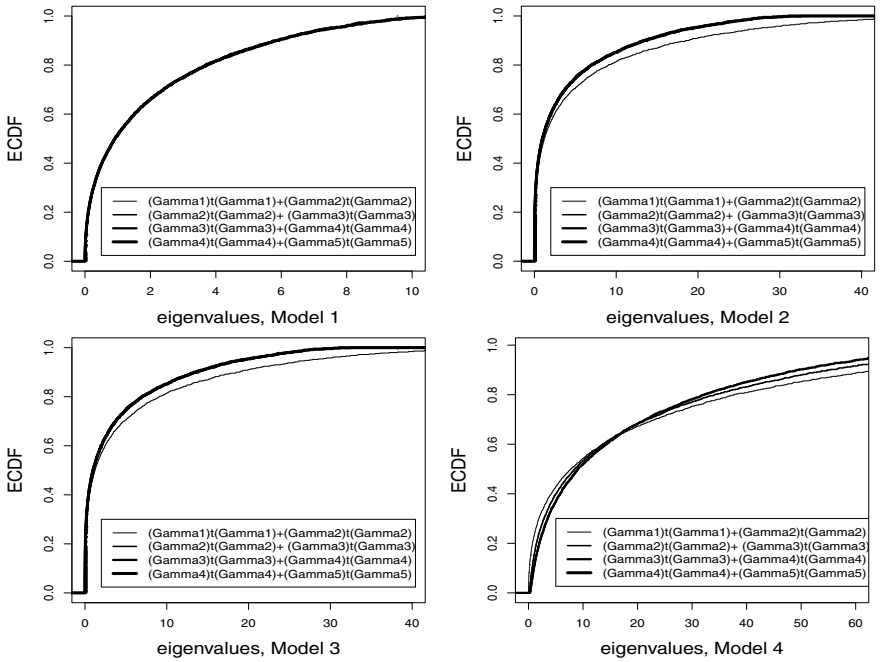


Fig. 11 ECDF of $\hat{\Pi}_{2u}$, $1 \leq u \leq 4$, $n = p = 500$

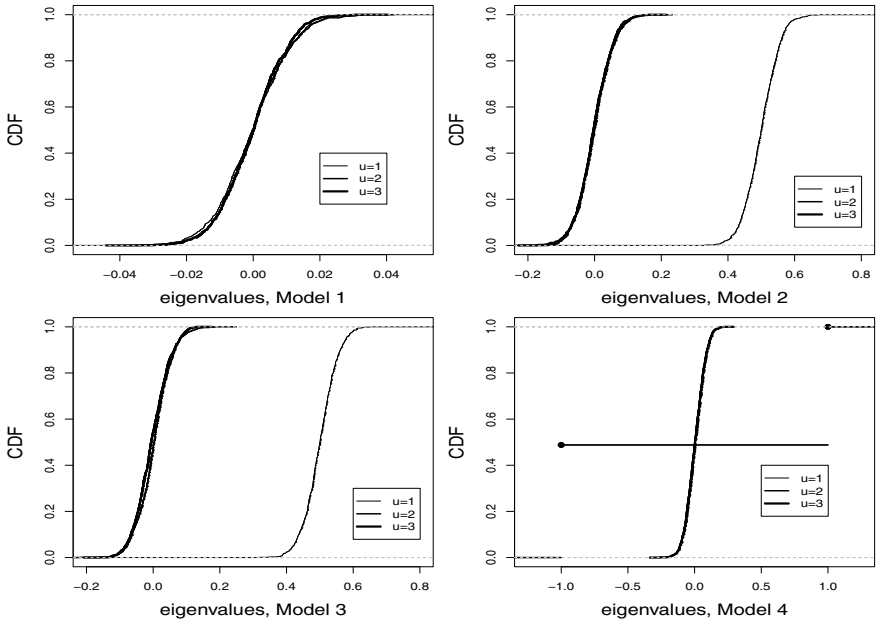


Fig. 12 ECDF of $\hat{\Pi}_{3u}$, $1 \leq u \leq 4$, $n = 500$, $p = n^{0.9}$

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Long-Time Behavior of Finite and Infinite Dimensional Reflected Brownian Motions



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1 Introduction

In this article, we will provide a survey of some results on the long-time behavior of reflected Brownian motions (RBM) in polyhedral domains. We will consider the settings of both finite and infinite dimensional reflected diffusions.

The simplest such model is a one-dimensional RBM $\{X(t)\}_{t\geq 0}$ defined as

$$X(t) = x + B(t) - \inf_{0\leq s\leq t} \{(x + B(s)) \wedge 0\}, t \geq 0, \ x \in \mathbb{R}_+, \tag{1}$$

where B is a standard Brownian motion given on some probability space. Occasionally we will write $X(t) = X(t; x)$ to emphasize dependence on the initial condition. There is an equivalent way to describe this process using the formulation of the Skorohod problem Skorokhod (1961).

The Skorohod problem on the positive real line $\mathbb{R}_+$ is to find for a given $\mathbf{y} \in \mathcal{D}([0, \infty) : \mathbb{R})$ (throughout, for a Polish space S, $\mathcal{D}([0, \infty) : S)$ will denote the space of functions from $[0, \infty)$ to S that are right continuous and have left limits(RCLL), equipped with the usual Skorohod topology), with $\mathbf{y}(0) \geq 0$, a pair of paths $\mathbf{x}, \mathbf{k} \in \mathcal{D}([0, \infty) : \mathbb{R})$ such that for all $t \geq 0$, $\mathbf{x}(t) = \mathbf{y}(t) + \mathbf{k}(t) \geq 0$, $\mathbf{k}(0) = 0$, $\mathbf{k}$ is non-

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decreasing, and $\mathbf{k}$ increases only at instants t when $\mathbf{x}(t) = 0$, namely

$$\int_{[0,\infty)} 1_{\{\mathbf{x}(s)>0\}} d\mathbf{k}(s) = 0.$$

The above problem has a unique solution for every $\mathbf{y} \in \mathcal{D}([0,\infty) : \mathbb{R})$ and we write $\mathbf{x} = \Gamma_1(\mathbf{y})$. Also, it is easily verified that X defined by (1) can equivalently be written as $X = \Gamma_1(x + B)$. The formulation of a Skorohod problem also allows one to define a generally reflected diffusion that on $(0,\infty)$ behaves as a diffusion with the infinitesimal generator (evaluated on smooth test functions f)

$$\mathcal{L}f(x) = \frac{1}{2}\sigma^2(x)f''(x) + b(x)f'(x), \ x \in (0,\infty)$$

for suitable coefficient functions σ and b. Such a reflected diffusion can be constructed by solving the stochastic equation

$$X(t) = \Gamma_1\left(x + \int_0^{\cdot} b(X(s))ds + \int_0^{\cdot} \sigma(X(s))dB(s)\right)(t), \ t \geq 0 \tag{2}$$

in a pathwise fashion (e.g. when b, σ are Lipschitz functions) (see Anderson & Orey 1976). In the special case when $\sigma(z) \equiv \sigma > 0$ and $b(z) = b \in \mathbb{R}$, X is referred to as a RBM with drift b and diffusion coefficient σ, or simply a (b,σ)-RBM. It is well known that a (b,σ)-RBM is positive recurrent if and only if $b < 0$ and the unique stationary distribution is an exponential distribution with rate $\lambda = 2|b|/\sigma^2$. In the general setting of (2), under conditions on the coefficients (e.g., b and σ Lipschitz, σ bounded, and $\sigma\sigma^T$ uniformly nondegenerate), it can be shown that if $\sup_{z\geq 0} b(z) < 0$, then the diffusion is positive recurrent, although this time an explicit form stationary distribution is not available. One can also show that the law at time t converges to the stationary distribution at a geometric rate in the total variation distance.

In this work, we are interested in the higher dimensional analogs of the above process. We will only consider reflected Brownian motions, or more generally reflected jump diffusions, in a finite or infinite non-negative orthant, namely, the state space will be $\mathbb{R}^d_+$ where $d \in \mathbb{N} \cup \{\infty\}$. Such multi-dimensional RBM arise from problems in stochastic networks Reiman (1984), Harrison and Williams (1987), applications in mathematical finance Fernholz and Karatzas (2009), scaling limits of interacting particle systems Karatzas et al. (2016), and in rank-based diffusion models for competing particles Karatzas et al. (2016), Sarantsev (2019). The next three sections will focus on the finite dimensional case, while the last two sections will consider infinite dimensional settings.

2 Finite Dimensional RBM

Consider for now $d < \infty$. Roughly speaking, a reflected Brownian motion in the non-negative orthant $\mathbb{R}^d_+$ behaves like an ordinary Brownian motion (with drift) in the interior of the orthant, and when it reaches the boundary of the domain, it is instantaneously pushed back, with the minimal force needed to keep it inside the domain, applied in some pre-specified directions of constraint. As in the one-dimensional case, a natural way to give a rigorous formulation of such an object is through a Skorohod problem.

Fix vectors $r^1, \ldots, r^d \in \mathbb{R}^d$. Here the vector r^i gives the direction in which the state of the RBM is pushed when it is about to exit the domain $\mathbb{R}^d_+$ from its ith face $F_i = \{x \in \mathbb{R}^d_+ : x_i = 0\}$. We refer to the $d \times d$ matrix $R = [r^1, \ldots, r^d]$ as the reflection matrix. For each $x \in \partial\mathbb{R}^d_+ = \cup_{i=1}^d F_i$, the set $r(x)$ of directions of constraints is defined as

$$r(x) := \left\{ Rq : q = (q_1, \ldots, q_d)' \in \mathbb{R}^d_+,\ q \cdot \mathbf{1} = 1,\ \text{and } q_i > 0 \text{ only if } x_i = 0 \right\}, \tag{3}$$

where $\mathbf{1} = (1, 1, \ldots)'$. The set $r(x)$ represents the collection of directions available for pushing the state of the process at the boundary point x. Note that if x has exactly one coordinate 0, the set $r(x)$ is a singleton.

The Skorohod problem associated with the domain $\mathbb{R}^d_+$ and the reflection matrix R is defined as follows. Given $\psi \in \mathcal{D}([0, \infty) : \mathbb{R}^d)$ such that $\psi(0) \in \mathbb{R}^d_+$, we say a pair of trajectories $\phi, \eta \in \mathcal{D}([0, \infty) : \mathbb{R}^d)$ solve the **Skorohod problem** for ψ with respect to reflection matrix R, if and only if $\eta(0) = 0$ and for all $t \geq 0$: (i) $\phi(t) = \psi(t) + \eta(t)$; (ii) $\phi(t) \in \mathbb{R}^d_+$; (iii) The total variation (with respect to the Euclidean norm on $\mathbb{R}^d$) of η on $[0, t]$, denoted as $|\eta|(t)$ is finite; (iv) $|\eta|(t) = \int_{[0,t]} 1_{\{\phi(s) \in \partial\mathbb{R}^d_+\}} d|\eta|(s)$; (v) there is a Borel measurable function $\gamma : [0, \infty) \to \mathbb{R}^d$ such that $\gamma(t) \in r(\phi(t))$, $d|\eta|$-a.e., and $\eta(t) = \int_{[0,t]} \gamma(s) d|\eta|(s)$. The trajectory ϕ can be viewed as the reflected version of ψ that stays in $\mathbb{R}^d_+$ at all times. Property (iv) says that the pushing term η is activated only when the path is about to exit the domain while property (v) ensures that the push is applied in the permissible directions of constraint, as specified by the reflection matrix R.

On the domain $D \subset D_0 := \{\phi \in \mathcal{D}([0, \infty) : \mathbb{R}^d) : \phi(0) \in \mathbb{R}^d_+\}$ on which there is a unique solution to the Skorohod problem we define the **Skorohod map** (SM) Γ as $\Gamma(\psi) = \phi$, if $(\phi, \psi - \phi)$ is the unique solution of the Skorohod problem.

In many problems of interest, it turns out that the Skorohod map is quite well-behaved in the sense that the following property is satisfied.

Property 1 *The Skorohod map is well defined on all of D_0 and the SM is Lipschitz continuous in the following sense: There exists a $K \in (0, \infty)$ such that for all $\phi_1, \phi_2 \in D_0$*

$$\sup_{0 \leq t < \infty} \|\Gamma(\phi_1)(t) - \Gamma(\phi_2)(t)\| \leq K \sup_{0 \leq t < \infty} \|\phi_1(t) - \phi_2(t)\|.$$

We refer the reader to Dupuis and Ishii (1991), Dupuis and Ramanan (1999) for sufficient conditions under which the above property holds. One important class of examples that arise naturally in many stochastic network models Reiman (1984), Harrison and Williams (1987) and also in the study of gaps between $d+1$ competing particles in rank-based diffusions (e.g. Karatzas et al., 2016; Sarantsev, 2019), and where the above property is satisfied, is the so-called **Harrison-Reiman** class Harrison and Reiman (1981). This corresponds to the family of models for which the matrix $P := I - R^T$ is substochastic (non-negative entries and row sums bounded above by 1) and transient ($P^n \to 0$ as $n \to \infty$). This family will be the focus of much of our discussion in subsequent sections.

When Property 1 holds, one can solve pathwise, as in the one-dimensional case, stochastic equations of the form

$$X(t) = \Gamma\left(x + \int_0^{\cdot} b(X(s))ds + \int_0^{\cdot} \sigma(X(s))dB(s)\right)(t), \ t \geq 0, \tag{4}$$

where $b : \mathbb{R}^d_+ \to \mathbb{R}^d$ and $\sigma : \mathbb{R}^d_+ \to \mathbb{R}^{d\times k}$ are suitable coefficient functions (e.g. Lipschitz continuous) and B is a standard k dimensional Brownian motion on some probability space (cf. Dupuis and Ishii, 1991). The solution in the case when $b(x) \equiv \mu$ and $\sigma(x) \equiv \mathbf{D}$, where $\mu \in \mathbb{R}^d$ and $\mathbf{D}$ is a $d \times d$ matrix, will be of particular interest here and will be referred to as the $(\mu, \mathbf{D})$-RBM with reflection matrix R. In later sections, we will also use the abbreviation RBM(μ, Σ, R), where $\Sigma = \mathbf{D}\mathbf{D}^T$.

The solution of (4) can be written in a somewhat more explicit form using boundary local times. For example, when $X(t; x)$ is a $(\mu, \mathbf{D})$-RBM with reflection matrix R starting from x, it can be represented as the solution of the equation

$$X(t; \mathbf{x}) = \mathbf{x} + \mathbf{D}B(t) + \mu t + RL(t), \tag{5}$$

where L, referred to as the local time process, is a d-dimensional non-decreasing continuous process satisfying

$$L(0) = 0, \quad \int_0^t X_i(s; \mathbf{x})dL_i(s) = 0 \text{ for all } t > 0 \text{ and } 1 \leq i \leq d. \tag{6}$$

It turns out that there are important applications arising from stochastic network theory (see e.g. Williams, 1998) that lead to well-defined Brownian motions constrained in the positive orthant according to a certain reflection mechanism, but the associated Skorohod problem is not well-posed for a typical path in $\mathcal{D}([0, \infty) : \mathbb{R}^d_+)$ (in fact uniqueness of solution fails even for linear paths). In order to cover such situations the papers Taylor and Williams (1993); Reiman and Williams (1988) introduce and study **semimartingale reflecting Brownian motions** (SRBM).

Suppose that Σ is a $d \times d$ positive definite matrix. For $\mathbf{x} \in \mathbb{R}^d_+$, an SRBM associated with the data $(\mu, \mathbf{D}, R)$ that starts from $\mathbf{x}$ is continuous, $\{\mathcal{F}_t\}$-adapted d-dimensional process X, defined on some filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t\geq 0}, P)$ such that: (i) $X(t) = \mathbf{x} + \mathbf{D}B(t) + \mu t + RL(t) \in \mathbb{R}^d_+$ for all $t \geq 0$, P-

a.s., (ii) B is a d-dimensional standard $\{\mathcal{F}_t\}$-Brownian motion, (iii) L is an $\{\mathcal{F}_t\}$-adapted d-dimensional process such that $L_i(0) = 0$ for $i = 1, \ldots, d$, P-a.s. For each $i = 1, \ldots, d$, L_i is continuous, non-decreasing and $\int_0^t 1_{\{X_i(s)\neq 0\}} dL_i(s) = 0$ for all $t \geq 0$, P-a.s.

A key condition in this context is that the matrix R is **completely-$\mathcal{S}$** (cf. Reiman & Willianms 1988), namely, for every $k \times k$ principal submatrix G of R, there is a k-dimensional vector v_G such that $v_G \geq 0$ and $Rv_G > 0$. Roughly speaking, the conditions say that, at each point x on the boundary, there is a permissible direction of reflection, namely, a $r \in r(x)$, that points inwards. In Reiman and Williams (1988, Theorem 2), it was shown that a necessary condition for the existence of an SRBM is that the reflection matrix R is a completely-$\mathcal{S}$ matrix. The paper Taylor and Williams (1993) shows that the condition is sufficient as well for the (weak) existence and uniqueness of SRBM to hold. Under this condition, the collection $\{P_{\mathbf{x}}\}_{\mathbf{x}\in\mathbb{R}^d_+}$, where $P_{\mathbf{x}}$ is the law of the SRBM starting from $\mathbf{x}$, forms a $\mathbb{R}^d_+$-valued strong Markov process.

2.1 Ergodicity.

The first basic result on long-time behavior of RBM in $\mathbb{R}^d_+$ is due to Harrison and Williams (1987) which treats the Harrison-Reiman class of RBM. Suppose that the following holds ((A1) is simply a restatement of the Harrison-Reiman class.)

Assumptions:

(A1) The matrix $P := I - R^T$ is substochastic and transient.
(A2) $\mathbf{b} := -R^{-1}\mu > 0$.
(A3) The matrix $\Sigma = \mathbf{D}\mathbf{D}^T$ is positive definite.

It was shown in Harrison and Williams (1987) that, when the above assumptions (A1)–(A3) hold, then the $(\mu, \mathbf{D})$-RBM is positive recurrent and consequently has a unique stationary distribution. Assumption (A2) is the well-known 'stability condition' which is sufficient for the existence of a stationary measure Harrison and Williams (1987, Sect. 6). The condition is almost necessary, in that, if $\mathbf{b}_i < 0$ for some i, then the RBM is transient Budhiraja and Dupuis (1999). The matrix $\Sigma = DD^T$ gives the covariance matrix associated with the driving diffusion of (5) and (A3) guarantees that this diffusion process is elliptic. Ellipticity gives the uniqueness of the stationary distribution Harrison and Williams (1987). Under additional conditions on R and $\mathbf{D}$, the paper Harrison and Williams (1987) shows that the stationary distribution takes an explicit and simple form given as a product of exponential distributions; however, in general, there is little one can say about this stationary distribution other than some properties of its tail (see, e.g., the next section and Budhiraja & Lee 2007).

The paper Atar et al. (2001) studied long-time behavior of constrained reflected diffusions of the form in (4). The basic assumptions are that Property 1 holds and the coefficients b, σ satisfy the following condition.

Assumptions:

(D1) b and σ are Lipschitz maps.
(D2) σ is bounded.
(D3) The matrix $\sigma\sigma^T$ is uniformly nondegenerate.

In addition, one needs a suitable stability condition. The key stability condition in Atar et al. (2001) is formulated in terms of the cone

$$\mathcal{C} := \left\{ -\sum_{i=1}^{d} \alpha_i r^i : \alpha_i \geq 0, i \in \{1, \ldots, d\} \right\}.$$

This cone $\mathcal{C}$ was introduced in Budhiraja and Dupuis (1999) to characterize stability properties of deterministic paths arising from certain law of large number limits, under the long-time scaling, of a $(\mu, \mathbf{D})$-RBM for which the associated reflection matrix R satisfies Property 1. The main result there showed that if $\mu \in \mathcal{C}^o$ then these deterministic paths are all attracted to the origin (namely, the paths converge to 0 as time becomes large) whereas if $\mu \in \mathcal{C}^c$ then these paths diverge to ∞. Finally, when $\mu \in \partial\mathcal{C}$, all solutions of the deterministic model are bounded, and for at least one initial condition the corresponding trajectory is not attracted to the origin. This stability characterization was instrumental in the study of the stability of constrained reflected diffusions in Atar et al. (2001). Let, for $\delta \in (0, \infty)$,

$$\mathcal{C}(\delta) := \{v \in \mathcal{C} : \text{dist}(v, \partial\mathcal{C}) \geq \delta\}.$$

The key stability assumption on the constrained diffusion model (4) made in Atar et al. (2001) is the following.

Assumption ($\mathcal{C}$-delta): There exists a $\delta \in (0, \infty)$ and a bounded set $A \subseteq \mathbb{R}^d_+$ such that for all $x \in \mathbb{R}^d_+ \backslash A$, $b(x) \in \mathcal{C}(\delta)$.

Under this stability assumption, together with assumptions (D1)–(D3) noted previously, Atar et al. (2001, Theorem 2.2) shows that the Markov process described by the constrained diffusion model (4) is positive recurrent and has a unique stationary distribution. The main proof idea there is to study the stability properties of a certain deterministic collection of controlled reflected paths of the form

$$\psi_v(t; \mathbf{x}) \doteq \Gamma(\mathbf{x} + \int_0^{\cdot} v(s)ds)(t), \ t \geq 0$$

where $\{v(t)\}_{t\geq 0}$ is a locally integrable function that takes values in $\mathcal{C}(\delta)$ for all $t \geq 0$. Denote by $T(\mathbf{x})$ the supremum of the hitting times to zero, taken over all such controlled paths, starting from $\mathbf{x}$. Then, the analysis of Budhiraja and Dupuis (1999) gives an explicit bound on $T(\cdot)$ as, $T(\mathbf{x}) \leq C\|\mathbf{x}\|/\delta$ for some $C \in (0, \infty)$ and all $\mathbf{x} \in \mathbb{R}^d_+$. The function $T(\cdot)$ turns out to be a natural Lyapunov function for the problem, although it lacks the required C^2-property for implementing the usual Itô's

formula based methods for Lyapunov functions for diffusion processes. Although not C^2, the function is nevertheless Lipschitz and that is enough to deduce the required uniform in time moment estimates needed to prove the main result.

The cone $\mathcal{C}$ can also be used to give conditions for stability of constrained Lévy processes and more general constrained jump-diffusions in $\mathbb{R}^d_+$ associated with a reflection matrix R for which Property 1 holds. Such a jump-diffusion is given as a solution of an equation of the form

$$X(t) = \Gamma\left(x + \int_0^{\cdot} b(X(s))ds + \int_0^{\cdot} \sigma(X(s))dB(s) + J(\cdot)\right)(t), \ t \geq 0, \qquad (7)$$

where

$$J(t) = \int_{[0,t]\times\mathbb{R}^n} h(\psi(X(s-), z))[N(ds, dz) - \lambda(dz)ds] + \int_{[0,t]\times\mathbb{R}^n} h'(\psi(X(s-), z))N(ds, dz),$$

N is a Poisson random measure (PRM) on $\mathbb{R}_+ \times \mathbb{R}^n$, with intensity measure $ds\ \lambda(dz)$, λ is a σ-finite measure on $\mathbb{R}^n$, b, σ satisfy (D1)-(D3) as before, $\psi : \mathbb{R}^d_+ \times \mathbb{R}^n \to \mathbb{R}^d$ is a suitable coefficient map, $h : \mathbb{R}^d \to \mathbb{R}^d$ is a nice truncation function (see Atar & Budhiraja 2002) and $h'(\mathbf{x}) = \mathbf{x} - h(\mathbf{x})$ for $\mathbf{x} \in \mathbb{R}^d$.

The work Atar and Budhiraja (2002) shows that the cone $\mathcal{C}$ can be used to study stability for a very general class of constrained jump-diffusions of the form in (7) for which only the first moment is given to be finite. Recall that, in the case of constrained diffusions as in (2), the condition for stability was Assumption $\mathcal{C}$-delta on the drift vector field $b(\cdot)$. Here, in the case of a jump-diffusion, the natural definition of the drift vector field is $\tilde{\beta} := \mathcal{L}\mathrm{id}$, where $\mathcal{L}$ denotes the generator of the 'unconstrained' jump diffusion away from the boundary, and id denotes the identity mapping on $\mathbb{R}^d$. In the special case of a Lévy process with finite mean, the drift is simply $\tilde{\beta}(\mathbf{x}) = E_{\mathbf{x}}(X(1)) - \mathbf{x}$ (which is independent of $\mathbf{x}$) where $E_{\mathbf{x}}$ denotes the expectation under which X starts from $\mathbf{x}$. The basic stability assumption in Atar and Budhiraja (2002) is that the range of $\tilde{\beta}$ is contained in $\cup_{k\in\mathbb{N}} k\mathcal{C}_1$ where $\mathcal{C}_1$ is a compact subset of the interior of $\mathcal{C}$. Under this assumption it is shown in Atar and Budhiraja (2002) that there exists a compact set A such that for any compact $C \subset \mathbb{R}^d_+$, $\sup_{\mathbf{x}\in C} E_{\mathbf{x}}(\tau_A) < \infty$, where τ_A is the first time X hits A. The proof of this result is based on the construction of a Lyapunov function, and on a detailed separate analysis of small and large jumps of the Markov process. As another consequence of the existence of a Lyapunov function, it is shown in Atar and Budhiraja (2002) that X is bounded in probability. From the Feller property of the process, it then follows that it admits at least one invariant measure. Finally, under further additional communicability conditions, it follows that the Markov process is positive Harris recurrent and admits a unique invariant measure (see Atar & Budhiraja 2002).

Up to now, stability results discussed in this section were for a setting where Property 1 is satisfied. In the setting of a $(\mu, \mathbf{D}, R)$-SRBM, under the completely-$\mathcal{S}$ condition on R, the ergodicity behavior has been studied in Dupuis and Williams (1994). In this work, the main stability condition is formulated in terms of constrained

versions of the linear trajectory $\eta(t;\mathbf{x}) = \mathbf{x} + \mu t, t \geq 0, \mathbf{x} \in \mathbb{R}^d_+$. Since the associated Skorohod problem is not necessarily well-posed in the setting of Dupuis and Williams (1994), there may be multiple constrained versions of $\eta(t;\mathbf{x})$ that are consistent with the reflection mechanism specified by R. The paper Dupuis and Williams (1994) shows that if all these constrained paths $\theta(t;\mathbf{x})$, for all initial conditions $\mathbf{x}$ are attracted to the origin (namely, $\theta(t;\mathbf{x}) \to 0$ as $t \to \infty$) then the SRBM is positive recurrent and has a unique stationary distribution. The basic idea in the proof is to construct a suitable C^2-Lyapunov function W. A key property of this Lyapunov function, in addition to certain conditions on directional derivatives of W at the boundary points, is that, for some $c > 0, \nabla W(\mathbf{x}) \cdot \mu \leq -c < 0$ for all $\mathbf{x} \in \mathbb{R}^d_+ \setminus \{0\}$. This property allows one to construct suitable supermartingales from which uniform in time moment estimates can be obtained. In fact, this Lyapunov function was also the inspiration for the Lyapunov function, for the general state-dependent drift and a jump-diffusion setting, constructed in Atar and Budhiraja (2002). There, due to the state dependence, the Lyapunov function $\tilde{W}$ needs to satisfy the property $\nabla \tilde{W} \cdot v \leq -c < 0$, uniformly, for all v in the range of the drift $\tilde{\beta} = \mathcal{L}\mathrm{id}$ which makes the construction more involved.

2.2 Geometric Ergodicity

The above results focused on the existence and uniqueness of stationary distributions and positive recurrence. A key question of interest is how long it takes for the RBM to start from arbitrary initial distributions to approach stationarity (rate of convergence). These questions were studied in Budhiraja and Lee (2007). This work studied $(\mu, \mathbf{D}, R)$-SRBM models where R is completely-$\mathcal{S}$ that satisfies the stability condition of Dupuis and Williams (1994), and constrained diffusion of the form in (4) with coefficients satisfying (D1)-(D3), the associated reflection matrix R satisfying Property 1, and drift satisfying the stability condition ($\mathcal{C}$-delta). For both of these settings Budhiraja and Lee (2007), constructs a suitable exponentially growing Lyapunov function V and establishes that the processes are V-uniformly ergodic. The property of V-uniform ergodicity says, in particular, that the law of the process at any fixed time t converges to the stationary distribution, at an exponential rate, in the total variation distance as $t \to \infty$. In fact, since V is exponentially growing, from this convergence, one can also deduce convergence of integrals of unbounded functions (with possible exponential growth) with respect to the law at time t, to the corresponding integrals with respect to the stationary distribution, at an exponential rate, as $t \to \infty$. The result of V-uniform ergodicity is also used in Budhiraja and Lee (2007) to prove that the unique invariant measures of the SRBM and of the reflected diffusions in (4) admit finite moment generating functions in suitable neighborhoods of zero. As other consequences of this ergodicity, one can obtain uniform (in time and initial condition in a compact set) estimates on exponential moments of these constrained processes. One can also give growth estimates of polynomial moments of the process as a function of the initial condition and obtain functional central limit theo-

rems for processes $\xi_n(t) := \frac{1}{\sqrt{n}} \left(\int_0^{nt} \left[F(X(s)) - \int F d\pi \right] \mathrm{d}s \right)$, as $n \to \infty$, where π is the unique invariant measure and F is allowed to have exponential growth, and characterize the asymptotic variance via the solution of a related Poisson equation (see Budhiraja & Lee 2007 for these results). The key steps in the proofs are the construction of a suitable Lyapunov function and establishing a minorization condition on a sufficiently large compact set (referred to as a 'small set'). The Lyapunov function provides good control on the exponential moments of the return times to the small set while the minorization condition implies the existence of abstract couplings of two copies of the process (via construction of 'pseudo-atoms' as described in Chap. 5 of Meyn & Tweedie 2012) which have a positive chance of coalescing inside the small set. Together, they furnish exponential rates of convergence (in a weighted total variation distance).

Other than Budhiraja and Lee (2007), only a few works obtain quantitative rates of convergence in some special cases. The paper Ichiba et al. (2013b) obtains explicit convergence rates for time averages of bounded functionals of the state process for a class of reversible rank-based diffusions with explicit stationary measures using Dirichlet form techniques. The setting of one-dimensional RBM is considered in Wang and Glynn (2014) where (among other results) an estimate on the spectral gap is provided as a function of the drift and the diffusion coefficient.

3 Parameter and Dimension Dependence on Rates of Convergence

The results of Budhiraja and Lee (2007), due to the somewhat implicit treatment of the process inside the small set, provide convergence rates that shed little light on how they qualitatively depend on the system parameters or the state dimension. In this section and the next, the focus is on studying, how the convergence rate to stationarity depends on the initial distribution, diffusion parameters, and the underlying dimension d? Moreover, it has been empirically observed that for certain types of RBM with parameters $(\mu^{(d)}, \Sigma^{(d)}, R^{(d)})$ indexed by the dimension $d \geq 1$, for any fixed $k \in \mathbb{N}$, the marginal distribution of the first k-coordinates approaches the stationary marginal at a *rate independent of* d. We will present results that identify conditions on $(\mu^{(d)}, \Sigma^{(d)}, R^{(d)})$ that give rise to this phenomenon, referred to as 'dimension-free' convergence, and give quantitative bounds on this rate. These results set the stage for studying ergodicity properties of *infinite dimensional reflected diffusions* which will be the focus of the last two sections. Among other things, a key challenge in studying long-time behavior of such diffusions lies in the fact that they can have multiple stationary distributions. Thus, even before making sense of rates of convergence, one needs to understand which stationary distribution the process will approach starting from a given initial distribution.

In a recent work, Blanchet and Chen (2020) obtains dimension-dependent bounds on rates of convergence in Wasserstein distance for a certain class of RBM whose

parameters satisfy certain 'uniformity conditions' in dimension (see (BC1)–(BC3) in Sect. 3.3.1). Although the assumptions imposed are somewhat rigid, there is a key new idea there which is to replace the role of abstract couplings (that, for example, underlie the approach in Budhiraja & Lee 2007) with more tractable *synchronous couplings* (namely, couplings where the RBM starting from different points are driven by the same Brownian motion) to analyze the behavior of the RBM inside the small set. Using explicit couplings to obtain better convergence rate estimates is a relatively recent but developing area. See Bolley et al. (2010), Eberle (2015), Eberle et al. (2019a), and Eberle et al. (2019b) for such results for other classes of diffusions. In this Section, we present results from Banerjee and Budhiraja (2020), which, building on the work of Blanchet and Chen (2020), systematically develop the application of synchronous couplings and construct associated Lyapunov functions to obtain quantitative rates of convergence for any RBM satisfying Assumptions (A1)–(A3). These rates explicitly highlight the dependence of system parameters and dimension. Further, the obtained rates are significantly better when applied to the class of RBM considered in Blanchet and Chen (2020) (see Sect. 3.3.1).

3.1 Main Result

We now present the main result of Banerjee and Budhiraja (2020). Given probability measures μ and ν on $\mathbb{R}^d_+$, a probability measure γ on $\mathbb{R}^d_+ \times \mathbb{R}^d_+$ is said to be a coupling of μ and ν if $\gamma(\cdot \times \mathbb{R}^d_+) = \mu(\cdot)$ and $\gamma(\mathbb{R}^d_+ \times \cdot) = \nu(\cdot)$. The L^1-Wasserstein distance between two probability measures μ and ν on $\mathbb{R}^d_+$ is given by

$$W_1(\mu, \nu) = \inf \left\{ \int_{\mathbb{R}^d_+ \times \mathbb{R}^d_+} \|\mathbf{x} - \mathbf{y}\|_1 \gamma(\mathrm{d}\,\mathbf{x}, \mathrm{d}\,\mathbf{y}) \ : \gamma \text{ is a coupling of } \mu \text{ and } \nu \right\},$$

where for a vector $z \in \mathbb{R}^d$, $\|z\|_1 = \sum_{i=1}^d |z_i|$. We will denote the law of a random variable X by $\mathcal{L}(X)$. Recall that from Harrison and Williams (1987), under Assumptions (A1)–(A3), there is a unique stationary distribution of the RBM. Denote by $\mathbf{X}(\infty)$ a random vector sampled from this stationary distribution. Define the *relaxation time*, $t_{rel}(\mathbf{x})$ for the RBM starting from $\mathbf{x} \in \mathbb{R}^d_+$ as

$$t_{rel}(\mathbf{x}) := \inf\{t \geq 0 : W_1\left(\mathcal{L}(X(t; \mathbf{x})), \mathcal{L}(\mathbf{X}(\infty))\right) \leq 1/2\}.$$

We will abbreviate the parameters of the RBM as $\Theta := (\mu, \Sigma, R)$. Recall that these parameters are required to satisfy (A1)–(A3). We will quantify the rate of convergence to equilibrium in terms of the following functions of Θ, d. Define the *contraction coefficient*

$$n(R) := \inf\{n \geq 1 : \|P^n \mathbf{1}\|_\infty \leq 1/2\}, \tag{8}$$

where $\mathbf{1}$ is a d-dimensional vector of ones and for $u \in \mathbb{R}^d$, $\|u\|_\infty := \sup_{1\le i\le d} |u_i|$. By Assumption (A1), $n(R) < \infty$.

Fix $\kappa \in (0,\infty)$. For any $\mathbf{x} \in \mathbb{R}^d_+$, define $\|\mathbf{x}\|^*_\infty := \sup_{1\le i\le d} \sigma_i^{-1}\mathbf{x}_i$. Let

$$a(\Theta) := \sup_{1\le i\le d}\left[\frac{\sum_{j=1}^d (R^{-1})_{ij}\sigma_j}{b_i}\right], \qquad b(\Theta) := \sup_{1\le i\le d}\left[\frac{\sum_{j=1}^d (R^{-1})_{ij}\sigma_j}{\sigma_i}\right],$$

$$R_1(\Theta,d) := n(R)(1+a(\Theta)^2\log(2d)), \qquad R_2(\Theta) := a(\Theta)^2 b(\Theta),$$

$$C_1(\mathbf{x},\Theta) := 2\|\mathbf{x}\|_1 + a(\Theta)\sum_{i,j}(R^{-1})_{ij}\sigma_j,$$

$$C_2(\mathbf{x},\Theta,\kappa) := 2\|\mathbf{x}\|_1 e^{3(\kappa a(\Theta)b(\Theta))^{-1}\|\mathbf{x}\|^*_\infty} + a(\Theta)\left[2d(1+d)\left(\sum_{i,j}(R^{-1})_{ij}^2\right)\left(\sum_{j=1}^d \sigma_j^2\right)\right]^{1/2}.$$

Theorem 1 (Banerjee and Budhiraja, 2020) *There exists a* $t_0 \in (0,\infty)$ *and* D_1, $D_2 \in (0,\infty)$ *such that for every* $d \in \mathbb{N}$, $\mathbf{x} \in \mathbb{R}^d_+$, *every parameter choice* Θ, *and* $t \ge t_0\left(1+(a(\Theta))^2\log(2d)\right)$,

$$W_1\left(\mathcal{L}(X(t;\mathbf{x}),\mathcal{L}(\mathbf{X}(\infty))\right) \le \mathbb{E}(\|X(t;\mathbf{x}) - X(t;\mathbf{X}(\infty))\|_1)$$
$$\le C_1(\mathbf{x},\Theta)\left(2e^{-\frac{D_1 t}{R_1(\Theta,d)}} + e^{-\frac{t}{16D_2R_2(\Theta)}}\right) + C_2(\mathbf{x},\Theta,D_2)e^{-\frac{t}{8D_2R_2(\Theta)}}.$$

In particular, the relaxation time satisfies

$$t_{rel}(\mathbf{x}) \le \max\{D_1^{-1}R_1(\Theta,d)\log(8C_1(\mathbf{x},\Theta)) + 16D_2R_2(\Theta)\log[4(C_1(\mathbf{x},\Theta)+C_2(\mathbf{x},\Theta,D_2))],$$
$$t_0\left(1+(a(\Theta))^2\log(2d)\right)\}.$$

3.2 Outline of Approach

We now give a very broad outline of the approach taken in Banerjee and Budhiraja (2020) for proving Theorem 1.

(i) As mentioned before, the key idea is to use synchronous couplings, namely, coupled RBMs from different initial states but driven by the same Brownian motion. An important feature of such couplings is that the L^1-distance between the two processes $X(\cdot;\mathbf{0})$ and $X(\cdot;\mathbf{x})$ is non-increasing in time. The rate of decay of this L^1-distance is quantified in terms of the contraction coefficient defined in (8). Roughly speaking, the process is observed at random times $\{\eta^k(\mathbf{x}) : k \ge 1\}$ such that all the coordinates of $X(\cdot;\mathbf{x})$ have hit zero at least once between successive $\eta^k(\mathbf{x})$'s. It turns out that the L^1-distance between the synchronously coupled processes decreases by a factor of $1/2$ by time $\eta^{n(R)}(\mathbf{x})$. Hence, to obtain bounds on L^1-Wasserstein distance, it suffices to estimate how many of these $\eta^k(\mathbf{x})$'s are smaller than some given large time t.

(ii) For any $\mathbf{v} > 0$ in $\mathbb{R}^d$ satisfying $R^{-1}\mathbf{v} \leq \mathbf{b}$, the RBM with oblique reflection in (5) can be dominated in a suitable manner by a normally reflected Brownian motion with drift $-\mathbf{v}$ and the same driving Brownian motion. One can construct appropriate Lyapunov functions and associated small sets for this dominating process by analyzing weighted sup-norms of its coordinates. This can be then used to estimate the number of times the coordinates of the dominating process hit zero by time t as a function of $\mathbf{v}$. This, in turn, lower bounds the number of $\eta^k(\mathbf{x})$'s for the original RBM less than t in terms of $\mathbf{v}$. Optimizing over feasible choices of $\mathbf{v}$ gives the result.

3.3 Examples

We now describe how Theorem 1 can be used to obtain bounds on the rate of convergence to equilibrium in two examples that are discussed in Sects. 3.3.1 and 3.3.2 below.

3.3.1 Blanchet-Chen RBM

This refers to the class of RBM under the set of assumptions in Blanchet and Chen (2020), which are:

(BC1) The matrix P is substochastic and there exist $\kappa > 0$ and $\beta \in (0, 1)$ not depending on the dimension d such that $\|\mathbf{1}^T P^n\|_\infty \leq \kappa(1-\beta)^n$ for all $n \geq 0$.
(BC2) There exists $\delta > 0$ independent of d such that $R^{-1}\mu < -\delta\mathbf{1}$.
(BC3) There exists $\sigma > 0$ independent of d such that $\sigma_i := \sqrt{\Sigma_{ii}}$ satisfies $\sigma^{-1} \leq \sigma_i \leq \sigma$ for every $1 \leq i \leq d$.

Under the above conditions Blanchet and Chen (2020) give a polynomial bound of $O(d^4(\log d)^2)$ on the relaxation time of the RBM. As shown in the following theorem, Theorem 1 gives a substantial improvement by establishing a polylogarithmic relaxation time of $O((\log d)^2)$.

Theorem 2 (Banerjee and Budhiraja, 2020) *Under Assumptions (BC1), (BC2), and (BC3), there exist positive constants E_1, E_2, E_3, E_4, t_1 such that for any $\mathbf{x} \in \mathbb{R}^d_+$, $t \geq t_1 \max\{\|\mathbf{x}\|_\infty, \log(2d)\}$,*

$$\mathbb{E}(\|X(t;\mathbf{x}) - X(t;\mathbf{X}(\infty))\|_1) \leq 2\left(2\|\mathbf{x}\|_1 + E_1 d^2\right) e^{-E_2 t/\log(2d)} + \left(4\|\mathbf{x}\|_1 + E_1 d^2\right) e^{-E_4 t/2} + E_3 d^2 e^{-E_4 t}.$$

In particular, the relaxation time satisfies

$$t_{rel}(\mathbf{x}) \leq \max\left\{E_2^{-1}\log\left[8\left(2\|\mathbf{x}\|_1 + E_1 d^2\right)\right]\log(2d) + E_4^{-1}\left[2\log\left[8\left(4\|\mathbf{x}\|_1 + E_1 d^2\right)\right] + \log(8E_3 d^2)\right],\right.$$
$$\left. t_1 \max\{\|\mathbf{x}\|_\infty, \log(2d)\}\right\}.$$

3.3.2 Gap Process of Rank-Based Diffusions

Rank-based diffusions are interacting particle systems where the drift and diffusion coefficient of each particle depend on its rank. Mathematically, they are represented by the SDE:

$$dY_i(t) = \left(\sum_{j=0}^{d} \delta_j \mathbf{1}_{[Y_i(t)=Y_{(j)}(t)]}\right) dt + \left(\sum_{j=0}^{d} \sigma_j \mathbf{1}_{[Y_i(t)=Y_{(j)}(t)]}\right) dW_i(t) \tag{9}$$

for $0 \le i \le d$, where $\{Y_{(j)}(t) : t \ge 0\}$ denotes the trajectory of the rank j particle as a function of time t ($Y_{(0)}(t) \le \cdots \le Y_{(d)}(t)$ for all $t \ge 0$), δ_j, σ_j denote the drift and diffusion coefficients of the rank j particle, and W_i, $0 \le i \le d$, are mutually independent standard one-dimensional Brownian motions. We will assume throughout that $\sigma_i > 0$ for all $0 \le i \le d$. Rank-based diffusions have been proposed and extensively studied as models for problems in finance and economics. A special case is the Atlas model Fernholz (2002), where the minimum particle (i.e., the particle with rank 1) is a Brownian motion with positive drift and the remaining particles are Brownian motions without drift (i.e., $\delta_i = 0$ for all $i \ge 1$). The general setting considered in (9) was introduced in Banner et al. (2005). In order to study the long-time behavior, it is convenient to consider the gap process $Z = (Z_1, \ldots, Z_d)$, given by $Z_i = Y_{(i)} - Y_{(i-1)}$ for $1 \le i \le d$. The process $Z \equiv Z(t; \mathbf{z})$ is a RBM in $\mathbb{R}^d_+$ given as

$$Z(t; \mathbf{z}) = \mathbf{z} + \mathbf{D}B(t) + \mu t + RL(t)$$

where $\mathbf{z}$ is the initial gap sequence, B is a standard d-dimensional Brownian motion, $\mu_i = \delta_i - \delta_{i-1}$ for $1 \le i \le d$, $\mathbf{D} \in \mathbb{R}^{d\times d}$, L is the local time process associated with Z and R satisfies Assumption (A1), namely, it belongs to the Harrison-Reiman class. The covariance matrix $\Sigma = \mathbf{D}\mathbf{D}^T$ has entries $\Sigma_{ii} = \sigma_{i-1}^2 + \sigma_i^2$ for $1 \le i \le d$, $\Sigma_{i(i-1)} = -\sigma_{i-1}^2$ for $2 \le i \le d$, $\Sigma_{i(i+1)} = -\sigma_i^2$ for $1 \le i \le d-1$ and $\Sigma_{ij} = 0$ otherwise. In particular, (A3) is satisfied, namely, Σ is positive definite. Moreover, R is given explicitly as $R = I - P^T$, where P is the substochastic matrix given by $P_{i(i+1)} = P_{i(i-1)} = 1/2$ for all $2 \le i \le d-1$, $P_{12} = P_{d(d-1)} = 1/2$ and $P_{ij} = 0$ if $|i - j| \ge 2$. From Harrison and Williams (1987), the process is positive recurrent and has a unique stationary distribution if Assumption (A2) is satisfied, namely, $\mathbf{b} = -R^{-1}\mu > 0$, which is same as the following condition:

$$b_k = \sum_{i=1}^{k} (\delta_{i-1} - \overline{\delta}) > 0 \text{ for } 1 \le k \le d, \text{ where } \overline{\delta} = (d+1)^{-1} \sum_{j=0}^{d} \delta_j. \tag{10}$$

In the special case where

$$\sigma_i^2 - \sigma_{i-1}^2 = \sigma_1^2 - \sigma_0^2 \text{ for all } 1 \le i \le d, \tag{11}$$

the stationary distribution is explicit and takes the form $\mathcal{L}(\mathbf{Z}(\infty)) = \otimes_{k=1}^{d} \mathrm{Exp}(2b_k \left(\sigma_{k-1}^2 + \sigma_k^2\right)^{-1})$ (see Sect. 5 of Ichiba et al., 2011). For the general case (i.e., σ_i are strictly positive and (10) is satisfied), explicit formulas for stationary distribution are not available, however from Budhiraja and Lee (2007), the law of $Z(t;\mathbf{z})$ converges to the unique stationary distribution in (weighted) total variation distance at an (albeit implicit) exponential rate.

From Theorem 1, one can give the following bound on the rate of L^1-Wasserstein convergence of the gap process to $\mathbf{Z}(\infty)$. Note that we do not require reversibility or an explicit expression for the stationary measure.

Two key quantities appearing in the rate of convergence are

$$a^* := \sup_{1\le i\le d} \frac{i(d+1-i)}{b_i}, \quad \sigma = \left(\sup_{0\le i\le d} \sigma_i\right) \vee \left(\sup_{0\le i\le d} \sigma_i^{-1}\right) \tag{12}$$

where b_i are defined in (10) and σ_i is the standard deviation of the rank i particle (see (9)).

Theorem 3 (Banerjee and Budhiraja, 2020) *There exist positive constants F_1, F_2, F_3, F_4, t_2 such that for any $d \in \mathbb{N}$, $\mathbf{z} \in \mathbb{R}_+^d$ and any $t \ge t_2 \max\{\sigma^2 a^* \|\mathbf{z}\|_\infty, 1 + \sigma^2 a^{*2} \log(2d)\}$,*

$$\begin{aligned}\mathbb{E}(\|Z(t;\mathbf{z}) - Z(t;\mathbf{Z}(\infty))\|_1) &\le 2\left(2\|\mathbf{z}\|_1 + F_1\sigma^2 a^* d^3\right) e^{-F_2 t/\left[d^2(1+\sigma^2 a^{*2}\log(2d))\right]} \\ &+ \left(4\|\mathbf{z}\|_1 + F_1\sigma^2 a^* d^3\right) e^{-F_4 t/[2\sigma^4 a^{*2}(d+1)^2]} + F_3\sigma^2 a^* d^{7/2} e^{-F_4 t/[\sigma^4 a^{*2}(d+1)^2]}.\end{aligned}$$

In particular, the relaxation time satisfies

$$\begin{aligned}t_{rel}(\mathbf{z}) \le \max\Big\{ & F_2^{-1}\left[d^2(1+\sigma^2 a^{*2}\log(2d))\right] \log\left[8\left(2\|\mathbf{z}\|_1 + F_1\sigma^2 a^* d^3\right)\right] \\ & + F_4^{-1}\sigma^4 a^{*2}(d+1)^2\left[2\log\left[8\left(4\|\mathbf{z}\|_1 + F_1\sigma^2 a^* d^3\right)\right] + \log(8F_3\sigma^2 a^* d^{7/2})\right], \\ & t_2 \max\{\sigma^2 a^* \|\mathbf{z}\|_\infty, 1 + \sigma^2 a^{*2}\log(2d)\}\Big\}.\end{aligned}$$

Remark 1 The standard Atlas model Fernholz (2002) is a special case of (9) with $\delta_0 = 1$, $\delta_i = 0$ for all $i \ge 1$ and $\sigma_i = 1$ for all i. For this model, using (10), for any $k \ge 1$,

$$b_k = \sum_{i=1}^{k} (\delta_{i-1} - \bar{\delta}) = \frac{(d+1-k)}{d+1}$$

and

$$a^* := \sup_{1\le i\le d} \frac{i(d+1-i)}{b_i} = \sup_{1\le i\le d} i(d+1) = d(d+1), \quad \sigma = 1.$$

Using these in Theorem 3, one obtains positive constants G_1, G_2, G_3, G_4, t_3 such that for any $d \in \mathbb{N}$, $\mathbf{z} \in \mathbb{R}_+^d$ and any $t \ge t_3\{d^2\|\mathbf{z}\|_\infty, 1 + d^2\log(2d)\}$,

$$\mathbb{E}(\|Z(t;\mathbf{z}) - Z(t;\mathbf{Z}(\infty))\|_1) \le G_1\left(\|\mathbf{z}\|_1 + d^5\right) e^{-G_2 t/d^6\log(2d)} + G_3 d^{11/2} e^{-G_4 t/d^6}.$$

In particular, the relaxation time for the standard Atlas model is $O(d^6(\log d)^2)$ as $d \to \infty$.

4 Dimension-Free Convergence Rates for Local Functionals of RBM

Typically, growing dimension slows down the rate of convergence for the whole system, as is reflected in the bounds obtained in the previous section, but empirically one often observes a much faster convergence rate to equilibrium of *local statistics* of the system. In this Section, we describe results for a class of RBMs for which convergence rates of local statistics do not depend on the underlying dimension of the entire system. More precisely, consider a family of processes $X^{(d)} \sim \text{RBM}(\mu^{(d)}, \Sigma^{(d)}, R^{(d)})$ indexed by the dimension $d \geq 1$ (the superscript (d) is dropped subsequently for notational convenience). Conditions on $(\mu^{(d)}, \Sigma^{(d)}, R^{(d)})$ are sought under which, for any fixed $k \in \mathbb{N}$, the marginal distribution of the first k-coordinates approaches the stationary marginal at a rate independent of d. The results below are taken from Banerjee and Brown (2022), where this phenomenon is named *dimension-free local convergence*.

Mathematically, this is challenging as the local evolution is no longer Markovian and the techniques in Blanchet and Chen (2020), Banerjee and Budhiraja (2020) cannot be readily applied. A crucial observation is that certain *weighted L^1-distances* (see $\|\cdot\|_{1,\beta}$ defined in Sect. 4.1) between synchronously coupled RBMs show dimension-free contraction rates. The evolution of such weighted distances are tracked in time for synchronously coupled RBMs $X(\cdot; \mathbf{0})$ and $X(\cdot; \mathbf{x})$ for $\mathbf{x} \in \mathbb{R}^d_+$. It is shown in Banerjee and Brown (2022) that for this distance to decrease by a dimension-free factor of its original value, only a subset of coordinates of $X(\cdot; \mathbf{x})$, whose cardinality depends on the value of the original distance (and not directly on the dimension d), need to hit zero. This is in contrast with the unweighted L^1-distance considered in Blanchet and Chen (2020), Banerjee and Budhiraja (2020) where all the coordinates need to hit zero to achieve such a contraction, thereby slowing down the convergence rate. Consequently, by tracking the hitting times to zero of a time-dependent number of coordinates, one achieves dimension-free convergence rates in this weighted L^1-distance as stated in Theorem 4. This, in turn, gives dimension-free local convergence as is made precise in (16). In Sect. 4.4, an application of Theorem 4 is presented for the gap process of a rank-based interacting particle system with asymmetric collisions to obtain explicit rates exhibiting dimension-free local convergence.

4.1 A Weighted L^1-Distance Governing Dimension-Free Local Convergence

The investigation of dimension-free convergence relies on the analysis of the weighted L^1-distance $\|X(\cdot;\mathbf{x}) - X(\cdot;\mathbf{X}(\infty))\|_{1,\beta} := \sum_{i=1}^d \beta^i |X_i(\cdot;\mathbf{x}) - X_i(\cdot;\mathbf{X}(\infty))|$ in time, for appropriate choices of $\beta \in (0,1)$. Towards this end, the following functionals play a key role:

$$u_\beta(\mathbf{x},t) = \|R^{-1}\left(X(t;\mathbf{x}) - X(t;\mathbf{0})\right)\|_{1,\beta} := \sum_{i=1}^d \beta^i \left|\left[R^{-1}\left(X(t;\mathbf{x}) - X(t;\mathbf{0})\right)\right]_i\right|, \quad (13)$$

$$u_{\pi,\beta}(t) = u_\beta(\mathbf{X}(\infty),t), \qquad t \geq 0. \quad (14)$$

In the following, when β is clear from context, we will suppress dependence on β and write u for u_β and u_π for $u_{\pi,\beta}$. The above functionals are convenient because the vector $R^{-1}\left(X(t;\mathbf{x}) - X(t;\mathbf{0})\right)$ is coordinate wise non-negative and non-increasing in time. Moreover, $R^{-1}\left(X(t;\mathbf{x}) - X(t;\mathbf{0})\right) \geq X(t;\mathbf{x}) - X(t;\mathbf{0}) \geq 0$ for all $t \geq 0$. This fact and the triangle inequality can be used to show for any $\mathbf{x} \in \mathbb{R}^d_+$, $t \geq 0$,

$$\|\left(X(t;\mathbf{x}) - X(t;\mathbf{X}(\infty))\right)\|_{1,\beta} \leq u(\mathbf{x},t) + u_\pi(t),$$

where the bound on the right hand side is non-increasing in time. Due to the monotonicity of the bound, it suffices to find 'events' along the trajectory of the coupled processes that lead to a reduction in this bound by a dimension-independent factor. Using this idea, conditions are obtained under which there exists a d-independent $\beta \in (0,1)$ and a function $f : \mathbb{R}_+ \mapsto \mathbb{R}_+$ not depending on the dimension d such that $f(t) \to 0$ as $t \to \infty$ and, for any $\mathbf{x}$ in an appropriate subset $\mathcal{S}$ of $\mathbb{R}^d_+$,

$$\mathbb{E}\left[\|\left(X(t;\mathbf{x}) - X(t;\mathbf{X}(\infty))\right)\|_{1,\beta}\right] \leq \mathbb{E}\left[u(\mathbf{x},t) + u_\pi(t)\right] \leq Cf(t), \qquad t \geq t_0, \quad (15)$$

where $C, t_0 \in (0,\infty)$ are constants not depending on d (but can depend on $\mathbf{x}$). This, in particular, gives dimension-free local convergence in the following sense: For any $k \in \{1,\ldots,d\}$, consider any function $\phi : \mathbb{R}^k_+ \mapsto [0,\infty)$ which is Lipschitz, i.e., there exists $L_\phi > 0$ such that

$$|\phi(\mathbf{u}) - \phi(\mathbf{v})| \leq L_\phi \|\mathbf{u} - \mathbf{v}\|_1, \qquad \mathbf{u},\mathbf{v} \in \mathbb{R}^k_+.$$

Recall that $\mathcal{L}(Z)$ denotes the law of a random variable Z and W_1 denotes the associated L^1-Wasserstein distance. Then, (15) implies for $\mathbf{x} \in \mathcal{S}$,

$$\begin{aligned} W_1\left(\mathcal{L}(\phi(X|_k(t;\mathbf{x}))), \mathcal{L}(\phi(\mathbf{X}|_k(\infty)))\right) &\leq \mathbb{E}\left[|\phi(X|_k(t;\mathbf{x})) - \phi(X|_k(t;\mathbf{X}(\infty)))|\right] \\ &\leq C\beta^{-k}L_\phi f(t), \qquad t \geq t_0, \end{aligned} \quad (16)$$

where for $\mathbf{x} = (x_1,\ldots,x_d)' \in \mathbb{R}^d_+$, $\mathbf{x}|_k = (x_1,\ldots,x_k)$.

4.2 Parameters and Assumptions

We now define the parameters that govern dimension-free local convergence which, in turn, are defined in terms of the original model parameters (μ, Σ, R) of the associated RBM. We abbreviate $\sigma_i = \sqrt{\Sigma_{ii}}$, $i = 1, \ldots, d$. Define for $1 \leq k \leq d$,

$$\mathbf{b}^{(k)} := -(R|_k)^{-1}\mu|_k, \qquad \mathbf{b} = \mathbf{b}^{(d)},$$

$$\underline{b}^{(k)} := \min_{1\leq i\leq k} \mathbf{b}_i^{(k)}, \qquad a^{(k)} := \max_{1\leq i\leq k} \frac{1}{\mathbf{b}_i^{(k)}} \sum_{j=1}^{k} ((R|_k)^{-1})_{ij}\sigma_j, \tag{17}$$

where $\mu|_k$ denotes the restriction of μ onto first k coordinates and $(R|_k)^{-1}$ denotes the inverse of the principal $k \times k$ submatrix of R. To get a sense of why these parameters are crucial, recall that our underlying strategy is to obtain contraction rates of $u(\mathbf{x}, \cdot)$ defined in (13) by estimating the number of times a subset of the coordinates of $X(\cdot; \mathbf{x})$, say $\{X_1(\cdot; \mathbf{x}), \ldots, X_k(\cdot; \mathbf{x})\}$, $k \leq d$, hit zero. However, this subset does not evolve in a Markovian way. Thus, we use monotonicity properties of RBMs to couple this subset with a $\mathbb{R}_+^k$-valued reflected Brownian motion $\bar{X}(\mathbf{x}|_k, \cdot)$, started from $\mathbf{x}|_k$ and defined in terms of $\mu|_k$, $\Sigma|_k$, $R|_k$ and (a possible restriction of) the same Brownian motion driving $X(\cdot; \mathbf{x})$, such that $X_i(\mathbf{x}, t) \leq \bar{X}_i(\mathbf{x}|_k, t)$ for all $1 \leq i \leq k$. The analysis in Banerjee and Budhiraja (2020) shows that the parameters defined in (17) with $k = d$ can be used to precisely estimate the minimum number of times all coordinates of $X(\cdot; \mathbf{x})$ hit zero by time t as t grows. Applying this approach, for any $1 \leq k \leq d$, the parameters (17) can be used to quantify analogous hitting times for the process $\bar{X}(\cdot; \mathbf{x}|_k)$ which, by the above coupling, gives control over corresponding hitting times of $\{X_1(\cdot; \mathbf{x}), \ldots, X_k(\cdot; \mathbf{x})\}$.

Given below is a set of assumptions on the model parameters (μ, Σ, R) which guarantee dimension-free local convergence.

Assumptions DF: There exist d-independent constants $\underline{\sigma}, \overline{\sigma}, b_0 > 0$, $r^* \geq 0$, $M, C \geq 1$, $k_0 \in \{2, \ldots, d\}$ and $\alpha \in (0, 1)$ such that for all $d \geq k_0$,

I. $(R^{-1})_{ij} \leq C\alpha^{j-i}$ for $1 \leq i \leq j \leq d$,
II. $(R^{-1})_{ij} \leq M$ for $1 \leq i, j \leq d$,
III. $\underline{b}^{(k)} \geq b_0 k^{-r^*}$ for $k = k_0, \ldots, d$,
IV. $\sigma_i \in [\underline{\sigma}, \overline{\sigma}]$ for $1 \leq i \leq d$.

We explain why Assumptions DF are 'natural' in obtaining dimension-free local convergence. Since P is transient and substochastic, it can be associated with a killed Markov chain on $\{0\} \cup \{1, \ldots, d\}$ with transition matrix P on $\{1, \ldots, d\}$ and killed upon hitting 0 (i.e. probability of going from state $k \in \{1, \ldots, d\}$ to 0 is $1 - \sum_{l=1}^{d} P_{kl}$ and $P_{00} = 1$). Moreover, since P is transient and $R = I - P^T$, we have $R^{-1} = \sum_{n=0}^{\infty} (P^T)^n$. This representation shows that $(R^{-1})_{ij}$ is the expected number of visits to site i starting from j of this killed Markov chain. For fixed $\mathbf{x} \in \mathbb{R}_+^d$ and $k << d$, consider a local statistic of the form $\phi(X|_k(t; \mathbf{x}))$ as in (16). For this statistic

to stabilize faster than the whole system, we expect the influence of the far away coordinates $X|_j(\cdot;\mathbf{x})$, $j >> k$, to diminish in an appropriate sense as j increases. This influence is primarily manifested through the oblique reflection arising out of the R matrix via the above representation of R^{-1}. I of Assumptions DF quantifies this intuition by requiring that the expected number of visits to state i starting from state $j > i$ of the associated killed Markov chain decreases geometrically with $j - i$. This is the case, for example, when this Markov chain started from $j > i$ has a uniform 'drift' away from i towards the cemetery state. See Sect. 4.4. In more general cases, one can employ Lyapunov function type arguments Meyn and Tweedie (2012) to the underlying Markov chain to check I.

II above implies that the killed Markov chain starting from state j spends at most M expected time at any other site $i \in \{1, \ldots, d\}$ before it is absorbed in the cemetery state 0. This expected time, as calculations show, is intimately tied to decay rates of $\| (X(\cdot;\mathbf{x}) - X(\cdot;\mathbf{X}(\infty))) \|_{1,\beta}$.

As noted in Harrison and Williams (1987), Blanchet and Chen (2020), and Banerjee and Budhiraja (2020), the 'renormalized drift' vector $\mathbf{b}$ characterizes positive recurrence of the whole system. Assumption III above, imposes a power law type coordinate wise lower bound on the renormalized drift vector $\mathbf{b}^{(k)}$ of the projected system $X|_k(\cdot;\mathbf{x})$ as k grows. In particular, if $\underline{b}^{(k)}$ is uniformly lower bounded by b_0, one can take $r^* = 0$. Finally, assumption IV above is a quantitative 'uniform ellipticity' condition on the coordinates of the driving noise $\mathbf{D}B(\cdot)$.

4.3 Main Result

The following result gives explicit bounds on the decay of expectation of the weighted distance $\|X(\cdot;\mathbf{x}) - X(\cdot;\mathbf{X}(\infty))\|_{1,\sqrt{\alpha}}$ (α defined in I of Assumptions DF) with time, for RBMs satisfying Assumptions DF. We first define some constants that will appear in Theorem 4. They are needed to bound moments of certain weighted norms of the stationary random variable $\mathbf{X}(\infty)$.

Suppose Assumptions DF hold, with $k_0 \in \{2, \ldots, d\}$ and $\alpha \in (0, 1)$ defined therein. Set

$$L_1 := k_0^{r^*+1} + \sum_{i=k_0}^{d} i^{3+r^*} \alpha^{i/8}. \tag{18}$$

Also, for $B \in (0, \infty)$, define the set

$$\mathcal{S}(b, B) := \left\{ x \in \mathbb{R}_+^d : \sup_{1 \le i \le d} \underline{b}^{(i)} \|x|_i\|_\infty \le B \right\}. \tag{19}$$

Theorem 4 (Banerjee and Brown, 2022) *Let X be a (μ, Σ, R)-RBM. Suppose that Assumptions DF hold with $\alpha \in (0, 1)$ defined therein.*

Fix any $B \in (0, \infty)$. Then there exist constants $C_0, C_0', C_1 > 0$ not depending on d, r^* or B such that with $t_0' = t_0'(r^*) = C_0' (1 + r^*)^{8+4r^*}$ and $L_1, L_2, \mathcal{S}(b, B)$ as defined in (18)–(19), we have for any $\mathbf{x} \in \mathcal{S}(b, B)$ and any $d > t_0'^{1/(4+2r^*)}$,

$$\mathbb{E}\left[\|X(t;\mathbf{x}) - X(t;\mathbf{X}(\infty))\|_{1,\sqrt{\alpha}}\right] \leq \begin{cases} C_1\left(L_1\sqrt{1+t^{1/(4+2r^*)}} + \|\mathbf{x}\|_\infty \exp\left\{B/\underline{\sigma}^2\right\}\right) \exp\{-C_0 t^{1/(4+2r^*)}\}, & t_0' \leq t < d^{4+2r^*}, \\ C_1\left(L_1\sqrt{1+t^{1/(4+2r^*)}} + \|\mathbf{x}\|_\infty \exp\left\{B/\underline{\sigma}^2\right\}\right) \exp\left\{-C_0 \frac{t}{d^{3+2r^*}}\right\}, & t \geq d^{4+2r^*}. \end{cases} \tag{20}$$

Theorem 4 directly implies dimension-free bounds on $t \mapsto \|X(t;\mathbf{x}) - X(t;\mathbf{X}(\infty))\|_{1,\sqrt{\alpha}}$ in the sense of (15) which, in turn, produce dimension-free local convergence rates as given by (16). The multi-part bound in the theorem is designed to 'continuously interpolate' the *dimension-free stretched exponential bounds* with the *dimension-dependent exponential bounds* obtained in Banerjee and Budhiraja (2020).

4.4 Application to the Asymmetric Atlas Model

Consider the interacting particle system represented by the following SDE:

$$Y_{(k)}(t) = Y_{(k)}(0) + \mathbf{1}_{k=1} t + B_k^*(t) + pL_k^*(t) - qL_{k+1}^*(t), \quad t \geq 0, \tag{21}$$

for $0 \leq k \leq d$, $p \in (0, 1)$, $q = 1 - p$. Here, $L_0^*(\cdot) \equiv L_{d+1}^*(\cdot) \equiv 0$, and for $0 \leq k \leq d - 1$, $L_{k+1}^*(\cdot)$ is a continuous, non-decreasing, adapted process that denotes the collision local time between the kth and $(k + 1)$th coordinate processes of $Y_{(\cdot)}$, namely $L_{k+1}^*(0) = 0$ and $L_{k+1}^*(\cdot)$ can increase only when $Y_{(k)} = Y_{(k+1)}$. $B_k^*(\cdot)$, $0 \leq k \leq d$, are mutually independent standard one-dimensional Brownian motions. Each of the $d + 1$ ranked particles with trajectories given by $(Y_{(0)}(\cdot), \ldots, Y_{(d)}(\cdot))$ evolves as an independent Brownian motion (with the particle 0 having unit positive drift) when it is away from its neighboring particles, and interacts with its neighbors through possibly asymmetric collisions. The symmetric Atlas model, namely, the case $p = 1/2$, was introduced in Fernholz (2002) as a mathematical model for stochastic portfolio theory. The asymmetric Atlas model was introduced in Karatzas et al. (2016). It was shown that it arises as the scaling limit of numerous well-known interacting particle systems involving asymmetrically colliding random walks Karatzas et al. (2016, Sect. 3). Since then, this model has been extensively analyzed: see Karatzas et al. (2016); Ichiba et al. (2013b, a); Sarantsev (2017) and references therein.

The gaps between the particles, defined by $Z_i(\cdot) = Y_{(i)}(\cdot) - Y_{(i-1)}(\cdot)$, $1 \le i \le d$, evolve as an RBM(μ, Σ, R) with Σ given by $\Sigma_{ii} = 2$ for $i = 1, \ldots, d$, $\Sigma_{ij} = -1$ if $|i - j| = 1$, $\Sigma_{ij} = 0$ if $|i - j| > 1$, μ given by $\mu_1 = -1$, $\mu_j = 0$ for $j = 2, \ldots, d$, and $R = I - P^T$, where

$$P_{ij} = \begin{cases} p & j = i + 1, \\ 1 - p & j = i - 1, \\ 0 & \text{otherwise.} \end{cases} \tag{22}$$

Clearly, R is in the Harrison-Reiman class. The paper Banerjee and Brown (2022) establishes dimension-free local convergence for the gap process of the Asymmetric Atlas model when $p \in (1/2, 1)$. Recall that the reflection matrix $R = I - P^T$ is associated with a killed Markov chain. For the Asymmetric Atlas model, this Markov chain has a more natural description as a random walk on $\{0, 1, \ldots, d + 1\}$ which increases by one at each step with probability p and decreases by one with probability $1 - p$, and is killed when it hits either 0 or $d + 1$. Then for $1 \le i, j \le d$, $(R^{-1})_{ij}$ is the expected number of visits to i starting from j by this random walk before it hits 0 or $d + 1$. Since $p > 1 - p$, the random walk has a drift towards $d + 1$, which suggests I, II of Assumptions DF hold. This is confirmed by direct computation, which gives for $q = 1 - p$,

$$(R^{-1})_{ij} = \begin{cases} \frac{(q/p)^{j-i}}{p-q} \frac{\left(1-(q/p)^{i}\right)\left(1-(q/p)^{d+1-j}\right)}{1-(q/p)^{d+1}} \le \frac{(q/p)^{j-i}}{p-q} & 1 \le i \le j \le d, \\ \frac{(p/q)^{i-j}}{p-q} \frac{\left((p/q)^{j}-1\right)\left((p/q)^{d+1-i}-1\right)}{(p/q)^{d+1}-1} \le \frac{1}{p-q} & 1 \le j < i \le d. \end{cases} \tag{23}$$

Now I, II, and IV of Assumptions DF hold with $M = C = \frac{1}{p-q}$, $\alpha = \frac{q}{p}$ and $\underline{\sigma} = \overline{\sigma} = \sqrt{2}$. Furthermore, the restriction $P|_k$ is defined exactly as in (22) with k in place of d. Thus, $(R|_k)^{-1}$ is given by (23) with k in place of d, and $\mathbf{b}^{(k)} = -(R|_k)^{-1}\mu|_k$ is the first column of $(R|_k)^{-1}$. This entails,

$$\begin{aligned} \mathbf{b}_i^{(k)} &= \frac{(p/q)^{i-1}}{p-q} \frac{((p/q)-1)\left((p/q)^{k+1-i}-1\right)}{(p/q)^{k+1}-1} \ge \frac{1}{q}\left(\frac{p}{q}\right)^{k-1} \frac{((p/q)-1)}{(p/q)^{k+1}} \\ &= \frac{p-q}{p^2} =: b_0 > 0, \qquad 1 \le i \le k,\ 1 \le k \le d. \end{aligned} \tag{24}$$

Thus, $\underline{b}^{(k)} \ge b_0$ for all $1 \le k \le d$, uniformly in d. This shows that III of Assumptions DF holds with b_0 specified by (24) and $r^* = 0$. Moreover, it follows from the first equality in (24) that $\mathbf{b}_i^{(k)} \le p/(p-q)$ for all $1 \le k \le d$ and $1 \le i \le k$. Therefore, recalling the definition of $\mathcal{S}(b, \cdot)$ from (19), for any $\mathbf{z} \in \mathbb{R}_+^d$,

$$\mathbf{z} \in \mathcal{S}(b,\, p\|z\|_\infty/(2p-1)).$$

The above observations result in the next theorem, which follows directly from Theorem 4. These bounds imply dimension-free convergence rates in the sense of (15) and (16).

Theorem 5 (Banerjee and Brown, 2022) *Suppose Z is the RBM representing the gap process of the Asymmetric Atlas model with $p \in (1/2, 1)$. Then there exist constants $\bar{C}, \bar{C}_0, t_0' > 0$ depending on p but not on d such that for $d > t_0'$,*

$$\begin{aligned} &\mathbb{E}\left[\|Z(t;\mathbf{z}) - Z(t;\mathbf{Z}(\infty))\|_{1,\sqrt{\frac{1-p}{p}}}\right] \\ &\leq \begin{cases} \bar{C}\left(\sqrt{1+t^{1/4}} + \|\mathbf{z}\|_\infty e^{p\|\mathbf{z}\|_\infty/(4p-2)}\right) e^{-\bar{C}_0 t^{1/4}}, & t_0' \leq t < d^4, \\ \bar{C}\left(\sqrt{1+t^{1/4}} + \|\mathbf{z}\|_\infty e^{p\|\mathbf{z}\|_\infty/(4p-2)}\right) e^{-\bar{C}_0 t/d^3}, & t \geq d^4. \end{cases} \end{aligned} \tag{25}$$

We note that the law of $\mathbf{Z}(\infty)$ has an explicit product form in this case. See Sarantsev (2017, Proposition 2.1).

4.5 Polynomial Local Convergence Rates for Perturbations from Stationarity of the Atlas Model

A natural question is whether one can expect dimension-free local convergence when Assumptions DF do not hold. This is currently an open problem. For one example where one does not expect such dimension-free local convergence to hold, consider the Asymmetric Atlas model from Sect. 4.4 when $p < 1/2$. Observe from (21) that a larger change of momentum happens to the lower particle (smaller k) during each collision. Hence, effects of unstable gaps between far away particles can propagate in a more unabated fashion to the lower particles. Moreover, by analyzing the associated killed Markov chain, one concludes that it takes roughly linear time in d for the effect of the dth gap to propagate to the first gap. This heuristic reasoning suggests that one cannot expect any reasonable form of dimension-free local convergence in this case.

Now consider the 'critical' case: the (symmetric) Atlas model, namely, the case $p = 1/2$ in the model of Sect. 4.4. Note that here, $\mathbf{b} = -R^{-1}\mu = \{(R^{-1})_{i1}\}_{i=1}^d > 0$ and $\Sigma_{ii} = 2$ for all i. Therefore, there exists a unique stationary distribution. In fact, if $\mathbf{Z}(\infty)$ denotes the corresponding stationary distributed gap, it holds that Harrison and Williams (1987), Ichiba et al. (2011)

$$\mathbf{Z}(\infty) \sim \bigotimes_{i=1}^d \mathrm{Exp}\left(2\left(1 - \frac{i}{d+1}\right)\right). \tag{26}$$

In the symmetric Atlas model, the associated killed Markov chain behaves as a simple symmetric random walk away from the cemetery points, and lacks the strong drift

towards the cemetery points as formulated in I of Assumptions DF. Indeed, R^{-1} is given as (see, e.g., Banerjee & Budhiraja 2020, Proof of Theorem 4, or take $p \to 1/2$ in (23)):

$$(R^{-1})_{ij} = \begin{cases} 2i\left(1 - \frac{j}{d+1}\right) & 1 \le i \le j \le d, \\ 2j\left(1 - \frac{i}{d+1}\right) & 1 \le j < i \le d. \end{cases} \tag{27}$$

The above representation shows that R^{-1} violates I, II of Assumptions DF (consider, for example, $i = j = \lfloor d/2 \rfloor$). Hence, the techniques sketched above for obtaining dimension-free convergence fail to apply in this case. Instead, a different approach is taken in Banerjee and Brown (2022) which involves analyzing the long-term behavior of pathwise derivatives of the RBM in initial conditions. Using this analysis, *polynomial convergence rates* to stationarity in L^1-Wasserstein distance are obtained when the initial distribution of the gaps between particles is in an appropriate perturbation class of the stationary measure (see Banerjee and Brown, 2022, Sect. 3.1, Definition 1). We give a representative result obtained as a special case of the more general theorem Banerjee and Brown (2022, Theorem 4).

Theorem 6 (Banerjee and Brown, 2022) *Consider* $\mathbf{Y} = (Y_1, Y_2, \ldots)^T$ *where* $\{Y_i\}_{i \ge 1}$ *are independent random variables with* $Y_i \sim \text{Exp}(i^{1+\beta})$ *(exponential with mean* $i^{-(1+\beta)}$*), for some* $\beta > 0$*. Then there exist positive constants* C_1, C_2 *not depending only on* β *such that*

$$\mathbb{E}\left[\|Z(t; \mathbf{Z}(\infty) + \mathbf{Y}|_d) - Z(t; \mathbf{Z}(\infty))\|_1\right] \le \begin{cases} C_1 t^{-\frac{\beta}{1+\beta}\frac{3}{32}}, & t_0' \le t < d^{16/3}, \\ C_2 d \exp\left(-C_0 \frac{t}{d^6 \log(2d)}\right), & t \ge t_0'' \, d^4 \log(2d), \end{cases}$$

where $t_0' \in (0, \infty)$ *does not depend on* d *or* β*.*

The analysis of the derivative process is based on the analysis of a random walk in a random environment generated by the times and locations where the RBM hits faces of $\mathbb{R}^d_+$. We mention here that Blanchet et al. (2021) has recently used the derivative process to study convergence rates for RBMs satisfying certain strong uniformity conditions in dimension (which do not hold for the symmetric Atlas model).

Although currently there are no available lower bounds, it seems plausible (based on informal calculations using the associated killed Markov chain) that the true L^1-Wasserstein distance indeed decays polynomially for $t \le t(d)$, for some $t(d) \to \infty$ as $d \to \infty$. The polynomial rates of convergence to stationarity obtained in Banerjee and Burdzy (2021) for the Potlatch process on $\mathbb{Z}^k$, which (for $k = 1$) can be loosely thought of as a 'Poissonian version' of the gap process of the infinite Atlas model constructed in Pal and Pitman (2008), lend further evidence to this belief.

Moreover, dimension-free local convergence is not expected to hold for initial distributions 'too far away' from $\mathbf{Z}(\infty)$. This belief stems from the analogy with the *infinite* Atlas model (see Sect. 5) which has uncountably many product form stationary distributions $\pi_a := \bigotimes_{i=1}^{\infty} \text{Exp}(2 + ia)$, $a \ge 0$ Sarantsev and Tsai (2017), only one of which, namely, π_0, is a weak limit of the stationary distribution (26)

of the d-dimensional Atlas model (extended to a measure on $\mathbb{R}_+^\infty$ in an obvious manner) as $d \to \infty$. This leads to the heuristic that, for large d, the d-dimensional gap process with initial distribution 'close' to the projection (onto the first d coordinates) of π_a for some $a > 0$ spends a long time near this projection (which diverges as $d \to \infty$) before converging to (26). From this heuristic, one expects that dimension-free convergence rates for associated statistics can only be obtained if the initial gap distribution is 'close' to the stationary measure (26) in a certain sense.

Evidence for this heuristic is provided in the few available results on 'uniform in dimension' convergence rates for rank-based diffusions Jourdain and Malrieu (2008), Jourdain and Reygner (2013). In both these papers, under strong convexity assumptions on the drifts of the particles, dimension-free exponential ergodicity was proven for the joint density of the particle system when the initial distribution is close to the stationary distribution as quantified by the Dirichlet energy functional (see Jourdain and Malrieu, 2008, Theorem 2.12 and Jourdain and Reygner, 2013, Corollary 3.8). The symmetric Atlas model lacks such convexity in drift and hence, the dimension-free Poincaré inequality for the stationary density, that is crucial to the methods of Jourdain and Malrieu (2008), Jourdain and Reygner (2013), does not apply.

5 Domains of Attraction of the Infinite Atlas Model

In the remaining two sections of this review, we will discuss infinite dimensional reflected Brownian motions. The processes we consider arise from the study of the infinite Atlas model and its various rank based diffusion model variants. This section will consider the standard Atlas model, whereas in the next section, we consider a more general family of rank based diffusions.

The Atlas model describes the evolution of a countable collection of particles on the real line moving according to mutually independent Brownian motions and such that, at any point of time, the lowest particle has a unit upward drift. Letting $\{Y_i(\cdot) : i \in \mathbb{N}_0\}$ denote the state processes of the particles in the system, one can formally describe the evolution by the following system of SDE:

$$dY_i(s) = \mathbf{1}(Y_i(s) = Y_{(0)}(s))ds + dW_i(s),\ s \geq 0, i \in \mathbb{N}_0, \tag{28}$$

where $Y_{(0)}(t)$ is the state of the lowest particle at time t. Define

$$\mathcal{U} := \left\{ \boldsymbol{y} = (y_0, y_1, \ldots)' \in \mathbb{R}^\infty : \sum_{i=0}^{\infty} e^{-\alpha[(y_i)_+]^2} < \infty \text{ for all } \alpha > 0 \right\}. \tag{29}$$

Following Sarantsev (2017), a $\boldsymbol{y} \in \mathbb{R}^\infty$ is called *rankable* if there is a one-to-one map $\boldsymbol{p}$ from $\mathbb{N}_0$ onto itself such that $y_{\boldsymbol{p}(i)} \leq y_{\boldsymbol{p}(j)}$ whenever $i < j, i, j \in \mathbb{N}_0$. If $\boldsymbol{y}$ is rankable, we will write $y_{(i)} := y_{\boldsymbol{p}(i)},\ i \in \mathbb{N}_0$. It is easily seen that any $\boldsymbol{y} \in \mathcal{U}$ is locally

finite and hence rankable. It was shown in Sarantsev (2017, Theorem 3.2 and Lemma 3.4) that, if the distribution of the initial vector $\mathbf{Y}(0)$ of the particles in the infinite Atlas model is supported within the set $\mathcal{U}$, then the system (28) has a weak solution that is unique in law and the solution process $\mathbf{Y}(t) \in \mathcal{U}$ a.s. for all $t \geq 0$. Consequently, it can be shown that, almost surely, the ranking $(Y_0(t), Y_1(t), \dots) \mapsto (Y_{(0)}(t), Y_{(1)}(t), \dots)$ is well defined (where $Y_{(i)}(t)$ is the state of the $i+1$th particle from below) for all $t \geq 0$. We will assume throughout that $0 \leq Y_0(0) \leq Y_1(0) \leq Y_2(0) \leq \dots$. By calculations based on Tanaka's formula (see Sarantsev, 2017, Lemma 3.5), it follows that the processes defined by

$$B_i^*(t) := \sum_{j=0}^{\infty} \int_0^t \mathbf{1}(Y_j(s) = Y_{(i)}(s)) dW_j(s), \ i \in \mathbb{N}_0, t \geq 0, \tag{30}$$

are independent standard Brownian motions which can be used to write down the following stochastic differential equation for $\{Y_{(i)}(\cdot) : i \in \mathbb{N}_0\}$:

$$dY_{(i)}(t) = \mathbf{1}(i = 0)dt + dB_i^*(t) - \frac{1}{2}dL_{i+1}^*(t) + \frac{1}{2}dL_i^*(t), \ t \geq 0, i \in \mathbb{N}_0. \tag{31}$$

Here, $L_0^*(\cdot) \equiv 0$ and for $i \in \mathbb{N}$, $L_i^*(\cdot)$ denotes the local time of collision between the $(i-1)$th and ith particle.

The gap process in the infinite Atlas model is the $\mathbb{R}_+^\infty$-valued process $\mathbf{Z}(\cdot) = (Z_1(\cdot), Z_2(\cdot), \dots)$ defined by

$$Z_i(\cdot) := Y_{(i)}(\cdot) - Y_{(i-1)}(\cdot), \ i \in \mathbb{N}. \tag{32}$$

We will be primarily interested in the long-time behavior of the gap process.

It was shown in Pal and Pitman (2008) that the distribution $\pi := \otimes_{i=1}^\infty \text{Exp}(2)$ is a stationary distribution of the gap process $\mathbf{Z}(\cdot)$. It was also conjectured there that this is the unique stationary distribution of the gap process. Surprisingly, this was shown to be not true in Sarantsev and Tsai (2017) who gave an uncountable collection of product form stationary distributions defined as

$$\pi_a := \bigotimes_{i=1}^{\infty} \text{Exp}(2 + ia), \ a \geq 0. \tag{33}$$

The 'maximal' stationary distribution π_0 (in the sense of stochastic domination) is somewhat special in this collection as described below. For this reason, in what follows, we will occasionally using the notation π for the measure π_0.

For initial gap distribution ν, we will denote by $\hat{\nu}_t$ the probability law of $\mathbf{Z}(t)$. Obviously, when $\nu = \pi_a$, $\hat{\nu}_t = \pi_a$ for all $t \geq 0$. In general, if $\hat{\nu}_t$ converges weakly to one of the stationary measures $\pi_a, a \geq 0$, we say that ν is in the *Domain of Attraction* (DoA) of π_a. Characterizing the domain of attraction of the above collection of stationary measures is a challenging open problem. It was shown in Sarantsev (2017,

Theorem 4.7) using comparison techniques with finite dimensional Atlas models that if the law ν of the initial gaps $\mathbf{Z}(0)$ stochastically dominates $\pi = \pi_0$ (in a coordinate-wise sense), then ν is in DoA of π, i.e. the law of $\mathbf{Z}(t)$ converges weakly to π as $t \to \infty$. Recently, Dembo et al. (2019) established a significantly larger domain of attraction for π using relative entropy and Dirichlet form techniques. They showed that a ν is in DoA of π if the random vector $\mathbf{Z}(0)$ with distribution ν almost surely satisfies the following conditions for some $\beta \in [1, 2)$ and eventually non-decreasing sequence $\{\theta(m) : m \geq 1\}$ with $\inf_m\{\theta(m-1)/\theta(m)\} > 0$:

$$\limsup_{m\to\infty} \frac{1}{m^{\beta}\theta(m)} \sum_{j=1}^{m} Z_j(0) < \infty, \tag{34}$$

$$\limsup_{m\to\infty} \frac{1}{m^{\beta}\theta(m)} \sum_{j=1}^{m} (\log Z_j(0))_- < \infty, \tag{35}$$

$$\liminf_{m\to\infty} \frac{1}{m^{\beta^2/(1+\beta)}\theta(m)} \sum_{j=1}^{m} Z_j(0) = \infty, \tag{36}$$

with the additional requirement that $\theta(m) \geq \log m$, $m \geq 1$, if $\beta = 1$. This is a significant advance in our understanding of the domain of attraction properties of the infinite Atlas model. However, the conditions (34)–(36) involve upper and lower bounding the growth rate of the starting points $Y_m(0) = \sum_{j=1}^{m} Z_j(0)$ with respect to m in terms of a common parameter $\beta \in [1, 2)$, which partially restricts its applicability. In particular, they do not cover all initial gap distributions that stochastically dominate π (e.g., $Z_j(0) \sim e^{j^2}$), which were shown to be in DoA of π in Sarantsev (2017). Moreover, one expects that the properties of the first few gaps should not impact the domain of attraction property of a given vector of initial gaps, however, (35) is not satisfied if even one of the gaps is zero, which makes this condition somewhat unnatural. This condition arises from the relative entropy methods used in Dembo et al. (2019) (see, e.g., the estimate (3.5) therein) which make an important use of the fact that the gaps are non-zero.

Beyond these results, little is known about the domain of attraction of stationary measures of the infinite Atlas model. Especially, for the other stationary measures π_a, $a > 0$, there are no results about the domain of attraction. The fundamental challenge in investigating the latter question is that stability for initial distribution π_a for any $a > 0$ involves a delicate interplay between the drift of the lowest ranked particle and the 'downward pressure' from the cloud of higher particles which has an exponentially increasing density. In particular, one cannot use approximation techniques based on finite dimensional systems as was done in previous works (namely, Sarantsev, 2017 and Dembo et al., 2019). This can also be observed, as noted previously, upon recalling that the marginals of the stationary distribution of the d-dimensional Atlas model in (26) approach those of π_0 as $d \to \infty$ and so a finite dimensional approximation approach is designed to *select* this particular stationary distribution of the infinite Atlas model.

In Banerjee and Budhiraja (2022), we consider a somewhat weaker formulation of domain of attraction of the stationary measures π_a, $a \geq 0$. Specifically, for initial gap distribution ν, define for $t > 0$

$$\nu_t := \frac{1}{t} \int_0^t \hat{\nu}_s ds,$$

where $\hat{\nu}_t$ is as introduced in the paragraph below (33). We say that a ν is in the *Weak Domain of Attraction* (WDoA) of π_a for $a \geq 0$, if $\nu_t \to \pi_a$ weakly as $t \to \infty$. Note that, although the convergence of the time averaged laws ν_t to π_a (ergodic limit) is implied by the convergence of $\hat{\nu}_t$ to π_a (marginal time limit), the converse is not clear. However, if ν is in the WDoA of π_a for some $a > 0$, it is *necessarily not* in the DoA of $\pi = \pi_0$. The paper Banerjee and Budhiraja (2022) provides sufficient conditions for a measure ν to be in the WDoA of π_a for a general $a \geq 0$. The main results of Banerjee and Budhiraja (2022) are described below.

5.1 Main Results

Denote by $\mathcal{S}_0$ the class of probability measures on $\mathbb{R}_+^\infty$ such that the corresponding $\mathbb{R}_+^\infty$-valued random variable $\mathbf{Y}(0) = (Y_i(0))_{i \in \mathbb{N}_0}$ is supported on $\mathcal{U}$ and

$$0 = Y_0(0) \leq Y_1(0) \leq \cdots \text{ a.s.}$$

Given a measure $\gamma \in \mathcal{S}_0$, from weak existence and uniqueness, we can construct a filtered probability space $(\Omega, \mathcal{F}, \mathbb{P}, \{\mathcal{F}_t\}_{t \geq 0})$ on which are given mutually independent $\mathcal{F}_t$-Brownian motions $\{W_i(\cdot), i \in \mathbb{N}_0\}$ and $\mathcal{F}_t$ adapted continuous processes $\{Y_i(t), t \geq 0, i \in \mathbb{N}_0\}$ such that Y_i solve (28) with $\mathbb{P} \circ (\mathbf{Y}(0))^{-1} = \gamma$. On this space, the processes $Y_{(i)}$ and Z_i are well defined and the distribution $\mathbb{P} \circ (\mathbf{Z}(\cdot))^{-1}$ is uniquely determined from γ. Furthermore, $\mathbf{Z}$ is a Markov process with values in (a subset of) $\mathbb{R}_+^\infty$. Let

$$\mathcal{S} = \{\mathbb{P} \circ \mathbf{Z}(0)^{-1} : \mathbb{P} \circ \mathbf{Y}(0)^{-1} \in \mathcal{S}_0\}.$$

Note that $\mu \in \mathcal{S}$ if for some $\mathbb{R}_+^\infty$-valued random variable $\mathbf{Y}(0)$ with probability law in $\mathcal{S}_0$, the vector $(Y_1(0) - Y_0(0), Y_2(0) - Y_0(0), \dots)$ has probability law μ.

The first theorem gives a sufficient condition for a probability measure on $\mathbb{R}_+^\infty$ to be in the WDoA of π.

Theorem 7 (Banerjee and Budhiraja, 2022) *Suppose that the probability measure μ on $\mathbb{R}_+^\infty$ satisfies the following: there exists a coupling $(\mathbf{U}, \mathbf{V})$ of μ and π such that, almost surely,*

$$\liminf_{d \to \infty} \frac{1}{\sqrt{d}(\log d)} \sum_{i=1}^{d} U_i \wedge V_i = \infty. \tag{37}$$

Then $\mu \in \mathcal{S}$ and it belongs to the WDoA of π.

The following corollary gives some natural situations in which (37) holds.

Corollary 1 (Banerjee and Budhiraja (2022)) *Let μ be a probability measure on $\mathbb{R}_+^\infty$ and $\mathbf{U} \sim \mu$. Suppose one of the following conditions hold:*

(i) *μ is stochastically dominated by π and, almost surely,*

$$\liminf_{d\to\infty} \frac{1}{\sqrt{d}(\log d)} \sum_{i=1}^{d} U_i = \infty. \tag{38}$$

(ii) *For some a.s. finite random variable M*

$$\liminf_{d\to\infty} \frac{1}{\sqrt{d}(\log d)} \sum_{i=1}^{d} (U_i \wedge M) = \infty, \ \text{a.s.} \tag{39}$$

(iii) *$U_i = \lambda_i \Theta_i$, $i \in \mathbb{N}$, where $\{\Theta_i\}$ are iid non-negative random variables satisfying $\mathbb{P}(\Theta_1 > 0) > 0$, and $\{\lambda_i\}$ are positive deterministic real numbers satisfying one of the following:*

(a) $\liminf_{j\to\infty} \lambda_j > 0$,

(b) $\limsup_{j\to\infty} \lambda_j < \infty$ *and*

$$\liminf_{d\to\infty} \frac{1}{\sqrt{d}(\log d)} \sum_{i=1}^{d} \lambda_i = \infty. \tag{40}$$

Then (37) holds. Hence, in all the above cases, $\mu \in \mathcal{S}$ and is in the WDoA of π.

Remark 1 We make the following observations.

(a) Suppose that π is stochastically dominated by μ. Then (37) holds by the strong law of large numbers, and so Theorem 7 shows that $\mu \in \mathcal{S}$ and is in the WDoA of π. In fact in this case Sarantsev (2017, Theorem 4.7) shows that μ is in the DoA of π.

(b) Note that for any $C \in (0, \infty)$,

$$((2C) \wedge 1)[U_i \wedge (1/2)] \le U_i \wedge C \le ((2C) \vee 1)[U_i \wedge (1/2)].$$

Hence, if (39) holds for one a.s. finite random variable M, then it holds for every such random variable. In particular, if $\limsup_{n\to\infty} U_n < \infty$ a.s. and (38) is satisfied then (37) holds and so $\mu \in \mathcal{S}$ and is in the WDoA of π.

(c) The paper Dembo et al. (2019) notes two important settings, where conditions (34)–(36) are satisfied. These are: (i) for some $c \in [1, \infty)$, $\lambda_j \in [c^{-1}, c]$ for all $j \in \mathbb{N}$ and $Z_j(0) = \lambda_j \Theta_j$ where Θ_j are iid with finite mean and such that $E \log(\Theta_1)_- < \infty$, and (ii) $Z_j(0) = \lambda_j \Theta_j$ where Θ_j are iid exponential with mean 1 and, either $\lambda_d \downarrow 0$ and $\frac{1}{\sqrt{d} \log d} \sum_{i=1}^d \lambda_i \to \infty$, as $d \to \infty$, or $\lambda_d \uparrow \infty$ and $\limsup_{d\to\infty} \frac{1}{d^\beta} \sum_{i=1}^d \lambda_i < \infty$ for some $\beta < 2$. We note that conditions assumed in the above settings are substantially stronger than the one assumed in part (iii) of Corollary 1. However, Dembo et al. (2019) establishes the stronger marginal time convergence as opposed to a time averaged limit considered in Corollary 1. See part (g) on a related open problem.

(d) In the setting of (iii), the sequence $\{\lambda_i\}$ satisfies (40) if $(\log i)/\sqrt{i}\lambda_i \to 0$ as $i \to \infty$. Indeed, note that, for any $C > 0$, there exists $i_C \in \mathbb{N}$ such that for all $i \geq i_C$,

$$\sum_{j=i_C}^{i} \lambda_j \geq C \sum_{j=i_C}^{i} \frac{\log j}{\sqrt{j}} \geq C \int_{i_C}^{i+1} \frac{\log z}{\sqrt{z}} dz.$$

Using the fact that the primitive of $\log z/\sqrt{z}$ is $2\sqrt{z}(\log z - 2)$, $\{\lambda_j\}$ is seen to satisfy (40) and so, if in addition $\limsup_{j\to\infty} \lambda_j < \infty$, we have that the conditions of (iii) (b) are satisfied.

(e) We note that the conditions prescribed in Theorem 7 and Corollary 1 all involve quantifying how close together the particles can be on the average in the initial configuration of the Atlas model. In particular, we do not require any upper bounds on the sizes of U_i. This is in contrast with condition (34) assumed in Dembo et al. (2019) which requires the particles to be not too far apart on the average. Intuitively, one expects convergence to π to hold when initially the particles are not too densely packed and upper bounds on the average rate of growth of the initial spacings are somewhat unnatural. The result in Sarantsev (2017, Theorem 4.7) also points to this heuristic by showing weak convergence to π from all initial gap configurations that stochastically dominate π. We also note that Theorem 7 and Corollary 1 allow the first few gaps of the initial gap sequence to be arbitrary and do not require any condition analogous to (35) and can, in particular, allow gaps to be zero.

(f) We note that the conditions (34)–(36) of Dembo et al. (2019) do not imply (37). To see this, consider the deterministic sequence of initial gaps: $U_i = i^{-2/3}$ for $n^3 < i < (n+1)^3$ and $U_{n^3} = n$, for any $n \in \mathbb{N}$. It can be checked that, $\sum_{i=1}^d U_i$ grows like $d^{2/3}$ and $\sum_{i=1}^d (\log U_i)_-$ grows like $d \log d$ as $d \to \infty$. Thus, (34)–(36) of Dembo et al. (2019) hold with $\beta = 1$ and $\theta(d) = \log d$. However, almost surely, $\sum_{i=1}^d U_i \wedge V_i$ grows like $d^{1/3}$, and therefore, (37) is violated.

(g) It will be interesting to investigate whether the condition in Theorem 7, in fact, implies the stronger property that μ is in the DoA of π. We leave this as an open problem.

The next theorem provides sufficient conditions for a measure on $\mathbb{R}_+^\infty$ to be in the WDoA of one of the other stationary measures π_a, $a > 0$.

Theorem 8 (Banerjee and Budhiraja, 2022) *Fix $a > 0$. Suppose that the probability measure μ on $\mathbb{R}_+^\infty$ satisfies the following: There exists a coupling $(\mathbf{U}, \mathbf{V}_a)$ of μ and π_a such that, almost surely,*

$$\limsup_{d\to\infty} \frac{\log\log d}{\log d} \sum_{i=1}^{d} |V_{a,i} - U_i| = 0, \text{ and } \limsup_{d\to\infty} \frac{U_d}{d V_{a,d}} < \infty. \tag{41}$$

Then $\mu \in \mathcal{S}$ and it belongs to the WDoA of π_a.

We remark that Theorem 8 also holds for the case $a = 0$. Indeed, suppose that the condition (41) in the theorem holds for $a = 0$. Denoting the corresponding $\mathbf{V}_0$ as $\mathbf{V}$, we have that

$$\sum_{i=1}^{d} U_i \wedge V_i = \sum_{i=1}^{d} U_i \vee V_i - \sum_{i=1}^{d} |V_i - U_i| \geq \sum_{i=1}^{d} V_i - \sum_{i=1}^{d} |V_i - U_i|.$$

Using this and the law of large numbers for $\{V_i\}$, it follows that (37) holds, and thus from Theorem 7, the conclusion of Theorem 8 holds with $a = 0$. The reason we do not note the case $a = 0$ in the statement of Theorem 8 is because (37) is a strictly weaker assumption than (41) when $a = 0$. This is expected as for any initial gap distribution that stochastically dominates π, convergence to π holds (see Remark 1 (a)). However, for convergence to π_a for some $a > 0$, the initial gap distribution should be appropriately close to π_a (see Remark 2 (b) below).

The following corollary gives a set of random initial conditions for which (41) holds.

Corollary 2 *Fix $a > 0$. Let*

$$\mu := \bigotimes_{i=1}^{\infty} \mathrm{Exp}(2 + ia + \lambda_i), \tag{42}$$

where $\{\lambda_i\}$ are real numbers satisfying $\lambda_i > -(2 + ia)$ for all i and

$$-a < \liminf_{i\to\infty} \frac{\lambda_i}{i} \leq \limsup_{i\to\infty} \frac{\lambda_i}{i} < \infty. \tag{43}$$

Moreover, assume

$$\limsup_{d\to\infty} \frac{\log\log d}{\log d} \sum_{i=1}^{d} \frac{|\lambda_i|}{i^2} = 0. \tag{44}$$

Then (41) holds, and hence, $\mu \in \mathcal{S}$ and belongs to the WDoA of π_a.

Remark 2 We note the following:

(a) Clearly, there are measures μ of the form in (42) for which $\limsup_{i\to\infty} \frac{\lambda_i}{i} < \infty$, $\liminf_{i\to\infty} \frac{|\lambda_i|}{i} > 0$ (and so (43) holds) and which do not lie in the WDoA of π_a (for example, $\pi_{a'}$ for any $a' \neq a$). However, the corollary says that if $|\lambda_i|$ grows slightly slower than i then μ is indeed in the WDoA of π_a. More precisely, the convergence in (44) holds, in particular, if $\limsup_{i\to\infty} |\lambda_i| \log\log i / i = 0$. Indeed, note that for any $\delta > 0$, there exists sufficiently large $i_\delta \in \mathbb{N}$ such that for $i \geq i_\delta$,

$$\sum_{j=i_\delta}^{i} \frac{|\lambda_j|}{j^2} \leq \sum_{j=i_\delta}^{i} \frac{\delta_j}{j \log\log j} \leq \delta \int_{i_\delta - 1}^{i} \frac{1}{z \log\log z} dz = \delta \int_{\log(i_\delta - 1)}^{\log i} \frac{1}{\log w} dw. \tag{45}$$

As $w \mapsto \frac{1}{\log w}$ is a slowly varying function, by Karamata's theorem Mikosch (1999, Theorem 1.2.6 (a)),

$$\lim_{i\to\infty} \frac{\log\log i}{\log i} \int_{\log(i_0 - 1)}^{\log i} \frac{1}{\log w} dw = 1.$$

Thus, we conclude from (45) that for any $\delta > 0$,

$$\limsup_{i\to\infty} \frac{\log\log i}{\log i} \sum_{j=1}^{i} \frac{|\lambda_j|}{j^2} \leq \delta.$$

As $\delta > 0$ is arbitrary, (44) holds.

(b) Note that if $\mathbf{U} \sim \pi$, then for $a, a' > 0$,

$$V_{a,i} = \frac{2}{2 + ia} U_i, \;\; V_{a',i} = \frac{2}{2 + ia'} U_i, \;\; i \in \mathbb{N}$$

defines a coupling of π_a and $\pi_{a'}$. For this coupling,

$$\sum_{i=1}^{d} |V_{a,i} - V_{a',i}| \sim O(|a - a'| \log d) \text{ and } \limsup_{d\to\infty} \frac{V_{a',d}}{d V_{a,d}} = 0. \tag{46}$$

Since $\pi_{a'}$ is obviously not in WDoA of π_a for $a \neq a'$, the first property in (46) says that the first requirement in (41) is not far from what one expects. We conjecture that for μ to be in the WDoA of π_a, it is necessary that $(\log d)^{-1} \sum_{i=1}^{d} |V_{a,i} - U_i| \to 0$ as $d \to \infty$ for some coupling $(\mathbf{U}, \mathbf{V}_a)$ of μ and π_a. Theorem 8 says that if the above convergence to 0 holds at a rate faster than $1/(\log\log d)$ for some coupling of μ and π_a, then that (together with the second condition in (41)) is sufficient for μ to be in the WDoA of π_a.

Further, observe that if the second condition in (41) does not hold, then there are infinitely many $d \in \mathbb{N}$ such that the lowest $d+1$ particles are separated from the rest by a large gap. If one such gap is very large, then it could happen that the distribution of gaps between the $d+1$ particles stabilizes towards the unique stationary gap distribution $\pi^{(d)}$ of the corresponding finite Atlas model before the remaining particles have interacted with them. As $\pi^{(d)}$ converges weakly to π as $d \to \infty$, one does not expect in such situations the convergence (of time averaged laws) to π_a to hold for any $a > 0$.

5.2 Outline of Approach

Proofs of the above results are based on a pathwise approach to studying the long-time behavior of the infinite Atlas model using synchronous couplings, namely, two versions of the infinite Atlas model described in terms of the ordered particles via (31) and started from different initial conditions but driven by the same collection of Brownian motions. Central ingredients in the proofs are suitable estimates on the decay rate of the L^1 distance between the gaps in synchronously coupled ordered infinite Atlas models. These estimates are obtained by analyzing certain excursions of the difference of the coupled processes where each excursion ensures the contraction of the L^1 distance by a fixed deterministic amount. The key is to appropriately control the number and the lengths of such excursions. Certain monotonicity properties of synchronous couplings and a quantification of the influence of far away coordinates on the first few gaps also play a crucial role. Using these tools, the discrepancy between the gap processes of two synchronously coupled infinite Atlas models can be controlled in terms of associated gap processes when the starting configurations differ only in finitely many coordinates. The latter are more convenient to work with as for them the initial L^1 distance between the gap processes is finite, and the aforementioned excursion analysis can be applied to obtain the main results.

6 Extremal and Product Form invariant Distributions of Infinite Rank-Based Diffusions

The results described in the last section give us some understanding about the local stability structure of the set of invariant distributions of the infinite Atlas model. However, it does not say anything about the geometry of the (convex) set of invariant measures. In particular, a longstanding open problem is to characterize all extremal invariant measures (corner points of the convex set). In this section, we outline some recent results from Banerjee and Budhiraja (2022), which makes progress in this direction.

We will in fact consider the more general setting of an infinite rank-based diffusion where the particles move on the real line according to mutually independent Brownian motions, with the ith particle from the left getting a constant drift of g_{i-1}. The vector $\boldsymbol{g} := (g_0, g_1, \ldots)'$ is called the drift vector and we call this model the $\boldsymbol{g}$-*Atlas model*. Such particle systems were originally introduced in stochastic portfolio theory Fernholz (2002); Fernholz and Karatzas (2009) as models for stock growth evolution in equity markets and have been investigated extensively in recent years in several different directions. In particular, characterizations of such particle systems as scaling limits of jump processes with local interactions on integer lattices, such as the totally asymmetric simple exclusion process, have been studied in Karatzas et al. (2016). Various types of results for the asymptotic behavior of the empirical measure of the particle states have been studied, such as propagation of chaos, characterization of the associated McKean-Vlasov equation and nonlinear Markov processes Shkolnikov (2012), Jourdain and Malrieu (2008), large deviation principles Dembo et al. (2016), characterizing the asymptotic density profile and the trajectory of the leftmost particle via Stefan free-boundary problems Cabezas et al. (2019). These particle systems also have close connections with Aldous' 'Up the river' stochastic control problem Aldous (2002), recently solved in Tang and Tsai (2018). Results on wellposedness of the associated stochastic differential equations (in the weak and strong sense) and on absence of triple collisions (three particles at the same place at the same time) have been studied in Bass and Pardoux (1987), Ichiba et al. (2013a), Ichiba and Karatzas (2010), Sarantsev (2015), and Ichiba and Sarantsev (2017).

As in the case of the infinite Atlas model, the natural state space of the location of the particles in the more general system is also given by $\mathcal{U}$ defined in (29). For a sequence $\{W_i\}_{i\in\mathbb{N}_0}$ of mutually independent standard Brownian motions given on some filtered probability space, and $\boldsymbol{y} = (y_0, y_1, \ldots)' \in \mathcal{U}$, $\boldsymbol{g} = (g_0, g_1, \ldots)' \in \mathcal{D}$, consider the following system of equations.

$$dY_i(t) = \left[\sum_{k=0}^{\infty} \mathbf{1}(Y_i(t) = Y_{(k)}(t))g_k\right]dt + dW_i(t),\ t \geq 0, \tag{47}$$

with $Y_i(0) = y_i$, $i \in \mathbb{N}_0$. Write $\boldsymbol{Y}(\cdot) := (Y_0(\cdot), Y_1(\cdot), \ldots)'$.

It has been shown in Sarantsev (2017) that if $\boldsymbol{g} \in \mathcal{D}$, where

$$\mathcal{D} := \left\{\boldsymbol{g} = (g_0, g_1, \ldots)' \in \mathbb{R}^\infty : \sum_{i=0}^{\infty} g_i^2 < \infty\right\}, \tag{48}$$

then for any $\boldsymbol{y} \in \mathcal{U}$, $P(\mathbf{Y}(t) \in \mathcal{U} \text{ for all } t \geq 0) = 1$, and there is a unique weak solution $\boldsymbol{Y}(\cdot)$ to (47). Analogous to the standard Atlas model setting, the ordered process $\{Y_{(i)}(\cdot) : i \in \mathbb{N}_0\}$ can be described by the SDE:

$$dY_{(i)}(t) = g_i dt + dB_i^*(t) - \frac{1}{2}dL_{i+1}^*(t) + \frac{1}{2}dL_i^*(t),\ t \geq 0,\quad Y_{(i)}(0) = y_{(i)},\ i \in \mathbb{N}_0. \tag{49}$$

where $\{B_i^*(\cdot) : i \in \mathbb{N}_0\}$ are independent Brownian motions defined in (30), $L_0^*(\cdot) \equiv 0$ and for $i \in \mathbb{N}$, $L_i^*(\cdot)$ denotes the local time of collision between the $(i-1)$th and ith particles. The gap process for the $(\boldsymbol{g}, \boldsymbol{y})$-infinite Atlas model is the $\mathbb{R}_+^\infty$-valued process $\boldsymbol{Z}(\cdot) = (Z_1(\cdot), Z_2(\cdot), \ldots)'$ defined by

$$Z_i(\cdot) := Y_{(i)}(\cdot) - Y_{(i-1)}(\cdot), \ i \in \mathbb{N}. \tag{50}$$

The natural state space for the gap process is given by

$$\mathcal{V} := \{\boldsymbol{z} \in \mathbb{R}_+^\infty : \text{ for some } \boldsymbol{y} \in \mathcal{U}, \boldsymbol{z} = (y_{(1)} - y_{(0)}, y_{(2)} - y_{(1)}, \ldots)'\}. \tag{51}$$

For this setting, it is known from the work of Sarantsev and Tsai (2017) that, once more, the process associated with the gap sequence of the ranked particle system has a continuum of product form stationary distributions given as

$$\pi_a^{\boldsymbol{g}} := \otimes_{n=1}^\infty \text{Exp}(n(2\bar{g}_n + a)), \quad a > -2 \inf_{n \in \mathbb{N}} \bar{g}_n, \tag{52}$$

where $\bar{g}_n := \frac{1}{n}(g_0 + \cdots + g_{n-1})$. In the special case where $\boldsymbol{g} \in \mathcal{D}_1$, where

$$\mathcal{D}_1 := \{\boldsymbol{g} \in \mathcal{D} : \text{ there exist } N_1 < N_2 < \cdots \to \infty \text{ such that } \bar{g}_k > \bar{g}_{N_j}, \ k = 1, \ldots, N_j - 1, \text{ for all } j \geq 1\}, \tag{53}$$

$\pi_a^{\boldsymbol{g}}$ is also an invariant distribution for $a = -2 \inf_{n \in \mathbb{N}} \bar{g}_n = -2 \lim_{j \to \infty} \bar{g}_{N_j}$; in particular for the infinite Atlas model $\pi_a = \pi_a^{\boldsymbol{g}^1}$ is invariant for all $a \geq 0$.

Using Kakutani's theorem, it is easy to verify that, for different values of a, the probability measures $\pi_a^{\boldsymbol{g}}$ are mutually singular. These distributions are also special in that they have a product form structure. In particular, if the initial distribution of the gap process is chosen to be one of these distributions, then the laws of distinct gaps at any fixed time are independent despite these gaps having a highly correlated temporal evolution mechanism (see (49)–(50)).

Now we present our main results from Banerjee and Budhiraja (2022) which show that the above distributions are also extremal. Moreover, in some sense, these are the only product form invariant distributions.

6.1 Main Results

We begin with some notation. Let $\mathbf{X} := \mathcal{C}([0, \infty) : \mathbb{R}_+^\infty)$ which is equipped with the topology of local uniform convergence (with $\mathbb{R}_+^\infty$ equipped with the product topology). Define measurable maps $\{\theta_t\}_{t \geq 0}$ from $(\mathbf{X}, \mathcal{B}(\mathbf{X}))$ to itself as

$$\theta_t(\boldsymbol{Z})(s) := \boldsymbol{Z}(t+s), \ t \geq 0, s \geq 0, \ \boldsymbol{Z} \in \mathbf{X}.$$

Given $\boldsymbol{g} \in \mathcal{D}$ and $\boldsymbol{z} \in \mathcal{V}$, we denote the probability distribution of the gap process of the $\boldsymbol{g}$-Atlas model on $(\mathbf{X}, \mathcal{B}(\mathbf{X}))$, with initial gap sequence $\boldsymbol{z}$, by $\mathbb{P}^{\boldsymbol{g}}_{\boldsymbol{z}}$. Also, for $\gamma \in \mathcal{P}(\mathbb{R}^{\infty}_{+})$ supported on $\mathcal{V}$ (namely, $\gamma(\mathcal{V}) = 1$), let $\mathbb{P}^{\boldsymbol{g}}_{\gamma} := \int_{\mathbb{R}^{\infty}_{+}} \mathbb{P}^{\boldsymbol{g}}_{\boldsymbol{z}}\, \gamma(d\boldsymbol{z})$. The corresponding expectation operators will be denoted as $\mathbb{E}^{\boldsymbol{g}}_{\boldsymbol{z}}$ and $\mathbb{E}^{\boldsymbol{g}}_{\gamma}$, respectively. Denote by $\mathcal{I}^{\boldsymbol{g}}$ the collection of all invariant (stationary) probability measures of the gap process of the $\boldsymbol{g}$-Atlas model supported on $\mathcal{V}$, namely

$$\mathcal{I}^{\boldsymbol{g}} := \{\gamma \in \mathcal{P}(\mathbf{X}) : \gamma(\mathcal{V}^c) = 0, \text{ and } \mathbb{P}^{\boldsymbol{g}}_{\gamma} \circ \theta_t^{-1} = \mathbb{P}^{\boldsymbol{g}}_{\gamma} \text{ for all } t \geq 0\}.$$

Abusing notation, the canonical coordinate process on $(\mathbf{X}, \mathcal{B}(\mathbf{X}))$ will be denoted by $\{\boldsymbol{Z}(t)\}_{t\geq 0}$. Let $\mathcal{M}(\mathbb{R}^{\infty}_{+})$ be the collection of all real measurable maps on $\mathbb{R}^{\infty}_{+}$. For $f \in \mathcal{M}(\mathbb{R}^{\infty}_{+})$, $\boldsymbol{z} \in \mathbb{R}^{\infty}_{+}$ and $t \geq 0$ such that $\mathbb{E}^{\boldsymbol{g}}_{\boldsymbol{z}}(|f(\boldsymbol{Z}(t))|) < \infty$ we write

$$T^{\boldsymbol{g}}_t f(\boldsymbol{z}) := \mathbb{E}^{\boldsymbol{g}}_{\boldsymbol{z}}(f(\boldsymbol{Z}(t))).$$

For $\gamma \in \mathcal{P}(\mathbb{R}^{\infty}_{+})$ let $L^2(\gamma)$ be the collection of all measurable $\psi : \mathbb{R}^{\infty}_{+} \to \mathbb{R}$ such that $\int_{\mathbb{R}^{\infty}_{+}} |\psi(\boldsymbol{z})|^2 \gamma(d\boldsymbol{z}) < \infty$. We denote the inner-product and the norm on $L^2(\gamma)$ as $\langle \cdot, \cdot \rangle$ and $\| \cdot \|$ respectively. Note that for $\boldsymbol{g} \in \mathcal{D}$, $\gamma \in \mathcal{I}^{\boldsymbol{g}}$, and $\psi \in L^2(\gamma)$, $T^{\boldsymbol{g}}_t \psi$ is γ a.e. well defined and belongs to $L^2(\gamma)$. Furthermore, the collection $\{T^{\boldsymbol{g}}_t\}_{t\geq 0}$ defines a contraction semigroup on $L^2(\gamma)$, namely

$$T^{\boldsymbol{g}}_t T^{\boldsymbol{g}}_s \psi = T^{\boldsymbol{g}}_{t+s}\psi, \text{ and } \|T^{\boldsymbol{g}}_t \psi\| \leq \|\psi\| \text{ for all } s, t \geq 0 \text{ and } \psi \in L^2(\gamma).$$

We now recall the definition of extremality and ergodicity. Let, for $\boldsymbol{g}$ as above and $\gamma \in \mathcal{I}^{\boldsymbol{g}}$, $\mathbb{I}^{\boldsymbol{g}}_{\gamma}$ be the collection of all $T^{\boldsymbol{g}}_t$-invariant square integrable functions, namely,

$$\mathbb{I}^{\boldsymbol{g}}_{\gamma} := \{\psi \in L^2(\gamma) : T^{\boldsymbol{g}}_t \psi = \psi, \ \gamma \text{ a.s., for all } t \geq 0\}.$$

We denote the projection of a $\psi \in L^2(\gamma)$ on to the closed subspace $\mathbb{I}^{\boldsymbol{g}}_{\gamma}$ as $\hat{\psi}^{\boldsymbol{g}}_{\gamma}$.

Definition 1 Let $\boldsymbol{g} \in \mathcal{D}$. A $\nu \in \mathcal{I}^{\boldsymbol{g}}$ is said to be an extremal invariant distribution of the gap process of the $\boldsymbol{g}$-Atlas model if, whenever for some $\varepsilon \in (0, 1)$ and $\nu_1, \nu_2 \in \mathcal{I}^{\boldsymbol{g}}$ we have $\nu = \varepsilon\nu_1 + (1-\varepsilon)\nu_2$, then $\nu_1 = \nu_2 = \nu$. We denote the collection of all such measures by $\mathcal{I}^{\boldsymbol{g}}_e$.

We call $\nu \in \mathcal{I}^{\boldsymbol{g}}$ an ergodic probability measure for the invariant distribution of the gap process of the $\boldsymbol{g}$-Atlas model if for all $\psi \in L^2(\nu)$, $\hat{\psi}^{\boldsymbol{g}}_{\nu}$ is constant ν-a.s. We denote the collection of all such measures by $\mathcal{I}^{\boldsymbol{g}}_{er}$.

We note that (cf. proof of Lemma 1 below) if $\gamma \in \mathcal{I}^{\boldsymbol{g}}_{er}$, then for any $\psi \in L^2(\gamma)$, and γ a.e. $\boldsymbol{z}$,

$$\frac{1}{t}\int_0^t T^{\boldsymbol{g}}_s \psi(\boldsymbol{z}) ds \to \int_{\mathbb{R}^{\infty}_{+}} \psi(\boldsymbol{x})\gamma(\boldsymbol{x}), \text{ in } L^2(\gamma), \text{ as } t \to \infty.$$

The following result, which says that extremal invariant measures and ergodic invariant measures are the same, is standard.

Lemma 1 *Let $\boldsymbol{g} \in \mathcal{D}$. Then $\mathcal{I}_e^{\boldsymbol{g}} = \mathcal{I}_{er}^{\boldsymbol{g}}$. Let $\gamma \in \mathcal{I}^{\boldsymbol{g}}$ and suppose that for every bounded measurable $\psi : \mathbb{R}_+^\infty \to \mathbb{R}$, $\hat{\psi}_\gamma^{\boldsymbol{g}}$ is constant, γ a.s. Then $\gamma \in \mathcal{I}_e^{\boldsymbol{g}}$.*

The following is the first main result of Banerjee and Budhiraja (2022).

Theorem 9 (Banerjee and Budhiraja, 2022) *Let $\boldsymbol{g} \in \mathcal{D}$. Then, for every $a > -2 \inf_{n\in\mathbb{N}} \bar{g}_n$, $\pi_a^{\boldsymbol{g}} \in \mathcal{I}_e^{\boldsymbol{g}} = \mathcal{I}_{er}^{\boldsymbol{g}}$. Also, when $\boldsymbol{g} \in \mathcal{D}_1$, $\pi_a^{\boldsymbol{g}} \in \mathcal{I}_e^{\boldsymbol{g}} = \mathcal{I}_{er}^{\boldsymbol{g}}$ also for $a = -2 \inf_{n\in\mathbb{N}} \bar{g}_n$.*

The above theorem proves the extremality of the invariant measures $\pi_a^{\boldsymbol{g}}$ for suitable values of a. One can ask whether these are the only extremal invariant measures of the gap process of the $\boldsymbol{g}$-Atlas model supported on $\mathcal{V}$. The answer to this question when $\boldsymbol{g} = \mathbf{0}$ is affirmative from results of Ruzmaikina and Aizenman (2005, Theorem 4.2) , if one restricts to extremal measures satisfying certain integrability constraints on the denseness of particle configurations. For a general $\boldsymbol{g} \in \mathcal{D}$, this is currently a challenging open problem. However, the next result makes partial progress towards this goal by showing that for any $\boldsymbol{g} \in \mathcal{D}_1$ (and under a mild integrability condition), the collection $\{\pi_a^{\boldsymbol{g}}\}$ exhausts all the extremal product form invariant distributions. In fact, we prove the substantially stronger statement that the measures $\pi_a^{\boldsymbol{g}}$ are the only product form (extremal or not) invariant distributions under a mild integrability condition. Qualitatively, this result says that these measures are the only invariant distributions that preserve independence of the marginal laws of the gaps in time.

Theorem 10 (Banerjee and Budhiraja, 2022) *Let $\boldsymbol{g} \in \mathcal{D}_1$ and let $\pi \in \mathcal{I}^{\boldsymbol{g}}$ be a product measure, i.e., $\pi = \otimes_{i=1}^\infty \pi_i$ for some $\pi_i \in \mathcal{P}(\mathbb{R}_+)$, $i \in \mathbb{N}$. Suppose that*

$$\int_{\mathbb{R}_+^\infty} \left(\sum_{j=1}^\infty e^{-\frac{1}{4}(\sum_{l=1}^j z_l)^2} \right) \pi(dz) < \infty. \tag{54}$$

Then, for some $a \geq -2 \lim_{j\to\infty} \bar{g}_{N_j}$, $\pi = \pi_a^{\boldsymbol{g}}$.

Recall that $\mathcal{V}$ defined in (51) consists of $z \in \mathbb{R}_+^\infty$ for which $\sum_{j=1}^\infty e^{-\alpha(\sum_{l=1}^j z_l)^2} < \infty$ for all $\alpha > 0$. In comparison, condition (54) requires a finite expectation of $\sum_{j=1}^\infty e^{-\frac{1}{4}(\sum_{l=1}^j z_l)^2}$ when z is distributed as π. Roughly speaking, condition (54) puts a restriction on the rate of increase of the density of particles as one moves away from the lowest ranked particle.

6.2 Connection with Prior Work in Interacting Particle Systems

Questions about extremality and ergodicity of stationary distributions have been addressed previously in the context of interacting particle systems on countably infi-

nite graphs (see Liggett, 1976; Enrique, 1982; Sethuraman, 2001; Balázs et al., 2007, and references therein). However, in all these cases, the interactions are Poissonian which enables one to use the (explicit) generator of associated processes in an effective way. The interactions in rank based diffusions are very 'singular' owing to the local time-based dynamics (see (31)) and generator-based methods seem to be less tractable. Furthermore, unlike previous works, the state space for the gap process (i.e., $\mathbb{R}_+^\infty$) is not countable and has a non-smooth boundary, and the process has intricate interactions (oblique reflections) with the boundary. Hence, proving extremality requires new techniques. Our proofs are based on constructing appropriate couplings for these infinite dimensional diffusions which then allow us to prove suitable invariance properties and a certain 'directional strong Feller property'. Although our setting and methods are very different, at a high level, the approach we take is inspired by the papers Sethuraman (2001), Balázs et al. (2007).

Complete characterization of extremal invariant measures is, in general, a very hard problem. This problem has been completely solved in a few examples of interacting particle systems such as the simple exclusion process Liggett (1976) and the zero range process Enrique (1982) where the extremal probability measures are fully characterized as an explicit collection of certain product form measures. However, in these models the particle density associated with distinct extremal measures are scalar multiples of each other owing to certain homogeneity properties in the dynamics (see, for example, Enrique, 1982, Theorem 1.10). This, along with the Poissonian nature of the interactions, enables one to prove useful monotonicity properties of the 'synchronously coupled' dynamics (see Liggett, 1976, Sect. 2 and Enrique, 1982, Sect. 4) using generator methods that are crucial to the above characterization results. A key challenge in extending these methods to rank based diffusions of the form considered here is that, in addition to the singular local time interactions, the point process associated with the configuration of particles with gaps distributed as $\pi_a^{\boldsymbol{g}}$ has an intensity function $\rho(x)$, that grows exponentially as $x \to \infty$ when $a > 0$ and, due to a nonlinear dependence on a, lacks the scalar multiple property for distinct values of a. This is a direct consequence of the inhomogeneity of the topological interactions in our particle systems where the local stability in a certain region of the particle cloud is affected both by the density of particles in the neighborhood and their relative ranks in the full system. Moreover, unlike the above interacting particle systems, in rank based diffusions, when the initial gaps are given by a stationary distribution, the point process of particles is typically not stationary. This phenomenon, where the gaps are stationary while the associated point process is not, referred to as *quasi-stationarity* in Ruzmaikina and Aizenman (2005), is technically challenging. We note that this latter paper studies one setting where the intensity function grows exponentially and the particle density lacks the scalar multiple property for distinct values of a. However, their setting, in the context of rank based diffusions, corresponds to the case $\boldsymbol{g} = (0, 0, \ldots)'$, where the unordered particles behave like independent standard Brownian motions, and this fact is crucially exploited in Ruzmaikina and Aizenman (2005).

6.3 Proof Outline

We now make some comments on proof ideas. The key step in proving the extremality of $\pi_a^{\boldsymbol{g}}$ is to establish that any bounded measurable function ψ on $\mathbb{R}_+^\infty$ that is $\pi_a^{\boldsymbol{g}}$-a.e. invariant, under the action of the semigroup of the Markov process corresponding to the $\boldsymbol{g}$-Atlas gap sequence, is constant $\pi_a^{\boldsymbol{g}}$-a.e. If $\boldsymbol{g} = \boldsymbol{g}^1$ and $a = 0$, we have that $\pi_a^{\boldsymbol{g}} = \otimes_{i=1}^\infty \text{Exp}(2)$, and therefore the coordinate sequence $\{Z_i\}_{i=1}^\infty$ is iid under $\pi_a^{\boldsymbol{g}}$. In this case, from the Hewitt-Savage zero-one law, it suffices to show that ψ is $\pi_a^{\boldsymbol{g}}$-a.e. invariant under all finite permutations of the coordinates of $\mathbb{R}_+^\infty$. For this, in turn, it suffices to simply prove the above invariance property for transpositions of the ith and $(i+1)$th coordinates, for all $i \in \mathbb{N}$. For a general $\pi_a^{\boldsymbol{g}}$, the situation is a bit more involved as the coordinate sequence $\{Z_i\}_{i=1}^\infty$ is not iid any more. Nevertheless, from the scaling properties of Exponential distributions, it follows that, with $c_n := 2[n(2\bar{g}_n + a)]^{-1}$, the sequence $\{\tilde{Z}_n\}_{n\geq 1}$, defined as $\tilde{Z}_n = c_n^{-1} Z_n$, $n \in \mathbb{N}$, is iid under $\pi_a^{\boldsymbol{g}}$. In this case, in order to invoke the Hewitt-Savage zero-one law, one needs to argue that for each i the map ψ is $\pi_a^{\boldsymbol{g}}$-a.e. invariant under the transformation that takes the $(i, i+1)$ coordinates (z_i, z_{i+1}) to $(\frac{c_i}{c_{i+1}} z_{i+1}, \frac{c_{i+1}}{c_i} z_i)$ and keeps the remaining coordinates the same. Establishing this property is at the heart of the proof of Theorem 9. A key technical idea in the proof is the construction of a mirror coupling of the first $i+1$ Brownian motions, and synchronous coupling of the remaining Brownian motions, in the evolution of the ranked infinite $\boldsymbol{g}$-Atlas model corresponding to a pair of nearby initial configurations. Estimates on the probability of a successful coupling, before any of the first i-gap processes have hit zero or the lowest i-particles have interacted with the higher ranked particles (in a suitable sense), are some of the important ingredients in the proof.

The proof of Theorem 10 hinges on establishing a key identity for expectations, under the given product form invariant measure π, of certain integrals involving the state process of the ith gap and the collision local time for the $(j-1)$th and jth particle, for $i \neq j$. This identity is a consequence of the product form structure of π and basic results on local times of continuous semimartingales. By using the form of the dynamics of the $\boldsymbol{g}$-Atlas model, the above identity allows us to compute explicitly the Laplace transform of the ith coordinate projection of π, via Itô's formula, from which it is then readily seen that π must be $\pi_a^{\boldsymbol{g}}$ for a suitable value of a.

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A Survey of Some Recent Developments in Measures of Association



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In honor of friend and teacher Prof. Rajeeva L. Karandikar on the occasion of his 65th birthday.

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1 Introduction

Measuring associations between variables is one of the central goals of data analysis. Arguably, the three most popular classical measures of association are Pearson's correlation coefficient, Spearman's ρ, and Kendall's τ. Although these coefficients are powerful for detecting monotonic associations, a practical problem is that they are not effective for detecting associations that are not monotonic. There have been many proposals to address this deficiency of the classical coefficients (Josse & Holmes, 2016), such as the maximal correlation coefficient (Hirschfeld, 1935; Gebelein, 1941; Rényi, 1959; Breiman & Friedman, 1985), various coefficients based on joint cumulative distribution functions and ranks (Gamboa et al., 2018; Drton et al., 2020; Han et al., 2017; Bergsma & Dassios, 2014; Nandy et al., 2016; Weihs et al. 2016, 2018; Yanagimoto, 1970; Csörgő, 1985; Puri & Sen, 1971; Hoeffding, 1948; Blum et al., 1961; Romano, 1988; Rosenblatt, 1975; Deb & Sen, 2021; Wang et al. 2017), kernel-based methods (Pfister et al., 2018; Gretton et al., 2005a, 2007; Sen & Sen, 2014; Zhang et al., 2018), information theoretic coefficients (Kraskov et al., 2004;

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Linfoot, 1957; Reshef et al., 2011), coefficients based on copulas (Dette et al., 2013; Lopez-Paz et al., 2013; Sklar, 1959; Schweizer & Wolff, 1981; Zhang, 2019), and coefficients based on pairwise distances (Székely et al., 2007; Székely & Rizzo, 2009; Heller et al., 2013; Friedman & Rafsky, 1983; Lyons, 2013).

This survey is about some recent developments in this area, beginning with a new coefficient of correlation proposed by the author in the paper (Chatterjee, 2021). This coefficient has the following desirable features: (a) It has a simple expression, like the classical coefficients. (b) It is a consistent estimator of a measure of dependence which is 0 if and only if the variables are independent and 1 if and only if one is a measurable function of the other. (c) It has a simple asymptotic theory under the hypothesis of independence.

The new coefficient is defined as follows. Let (X, Y) be a pair of random variables defined on the same probability space, where Y is not a constant. Let $(X_1, Y_1), \ldots, (X_n, Y_n)$ be i.i.d. pairs of random variables with the same law as (X, Y), where $n \geq 2$. Rearrange the data as $(X_{(1)}, Y_{(1)}), \ldots, (X_{(n)}, Y_{(n)})$ such that $X_{(1)} \leq \cdots \leq X_{(n)}$. (Note that $Y_{(i)}$ is just the Y-value 'paired with' $X_{(i)}$ in the original data, and not the ith order statistic of the Y's.) If there are ties among the X_i's, then choose an increasing rearrangement as above by breaking ties uniformly at random. Let r_i be the number of j such that $Y_{(j)} \leq Y_{(i)}$, and let l_i to be the number of j such that $Y_{(j)} \geq Y_{(i)}$. Then define

$$\xi_n(X, Y) := 1 - \frac{n \sum_{i=1}^{n-1} |r_{i+1} - r_i|}{2 \sum_{i=1}^{n} l_i(n - l_i)}. \tag{1.1}$$

This is the correlation coefficient proposed in Chatterjee (2021). When there are no ties among the Y_i's, $l_1, \ldots, l_n$ is just a permutation of $1, \ldots, n$, and so the denominator in the above expression is just $n(n^2 - 1)/3$. The following theorem is the main consistency result for ξ_n.

Theorem 1.1 (Chatterjee, 2021) *If Y is not almost surely a constant, then as $n \to \infty$, $\xi_n(X, Y)$ converges almost surely to the deterministic limit*

$$\xi(X, Y) := \frac{\int \mathrm{Var}(\mathbb{E}(1_{\{Y \geq t\}}|X))d\mu(t)}{\int \mathrm{Var}(1_{\{Y \geq t\}})d\mu(t)}, \tag{1.2}$$

where μ is the law of Y. This limit belongs to the interval $[0, 1]$. *It is* 0 *if and only if X and Y are independent, and it is* 1 *if and only if there is a measurable function $f : \mathbb{R} \to \mathbb{R}$ such that $Y = f(X)$ almost surely.*

The limiting value $\xi(X, Y)$ appeared in the literature prior to Chatterjee (2021), in a paper of Dette et al. (2013) (see also Gamboa et al., 2018; Kong et al. 2019). The paper Dette et al. (2013) gave a copula-based estimator for $\xi(X, Y)$ when X and Y are continuous, that is consistent under smoothness assumptions on the copula and is computable in time $n^{5/3}$ for an optimal choice of tuning parameters.

Note that neither ξ_n nor the limiting value ξ is symmetric in X and Y. A symmetrized version of ξ_n can be constructed by taking the maximum of $\xi_n(X, Y)$ and $\xi_n(Y, X)$.

2 Why Does it Work?

The complete proof of Theorem 1.1 is available in the supplementary materials of Chatterjee (2021), and also in the arXiv version of the paper. It is not too hard to see why $\xi(X, Y)$ has the properties listed in Theorem 1.1, although filling in the details takes some work. The proof of convergence of $\xi_n(X, Y)$ to $\xi(X, Y)$ is less obvious. The following is a very rough sketch of the proof, reproduced from a similar discussion in Chatterjee (2021).

For simplicity, consider only the case of continuous X and Y, where the denominator in (1.2) is simply $n(n^2 - 1)/3$. First, note that by the Glivenko–Cantelli theorem, $r_i/n \approx F(Y_{(i)})$, where F is the cumulative distribution function of Y. Thus,

$$\xi_n(X, Y) \approx 1 - \frac{3}{n}\sum_{i=1}^{n} |F(Y_i) - F(Y_{N(i)})|, \tag{2.1}$$

where $N(i)$ is the unique index j such that X_j is immediately to the right of X_i if we arrange the X's in increasing order. If X_i is the rightmost value, define $N(i)$ arbitrarily; it does not matter since the contribution of a single term in the above sum is of order $1/n$. Next, observe that for any $x, y \in \mathbb{R}$,

$$|F(x) - F(y)| = \int (1_{\{t \le x\}} - 1_{\{t \le y\}})^2 d\mu(t), \tag{2.2}$$

where μ is the law of Y. This is true because the integrand is 1 between x and y and 0 outside.

Suppose that we condition on $X_1, \ldots, X_n$. Since X_i is likely to be very close to $X_{N(i)}$, the random variables Y_i and $Y_{N(i)}$ are likely to be approximately i.i.d. after this conditioning. This leads to the approximation

$$\begin{aligned}\mathbb{E}[(1_{\{t \le Y_i\}} - 1_{\{t \le Y_{N(i)}\}})^2 | X_1, \ldots, X_n] &\approx 2\mathrm{Var}(1_{\{t \le Y_i\}} | X_1, \ldots, X_n) \\ &= 2\mathrm{Var}(1_{\{t \le Y_i\}} | X_i).\end{aligned}$$

This gives

$$\begin{aligned}\mathbb{E}(1_{\{t \le Y_i\}} - 1_{\{t \le Y_{N(i)}\}})^2 &\approx 2\mathbb{E}[\mathrm{Var}(1_{\{t \le Y\}} | X)] \\ &= 2\mathrm{Var}(1_{\{t \le Y\}}) - 2\mathrm{Var}(\mathbb{E}(1_{\{t \le Y\}} | X)).\end{aligned}$$

Combining this with (2.2), we get

$$\mathbb{E}|F(Y_i) - F(Y_{N(i)})| \approx \int 2[\mathrm{Var}(1_{\{t \le Y\}}) - \mathrm{Var}(\mathbb{E}(1_{\{t \le Y\}}|X))]d\mu(t).$$

But note that $\mathrm{Var}(1_{\{t \le Y\}}) = F(t)(1 - F(t))$, and $F(Y) \sim \mathrm{Uniform}[0, 1]$. Thus,

$$\int \mathrm{Var}(1_{\{t \le Y\}})d\mu(t) = \int F(t)(1 - F(t))d\mu(t) = \int_0^1 x(1 - x)dx = \frac{1}{6}.$$

Therefore by (2.1),

$$\mathbb{E}(\xi_n(X, Y)) \approx 6 \int \mathrm{Var}(\mathbb{E}(1_{\{t \le Y\}}|X))d\mu(t) = \xi(X, Y),$$

where the last identity holds because $\int \mathrm{Var}(1_{\{t \le Y\}})d\mu(t) = 1/6$, as shown above. This establishes the convergence of $\mathbb{E}(\xi_n(X, Y))$ to $\xi(X, Y)$. Concentration inequalities are then used to show that $\xi_n(X, Y) - \mathbb{E}(\xi_n(X, Y)) \to 0$ almost surely.

3 Asymptotic Distribution

Let X, Y and ξ_n be as in the previous section. For each $t \in \mathbb{R}$, let $F(t) := \mathbb{P}(Y \le t)$ and $G(t) := \mathbb{P}(Y \ge t)$. Let $\phi(y, y') := \min\{F(y), F(y')\}$. Define

$$\tau^2 = \frac{\mathbb{E}\phi(Y_1, Y_2)^2 - 2\mathbb{E}(\phi(Y_1, Y_2)\phi(Y_1, Y_3)) + (\mathbb{E}\phi(Y_1, Y_2))^2}{(\mathbb{E}G(Y)(1 - G(Y)))^2}, \tag{3.1}$$

where Y_1, Y_2, Y_3 are independent copies of Y. The following theorem gives the limiting distribution of ξ_n under the null hypothesis that X and Y are independent.

Theorem 3.1 (Chatterjee, 2021) *Suppose that X and Y are independent. Then $\sqrt{n}\xi_n(X, Y)$ converges to $N(0, \tau^2)$ in distribution as $n \to \infty$, where τ^2 is given by the formula (3.1) stated above. The number τ^2 is strictly positive if Y is not a constant, and equals $2/5$ if Y is continuous.*

The reason why τ^2 does not depend on the law of Y if Y is continuous is that in this case $F(Y)$ and $G(Y)$ are Uniform[0, 1] random variables, which implies that the expectations in (3.1) do not depend on the law of Y. If Y is not continuous, then τ^2 may depend on the law of Y. For example, it is not hard to show that if Y is a Bernoulli$(1/2)$ random variable, then $\tau^2 = 1$. Fortunately, if Y is not continuous, there is a simple way to estimate τ^2 from the data using the estimator

$$\hat{\tau}_n^2 = \frac{a_n - 2b_n + c_n^2}{d_n^2},$$

where a_n, b_n, c_n, and d_n are defined as follows. For each i, let

$$R(i) := \#\{j : Y_j \le Y_i\}, \quad L(i) := \#\{j : Y_j \ge Y_i\}. \tag{3.2}$$

(Note that $R(i)$ and $L(i)$ are different than r_i and l_i defined earlier.) Let $u_1 \le u_2 \le \cdots \le u_n$ be an increasing rearrangement of $R(1), \ldots, R(n)$. Let $v_i := \sum_{j=1}^{i} u_j$ for $i = 1, \ldots, n$. Define

$$a_n := \frac{1}{n^4}\sum_{i=1}^{n}(2n - 2i + 1)u_i^2, \quad b_n := \frac{1}{n^5}\sum_{i=1}^{n}(v_i + (n-i)u_i)^2,$$
$$c_n := \frac{1}{n^3}\sum_{i=1}^{n}(2n - 2i + 1)u_i, \quad d_n := \frac{1}{n^3}\sum_{i=1}^{n}L(i)(n - L(i)).$$

Then the following holds.

Theorem 3.2 (Chatterjee, 2021) *The estimator $\hat{\tau}_n^2$ can be computed in time $O(n \log n)$, and converges to τ^2 almost surely as $n \to \infty$.*

The question of proving a central limit theorem for ξ_n in the absence of independence is much more difficult than the independent case. This was left as an open question in Chatterjee (2021) and recently resolved in complete generality by Lin and Han (2022), following earlier proofs of Deb et al. (2020) and Shi et al. (2021b) under additional assumptions. Lin and Han (2022) also give a consistent estimator of the asymptotic variance of ξ_n in the absence of independence, solving another question that was left open in Chatterjee (2021). A central limit theorem for the symmetrized version of ξ_n (defined as the maximum of $\xi_n(X, Y)$ and $\xi_n(Y, X)$) under the hypothesis of independence was proved by Zhang (2022).

4 Power for Testing Independence

A deficiency of ξ_n, as already pointed out in Chatterjee (2021) through simulated examples, is that it has low power for testing independence against 'standard' alternatives, such as linear or monotone associations. This was theoretically confirmed by Shi et al. (2022a), where it was shown that the test of independence using ξ_n is rate-suboptimal against a family of local alternatives, whereas three other nonparametric tests of independence proposed in Hoeffding (1948); Blum et al. (1961); Bergsma and Dassios (2014); Yanagimoto (1970) are rate-optimal. Like ξ_n, the three competing test statistics considered in Shi et al. (2022a) are also computable in $O(n \log n)$ time. Similar results were obtained for a different type of competing test statistic by Cao and Bickel (2020).

A more detailed analysis of the power properties of ξ_n was carried out by Auddy et al. (2021), where the asymptotic distribution of ξ_n under any changing sequence of alternatives converging to the null hypothesis of independence was computed.

This analysis yielded exact detection thresholds and limiting power under natural alternatives converging to the null, such as mixture models, rotation models, and noisy nonparametric regression. The detection boundary lies at distance $n^{-1/4}$ from the null, instead of the more standard $n^{-1/2}$. This is similar to the power properties of other 'graph-based' statistics for testing independence, such as the Friedman–Rafsky statistic (Friedman & Rafsky, 1983; Bhattacharya, 2019).

A proposal for 'boosting' the power of ξ_n for testing independence, by incorporating multiple nearby ranks instead of only the nearest ones, was recently proposed by Lin and Han (2022). This modified estimator was shown to attain near-optimal rates of power against certain classes of alternative hypotheses.

The conceptual reason behind the absence of local power of statistics such as ξ_n was explained by Bickel (2022). An interesting question that remains unexplained is the following. It is seen in simulations that although ξ_n has low power for testing independence against standard alternatives such as linear and monotone, it becomes more powerful as the signal starts to get more and more oscillatory (Chatterjee, 2021). This gives an advantage over other coefficients in applications where oscillatory signals arise naturally (Chen, 2020; Sadeghi, 2022), and suggests that ξ_n may be efficient for certain kinds of local alternatives. No result of this sort has yet been proven.

5 Multivariate Extensions

Many methods have been proposed for testing independence nonparametrically in the multivariate setting. This includes classical tests (Puri et al., 1970; Gretton et al., 2005b; Székely et al., 2007) as well as a flurry of recent ones proposed in the last ten years (Heller et al., 2012, 2013; Heller & Heller, 2016; Zhu et al., 2017; Weihs et al., 2018; Kim et al., 2020; Deb & Sen, 2021; Shi et al., 2022b, c; Berrett et al., 2021; Deb et al., 2020).

Most of these papers are concerned only with testing independence, and not with measuring the strength of dependence as measured by a correlation coefficient such as ξ_n. Unfortunately, the ξ_n coefficient and many other popular univariate coefficients do not readily generalize to the multivariate setting because they are based on ranks. Statisticians have started taking a new look at this old problem in recent years by considering a multivariate notion of rank defined using optimal transport. Roughly speaking, the idea is as follows. Let ν be a 'reference measure' in $\mathbb{R}^d$, akin to the uniform distribution on $[0, 1]$ in $\mathbb{R}$. Given any probability measure μ in $\mathbb{R}^d$, let $F^\mu : \mathbb{R}^d \to \mathbb{R}^d$ be the map that 'optimally transports' μ to ν—that is, if $X \sim \mu$, then $F^\mu(X) \sim \nu$, and F^μ minimizes $\mathbb{E}\|X - F(X)\|^2$ among all such F. By a theorem of McCann (1995), such a map exists and is unique if μ and ν are both absolutely continuous with respect to Lebesgue measure. For example, when $d = 1$, F^μ is just the cumulative distribution function of X, which transforms X into a Uniform$[0, 1]$ random variable. For properties of this map, see, e.g., Figalli (2018) and Hallin et al. (2021).

The above idea suggests a natural definition of multivariate rank. If $X_1, \ldots, X_n$ are i.i.d. samples from μ, one can try to estimate F^μ using this data. Let F_n^μ be an estimate. Then $F_n^\mu(X_i)$ can act as a 'multivariate rank' of X_i among $X_1, \ldots, X_n$, divided by n. Since $F^\mu(X_i) \sim \nu$, we can then assume that $F_n^\mu(X_i)$ is approximately distributed according to ν, and then try to test for independence of random vectors using a test for independence that works when the marginal distributions are both ν. This idea has been made precise in a number of recent works in the statistics literature, such as Chernozhukov et al. (2017), Deb et al. (2020), Deb and Sen (2021), Hallin et al. (2021), Manole et al. (2021), Shi et al. (2021a, 2022b, c), Ghosal and Sen (2022), and Mordant and Segers (2022). For a survey and further discussions, see Han (2021).

A direct generalization of ξ_n to higher dimensional spaces has not been proposed so far, although the variant proposed in Deb et al. (2020) satisfies the same properties as ξ_n provided that the space on which Y takes values admits a nonnegative definite kernel and the space on which X takes values has a metric. This covers most spaces that arise in practice. There are a couple of other generalizations, proposed by Azadkia and Chatterjee (2021) and Gamboa et al. (2022), on measuring the dependence between a univariate random variable Y and a random vector $\mathbf{X}$. The coefficient proposed in Azadkia and Chatterjee (2021) (discussed in detail in Sect. 6) is based on a generalization of the ideas behind the construction of ξ_n. The coefficient proposed in Gamboa et al. (2022) combines the construction of ξ_n with ideas from the theory of Sobol indices. A new contribution of the present paper is a simple generalization of ξ_n to standard Borel spaces, which has been overlooked in the literature until now. This is presented in Sect. 8.

6 Measuring Conditional Dependence

The problem of measuring conditional dependence has received less attention than the problem of measuring unconditional dependence, partly because it is a more difficult task. Nonparametric conditional independence can be tested for discrete data using the classical Cochran–Mantel–Haenszel test (Cochran, 1954; Mantel & Haenszel, 1959), which can be adapted for continuous random variables by binning the data (Huang, 2010) or using kernels (Fukumizu et al., 2007; Zhang et al., 2012; Strobl et al., 2019; Doran et al., 2014; Sen et al., 2017). Besides these, there are methods based on estimating conditional cumulative distribution functions (Linton & Gozalo, 1997; Patra et al., 2016), conditional characteristic functions (Su & White, 2007; Ke & Yin, 2019), conditional probability density functions (Su & White, 2008), empirical likelihood (Su & White, 2014), mutual information and entropy (Runge, 2018; Joe, 1989; Póczos & Schneider, 2012), copulas (Bergsma, 2004; Song, 2009; Veraverbeke et al., 2011), distance correlation (Wang et al., 2015; Fan et al., 2020; Székely & Rizzo, 2014), and other approaches (Seth & Príncipe, 2012). A number of interesting ideas based on resampling and permutation tests have been proposed in recent years (Candès et al., 2018; Sen et al., 2017; Berrett et al., 2020).

In Azadkia and Chatterjee (2021), a new coefficient of conditional dependence was proposed, based on the ideas behind the ξ-coefficient defined in Chatterjee (2021). Like the ξ-coefficient, this one also has a long list of desirable features, such as being fully nonparametric and working under minimal assumptions. The coefficient is defined as follows.

Let Y be a random variable and $\mathbf{X} = (X_1, \ldots, X_p)$ and $\mathbf{Z} = (Z_1, \ldots, Z_q)$ be random vectors, all defined on the same probability space. Here $q \geq 1$ and $p \geq 0$. The value $p = 0$ means that $\mathbf{X}$ has no components at all. Let μ be the law of Y. The following quantity was proposed in Azadkia and Chatterjee (2021) as a measure of the degree of conditional dependence of Y and $\mathbf{Z}$ given $\mathbf{X}$:

$$T = T(Y, \mathbf{Z}|\mathbf{X}) := \frac{\int \mathbb{E}(\mathrm{Var}(\mathbb{P}(Y \geq t|\mathbf{Z}, \mathbf{X})|\mathbf{X}))d\mu(t)}{\int \mathbb{E}(\mathrm{Var}(1_{\{Y \geq t\}}|\mathbf{X}))d\mu(t)}. \tag{6.1}$$

If the denominator equals zero, T is undefined. If $p = 0$, then $\mathbf{X}$ has no components, and the conditional expectations and variances given $\mathbf{X}$ should be interpreted as unconditional expectations and variances. In this case, we write $T(Y, \mathbf{Z})$ instead of $T(Y, \mathbf{Z}|\mathbf{X})$. Note that T is a generalization of the statistic ξ appearing in Theorem 1.1. The following theorem summarizes the main properties of T.

Theorem 6.1 (Azadkia & Chatterjee, 2021) *Suppose that Y is not almost surely equal to a measurable function of $\mathbf{X}$ (when $p = 0$, this means that Y is not almost surely a constant). Then T is well-defined and $0 \leq T \leq 1$. Moreover, $T = 0$ if and only if Y and $\mathbf{Z}$ are conditionally independent given $\mathbf{X}$, and $T = 1$ if and only if Y is almost surely equal to a measurable function of $\mathbf{Z}$ given $\mathbf{X}$. When $p = 0$, conditional independence given $\mathbf{X}$ simply means unconditional independence.*

Now suppose we have data consisting of n i.i.d. copies $(Y_1, \mathbf{X}_1, \mathbf{Z}_1), \ldots, (Y_n, \mathbf{X}_n, \mathbf{Z}_n)$ of the triple $(Y, \mathbf{X}, \mathbf{Z})$, where $n \geq 2$. For each i, let $N(i)$ be the index j such that $\mathbf{X}_j$ is the nearest neighbor of $\mathbf{X}_i$ with respect to the Euclidean metric on $\mathbb{R}^p$, where ties are broken uniformly at random. Let $M(i)$ be the index j such that $(\mathbf{X}_j, \mathbf{Z}_j)$ is the nearest neighbor of $(\mathbf{X}_i, \mathbf{Z}_i)$ in $\mathbb{R}^{p+q}$, again with ties broken uniformly at random. Let R_i be the rank of Y_i, that is, the number of j such that $Y_j \leq Y_i$. If $p \geq 1$, define

$$T_n = T_n(Y, \mathbf{Z}|\mathbf{X}) := \frac{\sum_{i=1}^n (\min\{R_i, R_{M(i)}\} - \min\{R_i, R_{N(i)}\})}{\sum_{i=1}^n (R_i - \min\{R_i, R_{N(i)}\})}.$$

If $p = 0$, let L_i be the number of j such that $Y_j \geq Y_i$, let $M(i)$ denote the j such that $\mathbf{Z}_j$ is the nearest neighbor of $\mathbf{Z}_i$ (ties broken uniformly at random), and let

$$T_n = T_n(Y, \mathbf{Z}) := \frac{\sum_{i=1}^n (n \min\{R_i, R_{M(i)}\} - L_i^2)}{\sum_{i=1}^n L_i(n - L_i)}.$$

In both cases, T_n is undefined if the denominator is zero. The following theorem shows that T_n is a consistent estimator of T.

Theorem 6.2 (Azadkia & Chatterjee, 2021) *Suppose that Y is not almost surely equal to a measurable function of* $\mathbf{X}$. *Then as $n \to \infty$, $T_n \to T$ almost surely.*

For various other properties of T_n, such as rate of convergence, performance in simulations and real data, etc., see Azadkia and Chatterjee (2021). One problem that was left unsolved in Azadkia and Chatterjee (2021) was the question of proving a central limit theorem for T_n under the null hypothesis, which is crucial for carrying out tests for conditional independence. This question was partially resolved by Shi et al. (2021b), who proved a central limit theorem for T_n under the assumption that Y is independent of $(\mathbf{X}, \mathbf{Z})$. An improved version of this result was proved recently by Lin and Han (2022). A version for data supported on manifolds was proved by Han and Huang (2022).

The paper of Shi et al. (2021b) also develops the 'conditional randomization test' (CRT) framework of Candès et al. (2018) to test conditional independence using T_n, and find that T_n, like ξ_n is an inefficient test statistic. To address this concern, an improved generalization of T_n, called 'kernel partial correlation' (KPC), was proposed by Huang et al. (2020). Unlike T_n, KPC has the flexibility to use more than one nearest neighbor, which gives it better power properties.

Note that by the above theorems, a test of conditional independence based on T_n is consistent against all alternatives. The problem is that in the absence of an asymptotic theory for T_n, it is difficult to control the significance level of such a test. This is in fact an impossible problem, according to a recent result of Shah and Peters (2020) that proves the hardness of conditional independence testing in the absence of smoothness assumptions. Assuming some degree of smoothness, minimax optimal conditional independence tests were recently constructed by Neykov et al. (2021) and Kim et al. (2021).

7 Application to Nonparametric Variable Selection

The commonly used variable selection methods in the statistics literature rely on linear or additive models. This includes classical methods (Breiman, 1995; George & McCulloch, 1993; Chen & Donoho, 1994; Tibshirani, 1996; Efron et al., 2004; Friedman, 1991; Hastie et al., 2009; Miller, 2002) as well as modern ones (Candès & Tao, 2007; Zou, 2006; Zou & Hastie, 2005; Yuan & Lin, 2006; Fan & Li, 2001; Ravikumar et al., 2009). These methods are powerful and widely used in practice. However, they sometimes run into problems when significant interaction effects or nonlinearities are present. Such problems can be overcome by model-free methods (Candès et al., 2018; Ho, 1998; Amit & Geman, 1997; Breiman, 1996, 2001; Freund & Schapir, 1996; Hastie et al., 2009; Breiman et al., 1984; Battiti, 1994; Vergara & Estévez, 2014). On the flip side, the theoretical foundations of model-free methods are usually weaker than those of model-based methods.

In an attempt to combine the best of both worlds, a new method of variable selection called Feature Ordering by Conditional Independence (FOCI), was proposed in

Azadkia and Chatterjee (2021). This method uses the conditional dependence coefficient T_n described in the previous section in a stepwise fashion, as follows. Let Y be the response variable and let $\mathbf{X} = (X_j)_{1 \le j \le p}$ be the set of predictors. The data consists of n i.i.d. copies of $(Y, \mathbf{X})$. First, choose j_1 to be the index j that maximizes $T_n(Y, X_j)$. Having obtained $j_1, \ldots, j_k$, choose j_{k+1} to be the index $j \notin \{j_1, \ldots, j_k\}$ that maximizes $T_n(Y, X_j | X_{j_1}, \ldots, X_{j_k})$. Continue like this until arriving at the first k such that $T_n(Y, X_{j_{k+1}} | X_{j_1}, \ldots, X_{j_k}) \le 0$, and then declare the chosen subset to be $\hat{S} := \{j_1, \ldots, j_k\}$. If there is no such k, define $\hat{S}$ to be the whole set of variables. It may also happen that $T_n(Y, X_{j_1}) \le 0$. In that case, declare $\hat{S}$ to be empty. Note that this variable selection procedure involves no choice of tuning parameters, which may be an advantage in practice.

It was shown in Azadkia and Chatterjee (2021) that under mild conditions, the method selects a 'correct' set of variables with high probability. More precisely, it was shown that with high probability, the set $\hat{S}$ selected by FOCI has the property that Y and $(X_j)_{j \notin S}$ are conditionally independent given $(X_j)_{j \in \hat{S}}$. In other words, all the information about Y that one can get from $\mathbf{X}$ is contained in $(X_j)_{j \in \hat{S}}$. For further properties of FOCI and its performance in simulations and real data sets, see Azadkia and Chatterjee (2021).

An improved version of FOCI called KFOCI ('Kernel FOCI') was proposed by Huang et al. (2020). An application of FOCI to causal inference, via an algorithm named DAG-FOCI, was introduced in Azadkia et al. (2021). For another application to causal inference, see Chatterjee and Vidyasagar (2022).

8 A New Proposal: Generalization to Standard Borel Spaces

In this section, a simple but wide ranging generalization of ξ_n is proposed. In hindsight, this generalization seems obvious, but somehow this was overlooked both in the original paper Chatterjee (2021) as well as in the subsequent developments listed in Sect. 5.

Recall that two measurable spaces are said to be isomorphic to each other if there is a bijection between the two spaces which is measurable and whose inverse is also measurable. Recall that a *standard Borel space* is a measurable space that is isomorphic to a Borel subset of a Polish space (Srivastava, 1998, Chap. 3). In particular, every Borel subset of every Polish space is a standard Borel space. The Borel isomorphism theorem says that any uncountable standard Borel space is isomorphic to the real line (see Rao & Srivastava, 1994 for an elementary proof). In particular, if $\mathcal{X}$ is any standard Borel space, there is a measurable map $\varphi : \mathcal{X} \to \mathbb{R}$ such that φ is injective, $\varphi(\mathcal{X})$ is Borel, and φ^{-1} is measurable on $\varphi(\mathcal{X})$. We will say that φ is an isomorphism between $\mathcal{X}$ and a Borel subset of the real line, or simply, a 'Borel isomorphism'.

Now let $\mathcal{X}$ and $\mathcal{Y}$ be two standard Borel spaces. Let φ be a Borel isomorphism of $\mathcal{X}$ and ψ be a Borel isomorphism of $\mathcal{Y}$. Let (X, Y) be an $\mathcal{X} \times \mathcal{Y}$-valued pair of random variables, and let $(X_1, Y_1), \ldots, (X_n, Y_n)$ be i.i.d. copies of (X, Y). Let $X' := \varphi(X)$ and $Y' := \psi(Y)$, so that (X', Y') is a pair of real-valued random variables. Let $X'_i := \varphi(X_i)$ and $Y'_i := \psi(Y_i)$ for each i. Finally, define

$$\xi_n(X, Y) := \xi_n(X', Y'),$$

where $\xi_n(X', Y')$ is defined using $(X'_1, Y'_1), \ldots, (X'_n, Y'_n)$ as in equation (1.1). Note that the definition of $\xi_n(X, Y)$ depends on our choices of φ and ψ. Different choices of isomorphisms would lead to different definitions of ξ_n. The following theorem generalizes Theorem 1.1.

Theorem 8.1 *If Y is not almost surely a constant, then as $n \to \infty$, $\xi_n(X, Y)$ converges almost surely to the deterministic limit $\xi(X, Y)$, which equals $\xi(X', Y')$, defined as in (1.2) with X' and Y' in place of X and Y. This limit belongs to the interval $[0, 1]$. It is 0 if and only if X and Y are independent, and it is 1 if and only if there is a measurable function $f : \mathbb{R} \to \mathbb{R}$ such that $Y = f(X)$ almost surely. Moreover, the asymptotic distribution of $\sqrt{n}\xi_n(X, Y)$ under the hypothesis of independence, as given by Theorems 3.1 and 3.2, also holds, provided that τ and $\hat{\tau}_n$ are computed using X'_i and Y'_i instead of X_i and Y_i.*

Proof The convergence is clear from Theorem 1.1. Also, by Theorem 1.1, $\xi(X', Y') = 0$ if and only if X' and Y' are independent, and $\xi(X', Y') = 1$ if and only if Y' is a measurable function of X'. Since $X = \varphi^{-1}(X')$ and $Y = \psi^{-1}(Y')$, it follows that X and Y are independent if and only if X' and Y' are independent. For the same reason, Y is a measurable function of X almost surely if and only if Y' is a measurable function of X' almost surely. Note that Y is not almost surely a constant if and only if Y' is not almost surely a constant. Lastly, since $\xi_n(X, Y)$ is just $\xi_n(X', Y')$, any result about the asymptotic distribution of $\xi_n(X', Y')$, including Theorems 3.1 and 3.2 of this draft, can be transferred to $\xi_n(X, Y)$. □

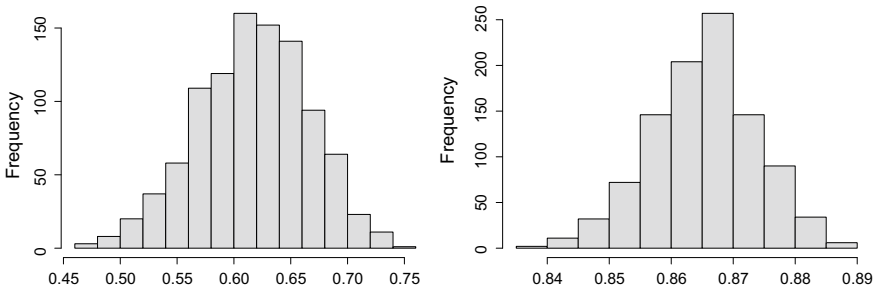
Just like the univariate case, the generalized ξ_n has the advantage of working under zero assumptions and having a simple asymptotic theory, as shown by Theorem 8.1. On the other hand, just like the univariate coefficient, one can expect the generalized coefficient to also suffer from low power for testing independence.

Theorem 8.1 is a nice, clean result, but to implement the idea in practice, one needs to work with actual Borel isomorphisms. Here is an example of a Borel isomorphism between $\mathbb{R}^d$ and a Borel subset of $\mathbb{R}$. Take any $x = (x_1, \ldots, x_d) \in \mathbb{R}^d$. Let

$$a_{i,1} \cdots a_{i,k_i} . b_{i,1} b_{i,2} \cdots$$

be the binary expansion of $|x_i|$. Filling in extra 0's at the beginning if necessary, let us assume that $k_1 = \cdots = k_d = k$. Then, let us 'interlace' the digits to get the number

$$a_{1,1} a_{2,1} \cdots a_{d,1} a_{1,2} a_{2,2} \cdots a_{d,2} \cdots a_{1,k} a_{2,k} \cdots a_{d,k} . b_{1,1} b_{2,1} \cdots b_{d,1} b_{1,2} b_{2,2} \cdots b_{d,2} \cdots .$$



(a) $n = 100$ (b) $n = 1000$

Fig. 1 Histograms of values of $\xi_n(X, Y)$ when X and Y are the polar and Cartesian coordinates of random points from the unit sphere in $\mathbb{R}^3$ (Example 8.1)

This is an encoding of the n-tuple $(|x_1|, \ldots, |x_d|)$. But we also want to encode the signs of the x_i's. Let $c_i = 1$ if $x_i \geq 0$ and 0 if $x_i < 0$. Sticking $1c_1c_2 \cdots c_d$ in front of the above list, we get an encoding of the vector x as a real number. (The 1 in front ensures there is no ambiguity arising from some of the leading c_i's being 0.) It is easy to verify that this mapping is measurable, injective, and its inverse (defined by its range) is also measurable.

Numerical simulations with ξ_n computed using the above scheme produced satisfactory results. Some examples are as follows. The examples show one potential problem with using statistics such as ξ_n in a high-dimensional setting: The bias may be quite large (even though the standard deviation is small), resulting in slow convergence of $\xi_n(X, Y)$ to $\xi(X, Y)$.

Example 8.1 (*Points on a sphere*) Non-uniform random points were generated on the unit sphere in $\mathbb{R}^3$ by drawing $\phi_1, \ldots, \phi_n$ uniformly from $[-\pi, \pi]$, drawing $\theta_1, \ldots, \theta_n$ uniformly from $[0, 2\pi]$, and defining

$$x_i = \sin\phi_i \cos\theta_i, \quad y_i = \sin\phi_i \sin\theta_i, \quad z_i = \cos\phi_i.$$

Taking $X_i = (\phi_i, \theta_i)$ and $Y_i = (x_i, y_i, z_i)$, $\xi_n(X, Y)$ was computed for $n = 100$ and $n = 1000$. One thousand simulations were done for each n. The histograms of the values of ξ_n are displayed in Fig. 1. For $n = 100$, $\xi_n(X, Y)$ had a mean value of 0.617 with a standard deviation of 0.049. For $n = 1000$, the mean value was 0.865 and the standard deviation was 0.009. Simulations were also done for $n = 10000$, where the mean value of $\xi_n(X, Y)$ turned out to be 0.957 and the standard deviation was 0.002. The slow convergence due to large bias is clearly apparent in this example.

Example 8.2 (*Points on a sphere plus noise*) This is the same example as the previous one, except that independent random noises were added to x_i, y_i and z_i. The noise variables were taken to be normal with mean zero and standard deviation σ, for various values of σ. Table 1 displays the means and standard deviations of $\xi_n(X, Y)$ in 1000 simulations, for $n = 100$ and $n = 1000$, and $\sigma = 0.01$, $\sigma = 0.05$ and $\sigma = 0.1$.

Table 1 Means and standard deviations of $\xi_n(X, Y)$ in 1000 simulations, when X = polar coordinates of a random point on a sphere and Y = Cartesian coordinates of the point plus independent $N(0, \sigma^2)$ errors in each coordinate (Example 8.2)

	σ		
n	0.01	0.05	0.1
100	0.545 (0.062)	0.440 (0.071)	0.357 (0.074)
1000	0.783 (0.015)	0.658 (0.020)	0.543 (0.023)

Example 8.3 (*Marginal independence versus joint dependence*) In this example, we have a pair of random variables $Y = (a, b)$ which is a function of a 4-tuple $X = (u, v, w, z)$, but u, v, w and z are marginally independent of Y_i. The variables are constructed as follows. Let a, b and c be independent Uniform[0, 1] random variables. Let

$$u = a + b + c \pmod 1, \quad v = \frac{a}{2} + \frac{b}{2} + c \pmod 1,$$
$$w = \frac{4a}{3} + \frac{2b}{3} + c \pmod 1, \quad z = \frac{2a}{3} + \frac{b}{3} + c \pmod 1.$$

Here $x \pmod 1$ denotes the 'remainder modulo 1' of a real number x. It is easy to see that individually, u, v, w and z are independent of $Y = (a, b)$, since the operation of adding the uniform random variable c and taking the remainder modulo 1 erases all information about the quantity to which c is added. However, Y can be recovered from the 4-tuple $X = (u, v, w, z)$ because $u - v = (a + b)/2 \pmod 1 = (a + b)/2$ and $w - z = (2a + b)/3 \pmod 1 = (2a + b)/3$, where the second identity holds in each case because $(a + b)/2 \in [0, 1]$ and $(2a + b)/3 \in [0, 1]$. With the above definitions, n i.i.d. copies of (X, Y) were sampled, and $\xi_n(X, Y)$ was computed, along with the asymptotic P-value for testing independence. This was repeated one thousand times. The average values of the coefficients and the P-values for $n = 100$ and $n = 1000$ are reported in Table 2. The table shows that even for n as small as 100, the hypothesis that X and Y are independent is rejected with average P-value 0.001, whereas the average P-value for the hypothesis and u and Y are independent is 0.491. On the other hand, even for n as large as 1000, the average P-value for the hypothesis that u and Y are independent is 0.495, whereas the average P-value for the hypothesis that X and Y are independent is 0.000.

The above examples indicate that it may not be 'crazy' to use the generalized version of ξ_n as a measure of association in the multivariate setting and beyond. Further investigations, through applications in simulated and real data, would be necessary to arrive at a concrete verdict.

The generalized version of ξ_n can also be used to define a coefficient of conditional dependence for random variables taking values in standard Borel spaces, as follows. Let $\mathcal{X}$, $\mathcal{Y}$ and $\mathcal{Z}$ be standard Borel spaces, and X, Y and Z be random variables taking values in $\mathcal{X}$, $\mathcal{Y}$ and $\mathcal{Z}$, respectively. Let W denote the $\mathcal{X} \times \mathcal{Z}$-valued random

Table 2 Average values of $\xi_n(u, Y)$ and $\xi_n(X, Y)$ and average P-values for testing independence in Example 8.3

	Average value		Average P-value	
n	$\xi_n(u, Y)$	$\xi_n(X, Y)$	$H_0 : u \perp\!\!\!\perp Y$	$H_0 : X \perp\!\!\!\perp Y$
100	0.002	0.313	0.491	0.001
1000	0.001	0.581	0.495	0.000

variable (X, Z). Using Borel isomorphisms, let us define the coefficients $\xi_n(X, Y)$ and $\xi_n(W, Y)$ as before, based on i.i.d. data $(X_1, Y_1, Z_1), \ldots, (X_n, Y_n, Z_n)$. One can then define a coefficient of conditional dependence between Y and Z given X as

$$\xi_n(Z, Y|X) := \frac{\xi_n(W, Y) - \xi_n(X, Y)}{1 - \xi_n(X, Y)},$$

leaving it undefined if the denominator is zero. The following theorem justifies its use as a measure of conditional dependence.

Theorem 8.2 *Suppose that Y is not almost surely equal to a measurable function of X. Then as $n \to \infty$, $\xi_n(Z, Y|X)$ converges almost surely to a deterministic limit $\xi(Z, Y|X) \in [0, 1]$, which is 0 if and only if Y and Z are independent given X, and 1 if and only if Y is almost surely equal to a measurable function of Z given X.*

Proof Let φ, ψ, η and δ be the Borel isomorphisms of $\mathcal{X}$, $\mathcal{Y}$, $\mathcal{Z}$ and $\mathcal{X} \times \mathcal{Z}$ that we are using in our definitions of ξ_n. Let $X' := \varphi(X)$, $Y' := \psi(Y)$, $Z' := \eta(Z)$ and $W' := \delta(W)$. By Theorem 1.1 and the assumption that Y is not almost surely a measurable function of X, we get that $\xi_n(Z, Y|X)$ converges almost surely to

$$\xi(Z, Y|X) := \frac{\xi(W', Y') - \xi(X', Y')}{1 - \xi(X', Y')}.$$

Let μ denote the law of Y'. Then

$$\begin{aligned}
\xi(W', Y') - \xi(X', Y') &= \frac{\int \mathrm{Var}(\mathbb{P}(Y' \geq t|W'))d\mu(t) - \int \mathrm{Var}(\mathbb{P}(Y' \geq t|X'))d\mu(t)}{\int \mathrm{Var}(1_{\{Y' \geq t\}})d\mu(t)} \\
&= \frac{\int \mathrm{Var}(\mathbb{P}(Y' \geq t|X', Z'))d\mu(t) - \int \mathrm{Var}(\mathbb{P}(Y' \geq t|X'))d\mu(t)}{\int \mathrm{Var}(1_{\{Y' \geq t\}})d\mu(t)} \\
&= \frac{\int \mathbb{E}(\mathrm{Var}(\mathbb{P}(Y' \geq t|X', Z')|X')d\mu(t)}{\int \mathrm{Var}(1_{\{Y' \geq t\}})d\mu(t)}.
\end{aligned}$$

Similarly,

$$1 - \xi(X', Y') = \frac{\int \mathbb{E}(\mathrm{Var}(1_{\{Y' \geq t\}}|X'))d\mu(t)}{\int \mathrm{Var}(1_{\{Y' \geq t\}})d\mu(t)}.$$

From the above expressions, we see that $\xi(Z, Y|X)$ is nothing but the quantity $T(Y, Z|X)$ displayed in equation (6.1). The claims of the theorem now follow from Theorem 6.1. □

9 R Packages

An R package for calculating ξ_n, as well as the generalized coefficient proposed in this survey, and P-values for testing independence—named XICOR—is available on CRAN (Chatterjee & Holmes, 2020). An R package for calculating the conditional dependence coefficient T_n and implementing the FOCI algorithm, called FOCI, is also available (Azadkia et al., 2020). The KFOCI algorithm of Huang et al. (2020) is implemented in an R package by the same name (Huang et al., 2022).

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Fractal Structure in the Directed Landscape



Shirshendu Ganguly and Milind Hegde

Dedicated to Professor Rajeeva L. Karandikar on his 65th birthday.

1 Introduction

A broad area of much recent research in probability, pursued in several distinct directions, is *random geometry* and *random metric spaces*, with much of the focus on the planar case (planarity in fact plays an important role). The objects of interest include distance minimizing paths—*geodesics*—as well as the metric itself. The underlying disorder leads to certain key differences from standard models of geometry, such as those arising from Euclidean spaces, some of which will play central roles in the remainder of this article.

Most of these models are expected to exhibit certain approximate scale invariance features which should become exact once an appropriate limit is taken. Such invariance properties often manifest themselves in intricate *fractal behaviour*, the topic reviewed in this article. For the uninitiated reader, this essentially refers to various patterns of self-similarity ubiquitous in nature (Husain et al. 2022a, b). While this survey covers rather modern developments, the general theme is a fairly familiar

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story in probability theory, where, for example, a similar line of consequences arises in Brownian motion, a universal scaling limit of one dimensional stochastic process; except that we are now dealing with significantly more complex high-dimensional random objects. At first glance, mathematical analysis of such properties may seem a daunting task. Nonetheless, there have been quite a few recent developments, and it turns out that many parallels and inspirations can be drawn from results about Brownian motion.

There are two broad classes of models of random planar geometry that have featured in most of the investigations. A first choice might be to choose a metric in some sense uniformly (with the base space fixed). Though highly nontrivial, this can indeed be made sense of. However, in this article we will consider a seemingly much simpler class of models which nevertheless leads to an equally rich theory. Perhaps the easiest example in this class consists of distorting the edge lengths in the Euclidean lattice by random weights. In other words, each edge is assigned an i.i.d. non-negative random variable which is its new length. Then the resulting graph metric is a random metric. This class of models, which is known as first passage percolation (FPP) was first introduced by Hammersley and Welsh to model fluid flow in a porous medium Hammersley and Welsh (1965), the interested reader is encouraged to look at the survey Auffinger et al. (2014).

It is worth remarking that while in the above example the underlying noise that drives the randomness is independent, the earlier mentioned example of a "uniform" metric corresponds to a logarithmically correlated noise field. The central object of study in this direction is known as Liouville quantum gravity, for which there are many recent surveys available; we point the reader to Gwynne (2019), Ding et al. (2021).

The case of interest for us, where the noise is independent (in fact, i.i.d.) falls into the Kardar-Parisi-Zhang universality class, a class of stochastic growth models (with the metric balls modeling growth) which are all believed to exhibit universal behaviour (akin to the universality of the central limit theorem). However, as alluded to earlier, we will be concerned with recent advances in understanding the geometric and fractal properties of some purportedly universal limiting objects, rather than aspects of universality itself.

1.1 Fractality in Brownian Motion

Before turning to fractal behaviour in the aforementioned more complex settings, to offer a flavor of the kind of results we will be interested in, we start by recalling an abridged history of fractal properties in (one-dimensional) Brownian motion. The first and perhaps most classical instance of fractality usually encountered is that of its zero set: it is a perfect (closed, with no isolated points) set whose Hausdorff dimension (a heuristic definition is given in Sect. 3) is almost surely $\frac{1}{2}$. The basic fact that allows fractal behaviour to arise is that B is a stochastic process whose index set (i.e., its domain $[0, \infty)$, representing time) is *uncountable*. Thus, events which have

probability zero at any fixed time (eg., $\mathbb{P}(B(t) = 0) = 0$ for any fixed $t > 0$) can nevertheless arise at a random exceptional time; the one-point probability estimate can only be extended to cover a countable set of times by a union bound, but cannot cover the entire domain. In all of the examples of fractality we will encounter in random metric spaces, this basic mechanism will be at play: events which almost surely do not occur at any typical location will almost surely happen at random exceptional locations, and the set of these exceptional locations will be fractal.

A second interesting point is that the fractal properties of the random set of exceptional locations will often be *deterministic*, just as the Hausdorff dimension of the Brownian zero set is deterministically $\frac{1}{2}$ almost surely. This can be understood as the manifestation of a kind of zero-one law which results from the approximate independence of these stochastic processes across different scales. After all, the question of fractal structure of these sets is exactly a question of the behaviour of the process across all small scales.

While one can formulate approximate statements for discrete random walks, ultimately fractal properties exist in a direct sense only in Brownian motion, their scaling limit. Similarly while there are many discrete models of random planar metric spaces with i.i.d. noise, such as FPP above, we will need a continuum object to investigate fractal properties. This is described in the next section. (It is worth pointing out that beyond the focus of this article on geodesic geometry, fractal properties have also been investigated in closely related models in the KPZ class in a number of recent studies, for example Corwin et al. (2021), Dauvergne (2022), Seppäläinen and Sorensen (2021), Das (2022), Das and Ghosal (2021), but in view of brevity we do not discuss these here).

2 The Directed Landscape

While we started our discussion with a model of a random metric space, most of the results are in fact about a related variant which is directed, and admits remarkable "integrable" properties allowing access to algebraic inputs. This is in contrast to the general intractableness of FPP. These directed models go by the name of last passage percolation (LPP) where instead of length minimizing paths as in FPP, one considers directed length maximizing paths (which are usually termed as geodesics as well). The scaling limit in this setting is what we will work with, which as a consequence of the above, already has an in-built direction, and is called the *directed landscape*, first constructed in the breakthrough paper Dauvergne et al. (2022).

The directed landscape $\mathcal{L} : \mathbb{R}^4_\uparrow \to \mathbb{R}$ is a random continuous function, where $\mathbb{R}^4_\uparrow = \{(x, s;\ y, t) : x, y \in \mathbb{R}, 0 \leq s < t < \infty\}$; we will think of (x, s) and (y, t) as pairs of points, and $\mathcal{L}(x, s;\ y, t)$ as a sort of "distance" between such points. That we require $s \leq t$ is a manifestation of the already alluded to the directedness in the models expected to converge to $\mathcal{L}$. Further, $\mathcal{L}$ satisfies the reverse triangle inequality owing to the length maximization property: for $(p; r), (q; r), (p; q) \in \mathbb{R}^4_\uparrow$,

$$\mathcal{L}(p;r) \geq \mathcal{L}(p;q) + \mathcal{L}(q;r).$$

In fact, this follows from a *metric composition* law that $\mathcal{L}$ possesses: for $(x, r;\ z, t) \in \mathbb{R}^4_{\uparrow}$ and any $s \in (r, t)$,

$$\mathcal{L}(x, r;\ z, t) = \max_{y \in \mathbb{R}} \mathcal{L}(x, r;\ y, s) + \mathcal{L}(y, s;\ z, t).$$

We should emphasize that there are currently only a handful of LPP models which can be shown to converge in some sense to the directed landscape.

As one would expect of any object which is a universal scaling limit (think again of Brownian motion), $\mathcal{L}$ possesses a scale invariance property as well as many symmetries, governed by the now well-known KPZ scaling exponents of $\frac{1}{3}$ and $\frac{2}{3}$, where $\frac{1}{3}$ describes the length fluctuation scale and $\frac{2}{3}$ the spatial correlation scale. The precise statement of scale invariance (along with a convenient translation invariance) is that, for fixed $t, s > 0$ and as a process in $x, y \in \mathbb{R}$,

$$\mathcal{L}(x, t;\ y, t+s) \stackrel{d}{=} s^{-1/3}\mathcal{L}(xs^{-2/3}, 0;\ ys^{-2/3}, 1).$$

The $\frac{1}{3}$ and $\frac{2}{3}$ should be contrasted with the scaling exponent of $\frac{1}{2}$ in the Gaussian universality class (we note in passing that $\frac{2}{3} = 1 - \frac{1}{3}$, while $\frac{1}{2} = 1 - \frac{1}{2}$, as a sort of very heuristic reason for why there are two exponents instead of one in the KPZ class). These scaling exponents will largely determine the values of fractal dimensions of various random exceptional sets we will consider.

2.1 Geodesics in $\mathcal{L}$

While the existence of geodesics is usually straightforward to show in the pre-limiting models, it is not a priori obvious that this property persists in the limit, i.e., that $\mathcal{L}$ is a geodesic space. Further, note that the definition of $\mathcal{L}$ does not explicitly contain an underlying noise field (unlike in the prelimit where the noise was i.i.d.). For this reason, the length (which we will also often refer to as weight, in keeping with common terminology in the literature) $\|\pi\|_{\mathcal{L}}$ of a continuous path $\pi : [s, t] \to \mathbb{R}$ under $\mathcal{L}$ is defined as

$$\|\pi\|_{\mathcal{L}} = \inf_{k \in \mathbb{N}} \inf_{s = t_0 < t_1 < \ldots < t_k = t} \sum_{i=1}^{k} \mathcal{L}\big(\pi(t_{i-1}), t_{i-1};\ \pi(t_i), t_i\big).$$

Given the above, a path $\pi : [s, t] \to \mathbb{R}$ is said to be a *geodesic* from $(\pi(s), s)$ to $(\pi(t), t)$ if $\|\pi\|_{\mathcal{L}} = \mathcal{L}(\pi(s), s;\ \pi(t), t)$.

It is a non-trivial fact, proved in Dauvergne et al. (2022), that, almost surely, geodesics exist between all pairs of points and that, almost surely for any given pair of points, the geodesic is unique.

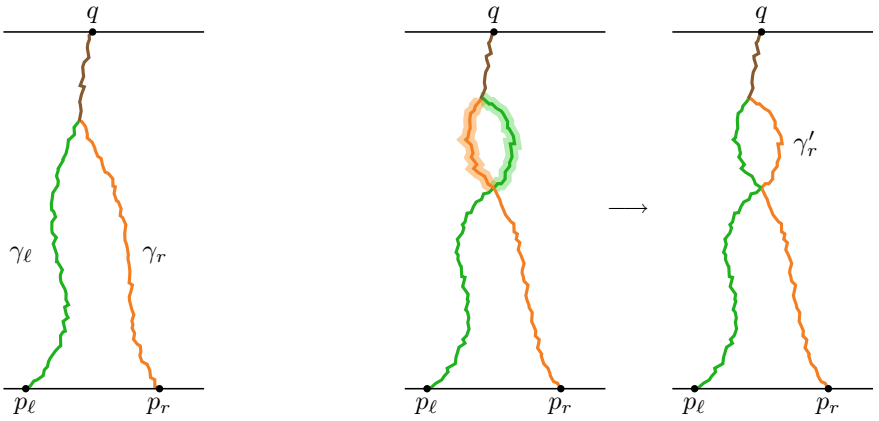


Fig. 1 In the left panel we see that a geodesic γ_ℓ whose endpoints are to the left of that of another γ_r will lie to the left of it for its entire journey; the two geodesics may, however, coincide. This is known as geodesic ordering. The right panel depicts the argument for this ordering: if it did not hold and γ_ℓ crossed over γ_r, then a "bubble" would be formed between the two geodesics. Then we can construct new geodesics γ_ℓ' and γ_r' (between the same respective endpoints as the originals) by swapping which geodesic each of the sides of the bubble belong to, as shown. This contradicts the almost sure uniqueness of geodesics between fixed endpoints

One aspect of understanding the random geometry of $\mathcal{L}$ is the joint behaviour of multiple geodesics between different end points. The simplest, of course, is the case of two geodesics. Let us start with an easy consequence of planarity.

Let p_ℓ, p_r, q be three points such that $(p_\ell; q), (p_r; q) \in \mathbb{R}^4_\uparrow$ and p_ℓ is to the left of p_r (see the left panel of Fig. 1). Let γ_ℓ and γ_r respectively be the geodesics from p_ℓ and p_r to q. We work on the probability one event where these geodesics are unique. We claim that γ_ℓ is to the left of γ_r, i.e., $\gamma_\ell(s) \leq \gamma_r(s)$ for any s which belongs to the domains of both paths; in particular, while the two can touch, they cannot cross. We will refer to this as the *ordering* of geodesics.

The proof is simple. Suppose γ_r crosses over γ_ℓ at some height. Because the paths are continuous and because of the ordering of the endpoints (along with planarity), γ_r must touch γ_ℓ at some later height. Look at both geodesics on the vertical interval between these two heights; it consists of two paths *with the same endpoints*; see the right panel of Fig. 1. The left path is a subset of γ_r while the right is a subset of γ_ℓ. Because both geodesics are length maximizers, the left and right paths must have the same lengths. Because they share the same endpoints on this interval, we can define a new path γ_r' which equals γ_r except in between the two heights, where it equals the *right* path—in short, we have swapped the two paths on the mentioned interval of heights. Then γ_r' is a path with the same endpoints and the same length as γ_r, which means γ_r' is also a geodesic, which contradicts the uniqueness of γ_r.

So we know that geodesic whose endpoints are ordered cannot cross—but they can touch. What is the structure of this touching? Reviewing the above argument more closely shows that, again on the almost sure event that the two geodesics are

unique, the set of heights where the two geodesics touch must be an interval; if not, we would again have a "bubble" which could be used to create distinct geodesics by a swapping operation; this observation applies even if the endpoints are not ordered. Thus, if two geodesics meet at some point, they will remain together for some interval of time, before separating and never meeting again (or they may never separate, if they have a common destination). This is called *geodesic coalescence*, induced by the propensity of the geodesics to pass through the atypically large part of environment, a key distinction from Euclidean geometry where geodesics are straight lines and hence have trivial intersection patterns.

3 Geodesic Coalescence in $\mathcal{L}$

Instead of directly trying to understand geodesic coalescence in full, it will prove fruitful to consider what can be thought of as a lower dimensional projection. Concretely, we focus on the case where geodesic have fixed distinct starting points, say $(-1, 0)$ and $(1, 0)$, and a common ending point $(x, 1)$.

Because the geodesic certainly meet at height 1, it follows from the above discussion that the set of times where they meet must be an interval $[y_x, 1]$ (that the interval is closed follows from the continuity of the paths). It can be shown that, for any given $x \in \mathbb{R}$, with probability 1, $y_x < 1$, i.e., there is a non-trivial interval of coalescence. Therefore, it is natural to ask if there exist random exceptional values of x where $y_x = 1$, i.e., endpoints such that the geodesics remain disjoint for their entire lifetimes except for the final instant? We will say that such geodesics are disjoint, ignoring the shared final point.

Much of the information pertaining to the above question turns out to be encoded by a projection of $\mathcal{L}$ called the *weight difference profile* $\mathcal{D} : \mathbb{R} \times (0, \infty) \to \mathbb{R}$ by

$$\mathcal{D}(x, t) = \mathcal{L}(1, 0;\ x, t) - \mathcal{L}(-1, 0;\ x, t).$$

This object (in fact, only its spatial marginal $x \mapsto \mathcal{D}(x, 1)$, which we will denote $\mathcal{D}_{\text{sp}}$) was first introduced in Basu et al. (2021). The expert reader might notice that this is closely related to, and can be thought of as a finite version of, *Busemann functions* from differential geometry which indeed have proven to be a valuable tool in studying FPP Hoffman (2008).

It was shown in Basu et al. (2021), Bates et al. (2022) that if x is such that the geodesics from $(-1, 0)$ and $(1, 0)$ to $(x, 1)$ are not disjoint, then $\mathcal{D}_{\text{sp}}$ is constant in some neighbourhood of x. Thus, the fact that any fixed x is almost surely such that the mentioned geodesics are not disjoint implies that $\mathcal{D}_{\text{sp}}$ is almost everywhere locally constant. Because the proofs of these statements rely only on geodesic coalescence, they also hold for the full process $(x, t) \mapsto \mathcal{D}(x, t)$ (the formal statement appears in the later article Ganguly and Zhang, 2022).

A curious fact, which is also a reasonably straightforward consequence of planarity, is that $\mathcal{D}_{\text{sp}}$ is non-decreasing. Thus $\mathcal{D}_{\text{sp}}$ is a random version of Cantor's func-

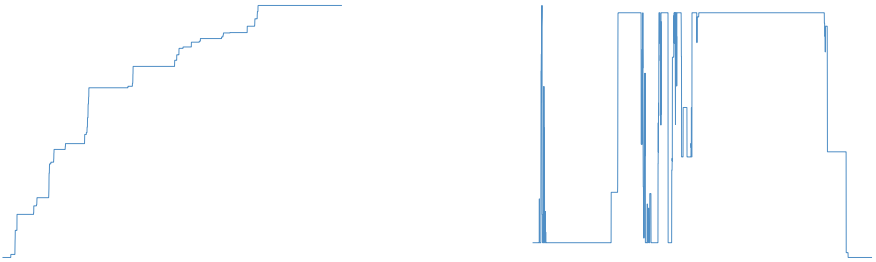


Fig. 2 In the left panel we have a simulation of $x \mapsto \mathcal{D}(x, 1)$, while on the right it is $t \mapsto \mathcal{D}(0, t)$. Note that the left process is non-decreasing, while the right one does not exhibit any monotonicity. Both simulations are done in the discrete model of Exponential last passage percolation, which is known to converge to $\mathcal{L}$

tion: almost everywhere constant, but nevertheless steadily increasing (a computation involving $\mathcal{L}$ shows that the growth rate is on average linear). (This statement does not hold for the full process, eg. for $t \mapsto \mathcal{D}(0, t)$, which will cause complications in the latter's study; we will discuss this a bit more later.) See Fig. 2.

The fact that the function is almost everywhere locally constant implies that the Lebesgue measure of the set of points of *non-constancy* (points where the function is not constant in any neighbourhood) is zero. A finer notion is needed to measure the size of these sets. A common choice is that of *fractal dimension*. While there are several definitions of fractal dimension available, a robust notion is the Hausdorff dimension. We do not include the somewhat technical definition, but remark that for purposes of intuition it is useful to keep in mind that the dimension of a subset $X \subseteq \mathbb{R}^n$ can be thought of the related number $\alpha > 0$ such that order $\varepsilon^{-\alpha}$ number of balls of radius ε are needed to cover X. For example, the Hausdorff dimension of a non-trivial interval is 1, of the unit square is 2, and so on.

It is well-known that the fractal dimension of the classical Cantor set, the set of points of non-constancy of the Cantor function, is $\log 2/\log 3$. The main result of Basu et al. (2021) determined the fractal dimension of the set of non-constancy points of $\mathcal{D}_{\text{sp}}$, which we will denote $\mathsf{NC}(\mathcal{D}_{\text{sp}})$.

Theorem 3.1 (Theorem 1.1 of *Basu et al. 2021*) *The Hausdorff dimension of* $\mathsf{NC}(\mathcal{D}_{\text{sp}})$ *is almost surely* $\frac{1}{2}$.

Let us now illustrate some of the ideas in Basu et al. (2021)'s proof of Theorem 3.1. As is usual while computing dimensions, one needs separate arguments for an upper bound and a lower bound on $\dim(\mathsf{NC}(\mathcal{D}_{\text{sp}}))$.

The upper bound. For the upper bound, we use the earlier mentioned fact that $\mathsf{NC}(\mathcal{D}_{\text{sp}})$ is a subset of the set Disj of x such that the two geodesics from $(-1, 0)$ and $(1, 0)$ to $(x, 1)$ are disjoint. (In fact, it was later shown in Bates et al. (2022) that the two sets coincide.) So it is enough to upper bound $\dim(\mathsf{Disj})$.

There are multiple broadly similar arguments to show this, but here we outline a perhaps more geometric strategy. Suppose there is a random $x^* \in \mathsf{Disj}$ (recall

that any fixed $x \notin \mathsf{Disj}$ a.s.), i.e., there are two disjoint geodesics from $(-1, 0)$ and $(1, 0)$ to $(x^*, 1)$. We claim that there must then be a random $y^* \in (-1, 1)$ such that there are *two* disjoint geodesics with endpoints in the intervals $[y^* - \varepsilon, y^* + \varepsilon] \times \{0\}$ and $[x^* - \varepsilon, x^* + \varepsilon] \times \{1\}$—so two distinct geodesics between arbitrarily close endpoints.

The argument essentially is a consequence of planar ordering of geodesics. Consider the family of geodesics ending at $(x^* - \varepsilon, 1)$ (which is now fixed) indexed by the starting point $(y, 0)$ as y varies. Picture dragging the starting point y from -1 rightwards and observing how the geodesic to $(x^* + \varepsilon, 1)$ changes. There must be some smallest y^* at which the latter geodesic first becomes disjoint from the initial geodesic $\gamma_{-1 \to x^* - \varepsilon}$ from $(-1, 0)$. Let γ^* be the label of this geodesic from $(y^*, 0)$ to $(x^* + \varepsilon, 1)$ which is disjoint from $\gamma_{-1 \to x^* - \varepsilon}$. It can then be shown that the geodesics from $(y^* \pm \varepsilon, 0)$ to $(x^* \pm \varepsilon, 1)$ are disjoint (assuming the geodesic mentioned are unique, which can be ensured by looking at endpoints on a deterministic fine mesh). We give some more details in the caption of Fig. 3.

So, given $x^* \in \mathsf{Disj}$, we have found two disjoint geodesics whose endpoints are within ε of each other. It turns out, somewhat remarkably, that probability of such non-coalescence can be computed. In fact, it was was proved in Hammond (2020) that, given two intervals I_1 and I_2 of length ε, the probability that there are two disjoint geodesics with endpoints in $I_1 \times 0$ and $I_2 \times 1$ is at most of order $\varepsilon^{3/2}$; let us call this event E_{I_1, I_2}. With this result in hand, let us divide up the horizontal line

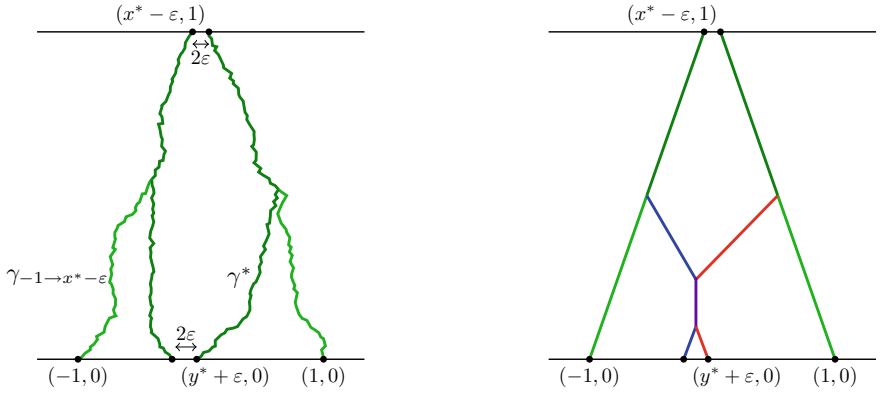


Fig. 3 In the left panel we depict objects such as $\gamma_{-1 \to x^* - \varepsilon}$ and γ^*. Recall that y^* is the left-most point on the bottom such that the geodesic from $(y^*, 0)$ to $(x^* + \varepsilon, 1)$ is disjoint from $\gamma_{-1 \to x^* - \varepsilon}$. In the right panel we consider the case that the path formed by blue+purple+blue+dark green when going bottom to top and γ^* (red+purple+red+dark green) are not disjoint, and derive a contradiction. It is immediate from the diagram that we can then construct a new path γ' from $(y^* - \varepsilon, 0)$ to $(x^* + \varepsilon, 1)$ given by blue+purple+red+dark green; this is a geodesic since each of the path fragments making it up are portions of geodesics. Further, γ' is disjoint from $\gamma_{-1 \to x^* - \varepsilon}$ because the blue and purple path fragments in the figure are part of a geodesic which coalesces with the latter geodesic only later. So we have a geodesic from $(y^* - \varepsilon, 0)$ to $(x^* + \varepsilon, 1)$ which is disjoint from $\gamma_{-1 \to x^* - \varepsilon}$, contradicting our definition of y^*

segments (of some fixed finite width at least 2, so that ± 1 are contained in them) at heights 0 and 1 into intervals of length ε. There are $O(\varepsilon^{-1})$ many such intervals on each line segment, so $O(\varepsilon^{-2})$ many pairs of intervals, one taken from each of the two line segments.

We know from the argument above that for any ε-interval I_1 on the upper segment that contains x^*, there is an ε-interval I_2 on the lower line segment such that E_{I_1,I_2} occurs. By the result from Hammond (2020), the expected number of pairs of such intervals is at most $\varepsilon^{-2} \times \varepsilon^{3/2} = \varepsilon^{-1/2}$. Thus the number of ε-intervals that contain a point like x^* is also at most $\varepsilon^{-1/2}$. This yields a Hausdorff dimension upper bound of $\frac{1}{2}$.

The lower bound. Generally, it is harder to give a lower bound on the Hausdorff dimension of a set than an upper bound. But here, surprisingly, a matching lower bound is easier to obtain, albeit relying on some key properties of the landscape. To outline it, we need to introduce a related process known as the parabolic Airy_2 process $\mathcal{P}_1$ (the reason for the subscript of 1 will be made clear in a few more pages): it can be defined as $\mathcal{P}_1(x) = \mathcal{L}(0, 0; x, 1)$, i.e., the weight profile in the directed landscape as a function of the varying ending point when the starting point is fixed.

Here the important fact about $\mathcal{P}_1$ is that it is *locally Brownian*. There are many ways in which this vague statement can be made precise (see for example Calvert et al. 2022; Corwin and Hammond 2014; Hammond 2022), but the one relevant here is that $\mathcal{P}_1$ is Hölder-$\frac{1}{2}^-$ Hammond (2019). As a result, $\mathcal{P}_1$ can vary over an interval of size ε by order $\varepsilon^{1/2}$.

Now $\mathcal{D}_{\text{sp}}(x) = \mathcal{L}(1, 0; x, 1) - \mathcal{L}(-1, 0; x, 1)$ can be regarded as the difference of a pair of coupled copies of $\mathcal{P}_1$ using the translation invariance of $\mathcal{L}$. Thus it too can vary over an interval of size ε by at most order $\varepsilon^{1/2}$.

The idea now is to divide up a given interval, say $[-1, 1]$, into sub-intervals of length ε^{-1} and ask for a lower bound on the number of these subintervals which contain a point of $\mathsf{NC}(\mathcal{D}_{\text{sp}})$, i.e., a point where $\mathcal{D}_{\text{sp}}$ increases. But since $\mathcal{D}_{\text{sp}}(1) - \mathcal{D}_{\text{sp}}(-1)$ is a unit order quantity, the control on the growth of $\mathcal{D}_{\text{sp}}$ from above implies that at least order $\varepsilon^{-1/2}$ many intervals are needed to obtain $O(1) = O(\varepsilon^{1/2} \times \varepsilon^{-1/2})$ growth. This yields the matching lower bound of $\frac{1}{2}$ for $\dim(\mathsf{NC}(\mathcal{D}_{\text{sp}}))$.

Note the crucial fact that $\mathcal{D}_{\text{sp}}$ is monotone, and so an increase in one of the sub-intervals can only add to and not cancel that in another. As mentioned, this feature will not be present when we look at the related situation of the temporal slice of $\mathcal{D}$ later.

3.1 The Weight Difference Profile and Brownian Local Time

With Theorem 3.1 stating that $\mathsf{NC}(\mathcal{D}_{\text{sp}})$ is a random fractal set of Hausdorff dimension $1/2$, one is naturally reminded of the already alluded to zero set of Brownian motion, i.e., $\{t > 0 : B(t) = 0\}$ where B is a standard Brownian motion started at 0 (see for example (Mörters and Peres, 2010, Theorem 4.24)). Associated is a natural

stochastic process which is non-decreasing, almost everywhere constant, and whose points of non-constancy are exactly the Brownian motion zero set: the Brownian local time $\mathfrak{L}$. It can be thought of as a measure of the amount of time B spends at zero, and can be precisely defined using this intuition:

$$\mathfrak{L}(t) = \lim_{\varepsilon \downarrow 0} \frac{1}{2\varepsilon} \int_0^t \mathbb{1}_{B(s) \in [-\varepsilon, \varepsilon]} \, \mathrm{d}s.$$

Amazingly, an identity of Lévy say that the local time has the same distribution as the running maximum process of Brownian motion. More precisely, if $M(t) = \sup_{0 \le s \le t} B(s)$, then, as processes,

$$\mathfrak{L} \stackrel{d}{=} M.$$

Now, because $\mathfrak{L}$ is non-negative and non-decreasing, it can be interpreted as the distribution function of a probability measure supported on the zero set of Brownian motion. A well-known technique to understand the Hausdorff dimension of a set is to construct a measure on it with certain mass distribution properties, and the argument for the Hausdorff dimension of the zero set of Brownian motion uses the measure defined via $\mathfrak{L}$, along with the distributional identity of $\mathfrak{L}$ with M for the analysis of the measure.

All of the above leads to an obvious question: is there a connection between $\mathcal{D}$ and $\mathfrak{L}$? It turns out that indeed there is.

Theorem 3.2 (Shirshendu Ganguly and Milind Hegde 2022; Dauvergne 2021) *Let* $[a, b] \subseteq \mathbb{R}$. *Then the law of* $\mathcal{D}_{\mathrm{sp}}(\cdot) - \mathcal{D}_{\mathrm{sp}}(a)|_{[a,b]}$ *is absolutely continuous to that of* $\mathfrak{L}(\cdot) - \mathfrak{L}(1)|_{1,b-a+1}$.

We need to consider the increment of $\mathcal{D}$ as the process is not non-negative by itself (unlike $\mathfrak{L}$), and we consider $\mathfrak{L}$ from the point 1 onward instead of zero because $\mathfrak{L}$ increases almost surely at 0, but $\mathcal{D}$ almost surely is locally constant at any fixed point.

Note that the Hausdorff dimension result of Theorem 3.1 is an immediate corollary of the just stated theorem, using the fact that the zero set of Brownian motion is almost surely $\frac{1}{2}$.

Theorem 3.2 is a global comparison in the sense that we consider the process on a unit order interval. One can also do a local comparison, in which we focus on the process in a neighborhood of a given point. For appropriate choice of (random) point, one obtains Brownian local time *exactly*, unlike the absolute continuity type of comparison made in Theorem 3.2.

More precisely, by a local comparison, we mean distributional limits of the form $\varepsilon^{-1/2}(\mathcal{D}_{\mathrm{sp}}(\tau + \varepsilon t) - \mathcal{D}_{\mathrm{sp}}(\tau))$ where τ is a random time. First, note that τ indeed needs to be random: at a deterministic time, $\mathcal{D}_{\mathrm{sp}}$ is almost surely locally constant, and so the local limit will be trivial. Further, for the same reason, τ needs to be almost surely a point of increase of $\mathcal{D}_{\mathrm{sp}}$ (i.e., τ is such that $\mathcal{D}_{\mathrm{sp}}(\tau + \varepsilon) > \mathcal{D}_{\mathrm{sp}}(\tau)$ for every $\varepsilon > 0$) for there to be any hope of obtaining $\mathfrak{L}$ in the limit.

Now, there are a number of ways of choosing a point of increase of $\mathcal{D}_{\text{sp}}$. For instance, we may fix $\lambda \in \mathbb{R}$ and consider the first point of increase τ_λ following λ. This would, in some sense, be a choice size-biased by the length of the flat portion preceding τ_λ; also, it excludes any of the "interior" points of increase of $\mathcal{D}_{\text{sp}}$ from being considered. Alternately, we could choose, for an interval $[c, d]$, a point $\xi_{[c,d]}$ *uniformly* from all the non-constant points of $\mathcal{D}_{\text{sp}}$ on $[c, d]$, by sampling from the probability measure on $[c, d]$ with distribution function $(\mathcal{D}_{\text{sp}}(\cdot) - \mathcal{D}_{\text{sp}}(c))/(\mathcal{D}_{\text{sp}}(d) - \mathcal{D}_{\text{sp}}(c))$, conditionally on the event that this is a non-zero measure (which is shown to hold with positive probability in Shirshendu Ganguly and Milind Hegde 2022). (Indeed, distributions at random times sampled according to such local times have been studied, for example, in the context of dynamical critical percolation Hammond et al. 2015.)

The local limit at both these types of random times is Brownian local time:

Theorem 3.3 (Shirshendu Ganguly and Milind Hegde 2022) *Let τ be equal to either τ_λ or $\xi_{[c,d]}$ (the second conditionally on $\mathsf{NC}(\mathcal{D}_{\text{sp}}) \cap [c, d] \neq \varnothing$) as above. Then,*

$$\lim_{\varepsilon \to 0} \varepsilon^{-1/2} \big(\mathcal{D}_{\text{sp}}(\tau + \varepsilon t) - \mathcal{D}_{\text{sp}}(\tau) \big) = \mathfrak{L}(t),$$

where the limit is in distribution in the topology of uniform convergence on compact sets of $[0, \infty)$.

In fact there are a couple more random times for which Theorem 3.3 is proved in Shirshendu Ganguly and Milind Hegde (2022), such as the first time $\mathcal{D}_{\text{sp}}$ hits a given deterministic level.

The arguments from Basu et al. (2021) provided no insight into the probabilistic structure of $\mathcal{D}_{\text{sp}}$ such as that stated in the last two theorems, and hence the proofs of these take a rather different route. We sketch some key features.

First recall the parabolic Airy$_2$ process $\mathcal{P}_1(\cdot) = \mathcal{L}(0, 0; \cdot, 1)$ introduced above. An important line of research within KPZ over the last decade has relied on embedding $\mathcal{P}_1$ in an ensemble of an infinite collection of random, continuous, non-intersecting curves. This ensemble is known as the parabolic Airy line ensemble and is denoted $\mathcal{P}$; $\mathcal{P}_1$ is the highest in terms of value as well as lowest-indexed curve in the ensemble. See Fig. 4.

The ensemble was constructed in Corwin and Hammond (2014). The important feature of $\mathcal{P}$, which is the reason it has proved a crucial tool within KPZ, is that it possesses an explicit resampling property in terms of non-intersecting Brownian bridges. This is known as the *Brownian Gibbs* property and it states the following for any fixed interval $[a, b]$ and $k \in \mathbb{N}$: the conditional distribution of $\mathcal{P}_1, \ldots, \mathcal{P}_k$ on $[a, b]$, conditional on the data of $\mathcal{P}_1, \ldots, \mathcal{P}_k$ outside $[a, b]$ and $\mathcal{P}_{k+1}, \mathcal{P}_{k+2}, \ldots$ on all of $\mathbb{R}$, is given by k independent rate *two* Brownian bridges $B_1, \ldots, B_k : [a, b] \to \mathbb{R}$, with $B_i(a) = \mathcal{P}_i(a)$ and $B_i(b) = \mathcal{P}_i(b)$, conditioned on not intersecting each other or the lower curve $\mathcal{P}_{k+1}$.

The curves below the top one in $\mathcal{P}$ also have an interpretation via the Robinson-Schensted-Knuth (RSK) correspondence from algebraic combinatorics, which we

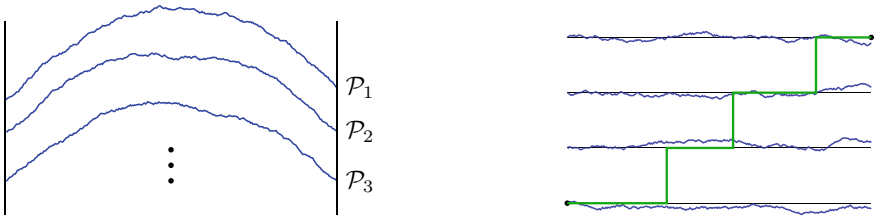


Fig. 4 On the left is a depiction of the parabolic Airy_2 line ensemble $\mathcal{P}$. On the right is an example of an up-right path in an environment given by $\mathcal{P}$ (shown without parabolic curvature for convenience)

will not get into, except mentioning that the latter is the key underlying device using which the earlier quoted estimate on the probability of non-coalescence was proven in Hammond (2020).

We will now state a rather remarkable and powerful statement which played a central role in the construction of the directed landscape $\mathcal{L}$ in Dauvergne et al. (2022). This involves considering last passage percolation across an environment defined by the parabolic Airy line ensemble. Analogous to the definition of geodesics in $\mathcal{L}$, we will consider a class of paths which are directed, assign a weight to each path in terms of the environment, and maximize the weight over all paths between given points. The class of directed paths are up-right paths; see the right panel of Fig. 4. An up-right path is assigned weight given by the summing the increments of $\mathcal{P}_i$ along the path, i.e., is the path lies on line i on the interval $[a_i, b_i]$ for each i, then its weight is $\sum \mathcal{P}_i(b_i) - \mathcal{P}_i(a_i)$.

The above mentioned last passage problem in $\mathcal{P}$ can be described by fixing a starting coordinate on the kth line (for some $k \geq 1$) and an ending coordinate on the first (i.e., top) line, and considering up-right paths between the two. The weight of a given path is given by the sum (over i) of increments of values of $\mathcal{P}_i$ along the interval that the path spends on the ith line. If the starting point is (y, k) and ending point is $(x, 1)$, we denote this by $\mathcal{P}[(y, k) \to (x, 1)]$.

It was shown in Dauvergne et al. (2022) that, roughly speaking, that $\mathcal{L}(0, y; x, 1)$ is equal to a certain last passage value in $\mathcal{P}$ where the starting point can be thought to be at $-\infty$, but in a way that depends on y, and the ending points is x on the top curve $\mathcal{P}_1$. This generalizes the earlier mentioned property that $\mathcal{L}(0, 0; x, 1) = \mathcal{P}_1(x)$ for all $x \in \mathbb{R}$, more precisely that there is a coupling between $\mathcal{L}$ and $\mathcal{P}$ such that that holds.

This basic representation of $\mathcal{L}$ in terms of a last passage problem through $\mathcal{P}$ is the underlying reason for the local time-like behaviour of $\mathcal{D}_{\text{sp}}$. More precisely, let us suppose that our two starting points for $\mathcal{D}_{\text{sp}}$ are 0 and 1 instead of -1 and 1; this is useful because $\mathcal{L}(0, 0; x, 1) = \mathcal{P}_1(x)$. We can handle the general case using stationarity properties of $\mathcal{L}$. Then we can write $\mathcal{D}_{\text{sp}}(x) = \mathcal{L}(1, 0; x, 1) - \mathcal{L}(0, 0; x, 1)$ as an infinite LPP problem in $\mathcal{P}$ which ends at x on the top line, minus $\mathcal{P}_1(x)$. (This representation of $\mathcal{L}$ as an infinite LPP problem in $\mathcal{P}$ is made precise in Sarkar and Virág 2021.)

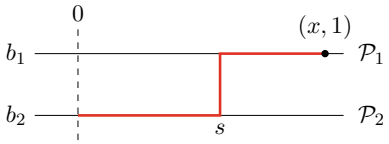


Fig. 5 A depiction of the toy example discussed in the text, with $b_i = \mathcal{P}_i(0)$. The red line jumps at the location s where the maximization $\max_{0\le s\le x}(\mathcal{P}_2(s) - \mathcal{P}_1(s))$ is achieved. In fact, one can think of the red line as a geodesic in the environment given by the parabolic Airy line ensemble $\mathcal{P}$

In this article, for simplicity, we assume a toy situation where we simply have two lines on the interval $[0, x]$ for some $x > 0$ and consider the LPP problem with two sources being $\{1, 2\} \times \{0\}$ pretending that they yield the values $\mathcal{L}(0, 0; x, 1)$ and $\mathcal{L}(1, 0; x, 1)$ respectively (Fig. 5). (The argument in Shirshendu Ganguly and Milind Hegde 2022 proceeds by reducing the original problem essentially to the above problem.)

This means that we can write

$$\mathcal{L}(1, 0; x, 1) = b_2 + \mathcal{P}[(0, 2) \to (x, 1)] = b_2 + \max_{0\le s\le x}\big[\mathcal{P}_1(x) - \mathcal{P}_1(s) + \mathcal{P}_2(s) - \mathcal{P}_2(0)\big].$$

where b_i is the value of $\mathcal{P}_i$ at 0 for $i = 1, 2$. Since $\mathcal{L}(0, 0; x, 1) = \mathcal{P}_1(x)$, we get that

$$\mathcal{D}_{\mathrm{sp}}(x) = b_2 + \max_{0\le s\le x}\big[\mathcal{P}_2(s) - \mathcal{P}_1(s)\big] - \mathcal{P}_2(0).$$

Putting in $x = 0$ shows that $\mathcal{D}(0) = b_2 - \mathcal{P}_1(0)$, so that

$$\mathcal{D}_{\mathrm{sp}}(x) - \mathcal{D}_{\mathrm{sp}}(0) = \max_{0\le s\le x}\big[\big(\mathcal{P}_2(s) - \mathcal{P}_2(0)\big) - \big(\mathcal{P}_1(s) - \mathcal{P}_1(0)\big)\big].$$

We now see the reason for the appearance of Brownian local time: the increments of curves of $\mathcal{P}$ are Brownian-like, in the sense that any finite collection on any compact interval are jointly absolutely continuous to independent rate two Brownian motions started from appropriate points, and the running maximum process of Brownian motion, by the identity of Lévy mentioned above, is exactly (in distribution) Brownian local time. The absolute continuity statement comparing $\mathcal{P}$ to Brownian motion is an easy consequence of the Brownian Gibbs property; see Corwin and Hammond (2014, Corollary 4.3).

3.2 Geodesic Local Time

While Brownian local time encodes the time that Brownian motion spends at zero, given all the discussions so far, it is natural to wonder if a counterpart exists for the geodesics in $\mathcal{L}$ which clearly forms another important class of random curves: for

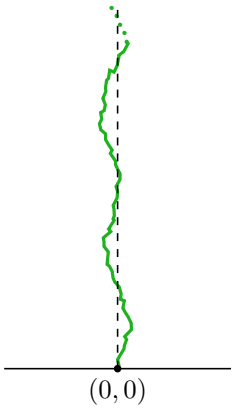


Fig. 6 A depiction of the semi-infinite geodesic in the 0 direction (i.e., directly vertical) starting from (0, 0)

example, does there exist a process, a *geodesic local time*, which measures the amount of time that a geodesic spends at zero? Besides being an independently interesting object of study, it has several applications as we will briefly allude to.

The geodesic local time was constructed recently in Ganguly and Zhang (2022). To discuss it more precisely, we need to introduce *semi-infinite geodesics*. Given a direction $r \in \mathbb{R}$, a semi-infinite geodesic from $(0, 0)$ in the direction r is a path $\pi_r : [0, \infty) \to \mathbb{R}$ such that (i) $t^{-1}\pi_r(t) \to r$ as $t \to \infty$ and (ii) for any $0 \leq s_1 < s_2 < \infty$, the restriction of π_r to $[s_1, s_2]$ is a geodesic from $(s_1, \pi_r(s_1))$ to $(s_2, \pi_r(s_2))$. See Fig. 6.

Semi-infinite geodesics are objects of interest in both first passage and last passage percolation. A related concept is a bi-infinite geodesic, which is a path defined on all of $\mathbb{R}$ instead of $[0, \infty)$ that shares the same property of all restrictions being geodesics. However, it is expected that (non-trivial) bi-infinite geodesics do not exist in first or last passage (non-trivial because in some models certain deterministic bi-infinite paths can be forced to satisfy the geodesic restriction property due to the definition of directed paths in the model), and this has been proven in certain integrable models of LPP Basu et al. (2018), Balázs et al. (2020) as well as in first passage percolation under some (unproven) hypotheses Damron and Hanson (2017).

Thus the existence of semi-infinite geodesics is not immediate. They do exist, however (unlike what is believed about bi-infinite geodesics), and this has been proven in a number of models, including the directed landscape Ganguly and Zhang (2022), Rahman and Virág (2021), Busani et al. (2022).

Let us consider the semi-infinite geodesic π_0 in the 0-direction. Analogous to $\mathfrak{L}$, we can search for a continuous stochastic process which is almost surely non-decreasing and whose points of non-constancy are exactly the times at which π_0 is at zero, i.e., the set of times $\{t > 0 : \pi_0(t) = 0\}$. This is the precise version of the geodesic local time we asked about above.

Theorem 3.4 (Ganguly and Zhang 2022) *Almost surely, for every $t > 0$ simultaneously, the limit*

$$\lim_{\varepsilon \downarrow 0} \frac{1}{2\varepsilon} \int_0^t \mathbb{1}_{\pi_0(s) \in [-\varepsilon, \varepsilon]} \, \mathrm{d}s$$

exists and is strictly positive.

In fact, it is further established in Ganguly and Zhang (2022) that the process is Hölder-$\frac{1}{3}^-$.

The proof of Theorem 3.4 proceeds differently compared to the construction of Brownian local time. This is necessitated by the basic fact that the underlying object of Brownian motion is a Markov process, while the semi-infinite geodesic π_0 is *not*; indeed, its value at any given height a priori depends on the entire infinite random environment.

Given the geodesic local time, one can try to understand the fractal structure of its set of points of non-constancy, analogous to finding the Hausdorff dimension of the zero set of Brownian motion using its local time.

Theorem 3.5 (Ganguly and Zhang 2022) *The set $\{t > 0 : \pi_0(t) = 0\}$ almost surely has Hausdorff dimension $\frac{1}{3}$.*

The value of the dimension can be understood in terms of the continuity properties of π_0. This is similar to how the fact that Brownian motion is Hölder-$\frac{1}{2}^-$ leads to the Hausdorff dimension of its zero set being $1 - \frac{1}{2} = \frac{1}{2}$. Geodesics in LPP have long been known to be Hölder-$\frac{2}{3}-$ Johansson (2000), Hammond and Sarkar (2020), Dauvergne et al. (2022), which suggests that the zero set of the geodesic should be expected to have Hausdorff dimension of $1 - \frac{2}{3} = \frac{1}{3}$ (note that these Hölder continuity results are for geodesics between deterministic points and so do not immediately transfer to semi-infinite geodesics, whose restrictions to finite intervals are only geodesics between random points).

3.3 The Full Spatio-Temporal Weight Difference Profile

The geodesic local time arose in Ganguly and Zhang (2022) in connection with another problem, that of understanding the fractal structure of the full difference profile process, i.e., which takes both spatial *and* temporal arguments.

As mentioned earlier, this two parameter weight difference profile shares the feature that it is almost everywhere constant with its spatial marginal that we have focused on thus far. However, unlike the spatial marginal, the temporal marginal $t \mapsto \mathcal{D}(x, t)$ for fixed $x \in \mathbb{R}$ is *not* monotone and the behavior is expected to be different; see Fig. 7. Recently in this direction, Ganguly and Zhang (2022) established the following fractal dimension statements:

Theorem 3.6 *Fix $x \in \mathbb{R}$. The set of points of non-constancy of $t \mapsto \mathcal{D}(x, t)$ is almost surely equal to $\frac{2}{3}$.*



Fig. 7 A simulation of the full difference profile $\mathcal{D}$ in a prelimiting model (Exponential LPP). Colours represent level sets. The strict colour gradation from left to right is a consequence of the monotonicity of the spatial marginal of $\mathcal{D}$, which, as can be seen by looking closely, is not present in the temporal direction

Note that the dimension is a more KPZ-esque multiple of $\frac{1}{3}$ rather than the Brownian-like $\frac{1}{2}$ for the spatial marginal. One can similarly ask for the fractal dimension of the non-constancy points of the entire spatio-temporal process:

Theorem 3.7 *The set of points of non-constancy Ganguly and Zhang (2022) of* $(x, t) \mapsto \mathcal{D}(x, t)$ *is almost surely equal to* $\frac{5}{3}$.

The arguments for the upper bounds of Theorems 3.6 and 3.7 are similar to the one for the upper bound in Theorem 3.1 via geometric considerations of geodesic disjointness. The lower bound, however, is much more difficult.

The basic problem is that, unlike in the spatial marginal of $\mathcal{D}$, the temporal and spatio-temporal versions are not monotone. Thus it is possible for the increment of $\mathcal{D}$ that occurs in one ε-interval to be of the opposite sign as that in some other ε-interval, leading to some degree of cancellation. In fact, an argument similar to the one outlined for Theorem 3.1 using Hölder continuity properties of $\mathcal{D}$ (this time in the temporal direction) would yield a lower bound of only $\frac{1}{3}$ in Theorem 3.6, precisely because cancellations have been ignored. Heuristically, this also explains why the true dimension is $\frac{2}{3}$: cancellations should follow a central limit theorem type effect, so that one expects the effective size after cancellations to be of order the square root of the sum of the absolute value of the increments; so to obtain the same overall change, if one needed $\varepsilon^{-1/3}$ many intervals to achieve a certain overall unit order increment when considering the sum of absolute values of the increments over ε-sub-intervals, then one should need $(\varepsilon^{-1/3})^2 = \varepsilon^{-2/3}$ many intervals after taking into account the cancellation effect, suggesting the true result of $\frac{2}{3}$.

3.4 Non-uniqueness of Semi-infinite Geodesics

The last topic we discuss in this survey concerns semi-infinite geodesics as introduced in the previous section. As we saw, we can define semi-infinite geodesics which go in the direction r for every $r \in \mathbb{R}$. We can also have the semi-infinite geodesics start from any point $p \in \mathbb{R}^2$ instead of just $(0, 0)$ in a completely analogous way.

It can be shown that semi-infinite geodesics exist in every direction and from every starting point simultaneously with probability one Busani et al. (2022) (it was previously shown in Rahman and Virág 2021 that they almost surely exist in all directions for a fixed point, or for all points in a fixed direction).

Once we have existence, the question of uniqueness arises. For any fixed direction, there is a unique semi-infinite geodesic in that direction Rahman and Virág (2021); more precisely, semi-infinite geodesics which emanate from any starting point in that fixed direction will all ultimately coalesce into a single one.

The fact that this statement was made for each fixed direction immediately raises the possibility that there exist random exceptional directions in which there exist multiple semi-infinite geodesics which do not ultimately coalesce. In fact, Busani et al. (2022) showed that there is a random countable dense set Ξ of directions such that this happens. For each direction $\xi \in \Xi$ and $p \in \mathbb{R}^2$, there exist at least two semi-infinite geodesics starting at p in the direction ξ which separate at some point and never meet again; note that they need not be disjoint because of this possibly shared initial portion.

Because Ξ is countable, it does not possess any fractal structure. However, fractal structure is present if one fixes a random direction in Ξ and then considers the set of starting points $p = (x, s)$ such that there are at least two *disjoint* semi-infinite geodesics starting at p in that direction, which we call $\mathfrak{S}_\xi \subseteq \mathbb{R}^2$. Busani et al. (2022) studies the slice $\mathfrak{S}_{\xi,s}$ of this set where s is fixed:

Theorem 3.8 *For any fixed $s \in \mathbb{R}$, for every $\xi \in \Xi$, $\mathfrak{S}_{\xi,s}$ almost surely has Hausdorff dimension $\frac{1}{2}$.*

This theorem has close connections to Theorem 3.1; indeed, Busani et al. (2022) approaches this set via Busemann functions, which are closely related to the weight difference profile $\mathcal{D}$.

4 Conclusion

The study of fractal structure within the directed landscape has opened up many avenues of research, and introduced new probabilistic objects such as the geodesic local time which are not yet deeply understood. One broad direction to be explored is to continue a theme touched upon in this survey, namely to establish analogues of

well-known properties of Brownian local time and its zero set for the geodesic local time and the set of heights when the semi-infinite geodesic is at zero. We similarly expect there to be a rich class of questions concerning other aspects of geodesic geometry that remain to be asked.

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Small Ball Probabilities for the Stochastic Heat Equation



Mathew Joseph

Dedicated to Prof. Rajeeva Karandikar on the occasion of his 65th birthday.

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In a general form, the small ball probability problem can be formulated as follows: Consider a real separable Banach space X and its associated Borel sigma algebra $\mathscr{X}$, and let μ be a probability measure on $(X, \mathscr{X})$. Given a Borel subset A of X which is convex and bounded, what is the asymptotics of $\mu(\epsilon A)$ as ϵ goes to 0? An example of such a problem is the probability that a standard Brownian motion stays in a small ball around 0 up to a fixed terminal time $T > 0$. More precisely given a standard Brownian motion $\{B_t\}_{0\le t\le T}$ on $\mathbf{R}^d$ up to a fixed time $T > 0$ the problems asks for the asymptotics of

$$P\left(\sup_{0\le t\le T} |B_t| \le \epsilon\right) \quad \text{as } \epsilon \downarrow 0. \tag{1}$$

This is a particular case of the general form stated above: X is the space $C([0, T], \mathbf{R}^d)$ equipped with the sup norm $\|\cdot\|_\infty$ and A is the set $\{f : \|f\|_\infty \le 1\}$. This problem was solved by Mogulskii (2024) who showed that

$$\log P\left(\sup_{0\le t\le T} |B_t| \le \epsilon\right) \sim \frac{-\pi^2}{8} \cdot \frac{T}{\epsilon^2} \quad \text{as } \epsilon \downarrow 0. \tag{2}$$

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By scaling arguments the above is equivalent to the probability that Brownian motion stays in the ball $\mathcal{B}(0, 1)$ for an unusally long time T/ϵ^2. Mogulskii used large deviations on the occupation measure of Bronwian motion to obtain (2).

Typically finding exact constants as in (2) is quite difficult, and indeed exact asymptotics have been found only in some special cases. In some cases the existence of the constant is known but the exact value of the constant has not been determined. For example in the case of fractional Brownian motions $X \sim \text{FBM}_\alpha,\ \alpha \in (0, 2)$

$$\epsilon^{2/\alpha} \log P\left(\sup_{0\le t\le T} |X_t| \le \epsilon\right) \xrightarrow{\epsilon\to 0} C_\alpha \tag{3}$$

for some *unknown* C_α (Li & Linde, 1998; Shao, 2003).

In case the existence of the small ball constant is not known, the next best thing we can hope for is lower and upper bounds. Let us give an example. The Hölder semi-norms

$$\|f\|_\beta := \sup_{0\le s\ne t\le 1} \frac{|f(t) - f(s)|}{|t - s|^\beta}$$

for $f : [0, 1] \to \mathbf{R}$ are a natural measure of the regularity of the function f, and one could ask for the probability that $\|X\|_\beta$ is exceptionally small. Since $X \sim \text{FBM}_\alpha$ is Hölder($\frac{\alpha}{2}-$), the question makes sense when $\beta < \frac{\alpha}{2}$. It was proved in Kuelbs (1995) that for $\beta < \alpha/2$,

$$-K_1\epsilon^{-\frac{2}{(\alpha-2\beta)}} \le \log P\left(\sup_{0\le s\ne t\le 1} \frac{|X_t - X_s|}{|t - s|^\beta} \le \epsilon\right) \le -K_2\epsilon^{-\frac{2}{(\alpha-2\beta)}},$$

for some positive constants K_1, K_2. It is an open question whether one can show the existence of a limiting constant as in (3).

We could consider small ball probabilities for processes in other norms (for e.g. L^p norm), and several such results have been proven. One could also ask about the small ball probabilities for processes indexed by sets in $\mathbf{R}^d$, for example (fractional) Brownian sheet. We refer the interested reader to the extensive survey Li and Shao (2001) for many of the known results. Let me point out that most of the results on small ball problems are for Gaussian processes, or for simple functionals of Gaussian processes.

Why would one be interested in small ball probabilities? It turns out that these are useful in obtaining law of iterated logarithm type results. For example, a general result states that for a centered Gaussian process $\{X(s), s \in [0, 1]\}$ starting at 0 and having stationary increments, a condition of the form

$$\exp(-c_2(h/x)^d) \le P\left(\sup_{s\in[0,h]} |X(s)| \le \sigma(x)\right) \le \exp(-c_1(h/x)^d)$$

(here $\sigma^2(x) = \mathbb{E}X(x)^2$) guarantees

$$1 \leq \liminf_{h\to 0} \frac{\sup_{s\in[0,h]} |X(s)|}{\sigma\left(h(c_1/\log\log(1/h))^{1/d}\right)} \text{ a.s.}, \quad \liminf_{h\to 0} \frac{\sup_{s\in[0,h]} |X(s)|}{\sigma\left(h(c_2/\log(1/h))^{1/d}\right)} \leq 1 \text{ a.s.}$$

Refer to Theorem 7.1 of Li and Shao (2001) for the exact statement. They are also useful for finding asymptotics of

$$P\left(\sup_{t\leq T} |X_t - \phi_t| \leq \epsilon\right) \quad \text{as } \epsilon \downarrow 0,$$

the probability that the process is close to a fixed function ϕ. Indeed, this is equivalent to finding the small ball probabilities of the shifted process $X_t - \phi_t$. For details and further applications see Li and Shao (2001).

As mentioned earlier, most of the results on the problem (1) have been for Gaussian processes. We now describe a recent result for a class of non-Gaussian processes: *stochastic partial differential equations* (SPDEs). Numerous physical phenomena are described by PDEs, and the addition of noise is natural as it takes into account the various randomness that may be present in the phenomena. While stochastic ODEs have been studied extensively for much of the twentieth century, stochastic PDEs (SPDEs) gained attention only in the end of the twentieth century. Much of the early motivation for SPDEs was from filtering theory and Dawson-Watanabe superprocesses (scaling limit of critical branching Brownian motions). There are different approaches to studying SPDEs; one approach is to view it as SDEs in functional spaces Da Prato and Zabczyk (2014) and another is the L^p theory approach of Krylov (2006). The approach that we follow is the *Walsh* approach (Walsh, 1986; Dalang et al., 2009) where we view the SPDE as a random field indexed by time and space.

We shall consider a subclass of such equations: the *stochastic heat equations* (SHEs). These equations have been of interest to physicists and mathematicians for several years now. Apart from being natural models to describe the flow of heat through material with impurities, they also turn out to describe the large scale behaviour of many interacting particle systems (Kardar et al., 1986). Recently Martin Hairer was awarded the Fields Medal for increasing our understanding of similar sorts of equations (Hairer, 2014). An introduction can be found in Dalang et al. (2009), Walsh (1986). Let $(u(t,x))_{t\geq 0,\, x\in[0,1]}$ be the solution of the equation

$$\begin{aligned} \partial_t u(t,x) &= \frac{1}{2}\partial_x^2 u(t,x) + \sigma\Big(t,x,u(t,x)\Big)\dot{W}(t,x), \qquad t \geq 0,\ x \in [0,1] \\ u(0,\cdot) &= u_0(\cdot), \end{aligned} \tag{SHE}$$

with periodic boundary conditions, where $u_0(\cdot)$ is the initial profile and $\dot{W}(t,x)$ is white noise defined as follows:

Definition 1 White noise $\dot{W}(t,x)$, $t \geq 0$, $x \in [0,1]$ is a distribution valued centered Gaussian process, such that for compactly supported continuous functions ϕ one has that $\int_{\mathbf{R}_+\times[0,1]} \phi(s,x)\dot{W}(s,x)dxds$ is a centered Gaussian random variable

with covariances

$$\text{Cov}\left(\int_{\mathbf{R}_+\times[0,1]} \phi(s,x)\dot{W}(s,x)dxds,\ \int_{\mathbf{R}_+\times[0,1]} \psi(s,x)\dot{W}(s,x)dxds\right)$$
$$= \int_{\mathbf{R}_+\times[0,1]} \phi(s,x)\psi(s,x)dxds$$

One can think of white noise as noise which is independent over different regions, but which is highly irregular (so irregular that it can only be defined as a distribution). Formally it is the derivative of Brownian sheet. Brownian sheet $W(t,x),\ t \geq 0, x \geq 0$ is a centered Gaussian random field with covariance given by $E[W(t,x)W(s,y)] = (s \wedge t) \cdot (x \wedge y)$. For a rectangle R with diagonal vertices $(s,y),\ (t,x),\ s \leq t,\ y \leq x$, we define $W(R) := W(t,x) - W(s,x) - W(t,y) + W(s,y)$. It can be checked that $W(R)$ is independent to $W(\tilde{R})$ if $R, \tilde{R}$ are disjoint rectangles, and $W(R) \overset{D}{=} W(\tilde{R})$ if R and $\tilde{R}$ have the same area.

By a solution to (SHE) we mean:

Definition 2 We say a random field $u = ((u(t,x)))_{t\geq 0,\, x\in[0,1]}$ is a solution to (SHE) if $u(t,x)$ is adapted to the filtration $\mathcal{F}_t = \sigma\,(W([a,b]\times[0,s]),\ s \leq t,\ [a,b] \subset [0,1])$ generated by white noise, and satisfies

$$u(t,x) = \Big(G_t * u_0\Big)(x) + N(t,x) \qquad \text{a.e. for each } t \geq 0,\ x \in [0,1], \tag{4}$$

where

$$G_t(x) = \sum_{k\in\mathbf{Z}} p_t(x+k),$$

and $p_t(x) = (2\pi t)^{-1/2}\exp(-x^2/2t)$ is the heat kernel on $\mathbf{R}$.

Note that $G_t(x)$ is the heat kernel on $[0,1]$ with periodic boundary conditions. Thus the first term in (4) is the solution of the heat equation (that is (SHE) with $\sigma \equiv 0$), and is a drift term. The second term in (4) is the random *noise term*

$$N(t,x) := \int_0^t \int_0^1 G_{t-s}(x-y)\sigma\,(s,y,u(s,y))\,\dot{W}(y,s)dyds.$$

We do not go into detail into how this stochastic integral is defined as it involves a lot of technicalities which are best learned from Walsh (1986). We will just say that there is an entire theory similar to that of stochastic integrals with respect to Brownian motion, and the basic idea is similar in that we first define stochastic integrals of *simple* processes and then extend it to *predictable* processes by a limiting procedure (see Walsh, 1986; Dalang et al., 2009). Note here though that by the Duhamel principle (4) is what a solution to (SHE) would satisfy if $\sigma(t,x,u(t,x))\dot{W}(t,x)$ were deterministic.

Throughout this paper we make the following assumptions on σ:

- There is a $\mathcal{D} > 0$ such that $|\sigma(t,x,u) - \sigma(t,x,v)| \leq \mathcal{D}|u-v|$ for all t, x, u, v.
- There are $\mathcal{C}_1, \mathcal{C}_2 > 0$ such that $\mathcal{C}_1 \leq |\sigma(t,x,u)| \leq \mathcal{C}_2$ for all t, x, u.

The first assumption (Lipschitz assumption on σ) is natural and gaurantees the existence of a solution to (SHE) (see Dalang et al., 2009; Walsh, 1986). The second assumption, although it appears quite restrictive, is actually quite general for the following result. We have

Theorem 1 (Athreya et al., 2021) *Fix $0 < \eta < 1$ and let $\|u_0\|_\infty \leq \epsilon/2$. Then there exists constants $C_1, \cdots, C_4$ dependent on $\mathcal{C}_1, \mathcal{C}_2$ and an ϵ_0 dependent additionally on $\mathcal{D}, \eta$ such that for all $\epsilon < \epsilon_0$*

$$C_1 \exp\left(-\frac{C_2 T}{\epsilon^{6+\eta}}\right) \leq P\left(\sup_{\substack{0\leq t\leq T\\ x\in[0,1]}} |u(t,x)| \leq \epsilon\right) \leq C_3 \exp\left(-\frac{C_4 T}{(1+\mathcal{D}^2)\epsilon^6}\right). \quad (5)$$

Remark 1 When $\mathcal{D} = 0$ (that is when σ does not depend on u) one can in fact take $\eta = 0$ in the left hand side of (5).

Remark 2 One can obtain similar bounds for

$$P\left(\sup_{\substack{0\leq t\leq T\\ x\in[0,1]}} |u(t,x) - h(t,x)| \leq \epsilon\right)$$

when h is a C^2 function. The constants $C_1, \cdots, C_4$ depend additionally on $H = \sup_{x\in[0,1], t\in[0,T]} \max[h(t,x), \partial_t h(t,x), \partial_x^2 h(t,x)]$.

1 Discussion of the Proof When $\sigma(t,x,u) = \sigma(t,x)$

In the case when σ is independent of the third variable, the solution u to (SHE) turns out to be Gaussian. We now provide a short discussion in this case, so that the reader can have a basic idea of how small ball problems can be solved in Gaussian cases.

Observe that when $t \approx 0$

$$\begin{aligned} E\left[N^2(t,x)\right] &\asymp \int_0^t \int_0^1 G_{t-s}^2(x-y)dyds \\ &\approx \int_0^t \int_{\mathbf{R}} p_{t-s}^2(x-y)dyds \\ &= C\sqrt{t} \end{aligned} \quad (6)$$

So one expects that $N(t,x)$ has fluctuations of order $t^{1/4}$ for any x. Now

$$P\left(\sup_{\substack{t\le\epsilon^4\\ x\in[0,1]}}|u(t,x)|\le\epsilon\right)\le P\left(\max_k|u(\epsilon^4,kc_0\epsilon^2)|\le\epsilon\right)$$

for any fixed $c_1>0$. Anderson's inequality Anderson (1955) states that for a centered Gaussian measure μ on $(\mathbf{R}^d,\mathcal{B}(\mathbf{R}^d))$ one has $\mu(A+x)\le\mu(A)$ for any convex symmetric A. Using this and the fact that N is a centered Gaussian process we obtain

$$\begin{aligned}P\left(\sup_{\substack{t\le\epsilon^4\\ x\in[0,1]}}|u(t,x)|\le\epsilon\right)&\le P\left(\max_k|N(\epsilon^4,kc_0\epsilon^2)|\le\epsilon\right)\\&\approx P\left(|N(\epsilon^4,kc_1\epsilon^2)|\le\epsilon\right)^{\frac{1}{c_0\epsilon^2}}.\end{aligned}$$

The last approximation requires an explanation. If the constant c_0 is chosen large enough, one can show that $N(\epsilon^4,kc_0\epsilon^2)$ and $N(\epsilon^4,lc_0\epsilon^2)$ are *approximately independent* if $k\neq l$. Now thanks to the argument (6) the term $P\left(|N(\epsilon^4,kc_0\epsilon^2)|\le\epsilon\right)$ is bounded away from 0 and 1. Therefore

$$P\left(\sup_{\substack{t\le\epsilon^4\\ x\in[0,1]}}|u(t,x)|\le\epsilon\right)\le\exp(-C/\epsilon^2)$$

for some C. Now Markov property of the process $u(t,\cdot)$ and the fact that we have ϵ^{-4} time intervals gives us an upper bound of $\exp(-C\epsilon^{-6})$.

Next we discuss the lower bound. Typically the lower bound is the harder part. Again we consider the time increment $[0,\epsilon^4]$. Following our rough computation (6) one can show using arguments used in the proof of Kolmogorov continuity theorem (p. 14 of Dalang et al., 2009) that for small $c_1>0$

$$P\left(\sup_{\substack{t\le\epsilon^4\\ x\in[0,c_1\epsilon^2]}}|N(t,x)|\le\lambda\epsilon\right)\ge1-e^{-C\lambda^2}$$

uniformly in ϵ, for some $C>0$. A fundamental theorem, only recently proved after being a conjecture for several decades is

Theorem 2 (Gaussian correlation inequality Royen, 2014) *Let μ be a centered Gaussian measure. Then for K, L symmetric convex sets one has*

$$\mu(K\cap L)\ge\mu(K)\mu(L).$$

The above theorem is a *game-changer* in the area of small ball probabilities. Till now, most of the proofs of lower bounds in this area involved finding subcases of the Gaussian correlation inequality suitable for the problem at hand. Now many of the previous results in this area can be significantly shortened. For the many implications of this fundamental inequality just do a Google or Wikipedia search.

Back to the lower bound. Thanks to the Gaussian correlation inequality one has

$$P\left(\sup_{\substack{t\le\epsilon^4\\ x\in[0,1]}}|N(t,x)|\le\frac{\epsilon}{2}\right)\ge\prod_k P\left(\sup_{\substack{t\le\epsilon^4\\ x\in[kc_1\epsilon^2,(k+1)c_1\epsilon^2]}}|N(t,x)|\le\frac{\epsilon}{2}\right)\ge\left(1-e^{-C\lambda^2}\right)^{\frac{1}{c_1\epsilon^2}}$$

In fact, what we actually need is a similar lower bound for $P\left(\sup_{\substack{t\le\epsilon^4\\ x\in[0,1]}}|u(t,x)|\le\frac{\epsilon}{2}\right)$. This can be proved by a Girsanov change of measure argument. This is a standard procedure for converting the problem to one without the drift (first term in (4)). For details we refer to the proof of Proposition 2.1(b) in Athreya et al. (2021). Finally to get a lower bound on $P\left(\sup_{\substack{t\le\epsilon^4\\ x\in[0,1]}}|u(t,x)|\le\frac{\epsilon}{2}\right)$ we divide the time interval $[0,T]$ into time increments of length ϵ^4 and use the Markov property of the process $u(t,\cdot)$.

Remark 3 The argument that we sketched above depends crucially on the fact that the field u is Gaussian in the case that $\sigma(t,x,u)=\sigma(t,x)$ is independent of the third variable. When σ also depends on u we use a perturbation argument to show that on the time interval $[0,\epsilon^4]$ we have that u is close to $u^{(g)}$, the solution to

$$\partial_t u^{(g)}=\frac{1}{2}\partial_x^2 u^{(g)}+\sigma\left(t,x,u_0(x)\right)\dot{W}(t,x).$$

After that we apply the Markov property of the process $u(t,\cdot)$ as above.

It is known that the solutions to (SHE) are Hölder $(\frac{1}{2}-)$ in space and Hölder $(\frac{1}{4}-)$ in time. Using techniques described in this paper one can also obtain bounds on the small ball probabilities of the Holder norms. More precisely

Theorem 3 (Foondun et al., 2022) *Let* $0<\theta<\frac{1}{2}$ *and* $0<\epsilon<1$. *Then for every* $\eta>0$ *there exists constants* $C_1,\cdots,C_4$ *dependent on* $\mathcal{C}_1,\mathcal{C}_2,\mathcal{D},\eta,\theta$ *such that*

$$C_1\exp\left(-\frac{C_2T}{\epsilon^{\frac{3}{\theta}+\eta}}\right)\le P\left(\sup_{\substack{t\le T\\ x\ne y\in[0,1]}}\frac{|u(t,x)-u(t,y)|}{|x-y|^{\frac{1}{2}-\theta}}\le\epsilon\right)\le C_3\exp\left(-\frac{C_4T}{\epsilon^{\frac{3}{\theta}}|\log\epsilon|^{\frac{3}{2}}}\right)$$

$$C_1\exp\left(-\frac{C_2T}{\epsilon^{\frac{3}{\theta}+\eta}}\right)\le P\left(\sup_{\substack{s\ne t\in[0,T]\\ x\in[0,1]}}\frac{|u(t,x)-u(s,x)|}{|t-s|^{\frac{1}{4}-\frac{\theta}{2}}}\le\epsilon\right)\le C_3\exp\left(-\frac{C_4T}{\epsilon^{\frac{3}{\theta}}|\log\epsilon|^{\frac{3}{2}}}\right)$$

The above gives bounds on the probability that (SHE) is unusually regular.

2 Final Comments

Here are some questions:

- Suppose we consider (SHE) on $\mathbf{R}$ rather than $[0, 1]$ with initial constant profile $u_0 \equiv 1$. The probability that $u_0(x)$ remains within $[1 - \epsilon, 1 + \epsilon]$ *at all* x at any future time $t > 0$ is 0. One can ask for bounds on the probability that $u_t(x) \in [1 - \epsilon, 1 + \epsilon]$, $x \in [0, 1]$, $t \in [0, T]$. One then has to control the effect of the solution from far away on the region $[0, 1]$. More interestingly what can said when $\sigma(t, x, u) = Cu$ (the parabolic Anderson model Carmona and Molchanov, 1994, this equation is connected to the KPZ equation Kardar et al., 1986; Hairer, 2013)?
- What if the initial profile is a measure, say $u_0 = \delta_0$? Can we say something about the probability that u stays close to the heat kernel $G_t(x)$ up to a fixed time $T > 0$?

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Directed Spanning Forest for Ginibre Ensemble Is a Tree



Kumarjit Saha

1 Introduction and Main Result

Directed random networks, where edges have a preferred direction of propagation, have long been an important class of models for understanding the large-scale behaviour of systems in a wide array of applications. These include, but are not limited to, transport networks, power grids, various kinds of social networks, different types of communication networks including wireless sensor networks, multicast communication networks, peer-to- peer networks and drainage networks, to name a few. For a partial overview of the literature, we refer the reader to Baccelli and Błaszczyszyn (2009), Baccelli and Błaszczyszyn (2010), Penrose (2003) and the references therein.

It has been empirically observed that river networks are structurally self-similar and satisfy various scaling laws. Studying drainage networks is a statistical approach to understand the unifying behaviour of river basins. Scheidegger (1967) was the first to introduce a directed network by imposing a preferential flow condition where each source is emptying to one of its two neighbours in a preferred direction. Howard (1971) removed the restriction of drainage to neighbouring sights only. Rodriguez-Iturbe and Rinaldo (1997) presented an excellent survey of development in this field.

In this note, we will consider drainage network models defined on the two-dimensional Euclidean plane only. Informally, such a model can be described in the following way. A random subset of $\mathbb{Z}^2$ or $\mathbb{R}^2$ is selected as a collection of 'source' vertices such that for any $l \in \mathbb{Z}$, the upper half-plane $\mathbb{H}^+(l) := \{\mathbf{x} \in \mathbb{R}^2 : \mathbf{x}(2) > l\}$ contains at least one source vertex almost surely. From each source, exactly one directed edge is drawn to another source vertex with a strictly higher y-coordinate, representing the flow of water from one source to another.

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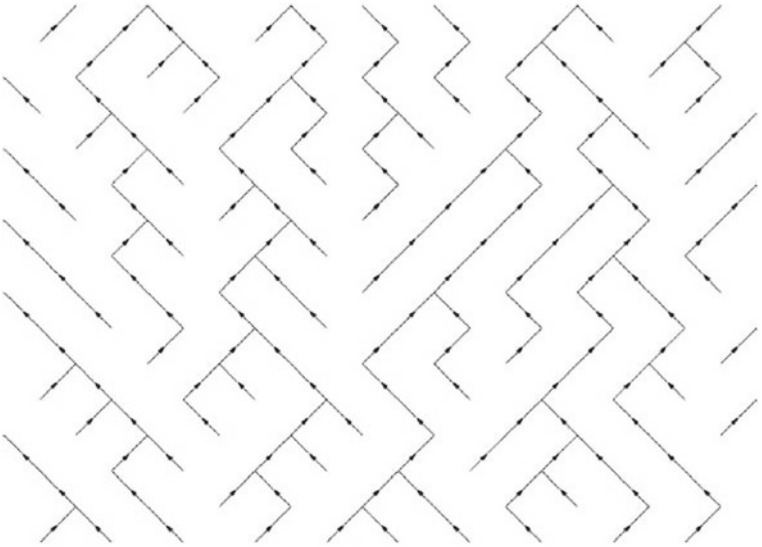


Fig. 1 Schidegger drainage network model consists of coalescing simple symmetric random walks in discrete time where the horizontal and the vertical coordinates play the role of space and time, respectively

The simplest drainage network model is the Scheidegger model introduced in Scheidegger (1967) where on the oriented lattice

$$\mathbb{Z}^2_{\text{even}} := (m, n) \in \mathbb{Z}^2 : m + n \text{ even } \},$$

each vertex acts as a source vertex and a source vertex (m, n) connects to one of its two neighbours at the next level, viz., $(m - 1, n + 1)$ or $(m + 1, n + 1)$ with equal probability independent of everything else. Mathematically, this model can be described as below.

Consider an i.i.d. collection of Rademacher r.v.'s $\{I_{(m,n)} : (m, n) \in \mathbb{Z}^2_{\text{even}}\}$. For $(m, n) \in \mathbb{Z}^2_{\text{even}}$ the next step is denoted by $h(m, n)$ which is defined as

$$h(m, n) := (m + I_{(m,n)}, n + 1).$$

The Scheidegger model is represented by the random graph G with vertex set $\mathbb{Z}^2_{\text{even}}$ and directed edge set $E := \{\langle (m, n), h(m, n)\rangle : (m, n) \in \mathbb{Z}^2_{\text{even}}\}$ (see Fig. 1).

Set $h^0(m, n) = (m, n)$ and for $k \geq 1$, the k'th successive step starting from (m, n) is denoted as $h^k(m, n) := h(h^{k1}(m, n))$. The flow starting from (m, n) is represented as a sequence of successive steps given by $\{h^k(m, n) : k \geq 0\}$. In Remark 1.1, we highlight some properties of this process of successive steps, which explains the simplicity of this network model. In what follows, for $\mathbf{x} \in \mathbb{R}^2$, the notation $\mathbf{x}(i)$ denotes the i-th coordinate of $\mathbf{x}$ for $i = 1, 2$.

Remark 1.1 We note that the $\mathbb{Z}$ valued process $\{h^k(m,n)(1) : k \geq 0\}$ gives a simple symmetric random walk starting at $m \in \mathbb{Z}$. Further, it follows that for $(m,n), (m',n) \in \mathbb{Z}^2_{\text{even}}$ with $m \neq m'$, the two random walks, $\{h^k(m,n)(1) : k \geq 0\}$ and $\{h^k(m',n)(1) : k \geq 0\}$, are independent till the time they meet and coalesce. Hence, the coalescing time of these two walks is finite almost surely. This explains the simplicity of this model and implies that the Scheidegger network is connected a.s.

Since then various other drainage network models (with complex dependencies in constructing edges) have been proposed and connectivity property of these networks have been established (see Gangopadhyay et al. (2004), Ferrari et al. (2004), Athreya et al. (2008), Coupier and Tran (2013),Roy et al. (2016), Coupier et al. (2020)). We should mention here that all these models have natural extensions for higher dimensions, and for most of these models, tree-forest dichotomy problems depending on dimensions have been studied. Though in this note, we will restrict to network models defined on the two-dimensional Euclidean plane only.

Regarding dependency involved in constructing network edges, we mention here about the directed spanning forest (DSF) network for a stationary ergodic locally finite point process $\mathcal{P}$. We need to introduce some notations first. For $l \in \mathbb{R}$ let $\mathbb{H}^+(l) := \{\mathbf{x} \in \mathbb{R}^2 : \mathbf{x}(2) \geq l\}$ denote the upper half-plane. Similarly, the lower half-plane is defined as $\mathbb{H}^-(l) := \{\mathbf{x} \in \mathbb{R}^2 : \mathbf{x}(2) \leq l\}$. For every $l \in R$, we assume that the set $\mathcal{P} \cap \mathbb{H}^+(l)$ is non-empty a.s. For $\mathbf{x} \in \mathbb{R}^2$, the DSF step $h(\mathbf{x}, \mathcal{P}) \in \mathcal{P}$ along the direction $e_y := (0,1)$ is the a.s. unique $\mathcal{P}$ point defined as

$$h(\mathbf{x}, \mathcal{P}) := \text{argmin}\{||\mathbf{y} - \mathbf{x}||_2 : \mathbf{y} \in \mathcal{P}, \mathbf{y}(2) > \mathbf{x}(2)\}.$$

In other words, $h(\mathbf{x}, \mathcal{P})$ is the a.s. unique nearest (w.r.t. Euclidean norm) $\mathcal{P}$ point to $\mathbf{x}$ with strictly higher y-coordinate. The Euclidean DSF (constructed based on the Euclidean norm) along the direction e_y for $\mathcal{P}$ is a directed network with vertex set $\mathcal{P}$ and edge set $\{\langle \mathbf{x}, h(\mathbf{x}, \mathcal{P})\rangle : \mathbf{x} \in \mathcal{P}\}$. In what follows, when the underlying point process $\mathcal{P}$ is clear from the context, we drop it from the notation and use $h(\mathbf{x})$ instead of $h(\mathbf{x}, \mathcal{P})$. The intricate dependencies caused by the construction of edges based on Euclidean distances make this model harder to study. It is important to observe that the process $\{h^k(\mathbf{x}) : k \geq 0\}$ is not even Markov. To explain this, we need to introduce some notations. For $\mathbf{x} \in \mathbb{R}^2$, $r > 0$ and $p \geq 1$, let $B_p(\mathbf{x}, r) := \{\mathbf{y} : ||\mathbf{y} - \mathbf{x}||_p < r\}$ denote the (open) ℓ_p ball of radius r centred at $\mathbf{x}$. The 'upper' half of this ball is given by $B_p^+(\mathbf{x}, r) := B_p(\mathbf{x}, r) \cap \mathbb{H}^+(\mathbf{x}(2))$. The definition of a DSF step ensures that given the DSF step $\langle \mathbf{x}, h(\mathbf{x})\rangle$ we must have

$$B_p^+(\mathbf{x}, ||\mathbf{x} - h(\mathbf{x})||_2) \cap \mathcal{P} = \emptyset.$$

Therefore, information about the past steps affects the distribution of the future DSF steps (see Fig. 2).

The study of the directed spanning forest (DSF) on the Poisson point process (PPP) was initiated in Baccelli and Bordenave (2007). Baccelli and Bordenave (2007)

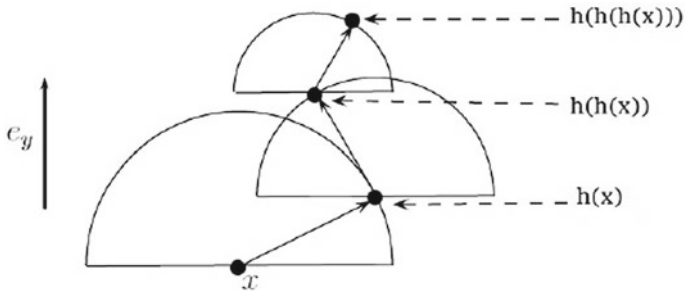


Fig. 2 This figure shows the first few (Euclidean) DSF steps starting from **x**

conjectured that the Poisson DSF is a.s. connected. This remained open for some time, and finally, Coupier and Tran (2013) proved that the (two-dimensional) Poisson DSF is connected a.s. In this context, it is useful to mention here that a discrete version of the directed spanning forest constructed on a random subset of $\mathbb{Z}^d$ for $d \geq 2$ was studied in Roy et al. (2016) and it was proved that the (two-dimensional) discrete DSF is connected a.s. The technique of Roy et al. (2016) is based on a renewal construction. We will not discuss this technique in this note as it is unlikely that a similar technique would be applicable for a strongly correlated point process like the Ginibre ensemble.

The authors of Coupier and Tran (2013) actually used a Burton-Keane type argument to establish connectivity of the Poisson DSF. In this note, we will discuss this argument in detail and modify it to prove the connectivity of the DSF constructed on the Ginibre ensemble point process. This is the main result of this note which has been stated in Theorem 1.2. To the best of author's knowledge, the connectivity property of the Ginibre DSF has not been explored in the literature.

Theorem 1.1 *The Ginibre DSF (constructed w.r.t. Euclidean norm) is connected a.s.*

It is important to note that unlike PPP for the Ginibre ensemble, we don't have the property of spatial independence, i.e., independence of distribution of points over disjoint regions. In fact, this is a strongly correlated point process and it satisfies the notion of 'rigidity' in the sense of Ghosh and Peres (2017) (see Theorem 2.1). Adaptation of Burton-Keane type argument for establishing connectivity of the Ginibre DSF is motivated from Ghosh et al. (2016) where Ghosh et. al. studied the percolation properties of this point process.

We would like to mention here that analysis of DSFs constructed on strongly correlated point processes is of immense theoretical and practical importance. In general, for a model of drainage network analysis, an important hypothesis is the random distribution of its nodes. Generally speaking, the distribution of nodes is often taken to be independent over disjoint regions, like the PPP and its variants (see, e.g., Meester and Roy (1996), Penrose (2003)). Such distribution of nodes is highly amenable to rigorous mathematical treatment, but is often limited to effectively

represent the ground realities. However, little is understood about the stochastic geometry of networks arising from strongly correlated point processes, principally because the tools and techniques overwhelmingly rely on the spatial independence property of the underlying point process.

In the next section, we describe the Ginibre ensemble briefly and its 'rigidity' and 'tolerance' properties. In Sect. 2, we discuss the Burton-Keane type argument given by Coupier and Tran for the Poisson DSF Coupier and Tran (2013). In Sect. 4, we describe adaptation of this argument for the Ginibre DSF.

2 The Ginibre Ensemble and Its Rigidity and Tolerance Properties

The Ginibre ensemble (in the finite dimensional setting) was introduced in the physics literature by Ginibre Ginibre (1965) as a model based on non-Hermitian random matrices. The infinite Ginibre ensemble is obtained as a weak limit of these finite dimensional point processes. Like the Poisson process, it is translation-invariant and ergodic under rigid motions of the plane. In fact, it is a determinantal point process with the determinantal kernel

$$K(z, w) = \sum_{j=0}^{\infty} \frac{z\overline{w}}{j!} = \exp^{z\overline{w}}$$

and the background measure is $\int \exp^{-|z|^2/2} d\ell(z)$. Here ℓ denotes the Lebesgue measure on the complex plane $\mathbb{C}$. Krishnapur (Krishnapur (2006), Theorem 3.0.5) has shown that the Ginibre ensemble is the unique determinantal point process on $\mathbb{C}$ that is isometry invariant, has a sesqui-holomorphic kernel (i.e., the determinantal kernel is holomorphic in the first variable and anti-holomorphic in the second), and is normalized to have unit intensity. In what follows, we denote the Ginibre ensemble point process as $\mathcal{G}$.

Let $\mathcal{D}$ be a bounded open set in $\mathbb{C}$. Let $\mathcal{S}$ denote the Polish space of locally finite point configurations on $\mathbb{C}$. The decomposition $\mathbb{C} = \mathcal{D} \cup \mathcal{D}^c$ induces a factorization $\mathcal{S} = \mathcal{S}_{\text{in}} \times \mathcal{S}_{\text{out}}$, where $\mathcal{S}_{\text{in}}$ and $\mathcal{S}_{\text{out}}$ are, respectively, the spaces of finite point configurations on $\mathcal{D}$ and locally finite point configurations on $\mathcal{D}^c$. This immediately leads to the natural decomposition $\Gamma = (\Gamma_{in}, \Gamma_{out}) for any \Gamma \in \mathcal{S}$, and consequently a decomposition of the point process Π as $\Pi = (\Pi_{\text{in}}, \Pi_{\text{out}})$. An important question is if we know the configuration of Π_{out}, what can we conclude about Π_{in}?

The answer is trivial for PPP because of the spatial independence property. But this question of spatial conditioning in point processes with strong correlation structures has attracted a fair amount of interest. In Ghosh and Peres (2017), the authors showed that in the Ginibre ensemble, points outside a bounded open set $\mathcal{D}$, with boundary of zero Lebesgue measure, determine the number of points in $\mathcal{D}$ exactly. The next

theorem is taken from Ghosh and Peres (2017) which explains the rigidity property of the Ginibre ensemble.

Theorem 2.1 *For the Ginibre ensemble, there is a measurable function* $N : \mathcal{S}_{out} \mapsto \mathbb{N} \cup \{0\}$ *such that a.s. we have*

$$\textit{Number of points in } \mathcal{S}_{in} = N(\mathcal{S}_{out}).$$

Theorem 2.1 clearly shows that $\mathcal{G}_{\text{in}}$ and $\mathcal{G}_{\text{out}}$ distributed over disjoint regions $\mathcal{D}_{\text{in}}$ and $\mathcal{D}_{\text{out}}$ are not independent, rather $\mathcal{G}_{\text{out}}$ completely determines the number of points in $\mathcal{G}_{\text{in}}$. This explains the 'rigidity' phenomena of the Ginibre ensemble. For more details, we refer the reader to Ghosh and Peres (2017) and Ghosh (2016). The next theorem quoted from Ghosh and Peres (2017) shows that $\mathcal{G}_{\text{out}}$ only determines the number of points in $\mathcal{G}_{\text{in}}$ and 'nothing more'. Given the configuration $\mathcal{G}_{\text{out}}$, we denote the conditional measure as $\rho(\cdot|\mathcal{G}_{\text{out}})$. Since a.s. the size of $\mathcal{G}_{\text{in}}$ equals $N(\mathcal{G}_{\text{out}})$, we can assume that each measure $\rho(\cdot|\mathcal{G}_{\text{out}})$ is supported on $\mathcal{D}^{N(\mathcal{G}_{\text{out}})}$.

Theorem 2.2 *For the Ginibre ensemble,* $\mathbb{P}$*-a.s. the measure* $\rho(\cdot|\mathcal{G}_{out})$ *and the Lebesgue measure* ℓ *on* $\mathcal{D}^{N(\mathcal{G}_{out})}$ *are mutually absolutely continuous.*

In the next section, we describe the connectivity property of the (two-dimensional) Poisson DSF.

3 Burton-Keane Argument for the Poisson DSF

We consider the DSF along the direction e_y where the underlying point process is given by a homogeneous PPP $\mathcal{S}$ of intensity $\lambda > 0$ (see (1)) and call it as the Poisson DSF. Coupier and Tran (2013) showed that the Poisson DSF is connected a.s.

Theorem 3.1 (Theorem 1 of Coupier and Tran 2013) *The Poisson DSF is connected with probability* 1.

The geometric complexity of the Euclidean norm particularly makes this network model difficult to study. In Fig. 2, we present an instance of the first few steps of the DSF process $\{h^k(\mathbf{x}) : k \geq 0\}$. As mentioned earlier, for any $k \geq 0$ interior of the 'semi-circular' region

$$B_2^+(h^k(\mathbf{x}), ||h^k(\mathbf{x}) - h^{k+1}(\mathbf{x})||_2)$$

must be free of Poisson points, and thereby, the process $\{h^k(\mathbf{x}) : k \geq 0\}$ is not Markov. Let $\mathcal{N}_{\mathcal{S}}$ denote the number of (infinite tree) components in the Poisson DSF. The ergodicity property of the PPP ensures that the r.v. $\mathcal{N}_{\mathcal{S}}$ is constant a.s. To prove Theorem 3.1, it is equivalent to show that

$$\mathbb{P}(\mathcal{N}_{\mathcal{S}} \geq 2) = 0. \tag{1}$$

Coupier and Tran (2013) proved Eq. (1) through contradiction. Steps for establishing the desired contradiction can be divided into two parts. The first part is general and holds for any translation-invariant point process with exponentially decaying hole probability and the second part of the argument uses the spatial independence property of PPP. In Sect. 3.1, we describe the general argument, and the specific part is being discussed in Sect. 3.1.

3.1 Common Part of the Argument

We need to introduce some notations. For $m, M \in \mathbb{N}$ let $C_{m,M}$ denote the rectangular region $C_{m,M} := [-m, m] \times [-M, M]$. For $\mathbf{x} \in \mathbb{R}^2$, the (infinite) DSF path, denoted by $\gamma^{\mathbf{x}}$, starting from $\mathbf{x}$ is obtained by linearly joining the successive steps $h^k(\mathbf{x})$ and $h^{k+1}(\mathbf{x})$ for $k \geq 0$. With a slight abuse of notation, we identify the DSF path $\gamma^{\mathbf{x}}$ with its trajectory too, as a subset of $\mathbb{R}^2$. The following remark highlights certain 'non-crossing' property of the DSF paths.

Remark 3.1 Any two DSF paths either never touch each other, i.e., they remain disjoint as subsets of $\mathbb{R}^2$ or they touch, and as soon as they touch, they coalesce. Hence, DSF paths are non-crossing in the sense that for any two DSF paths $\gamma^{\mathbf{x}}$ and $\gamma^{\mathbf{y}}$ and for all $s_1, s_2 \geq \mathbf{x}(2) \vee \mathbf{y}(2)$ a.s. we have

$$(\gamma^{\mathbf{x}}(s_1) - \gamma^{\mathbf{y}}(s_1))(\gamma^{\mathbf{x}}(s_2) - \gamma^{\mathbf{y}}(s_2)) \geq 0.$$

For $m, M \in \mathbb{N}$ let $F_{m,M}$ denote the event

$$F_{m,M} := \{\text{There exists } \mathbf{x} \in \mathcal{S} \cap C_{m,M} \text{ such that for all } \mathbf{y} \in S \cap (\mathbb{H}^-(M) \setminus C_{m,M}), \text{ we have } \gamma^{\mathbf{x}} \cap \gamma^{\mathbf{y}} = \emptyset\}.$$

The next lemma is the first step towards establishing the desired contradiction.

Lemma 3.1 *If $P(\mathcal{N}_{\mathcal{S}} \geq 2) > 0$, then we must have $P(F_{m,M}) > 0$ for some $m, M \in \mathbb{N}$.*

We will see that the proof of Lemma 3.1 uses the spatial independence property of PPP. For the time being, we postpone proving Lemma 3.1 and discuss how this lemma leads to a contradiction. We continue to work with the choice of m, M as in Lemma 3.1. We need to introduce some notations first. For $\mathbf{x} \in \mathbb{Z}^2$ let $C^{\mathbf{x}}_{m,M}$ denote the translated rectangular region $\mathbf{x} + C_{m,M}$. Similarly, $F^{\mathbf{x}}_{m,M}$ denotes the corresponding translated version of the event $F_{m,M}$.

For $L \in \mathbb{N}$, consider the following set of $(2L+1)^2$ many lattice points

$$\mathbb{Z}_{m,M,L} := \{(2im, 2jM) : |i|, |j| \leq L\} \subset \mathbb{Z}^2,$$

and the rectangular region, which is obtained by taking the union of rectangular boxes $C^x_{m,M}$ for $x \in \mathbb{Z}_{m,M,L}$, is given by

$$\mathcal{R}_{m,M,L} := \bigcup_{\mathbf{x}\in\mathbb{Z}_{m,M,L}} C^x_{m,M}.$$

Let $\eta_{m,M,L}$ denote the (random) number of the DSF edges crossing the boundary of the region $\mathcal{R}_{m,M,L}$. We observe that for any two lattice points $\mathbf{x}, \mathbf{y} \in \mathbb{Z}_{m,M,L}$, joint occurrence of the events $F^x_{m,M}$ and $F^y_{m,M}$ would imply existence of at least two disjoint (infinite) DSF paths both starting inside $\mathcal{R}_{m,M,L}$ and crossing the boundary of $\mathcal{R}_{m,M,L}$. Hence, a.s. we have

$$\eta_{m,M,L} \geq \sum_{\mathbf{x}\in\mathbb{Z}_{m,M,L}} \mathbf{1}_{F^{\mathbf{x}}_{m,M}},$$

where for any event A, the notation $\mathbf{1}_A$ denotes the corresponding indicator r.v. Taking expectation on both sides and using the translation invariance properties of the PPP, we get

$$\mathbb{E}(\eta_{m,M,L}) \geq \mathbb{E}\left(\sum_{\mathbf{x}\in\mathbb{Z}_{m,M,L}} \mathbf{1}_{F^{\mathbf{x}}_{m,M}}\right) = \sum_{\mathbf{x}\in\mathbb{Z}_{m,M,L}} \mathbb{P}(F^{\mathbf{x}}_{m,M}) = (2L+1)^2\mathbb{P}(F_{m,M}).$$

Hence, Eq. (1) gives $\mathbb{E}(\eta_{m,M,L}) \geq C_{m,M}(2L+1)^2$ for some $C_{m,M} > 0$ if Lemma 3.1 holds. This shows that under the assumption that Lemma 3.1 holds, $\mathbb{E}(\eta_{m,M,L})$ grows at least as L^2. This gives us the required contradiction as the expected number of edges crossing the boundary of $\mathcal{R}_{m,M,L}$ should be of an order close to the perimeter of the rectangle $\mathcal{R}_{m,M,L}$. The next lemma proves that $\mathbb{E}(\eta_{m,M,L})$ is at most of order of $L^{3/2}$ and establishes the required contradiction. This completes the proof of Theorem 3.1.

Lemma 3.2 *For any $m, M \in \mathbb{N}$, there exists $C = C(m, M) > 0$ such that for all large $L \in \mathbb{N}$ we have*

$$\mathbb{E}(\eta_{m,M,L}) \leq CL^{3/2}.$$

Proof We decompose the set of edges, going out of $\mathcal{R}_{m,M,L}$, into two sets: one whose length are more than $\sqrt{L}$ and the other one whose length are no more than $\sqrt{L}$. We show that the expected number of edges in both sets are of the order of $L^{3/2}$. It should be noted that the number of edges going out of $\mathcal{R}_{m,M,L}$ essentially depend on the length of the perimeter of the rectangle. We define $\eta^{\leq}_{m,M,L}$ as the number of edges going out of the region $\mathcal{R}_{m,M,L}$ which are shorter than $\sqrt{L}$. Similarly, $\eta^{\geq}_{m,M,L}$ denotes the number of edges going out of $\mathcal{R}_{m,M,L}$ which are longer than $\sqrt{L}$. For the edges which are shorter than $\sqrt{L}$, in order to go out of the region $\mathcal{R}_{m,M,L}$ it must start from a Poisson point in a rectangular strip of width $\sqrt{L}$ as shown in Fig. 3. Thus, the r.v. $\eta^{\leq}_{m,M,L}$ must be dominated by the number of points in the strip.

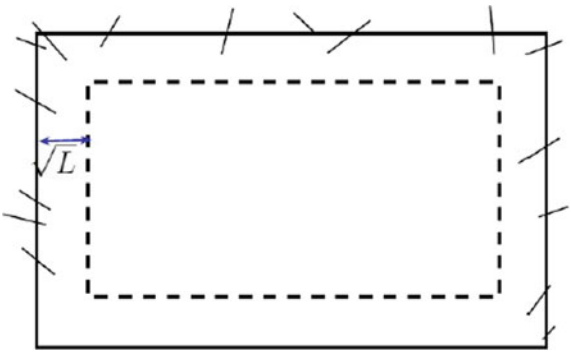


Fig. 3 Edges in $\eta^{\geq}_{m,M,L}$ start in a strip of width $\sqrt{L}$ along the perimeter of $\mathcal{R}_{m,M,L}$.Edges in $\eta^{\geq}_{m,M,L}$ start in a strip of width $\sqrt{L}$ along the perimeter of $\mathcal{R}_{m,M,L}$

Clearly, the expected number of points in the strip is given by the area, which is dominated by $C_1 L^{3/2}$ for some constant $C_1 = C_1(m, M)$. Hence, we have

$$\mathbb{E}(\eta^{\leq}_{m,M,L}) \leq C_1 L^{3/2}. \tag{2}$$

In order to bound the r.v. $\eta^{\geq}_{m,M,L}$, we note that any (Poisson) DSF edge of length bigger than $\sqrt{L}$ starts from a Poisson point $\mathbf{x} \in \mathcal{S} \cap \mathcal{R}_{m,M,L}$ for which interior of the semi-circular region $B_2^+(\mathbf{x}, \sqrt{L})$ must be free from Poisson points.

Let Z be the number of Poisson points in $\mathcal{R}_{m,M,L}$ such that for each point, the 'upper' semi-circular region of radius $\sqrt{L}$ contains no other Poisson points. Precisely, we define Z as

$$Z := \#\{\mathbf{y} \in \mathcal{S} \cap \mathcal{R}_{m,M,L} : B_2^+(\mathbf{y}, \sqrt{L}) \cap \mathcal{S} = \emptyset\},$$

where for any set A, the notation $\#A$ denotes cardinality of the set A. Therefore, we have

$$\eta^{\geq}_{m,M,L} \leq Z \text{ a.s.}$$

It is easy to obtain an upper bound for the expectation of Z. Indeed,

$$\mathbb{E}(\eta^{\geq}_{m,M,L}) \leq \mathbb{E}(Z) \leq C_2 L^2 \exp(-\pi\lambda L/2), \tag{3}$$

for some suitable choice of $C_2 = C_2(m, M) > 0$. Combining (2) and (3), we complete the proof.

Finally, in order to complete the proof of Theorem 3.1, we need to prove Lemma 3.1 and it is done in the next section.

3.2 Proof of Lemma 3.1

In this section, we present a proof of Lemma 3.1 for the Poisson DSF. We mentioned earlier that this part of the proof uses spatial independence property of PPP.

Let $T_{m,M} := [-m, m] \times \{M\}$ denote the 'top' boundary of the rectangular region $C_{m,M}$. In what follows, for any $A \subset \mathbb{R}^2$, the notation $\delta(A)$ denotes boundary of the set A. Let $E_{m,M}$ denote the event that

$$E_{m,M} := \{\text{There exist } \mathbf{x}, \mathbf{y} \in C_{m,M} \cap \mathcal{S} \text{ such that } \gamma^{\mathbf{x}} \cap \gamma^{\mathbf{y}} = \emptyset \text{ and } (\gamma^{\mathbf{x}} \cup \gamma^{\mathbf{y}}) \cap \delta(C_{m,M}) \subset T_{m,M}\}.$$

In other words, the event $E_{m,M}$ says that there exist two disjoint (infinite) DSF paths both starting inside the region $C_{m,M}$ and exiting the region through its top boundary only. The next lemma is the first step towards proving Lemma 3.1.

Lemma 3.3 *If $\mathbb{P}(\mathcal{N}_{\mathcal{S}} \geq 2) > 0$, then we have $\mathbb{P}(E_{m,M}) > 0$ for some $m, M \in \mathbb{N}$.*

Proof If $\mathbb{P}(\mathcal{N}_{\mathcal{S}} \geq 2) > 0$ then we have

$$\mathbb{P}(\mathcal{N}_{\mathcal{S}} \geq 2) = \lim_{M \to \infty} \mathbb{P}\big(\text{There exist } \mathbf{x}, \mathbf{y} \in C_{M,M} \cap \mathcal{S} \text{ with } \gamma^{\mathbf{x}} \cap \gamma^{\mathbf{y}} = \emptyset\big) > 0. \quad (4)$$

Clearly, Eq. (4) gives us that there exists $M \in \mathbb{N}$ such that

$$\mathbb{P}\big(\text{there exist } \mathbf{x}, \mathbf{y} \in C_{M,M} \cap \mathcal{S} \text{ with } \gamma^{\mathbf{x}} \cap \gamma^{\mathbf{y}} = \emptyset\big) > 0.$$

The other part follows from the observation that almost surely, parts of the trajectories $\gamma^{\mathbf{x}}$ and $\gamma^{\mathbf{y}}$ in between the horizontal lines $y = -M$ and $y = M$ are compact sets. Hence, we can chose $m \geq M$ large enough such that both the trajectories exit only through the top boundary of the region $C_{m,M}$.

The next lemma is a stronger version of Lemma 3.3. For small $\nu \in (0, 1/2)$ we define the event $E^{\nu}_{m,M}$ as

$$\begin{aligned} E^{\nu}_{m,M} := \big\{&\text{there exist } \mathbf{x}, \mathbf{y} \in C_{m,M} \cap \mathcal{S} \text{ such that } \gamma^{\mathbf{x}} \cap \gamma^{\mathbf{y}} = \emptyset \text{ and the following} \\ &\text{conditions are satisfied:} \\ &(i)(\gamma^{\mathbf{x}} \cup \gamma^{\mathbf{y}}) \cap \delta(C_{m,M}) \subset T_{m,M} \\ &(ii)([-m, m] \times [M - \nu, M]) \cap \mathcal{S} = \emptyset \text{ and} \\ &(iii) h(\mathbf{x})(2) \wedge h(\mathbf{y})(2) > M + \nu\big\}. \end{aligned}$$

Lemma 3.4 *If $\mathbb{P}(\mathcal{N}_{\mathcal{S}} \geq 2) > 0$, then we have $\mathbb{P}E^{\nu_{m,M}} > 0$ for some $m, M \in \mathbb{N}$ and $\nu \in (0, 1/2)$.*

Proof We have already shown that under the assumption $\mathbb{P}(\mathcal{N}_{\mathcal{S}} \geq 2) > 0$, by virtue of the previous lemma, we have $\mathbb{P}(E_{m,M}) > 0$ for some $m, M \in \mathbb{N}$. We observe that

the number of Poisson points inside the region $C_{m,M}$ (as in Lemma 3.3) is finite a.s. Further, the minimum distance of these points from the horizontal line $y = M$ is strictly positive a.s. This allows us to choose $\nu > 0$ appropriately such that the strip $[-m, m] \times [M - \nu, M]$ is free from Poisson points. Further, for Poisson points $\mathbf{x}, \mathbf{y}$ as in the definition of $E^{m,M}$ we have

$$h(\mathbf{x})(2) \wedge h(\mathbf{y})(2) > M \text{ a.s.}$$

Hence, we can update the choice of $\nu \in (0, 1/2)$ accordingly.

The next lemma is similar to Lemma 3.4 but it involves three paths instead of two. We define the event $A^{\nu}_{m,M}$ as

$$\begin{aligned} A^{\nu}_{m,M} := \big\{&\text{there exist } \mathbf{x}, \mathbf{y}, \mathbf{z} \in C_{m,M} \cap \mathcal{S} \text{ such that the following conditions are satisfied:} \\ &(i)(\gamma^{\mathbf{x}} \cap \gamma^{\mathbf{y}}) \cup (\gamma^{\mathbf{x}} \cap \gamma^{\mathbf{z}}) \cup (\gamma^{\mathbf{y}} \cap \gamma^{\mathbf{z}}) = \emptyset, \\ &(ii)(\gamma^{\mathbf{x}} \cup \gamma^{\mathbf{y}} \cup \gamma^{\mathbf{z}}) \cap \delta(C_{m,M}) \subset T_{m,M} \\ &(iii)([-m, m] \times [M - \nu, M]) \cap \mathcal{S} = \emptyset \text{ and} \\ &(iv) \min\{h(\mathbf{x})(2), h(\mathbf{y})(2), h(\mathbf{z})(2)\} > M + \nu\big\}. \end{aligned}$$

Lemma 3.5 *If $\mathbb{P}(\mathcal{N}_{\mathcal{S}} \geq 2) > 0$ then there exist $m, M \in \mathbb{N}$ and $\nu \in (0, 1/2)$ such that we have $\mathbb{P}(A^{\nu}_{m,M}) > 0$.*

Proof For $i \in \mathbb{Z}$ let $E^{\nu}_{m,M}(i)$ denote translated version of the event $E^{\nu}_{m,M}$ translated to the box $2mi + C_{m,M}$. We have

$$\mathbb{P}\big(\cup_{i \in \mathbb{Z}} E^{\nu}_{m,M}(i)\big) \leq 1. \tag{5}$$

The translation invariance property of the PPP ensures that $\mathbb{P}(E^{\nu}_{m,M}(i)) = \mathbb{P}(E^{\nu}_{m,M})$. Therefore, if all the pairwise interaction terms of the form $E^{\nu}_{m,M}(i) \cap E^{\nu}_{m,M}(j)$ for $i \neq j$ are of zero probability, then the R.H.S. in (5) becomes infinite. Hence, there exist $i, j \in \mathbb{Z}$ with $i \neq j$ such that

$$\mathbb{P}(E^{\nu}_{m,M}(i) \cap E^{\nu}_{m,M}(j)) > 0.$$

Set $m' := |2im| \vee |2jm| + m$. The non-crossing property of DSF paths, as mentioned in Remark 3.1, gives us the following event inclusion:

$$A^{\nu}_{m',M} \supseteq (E^{\nu}_{m,M}(i) \cap E^{\nu}_{m,M}(j)).$$

This follows from the fact that joint occurrence of the events $E^{\nu}_{m,M}(i)$ and $E^{\nu}_{m,M}(j)$ ensures existence of at least three disjoint DSF paths starting from the rectangular region $[-m', m'] \times [-M, M]$. For this part, we actually need the fact that in each of the rectangular boxes, both the disjoint DSF paths exit through the 'top' boundary only. This completes the proof. It is useful to mention that this part of the proof uses planar structure of the two-dimensional DSF constructed on the Euclidean plane.

In what follows, to simplify notation, we denote m' simply as m. Next for $R \in \mathbb{N}$ with $R > m \vee 10$, we define a stronger version of the event $A^{\nu}_{m,M}$ defined as

$$A^{\nu,R}_{m,M} := A^{\nu}_{m,M} \cap \big([-R, -m] \times [M-\nu, M] \cap \mathcal{S} \neq \emptyset\big) \cap \big([m, R] \times [M-\nu, M] \cap \mathcal{S} \neq \emptyset\big).$$

In other words, the event $A^{\nu,R}_{m,M}$ requires that in addition to the event $A^{\nu}_{m,M}$, both the strips $[-R, -m] \times [M-\nu, M]$ and $[m, R] \times [M-\nu, M]$ must contain Poisson points. We make sure that $R > 10$, which is needed in the proof of Lemma 3.1. We explain the reason for introducing this additional restriction in the definition of the event $A^{\nu,R}_{m,M}$ in Remark 3.2.

Remark 3.2 The crucial observation is that the occurrence of the event $A^{\nu,R}_{m,M}$ depends only on Poisson points in the region $[-R, -m] \times [M-\nu, M] \cup \mathbb{H}^{+}(M+?)$. We mentioned earlier that from the definition of the DSF step $h(\mathbf{w})$ it follows that

$$B^{+}_{2}(\mathbf{w}, ||\mathbf{w} - h(\mathbf{w})||_2) \cap \mathcal{S} = \emptyset \text{ (see Fig. 2).}$$

In the definition of the event $A^{\nu,R}_{m,M}$, the presence of Poisson points in both the regions $[-R, -m] \times [M-\nu, M]$ and $[m, R] \times [M-\nu, M]$ ensures that all the three semi-circular balls

$$B^{+}(\mathbf{x}, ||\mathbf{x} - h(\mathbf{x})||_2),\ B^{+}(\mathbf{y}, ||\mathbf{y} - h(\mathbf{y})||_2) \text{ and } B^{+}(\mathbf{z}, ||\mathbf{z} - h(\mathbf{z})||_2)$$

where $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathcal{S}$ are as in the definition of the event $A^{\nu,R}_{m,M}$ are free from Poisson points. Hence, these regions must be contained in the region

$$([-R, R] \times [-M, M+\nu]) \cup \mathbb{H}^{+}(M+\nu).$$

Therefore, the spatial independence property of PPP ensures that Poisson points in the region

$$\mathbb{H}^{-}(M+\nu) \setminus ([-R, R] \times [-M, M+\nu])$$

are distributed independently of the event $A^{\nu,R}_{m,M}$.

We will use Remark 3.2 to prove Lemma 3.1.

Lemma 3.6 *If* $\mathbb{P}(\mathcal{N}_{\mathcal{S}} \geq 2) > 0$*, then there exist* $m, M, R \in \mathbb{N}$ *and* $\nu \in (0, 1/2)$ *such that* $\mathbb{P}(A^{\nu,R}_{m,M}) > 0$.

Proof Earlier we showed that under the assumption $\mathbb{P}(\mathcal{N}_{\mathcal{S}} \geq 2) > 0$ we must have $\mathbb{P}(A^{\nu}_{m,M}) > 0$ for some $m, M \in \mathbb{N}$ and $\nu \in (0, 1/2)$. The proof of Lemma 3.6 just follows from the fact that both the infinite strips $(-\infty, -m] \times [M-\nu, M]$ and $[m, \infty) \times [M-\nu, M]$ must contain Poisson points. In fact, both the strips contain infinitely many Poisson points.

Observe that occurrence of the event $A_{m,M}^{\nu,R}$ ensures that there are at least three disjoint DSF paths. We assume that the Poisson points $\mathbf{x}, \mathbf{y}, \mathbf{z}$ as in the definition of event $A_{m,M}^{\nu,R}$ are such that we have

$$\gamma^{\mathbf{x}}(M) < \gamma^{\mathbf{y}}(M) < \gamma^{\mathbf{z}}(M).$$

The non-crossing property of DSF paths, as mentioned in Remark 3.1, ensures that the 'middle' path $\gamma^{\mathbf{y}}$ remains trapped in between $\gamma^{\mathbf{x}}$ and $\gamma^{\mathbf{z}}$ and stays in the middle always. Precisely, we have

$$\gamma^{\mathbf{x}}(t) \le \gamma^{\mathbf{y}}(t) < \gamma^{\mathbf{z}}(t) \text{ for all } t \ge M.$$

Further, the non-crossing property of DSF paths ensures that no DSF path starting inside $\mathbb{H}^-(M) \setminus C_{R,M}$ can touch the DSF path $\gamma^{\mathbf{y}}$ in the upper half-plane $\mathbb{H}^+(M)$ as $\gamma^{\mathbf{y}}$ stays in the middle of $\gamma^{\mathbf{x}}$ and $\gamma^{\mathbf{z}}$. Hence, in order to complete the proof of Lemma 3.1, we need to make a 'shield' so that any DSF path starting inside $\mathbb{H}^-(M) \setminus C_{R,M}$ cannot touch the path $\gamma^{\mathbf{y}}$ in the lower half-plane $\mathbb{H}^-(M)$. We explain the shield construction now and complete the proof of Lemma 3.1 thereby. In fact, the suggested shield construction would ensure that all the subsequent steps of each DSF path starting from a Poisson point inside $\mathbb{H}^-(M) \setminus C_{R,M}$ are out of the rectangular region $C_{R,M}$. This part of the proof uses spatial independence property of PPP.

Proof of Lemma 3.1: As mentioned earlier, Poisson points in the region

$$\mathbb{H}^-(M+\nu) \setminus \big([-R, R] \times [-M, M+\nu]\big)$$

are distributed independent of the event $A_{m,M}^{\nu,R}$, and hence, we can modify configurations of Poisson points in this region to create a shield. For the given choice of R, M, we consider a sequence of three nested ℓ_1 triangles:

$$\begin{aligned}
\nabla_1 &:= B_1^-((0, M), 3R),\\
\nabla_2 &:= B_1^-((0, M), 4R) \text{ and}\\
\nabla_3 &:= B_1^-((0, M), 5R).
\end{aligned}$$

It is not difficult to observe that this gives us a sequence of three isosceles nested triangles each having base on the line $y = M$. Clearly, we have

$$C_{R,M} \subset \Delta_1 \subset \nabla_2 \subset \nabla_3.$$

Next, we consider translated copies of the unit square $[-1/2, 1/2]^2$ centred at lattice points on the 'slanted' part of the boundary of the triangle ∇_2 and denote this collection as

$$I = I(R, M) := \big\{(\mathbf{w} + [-1/2, 1/2]^2) : \mathbf{w} \in \delta(\nabla_2) \cap \mathbb{Z}^2, \mathbf{w}(2) < M\big\}.$$

The corresponding region obtained by considering the union of squares in the collection $I(R, M)$ is denoted by

$$\mathcal{R} = \mathcal{R}(R, M) := \bigcup_{\mathbf{w}\delta(\nabla_2)\cap\mathbb{Z}^2, \mathbf{w}(2)<M} \big(\mathbf{w} + [-1/2, 1/2]^2\big).$$

By definition, $\mathcal{R}$ is contained in the lower half-plane $\{\mathbf{w} \in \mathbb{R}^2 : \mathbf{w}(2) \le M\}$. Let $B_{R,M}$ denote the event that

$$\begin{aligned} B_{R,M} := \Big\{ &\big(\nabla_3 \setminus (\mathcal{R} \cup C_{R,M})\big) \cap \mathcal{S} = \emptyset \text{ and every square of the form } \mathbf{w} + [-1/2, 1/2]^2 \\ &\text{in the set } I \text{ must contain at least one Poisson point} \Big\} \bigcap \\ &\Big((B_2^+((-4R, M), \nu) \cap \mathcal{S} \neq \emptyset) \cap (B_2^+((4R, M), \nu) \cap \mathcal{S} \neq \emptyset) \Big). \end{aligned}$$

Clearly, the event $B_{R,M}$ is of positive probability. We observe that the event $B_{R,M}$ involves Poisson points in the region $\mathbb{H}^+(M + \nu) \setminus \big([-R, R] \times [-M, M + \nu]\big)$ and Remark 3.2 ensures that the event $B_{R,M}$ is independent of the event $A_{m,M}^{\nu,R}$. Hence, under the assumption $\mathbb{P}(\mathcal{N}_{\mathcal{S}} \ge 2) > 0$, the event $A_{m,M}^{\nu,R} \cap B_{R,M}$ has a positive probability of occurrence.

We discuss the implication of occurrence of the event $A_{m,M}^{\nu,R} \cap B_{R,M}$ now. Consider a square $\mathbf{w} + [-1/2, 1/2]^2$ in the set I. We call the two squares $\big(\mathbf{w} + (\pm 1, 1)\big) + [-1/2, 1/2]^2$ as 'higher neighbouring' squares for the given square. We observe that at least one of them belongs to the set I. Since, any two points in a square and in one of its higher neighbouring squares are within Euclidean distance $2\sqrt{2}$ and the region $\big(\nabla_3 \setminus (\mathcal{R} \cup C_{R,M})\big)$ is free from Poisson points, the choice of $R(> 10)$ ensures that a Poisson point in a square $\mathbf{w} + [-1/2, 1/2]^2$ in the set I with $\mathbf{w}(2) \le M - 2$ must connect to some other Poisson point in the same square or in a (higher) neighbouring square in the set I. By the same logic, the small value of $\nu \in (0, 1/2)$ makes it sure that the subsequent steps of the DSF path starting from a Poisson point in the square $(4R, M - 1) + [-1/2, 1/2]^2$ must be inside the same square or in the upper half-plane $\mathbb{H}^+(M)$. The same argument holds for DSF paths starting from Poisson points in the square $(-4R, M - 1) + [-1/2, 1/2]^2$. Hence, on the event $A_{m,M}^{\nu,R} \cap B_{R,M}$, subsequent steps of DSF paths starting from Poisson points inside the region $\mathcal{R}$ must be outside the rectangle $C_{R,M}$. Further, a Poisson point in the lower half-plane $\mathbb{H}^-(M)$ outside the region $\mathcal{R}$ must be outside the region ∇_3 too, and if it takes step inside ∇_3, then it must connect to a Poisson point inside $\mathcal{R}$. Because of the reason explained earlier, subsequent steps of the DSF path starting from it must be outside the rectangle $C_{R,M}$. Hence, on the event $A_{m,M}^{\nu,R} \cap B_{R,M}$ none of the DSF paths starting from Poisson points in $\mathbb{H}^-(M) \setminus C_{R,M}$ can touch a DSF path starting from $C_{R,M}$ inside the lower half-plane $\mathbb{H}^-(M)$. Hence, we have the following event inclusion:

$$A_{m,M}^{\nu,R} \cap B_{R,M} \subset F_{m,M},$$

where the event $F_{m,M}$ is defined as in Lemma 3.1. Therefore, we have shown that under the assumption $\mathbb{P}(\mathcal{N}_{\mathcal{S}} \geq 2) > 0$ we have $\mathbb{P}(A^{\nu,R}_{m,M} \cap B_{R,M}) > 0$ as well. This completes the proof.

Remark 3.3 One may consider high dimensional extensions of the Poisson DSF and want to explore the tree-forest dichotomy behaviour depending on dimensions. The Burton- Keane argument of Coupier and Tran (2013) crucially uses the planarity structure of two-dimensional Euclidean plane and cannot be used for higher dimensions. Further, Theorem 3.1 shows that the coalescing time for two DSF paths is a.s. finite but it does not say anything about its tail decay. In a later work, Coupier et. al. Coupier et al. (2020) used a renewal technique and explored tail decay of coalescing time for two DSF paths.

4 Burton-Keane Argument for the Ginibre DSF and Proof of Theorem 1.1

In this section, we consider the DSF for the Ginibre point process and prove that it is connected a.s. The proof of Theorem 1.1 follows from a similar adaptation of a Burton-Keane type argument. It is important to note that the proof of Lemma 3.1 for the Poisson DSF, more specifically, the 'shield' construction part requires the spatial independence property of PPP. We don't have this property for a strongly correlated point process like the Ginibre ensemble. However, as mentioned in Theorems 2.1 and 2.2, we have a precise understanding of the conditional distribution of points of $\mathcal{G}$ inside a domain given the points of $\mathcal{G}$ outside. Using this understanding, we will propose a 'shield' construction for the Ginibre ensemble. The proposed method of shield construction for the Ginibre ensemble is motivated from Ghosh et al. (2016). This completes the proof of Theorem 1.1.

Proof of Theorem 1.1: The same argument as in Theorem 3.1 gives us that to prove Theorem 1.1, it suffices to prove a version of Lemma 3.1 for the Ginibre DSF. Let $\mathcal{N}_{\mathcal{G}}$ denote the number of components for the Ginibre DSF. As argued earlier, to prove Theorem 3.1, it is equivalent to show that $\mathbb{P}(\mathcal{N}_{\mathcal{G}} \geq 2) = 0$. Towards that for $m, M \in \mathbb{N}$ let $A_{m,M}$ denote the event that

$$A_{m,M} := \{\text{there exist } \mathbf{x}, \mathbf{y}, \mathbf{z} \in C_{m,M} \cap \mathcal{G} \text{ with } (\gamma^{\mathbf{x}} \cup \gamma^{\mathbf{y}} \cup \gamma^{\mathbf{z}}) \cap \delta(C_{m,M}) \subset T_{m,M} \\ \text{and } (\gamma^{\mathbf{x}} \cap \gamma^{\mathbf{y}}) \cup (\gamma^{\mathbf{x}} \cap \gamma^{\mathbf{z}}) \cup (\gamma^{\mathbf{z}} \cap \gamma^{\mathbf{y}}) = \emptyset\},$$

where with a slight abuse of notations, $\gamma^{\mathbf{x}}$ denotes the Ginibre DSF path starting from $\mathbf{x} \in \mathcal{G}$. In other words, the event $A_{m,M}$ implies that the rectangular region $C_{m,M}$ contains at least three disjoint Ginibre DSF paths all are starting inside the region $C_{m,M}$ and exiting the region $C_{m,M}$ through its 'top' boundary only. The same argument as in the case of the Poisson DSF gives us that, under the assumption $\mathbb{P}(\mathcal{N}_{\mathcal{G}} \geq 2) > 0$ there exist $m, M \in \mathbb{N}$ such that $\mathbb{P}(A_{m,M})$ is positive. Based on this choice of $m, M \in \mathbb{N}$, we consider the ℓ_1 triangle

$$\nabla := B_1^-\big((0, M), m + 2M + 10\big).$$

It is not difficult to see that $C_{m,M} \subset \nabla$. For $R \in \mathbb{N}$ with $R \geq m + 2M + 20$ we consider the event $E = E(R)$ such that interior of the region

$$\Big(B_1^-\big((0, M), R\big) \cup B_1^+\big((-R, M), 1/2\big) \cup B_1^+\big((R, M), 1/2\big)\Big) \setminus C_{m,M}$$

contains at least $2(m + 2M + 11)$ many points from the Ginibre ensemble. By ergodicity of $\mathcal{G}$ we have that $P(E) > 0$ when R is large enough. We denote union of these regions as

$$\Delta := B_1^-\big((0, M), R\big) \cup B_2^+\big((-R, M), 1/2\big) \cup B_2^+\big((R, M), 1/2\big)$$

and configuration of points of $\mathcal{G}$ inside Δ by ξ and those outside Δ by ω. Let the number of points in Δ be denoted by $N(\Delta)$. By construction we have that the region ∇ is well inside the region $B_1^-((0, M), R)$. As in the case of the Poisson DSF, we consider the following set of unit squares centred at lattice points on the slanted boundaries of the region ∇ given by

$$I := \{\mathbf{w} + [-1/2, 1/2]^2 : \mathbf{w} \in \delta(\nabla) \cap \mathbb{Z}^2, \mathbf{w}(2) < M\}.$$

The region obtained by taking union of the squares in the set I is denoted as

$$\mathcal{R} := \bigcup_{\mathbf{w} \in \delta(\nabla) \cap \mathbf{z}^2, \mathbf{w}(2) < M} \big(w + [-1/2, 1/2]^2\big).$$

We define an event E' as follows: corresponding to every configuration (ξ, ω) in E, we define a new configuration (ξ', ω) where ξ' is obtained by moving all the Ginibre points in the region $\Delta \setminus C_{m,M}$, i.e., changing the configuration of ξ so that the following conditions are satisfied:

(i) Each unit square in the set I contains at least one point from $\mathcal{G}$.
(ii) Each of the two rectangles $B_2^+((-R, M), 1/2)$ and $B_2^+((R, M), 1/2)$ contains at least one point from $\mathcal{G}$.
(iii) The region $B_1^-\big((0, M), R\big) \setminus (C_{m,M} \cup \mathcal{R})$ does not have any point from $\mathcal{G}$.

Note that the configuration ξ inside the rectangle $C_{m,M}$ remains unchanged. We observe that for items (i) and (ii) it is sufficient to have $2(m + 2M + 10) + 2$ many $\mathcal{G}$ points in the region $\Delta \setminus C_{m,M}$. This motivates the choice of the event E.

As observed earlier, any two points in two neighbouring squares in the set I have at most Euclidean distance $2\sqrt{2}$ in between them. This ensures that in configuration ξ', the DSF edge starting from a $\mathcal{G}$ point inside a square in the set I connects to another $\mathcal{G}$ point either in the same square or in a higher neighbouring square in the set I only. The presence of $\mathcal{G}$ points inside the regions, $B_2^+((-R, M), 1/2)$ and $B_2^+((R, M), 1/2)$, ensures that eventually such a path must go out of the lower half-plane $\mathbb{H}^-(M)$

without entering the concerned rectangle $C_{m,M}$. Note that in configuration ξ', all $\mathcal{G}$ points inside the region $\Delta \setminus C_{m,M}$ are inside $\mathcal{R}$ and we showed that the DSF paths starting from points inside $\mathcal{R}$ cannot enter inside the region $C_{m,M}$. Further, this ensures that in configuration ξ', if the DSF step starting from any $\mathcal{G}$ point in the region $\mathbb{H}^-(M) \setminus \Delta$ connects to a $\mathcal{G}$ point inside Δ then it must be inside $\mathcal{R}$ too and subsequent steps of the resulting Ginibre DSF path fail to enter the rectangle $C_{m,M}$. Hence, in configuration ξ', we have a 'shield' which ensures that subsequent steps of the DSF paths starting from Ginibre points inside the region $\mathbb{H}^-(M) \setminus C_{m,M}$ must be outside the rectangle $C_{m,M}$.

Now consider $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathcal{G}$ as in the definition of the event $A_{m,M}$ and assume that

$$\gamma^{\mathbf{x}}(M) < \gamma^{\mathbf{y}}(M) < \gamma^{\mathbf{z}}(M).$$

In Remark 3.1, we mentioned about certain non-crossing property of Poisson DSF paths. Note that this property does not depend on the underlying point process and holds for Ginibre DSF paths too. This non-crossing property prevents DSF paths starting from Ginibre points in $\mathbb{H}^-(M) \setminus C_{m,M}$ to touch the DSF path $\gamma^{\mathbf{y}}$ inside the upper half-plane $\mathbb{H}^+(M)$. Hence, configuration (ξ', ω) satisfies conditions of the event $F_{m,M}$ (defined as in Lemma 3.1).

Finally, note that we can move each point in ξ' in sufficiently small disks around itself, resulting in new configurations $(\tilde{\xi}, \omega)$ such that connections of the (Ginibre) DSF paths starting inside $C_{m,M}$ as well as the number of points in Δ remain unaltered. The event E' consists of all such configurations $(\tilde{\xi}, \omega)$ as (ξ, ω) varies over all configurations in E. This explains that for each ω, the set of configurations $\{\tilde{\xi} : (\tilde{\xi}, \omega) \in E'\}$ constitutes an open subset of $\Delta^{N(\Delta)}$, when considered as a vector in the usual way. Since $P(E) > 0$, by Theorem 2.2 applied to the domain Δ, we also have $\mathbb{P}(E') > 0$. This completes the proof as we have already argued that

$$E' \subset F_{m,M}.$$

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Convergence of Stochastic Approximation via Martingale and Converse Lyapunov Methods

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In Honour of the Sixty-Fifth Birthday of Prof. Rajeeva L. Karandikar.

1 Introduction

The stochastic approximation (SA) algorithm, originally introduced by Robbins and Monro (1951), is a widely used method for finding a zero of a function $\mathbf{f} : \mathbb{R}^d \to \mathbb{R}^d$, when only noisy measurements of $\mathbf{f}$ are available. Since its introduction, it has been a workhorse of probability theory and has found many applications. An early survey can be found in Lai (2003), and a more recent and broader survey can be found in Franci and Grammatico (2022).

Here is a brief description of the algorithm. Suppose $\mathbf{f} : \mathbb{R}^d \to \mathbb{R}^d$, and it is desired to find a solution $\boldsymbol{\theta}^*$ to the equation $\mathbf{f}(\boldsymbol{\theta}) = \mathbf{0}$, when one can access only noisy measurements of $\mathbf{f}(\cdot)$. In broad terms, the algorithm proceeds as follows: One begins with an initial guess $\boldsymbol{\theta}_0$. In the original version of SA proposed in Robbins and Monro (1951), at time $t + 1 \geq 1$, one has access to a noisy measurement

$$\mathbf{y}_{t+1} = \mathbf{f}(\boldsymbol{\theta}_t) + \boldsymbol{\xi}_{t+1}, \tag{1}$$

where $\boldsymbol{\theta}_t$ is the current guess and $\boldsymbol{\xi}_{t+1}$ is the measurement error. Then the guess $\boldsymbol{\theta}_t$ is updated according to

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t + \alpha_t \mathbf{y}_{t+1}, \tag{2}$$

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where $\{\alpha_t\}_{t\geq 1}$ is a prespecified deterministic sequence of step sizes, with $\alpha_t \in (0, 1)$ for all t. It is desired to study the limit behavior of the sequence $\{\boldsymbol{\theta}_t\}$. Of particular interest is the situation where $\boldsymbol{\theta}^*$ is the only solution to the equation $f(\boldsymbol{\theta}) = \mathbf{0}$. We would like to find suitable conditions to ensure that the iterates $\{\boldsymbol{\theta}_t\}$ converge almost surely to $\boldsymbol{\theta}^*$.

In the original paper Robbins and Monro (1951), the step sizes α_t are positive and satisfy the conditions below, generally referred to as the Robbins–Monro conditions. The two conditions are often shown together, but we display them separately for reasons that will become clear later.

$$\sum_{t=1}^{\infty} \alpha_t^2 < \infty. \tag{3}$$

$$\sum_{t=1}^{\infty} \alpha_t = \infty. \tag{4}$$

In addition, various assumptions are imposed on the function $\mathbf{f}(\cdot)$, which have been relaxed by successive researchers. Until 2000, the available results had the following general format: *If* the sequence of iterates $\{\boldsymbol{\theta}_t\}$ is bounded almost surely, *then* $\boldsymbol{\theta}_t \to \boldsymbol{\theta}^*$ almost surely, provided some other conditions are satisfied. Some authors refer to the almost-sure boundedness of the iterates $\{\boldsymbol{\theta}_t\}$ as "stability," and $\boldsymbol{\theta}_t \to \boldsymbol{\theta}^*$ almost surely as "convergence." Thus, the typical result stated that stability plus other conditions imply convergence.

A major breakthrough was achieved in Borkar and Meyn (2000), in which the stability of the iterations is a conclusion and not a hypothesis. Thus, under suitable conditions on the function $\mathbf{f}$, both stability and convergence follow. In Borkar and Meyn (2000), the authors follow the well-established ODE method, whereby the sample paths of the recursion (2) are related to the (deterministic) solution trajectories of the ODE $\dot{\boldsymbol{\theta}} = \mathbf{f}(\boldsymbol{\theta})$. This method was introduced in Derevitskii and Fradkov (1974), Ljung (1977), and Kushner and Clark (1978). A very readable summary of this method is found in Métevier and Priouret (1984).

The present paper has the same objective, namely to make the stability of the iterations a conclusion and not a hypothesis. It is ironic that, as far back as 1965, there is a paper by Gladyshev (1965) that established both stability and convergence of the SA algorithm. This paper uses martingale methods, but is restricted to a special class of functions $\mathbf{f}$. On the other side, the proof is very short and elegant. Moreover, there is a clear "division of labor" whereby (3) leads to the almost-sure boundedness of the iterations, while the addition of (4) leads to convergence. This division of labor is not found in any other paper until now. Thus, the present author was motivated by a desire to extend the martingale methods of Gladyshev (1965) to the same class of functions $\mathbf{f}$ as are covered by the ODE method in Borkar and Meyn (2000), or perhaps a more general class. Subsequent to the publication of Gladyshev (1965), Robbins and Siegmund (1971) introduced a very general theorem that they called an "almost supermartingale convergence theorem," in which they rediscovered some of

the basic ideas in Gladyshev (1965). In Robbins and Siegmund (1971), the authors mention that they were unaware of Gladyshev (1965) at the time of writing their paper.

This is the starting point of the present paper. We first study the case where $\boldsymbol{\theta} = \boldsymbol{\theta}^*$, in addition to being the unique solution equilibrium of the associated ODE $\dot{\boldsymbol{\theta}} = \mathbf{f}(\boldsymbol{\theta})$. Then we prove that the iterations $\{\boldsymbol{\theta}_t\}$ converge to $\boldsymbol{\theta}^*$ provided there exists a suitable "Lyapunov function." This proof is based on the Robbins–Siegmund theorem in Robbins and Siegmund (1971). It is shown by example that our new theorem is more general than currently known theorems. However, the need to assume the existence of a suitable Lyapunov function is still a bottleneck. To overcome this, we strengthen the hypothesis to the statement that $\boldsymbol{\theta}^*$ is a globally *exponentially* stable equilibrium. Then we state and prove a new converse theorem on the existence of a Lyapunov function that has a globally bounded Hessian matrix. The converse theorem proved here builds on an earlier theorem in Corless and Glielmo (1998), which uses a significantly different Lyapunov function compared to its predecessors. This result is new and is possibly of independent interest within the stability theory community. In each case (namely, $\boldsymbol{\theta}^*$ is either globally asymptotically stable, or globally exponentially stable), the "division of labor" as in Gladyshev (1965) continues to hold: If (3) is satisfied, then the sequence of iterates $\{\boldsymbol{\theta}_t\}$ is bounded almost surely. Further, if (4) also holds, then $\boldsymbol{\theta}_t \to \boldsymbol{\theta}^*$ almost surely.

Shortly after the publication of Robbins and Monro (1951), a different stochastic approximation algorithm was introduced by Kiefer and Wolfowitz in Kiefer and Wolfowitz (1952) and analyzed further by Blum in Blum (1954). Suppose $J : \mathbb{R}^d \to \mathbb{R}$ is in $\mathcal{C}^1$, and the objective is to find a stationary point of J, that is, a solution of $\nabla J(\boldsymbol{\theta}) = \mathbf{0}$. If one has access to possibly noisy measurements of $\nabla J(\cdot)$, then this is the same as the problem studied in Robbins and Monro (1951). However, in Kiefer and Wolfowitz (1952), the gradient is approximated using first-order differences. Here is the problem formulation. Let $[d]$ denote $\{1, \cdots, d\}$, and let $\mathbf{e}_i \in \mathbb{R}^d$ denote the ith elementary unit vector. Let $\{c_t\}$ denote a sequence of "increments," and let $\boldsymbol{\theta}_t$ denote the current guess to the solution of $\nabla J(\boldsymbol{\theta}) = \mathbf{0}$. Now define $\mathbf{y}_{t+1}$ by

$$y_{t+1,i} = \frac{[J(\boldsymbol{\theta}_t - c_t\mathbf{e}_i) + \xi^-_{t+1,i}] - [J(\boldsymbol{\theta}_t + c_t\mathbf{e}_i) + \xi^+_{t+1,i}]}{2c_t}. \tag{5}$$

Thus, one systematically perturbs the ith component of the current guess $\boldsymbol{\theta}_t$ in both the positive and the negative direction by the increment c_t (which is the same for all coordinates i) and measures the value of J at these $2d$ perturbed vectors. This process is repeated for each of the d coordinates. Here, $\xi^-_{t+1,i}$ and $\xi^+_{t+1,i}$ denote, respectively, the measurement errors when the ith coordinate is perturbed in the negative and positive directions. The resulting vector $\mathbf{y}_{t+1}$ is an approximation to $-\nabla J(\boldsymbol{\theta}_t)$ and is used in (2) to compute the next iteration $\boldsymbol{\theta}_t$. With this definition of the measurement vector $\mathbf{y}_{t+1}$, $\boldsymbol{\theta}_t$ is updated to $\boldsymbol{\theta}_{t+1}$ in accordance with (2), where $\{\alpha_t\}$ is a predetermined sequence of step sizes as before. This approach is suggested in Kiefer and Wolfowitz (1952) for the scalar case ($d = 1$) and for the multivariable case

in Blum (1954), with slight changes.[1] Note that, even with noise-free measurements, $\mathbf{y}_{t+1}$ would only *approximately* equal $-\nabla J(\boldsymbol{\theta}_t)$. Thus, there are **two** sources of error in (5), namely the use of first-order differences to approximate the gradient and the presence of measurement noise vectors. If one were to rewrite (5) as

$$y_{t+1,i} = \frac{J(\boldsymbol{\theta}_t - c_t\mathbf{e}_i) - J(\boldsymbol{\theta}_t + c_t\mathbf{e}_i)}{2c_t} + \frac{\xi^-_{t+1,i} - \xi^+_{t+1,i}}{2c_t},$$

then as $c_t \to 0$, the first term approaches $-[\nabla J(\boldsymbol{\theta}_t)]_i$, but the variance of the second term approaches infinity if, as is common, the measurement errors remain bounded away from zero. Thus, the analysis of the stochastic approximation algorithm with approximate gradients is more delicate than with noisy gradients.

The convergence analysis in Blum (1954) is rather convoluted. In contrast, in the present paper, we use the Robbins–Siegmund theorem to analyze the approach in (5). For simplicity, it is assumed that $J(\cdot)$ is convex. After Blum (1954), the case where J is not convex is studied by Bertsekas and Tsitsiklis (2000). However, in that paper, as in other places, the almost-sure boundedness of the iterations is a hypothesis, not a conclusion. It is shown here that if the function J is convex and $\mathcal{C}^2$, has a unique global minimum which can be taken as $\boldsymbol{\theta}^* = \mathbf{0}$, and its Hessian matrix is uniformly bounded both above and below, then the two conditions

$$\sum_{t=0}^{\infty}\left(\frac{\alpha_t}{c_t}\right)^2 < \infty, \sum_{t=0}^{\infty}\alpha_t c_t < \infty \tag{6}$$

are sufficient to imply the almost-sure boundedness of the iterates. If in addition to (6), we have that

$$\sum_{t=0}^{\infty}\alpha_t = \infty,$$

(which is the same as (4)), then $\boldsymbol{\theta} \to \mathbf{0}$ almost surely as $t \to \infty$. Note that in Blum (1954), the author adds the condition that $c_t \to 0$ as $t \to \infty$. No such assumption is made here. Instead, it is assumed only that the sequence $\{c_t\}$ is bounded above. Therefore, just as (3) and (4) are together referred to as the Robbins–Monro conditions, it would be appropriate to refer to (6) and (4) as the Blum conditions. To shed some light on the Blum conditions, suppose that $\alpha_t = (t+1)^{-p}$ and $c_t = (t+1)^{-r}$. Then the Blum conditions are equivalent to $p + r < 1$ and $p - r > 1/2$.

[1] In Blum (1954), instead of $2d$ measurements, only $d + 1$ measurements are used.

2 Statements of Main Results

2.1 Preliminaries

To state our first theorem on the convergence of the SA algorithm, we begin by recalling some notations and concepts from nonlinear stability theory. The reader is referred to Vidyasagar (2002), Khalil (2002), and Hahn (1967) for further details.

Definition 1 A function $\phi : \mathbb{R}_+ \to \mathbb{R}_+$ is said to **belong to class** $\mathcal{K}$, denoted by $\phi \in \mathcal{K}$, if $\phi(0) = 0$, and $\phi(\cdot)$ is strictly increasing. A function $\phi \in \mathcal{K}$ is said to **belong to class** $\mathcal{KR}$, denoted by $\phi \in \mathcal{KR}$, if in addition, $\phi(r) \to \infty$ as $r \to \infty$. A function $\phi : \mathbb{R}_+ \to \mathbb{R}_+$ is said to **belong to class** $\mathcal{B}$, denoted by $\phi \in \mathcal{B}$, if $\phi(0) = 0$, and in addition, for all $0 < \epsilon < M < \infty$ we have that

$$\inf_{\epsilon \le r \le M} \phi(r) > 0. \tag{7}$$

Observe that every ϕ of class $\mathcal{K}$ also belongs to class $\mathcal{B}$. However, the converse is not true. Define

$$\phi(r) = \begin{cases} r, & \text{if } r \in [0, 1], \\ e^{-(r-1)}, & \text{if } r > 1. \end{cases}$$

Then ϕ belongs to Class $\mathcal{B}$. However, since $\phi(r) \to 0$ as $r \to \infty$, ϕ cannot be bounded below by any function of class $\mathcal{K}$.

Observe that if $\phi \in \mathcal{KR}$, then ϕ^{-1} exists and also belongs to $\mathcal{KR}$.

Next, we recall a standard concept from stability theory. Consider the ODE

$$\dot{\boldsymbol{\theta}} = f(\boldsymbol{\theta}), \tag{8}$$

and suppose $V : \mathbb{R}^d \to \mathbb{R}_+$ is $\mathcal{C}^1$. Let $\nabla V : \mathbb{R}^d \to \mathbb{R}^d$ denote the gradient of V, viewed as a column vector. Then the function $\dot{V} : \mathbb{R}^d \to \mathbb{R}$ defined by

$$\dot{V}(\boldsymbol{\theta}) = \langle \nabla V(\boldsymbol{\theta}), \mathbf{f}(\boldsymbol{\theta}) \rangle \tag{9}$$

is called the **derivative** of V along the trajectories of (8).

Next, for convenience, we state the various assumptions on the function $\mathbf{f} : \mathbb{R}^d \to \mathbb{R}^d$. Note that not every theorem requires every assumption.

(F1) The equation $\mathbf{f}(\boldsymbol{\theta}) = \mathbf{0}$ has a unique solution $\boldsymbol{\theta}^*$.

(F2) The function $\mathbf{f}$ is globally Lipschitz-continuous with constant L.

$$\|\mathbf{f}(\boldsymbol{\theta}) - \mathbf{f}(\boldsymbol{\phi})\|_2 \le L\|\boldsymbol{\theta} - \boldsymbol{\phi}\|_2, \ \forall \boldsymbol{\theta}, \boldsymbol{\phi} \in \mathbb{R}^d. \tag{10}$$

Note that, as a consequence of this assumption, for each $\boldsymbol{\theta} \in \mathbb{R}^d$ there is a unique function $\mathbf{s}(\cdot, \boldsymbol{\theta})$ that satisfies the ODE

$$\frac{d\mathbf{s}(t,\boldsymbol{\theta})}{dt} = \mathbf{f}(\mathbf{s}(t,\boldsymbol{\theta})), \mathbf{s}(0,\boldsymbol{\theta}) = \boldsymbol{\theta}. \tag{11}$$

(F2') The function $\mathbf{f}$ is twice continuously differentiable and is globally Lipschitz-continuous with constant L.

(F3) The equilibrium $\boldsymbol{\theta}^*$ of the ODE $\dot{\boldsymbol{\theta}} = \mathbf{f}(\boldsymbol{\theta})$ is globally exponentially stable. Thus there exist constants $\mu \geq 1, \gamma > 0$ such that

$$\|\mathbf{s}(t,\boldsymbol{\theta}) - \boldsymbol{\theta}^*\|_2 \leq \mu \|\boldsymbol{\theta} - \boldsymbol{\theta}^*\|_2 \exp(-\gamma t), \ \forall t \geq 0, \ \forall \boldsymbol{\theta} \in \mathbb{R}^d. \tag{12}$$

(F4) There is a finite constant K such that

$$\|\nabla^2 f_i(\boldsymbol{\theta})\|_S \cdot \|\boldsymbol{\theta} - \boldsymbol{\theta}^*\|_2 \leq K, \ \forall i \in [d], \ \forall \boldsymbol{\theta} \in \mathbb{R}^d, \tag{13}$$

where $[d]$ denotes the set $\{1, \ldots, d\}$, and $\|\cdot\|_S$ denotes the spectral norm of a matrix, i.e., its largest singular value. A consequence of (13) is that

$$\left| \frac{\partial^2 f_i(\boldsymbol{\theta})}{\partial \theta_j \partial \theta_k} \right| \cdot \|\boldsymbol{\theta} - \boldsymbol{\theta}^*\|_2 \leq K, \ \forall i, j, k \in [d], \ \forall \boldsymbol{\theta} \in \mathbb{R}^d. \tag{14}$$

Next, we come to the noise sequence $\{\boldsymbol{\xi}_t\}$ in (1). For a stochastic process $\{\mathbf{x}_t\}$, we use the notation $\mathbf{x}_\tau^t$ to denote the tuple $\mathbf{x}_\tau, \cdots, \mathbf{x}_t$, when $t \geq \tau$. Suppose the iterative sequence $\{\boldsymbol{\theta}_t\}$ is defined by the stochastic approximation algorithm (2). For $t \geq 1$, let $\mathcal{F}_t$ denote the σ-algebra generated by the tuples $\boldsymbol{\theta}_0^t, \boldsymbol{\xi}_1^t$ (note that there is no $\boldsymbol{\xi}_0$). Then it is assumed that $\{\boldsymbol{\xi}_t\}$ satisfies the following:

(N1) The conditional expectation

$$E(\boldsymbol{\xi}_{t+1}|\mathcal{F}_t) = 0, \ \text{a.s.} \tag{15}$$

This states that the process $\{\boldsymbol{\xi}_t\}$ is a martingale difference sequence with respect to the filtration $\mathcal{F}_t$; it also guarantees that the measurement $\mathbf{y}_{t+1}$ of $\mathbf{f}(\boldsymbol{\theta}_t)$ is unbiased.

(N2) There exists a constant $D \geq 0$ such that

$$E(\|\boldsymbol{\xi}_{t+1}\|_2^2|\mathcal{F}_t) \leq D(1 + \|\boldsymbol{\theta}_t - \boldsymbol{\theta}^*\|_2^2) \ \text{a.s.}, \ \forall t \geq 0. \tag{16}$$

2.2 Theorem Statements

Before stating theorems about stochastic approximation, we state a new theorem that gives a sufficient condition for global asymptotic stability. This theorem is not used *in this form* anywhere else in the paper, except in one example. However,

because it represents a weakening of the best-known sufficient conditions, it should be of interest to researchers in nonlinear stability theory. The definition of global asymptotic stability is quite standard and can be found in any standard reference, such as Hahn (1967), Vidyasagar (2002), and Khalil (2002).

Theorem 1 *Suppose Assumptions (F1) holds and that there exist a function* $V : \mathbb{R}^d \to \mathbb{R}_+$ *and functions* $\eta, \psi \in \mathcal{KR}$, $\phi \in \mathcal{B}$ *such that*

$$\eta(\|\boldsymbol{\theta} - \boldsymbol{\theta}^*\|_2) \leq V(\boldsymbol{\theta}) \leq \psi(\|\boldsymbol{\theta} - \boldsymbol{\theta}^*\|_2), \ \forall \boldsymbol{\theta} \in \mathbb{R}^d, \tag{17}$$

$$\dot{V}(\boldsymbol{\theta}) \leq -\phi(\|\boldsymbol{\theta} - \boldsymbol{\theta}^*\|_2), \ \forall \boldsymbol{\theta} \in \mathbb{R}^d. \tag{18}$$

Then $\boldsymbol{\theta}^*$ *is a globally asymptotically stable equilibrium of the ODE* (8).

Remark: The usual theorems on global asymptotic stability look similar, except that the function $\phi(\cdot)$ in (18) is assumed to belong to the Class $\mathcal{K}$, not class $\mathcal{B}$; see for example (Vidyasagar, 2002, Theorem 5.3.56). The change here is that the assumption on $\phi(\cdot)$ is weakened to $\phi \in \mathcal{B}$ from $\phi \in \mathcal{K}$.

We now state the first result on the almost-sure boundedness and convergence of the stochastic approximation algorithm (2).

Theorem 2 *Suppose Assumptions (F1)–(F2) and (N1)–(N2) hold. Suppose in addition that there exists a Lyapunov function* $V : \mathbb{R}^d \to \mathbb{R}_+$ *that satisfies the following conditions:*

1. *There exist constants* $a, b > 0$ *such that*

$$a\|\boldsymbol{\theta} - \boldsymbol{\theta}^*\|_2^2 \leq V(\boldsymbol{\theta}) \leq b\|\boldsymbol{\theta} - \boldsymbol{\theta}^*\|_2^2, \ \forall \boldsymbol{\theta} \in \mathbb{R}^d. \tag{19}$$

2. *There exists a function* $\phi \in \mathcal{B}$ *such that*

$$\dot{V}(\boldsymbol{\theta}) \leq -\phi(\|\boldsymbol{\theta} - \boldsymbol{\theta}^*\|), \ \forall \boldsymbol{\theta} \in \mathbb{R}^d. \tag{20}$$

3. *There is a finite constant* M *such that*

$$\|\nabla^2 V(\boldsymbol{\theta})\|_S \leq 2M, \ \forall \boldsymbol{\theta} \in \mathbb{R}^d. \tag{21}$$

Under these hypotheses,

1. *If the step size sequence* $\{\alpha_t\}$ *satisfies* (3)*, then the stochastic process* $\{\boldsymbol{\theta}_t\}$ *is bounded almost surely.*
2. *If the step size sequence* $\{\alpha_t\}$ *satisfies* (4) *in addition to* (3)*, then the stochastic process* $\{\boldsymbol{\theta}_t\}$ *converges almost surely to* $\boldsymbol{\theta}^*$ *as* $t \to \infty$.

Remark: The above theorem requires the existence of a suitable Lyapunov function V that satisfies various conditions. Therefore, verifying whether or not such a function exists is a bottleneck. It would be highly desirable to provide sufficient conditions that involve only the function $\mathbf{f}(\cdot)$ that guarantee the existence of a suitable Lyapunov function. As shown in Theorem 1, the conditions on V in Theorem 2 ensure that the equilibrium $\boldsymbol{\theta}^*$ of the ODE (8) is globally asymptotically stable. By strengthening the assumption to the global *exponential* stability of $\boldsymbol{\theta}^*$, and adding a few other assumptions, it is possible to prove the following "converse" Lyapunov theorem that establishes the existence of a suitable V function. Note that there is already a well-developed "converse Lyapunov theory" that establishes all the requirements on V, *except* for the global boundedness of the Hessian of V; see for example (Vidyasagar, 2002, Sect. 5.7) or (Hahn, 1967, Sects. 48–51). Therefore the contribution of Theorem 5 is in establishing that V has a globally bounded Hessian. This theorem is new and possibly of independent interest to researchers in nonlinear stability theory.

Theorem 3 *Suppose Assumptions (F1), (F2'), (F3), and (F4) hold. Under these hypotheses, there exists a $\mathcal{C}^2$ function $V : \mathbb{R}^d \to \mathbb{R}_+$ such that V and its derivative $\dot{V} : \mathbb{R}^d \to \mathbb{R}$ together satisfy the following conditions: There exist positive constants a, b, c and a finite constant M such that*

$$a\|\boldsymbol{\theta} - \boldsymbol{\theta}^*\|_2^2 \leq V(\boldsymbol{\theta}) \leq b\|\boldsymbol{\theta} - \boldsymbol{\theta}^*\|_2^2, \dot{V}(\boldsymbol{\theta}) \leq -c\|\boldsymbol{\theta} - \boldsymbol{\theta}^*\|_2^2, \ \forall \boldsymbol{\theta} \in \mathbb{R}^d. \tag{22}$$

$$\|\nabla^2 V(\boldsymbol{\theta})\|_S \leq 2M, \ \forall \boldsymbol{\theta} \in \mathbb{R}^d. \tag{23}$$

Combining Theorems 2 and 3 gives the following "self-contained" theorem:

Theorem 4 *Suppose Assumptions (F1), (F2'), (F3), and (F4) as well as Assumptions (N1)–(N2) hold. Under these hypotheses,*

1. *If the step size sequence $\{\alpha_t\}$ satisfies (3), then the stochastic process $\{\boldsymbol{\theta}_t\}$ is bounded almost surely.*
2. *If the step size sequence $\{\alpha_t\}$ satisfies (4) in addition to (3), then the stochastic process $\{\boldsymbol{\theta}_t\}$ converges almost surely to $\boldsymbol{\theta}^*$ as $t \to \infty$.*

The proof of this theorem is omitted, as it is a ready consequence of Theorems 2 and 3.

Now we study the stochastic approximation algorithm when $\mathbf{y}_{t+1}$ in (2) is defined as in (5). The objective function $J(\cdot)$ is assumed to satisfy the following conditions:

(J1) $J(\cdot)$ is convex, is in $\mathcal{C}^2$, and has a unique global minimizer $\boldsymbol{\theta}^*$, which can be taken as $\mathbf{0}$ by translating coordinates.

(J2) The Hessian matrix $\nabla^2 J$ is bounded both above and below. Specifically, suppose there exist finite constants $0 < h < M$ such that

$$h \leq \|\nabla^2 J(\boldsymbol{\theta})\|_S \leq \frac{M'}{2}, \ \forall \boldsymbol{\theta} \in \mathbb{R}^d, \tag{24}$$

where $\|\cdot\|_S$ denotes the spectral norm of a matrix, that is, the largest singular value.

Note that Assumption (J2) implies the following relationships:

$$\langle \boldsymbol{\theta}, \nabla J(\boldsymbol{\theta}) \rangle \geq h \|\boldsymbol{\theta}\|_2^2, \ \forall \boldsymbol{\theta} \in \mathbb{R}^d. \tag{25}$$

$$\|\nabla J(\boldsymbol{\theta})\|_2 \leq L \|\boldsymbol{\theta}\|_2, \ \forall \boldsymbol{\theta} \in \mathbb{R}^d. \tag{26}$$

Next we state the assumptions on the noise sequences $\{\boldsymbol{\xi}_t^-\}, \{\boldsymbol{\xi}_t^+\}$, which is analogs of assumptions (N1) and (N2). Let $\mathcal{F}_t$ denote the σ-algebra generated by $\boldsymbol{\theta}_0^t$, $\boldsymbol{\xi}_{+,1}^{+,t}$ and $\boldsymbol{\xi}_{-,1}^{-,t}$. Then

(N3) We have that

$$E(\xi_{t+1,i}^+ | \mathcal{F}_t) = 0, E(\xi_{t+1,i}^- | \mathcal{F}_t) = 0, \text{ a.s., } \forall i \in [d], \ \forall t \geq 0. \tag{27}$$

(N4) Suppose that $\boldsymbol{\xi}_t^+$ and $\boldsymbol{\xi}_t^-$ are independent for each t, and define $\boldsymbol{\xi}_t = \boldsymbol{\xi}_t^+ + \boldsymbol{\xi}_t^-$. There there is a finite constant D such that

$$E(\|\boldsymbol{\xi}_{t+1}\|_2^2 | \mathcal{F}_t) \leq D(1 + \|\boldsymbol{\theta}_t\|_2^2), \ \forall t. \tag{28}$$

Theorem 5 *Suppose $J : \mathbb{R}^d \to \mathbb{R}$ satisfies assumptions (J1) and (J2), and that $\{\boldsymbol{\theta}_t\}$ is defined by (2) where y_{t+1} is defined by (5), where the noise sequences satisfy assumptions (N3) and (N4). Under these conditions,*

1. *If the sequences $\{\alpha_t\}, \{c_t\}$ together satisfy (6), then $\{\boldsymbol{\theta}_t\}$ is bounded almost surely.*
2. *If, in addition, $\{\alpha_t\}$ satisfies (4), then $\boldsymbol{\theta}_t \to \mathbf{0}$ almost surely as $t \to \infty$.*

2.3 Discussion of the Theorems

So far as the author is able to determine, there is no predecessor to Theorem 2, which establishes both the almost-sure boundedness and the convergence of the SA algorithm. However, this theorem requires the existence of a suitable Lyapunov function.

The closest available results to Theorem 4 in the current literature are from Borkar and Meyn (2000). In that paper, it is also assumed that there is a unique solution $\boldsymbol{\theta}^*$ to the equation $\mathbf{f}(\boldsymbol{\theta}) = \mathbf{0}$. In addition, it is assumed that the functions

$$\mathbf{f}_r(\boldsymbol{\theta}) : \boldsymbol{\theta} \mapsto \frac{\mathbf{f}(r\boldsymbol{\theta})}{r}$$

converge uniformly over compact sets to a limit function $\mathbf{f}_\infty$ as $r \to \infty$, and that $\mathbf{0}$ is a globally exponentially stable equilibrium of the ODE $\dot{\boldsymbol{\theta}} = \mathbf{f}_\infty(\boldsymbol{\theta})$.[2] There is no requirement of the limit function $\mathbf{f}_\infty$ here. Instead, we have Assumption (F4), which requires that the spectral norm of the Hessian matrix of each component $f_i(\boldsymbol{\theta})$ decays at least as fast as $1/\|\boldsymbol{\theta} - \boldsymbol{\theta}^*\|_2$. Obviously, if $\mathbf{f} \in \mathcal{C}^2$, then the Hessian matrix $\nabla^2 f_i(\boldsymbol{\theta})$ is bounded over every compact set. Thus, Assumption (F4) is about the behavior of the Hessian $\nabla^2 f_i(\boldsymbol{\theta})$ when $\|\boldsymbol{\theta}\|_2$ is large. If $\mathbf{f}(\boldsymbol{\theta}) = A\boldsymbol{\theta}$ for some matrix A, then this assumption holds trivially. A slight extension is that if $\mathbf{f}(\cdot)$ is of the form

$$\mathbf{f}(\boldsymbol{\theta}) = A\boldsymbol{\theta} + \|\boldsymbol{\theta}\|_2^{-2r}\mathbf{h}(\boldsymbol{\theta}),$$

where every component of $\mathbf{h}$ has a bounded Hessian and $r \geq 1$, then again Assumption (A4) is satisfied. Note that the same can be said about Assumption (F2) as well. Because $\mathbf{f}$ is continuously differentiable, it is Lipschitz-continuous over every compact set; therefore, the essence of (F2) is about the behavior of the Jacobian matrix $\nabla\mathbf{f}(\boldsymbol{\theta})$ when $\|\boldsymbol{\theta}\|_2$ is large. Note that

$$\nabla^2\mathbf{f}_r(\boldsymbol{\theta}) = r\nabla^2\mathbf{f}(\boldsymbol{\theta}).$$

Hence, if the function $\mathbf{f}_\infty$ exists *and has a globally bounded Hessian*, then this is essentially the same as (F4). Thus, with this additional assumption, the present theorem includes the result of Borkar and Meyn (2000) as a special case.[3]

The existence of a Lyapunov function V that satisfies (22) is quite standard. Indeed, the usual choice is

$$V(\boldsymbol{\theta}) := \int_0^\infty \|\mathbf{s}(t, \boldsymbol{\theta})\|_2^2 dt. \tag{29}$$

However, for this choice of V, no conclusions can be drawn about the behavior of the gradient ∇V nor the Hessian $\nabla^2 V$. In Corless and Glielmo (1998), the authors introduce a completely different Lyapunov function of the form

$$V(\boldsymbol{\theta}) := \int_0^T e^{\kappa\tau}\|\mathbf{s}(\tau, \boldsymbol{\theta} - \boldsymbol{\theta}^*\|_2^2 \, d\tau, \tag{30}$$

where $0 < \kappa < \gamma$ is arbitrary, and T is any *finite* number such that

$$\frac{\ln \mu}{\gamma - \kappa} \leq T < \infty,$$

[2] Actually, in Borkar and Meyn (2000), it is assumed only that $\mathbf{0}$ is globally asymptotically stable. However, due to the scale-invariance of the function $\mathbf{f}_\infty$, this implies that actually $\mathbf{0}$ is globally exponentially stable. Moreover, even if the original equilibrium $\boldsymbol{\theta}^* \neq \mathbf{0}$, the equilibrium of the scale-free limit function $\mathbf{f}_\infty$ can only be $\mathbf{0}$.

[3] The author is grateful to Prof. Sean Meyn for this observation.

where μ, γ are defined in (12). For this choice of Lyapunov function, it is shown in Corless and Glielmo (1998) that there exists a finite constant L' such that

$$\|\nabla V(\boldsymbol{\theta})\|_2 \leq L' \|\boldsymbol{\theta} - \boldsymbol{\theta}^*\|_2. \tag{31}$$

Now Theorem 3 extends the theory further by showing that, if (F4) holds, then $\nabla^2 V$ is globally bounded. This in turn implies (31). As we shall see, (F4) is the key assumption that allows us to extend the converse Lyapunov theory of Corless and Glielmo (1998) and prove that V has a globally bounded Hessian.

3 Proof of Theorem 1

Proof Let $\boldsymbol{\theta}(\cdot)$ denote a solution trajectory of the ODE (8). Then (18) implies that $V(\boldsymbol{\theta}(t))$ is a nonincreasing function of t, and therefore has a limit as $t \to \infty$, call it V_∞. It is now shown that $V_\infty = 0$. To see this, suppose that $V_\infty > 0$. Then the right-side bound in (17) implies that

$$\|\boldsymbol{\theta} - \boldsymbol{\theta}^*\|_2 \geq \psi^{-1}(V_\infty) =: c_1 > 0, \ \forall t,$$

while the left-side bound in (17) implies that

$$\|\boldsymbol{\theta} - \boldsymbol{\theta}^*\|_2 \leq \eta^{-1}(V(\boldsymbol{\theta}(0))) =: c_2 < \infty.$$

In turn, this implies that

$$-\dot{V}(\boldsymbol{\theta}(t)) \geq \inf_{c_1 \leq r \leq c_2} \phi(r) > 0,$$

because $\phi(\cdot)$ is a function of class $\mathcal{B}$. This is a contradiction because

$$V(\boldsymbol{\theta}(t)) = V(\boldsymbol{\theta}(0)) + \int_0^t \dot{V}(\boldsymbol{\theta}(\tau))\, d\tau,$$

and if $-\dot{V}(\boldsymbol{\theta}(\tau))$ is bounded away from zero, then eventually $V(\boldsymbol{\theta}(t))$ would become negative. Therefore $V_\infty = 0$, and $V(\boldsymbol{\theta}(t)) \to 0$ as $t \to \infty$. Now the left inequality in (17) shows that $\|\boldsymbol{\theta}(t)\|_2 \to 0$ as $t \to \infty$. □

4 Proof of Theorem 2

The proof of Theorem 2 is based on the following well-known theorem from Robbins and Siegmund (1971). The original reference Robbins and Siegmund (1971) is somewhat inaccessible. However, the same theorem is stated as Lemma 2 in

Benveniste et al. (1990, Sect. 5.2). A recent survey of many results along similar lines is found in Franci and Grammatico (2022).

Theorem 6 *Suppose* $\{z_t\}, \{\eta_t\}, \{\gamma_t\}, \{\psi_t\}$ *are nonnegative stochastic processes adapted to some filtration* $\{\mathcal{F}_t\}$ *that satisfy*

$$E(z_{t+1}|\mathcal{F}_t) \leq (1+\eta_t)z_t + \gamma_t - \psi_t \text{ a.s., } \forall t. \tag{32}$$

Define the set $\Omega_0 \subseteq \Omega$ *by*

$$\Omega_0 := \{\omega : \sum_{t=0}^{\infty} \eta_t(\omega) < \infty\} \cap \{\omega : \sum_{t=0}^{\infty} \gamma_t(\omega) < \infty\}. \tag{33}$$

Then, for all[4] $\omega \in \Omega_0$*, we have that (i)* $\lim_{t\to\infty} z_t(\omega)$ *exists and is finite, and (ii)*

$$\sum_{t=0}^{\infty} \psi_t(\omega) < \infty, \ \forall \omega \in \Omega_0. \tag{34}$$

In particular, if $P(\Omega_0) = 1$*, then* $\{z_t\}$ *is bounded almost surely, in the sense that*

$$P\{\omega \in \Omega : \sup_t z_t(\omega) < \infty\} = 1, \tag{35}$$

and

$$\sum_{t=0}^{\infty} \psi_t(\omega) < \infty \ \forall \omega \in \Omega_0. \tag{36}$$

Note that, while applying the above theorem, it is always assumed that $P(\Omega_0) = 1$.

Proof (Of Theorem 2:) We begin by observing that, by Taylor's theorem, we have

$$V(\boldsymbol{\theta}+\boldsymbol{\eta}) = V(\boldsymbol{\theta}) + \langle \nabla V(\boldsymbol{\theta}), \boldsymbol{\eta}\rangle + \frac{1}{2}\langle \boldsymbol{\eta}, \nabla^2 V(\boldsymbol{\theta}+\lambda\boldsymbol{\eta})\boldsymbol{\eta}\rangle,$$

for some $\lambda \in [0, 1]$. Since $\|\nabla^2 V(\boldsymbol{\theta}+\lambda\boldsymbol{\eta})\|_S \leq 2M$, it follows that

$$V(\boldsymbol{\theta}+\boldsymbol{\eta}) \leq V(\boldsymbol{\theta}) + \langle \nabla V(\boldsymbol{\theta}), \boldsymbol{\eta}\rangle + M\|\boldsymbol{\eta}\|_2^2, \ \forall \boldsymbol{\theta}, \boldsymbol{\eta} \in \mathbb{R}^d.$$

Now apply the above bound with $\boldsymbol{\theta} = \boldsymbol{\theta}_t$ and $\boldsymbol{\eta} = \alpha_t \mathbf{f}(\boldsymbol{\theta}_t) + \alpha_t \boldsymbol{\xi}_{t+1}$. This gives

$$\begin{aligned} V(\boldsymbol{\theta}_{t+1}) &\leq V(\boldsymbol{\theta}_t) + \alpha_t \langle \nabla V(\boldsymbol{\theta}), \mathbf{f}(\boldsymbol{\theta}_t)\rangle + \alpha_t \langle \nabla V(\boldsymbol{\theta}_t), \boldsymbol{\xi}_{t+1}\rangle + \alpha_t^2 M\|\mathbf{f}(\boldsymbol{\theta}_t) + \boldsymbol{\xi}_{t+1}\|_2^2 \\ &= V(\boldsymbol{\theta}_t) + \alpha_t \dot{V}(\boldsymbol{\theta}_t) + \alpha_t \langle \nabla V(\boldsymbol{\theta}_t), \boldsymbol{\xi}_{t+1}\rangle + \alpha_t^2 M[\|\mathbf{f}(\boldsymbol{\theta}_t)\|_2^2 + \|\boldsymbol{\xi}_{t+1}\|_2^2 + 2\langle \mathbf{f}(\boldsymbol{\theta}_t, \boldsymbol{\xi}_{t+1})\rangle]. \end{aligned}$$

[4] Here and elsewhere, "for all" really means "for almost all."

Now we bound $E(V(\boldsymbol{\theta}_{t+1})|\mathcal{F}_t)$ and make use of the assumptions (F1), (F2), (N1), (N2) and the bounds in (19). This gives successively

$$\begin{aligned} E(V(\boldsymbol{\theta}_{t+1}))|\mathcal{F}_t &\leq V(\boldsymbol{\theta}_t) + \alpha_t^2 M[L^2\|\boldsymbol{\theta} - \boldsymbol{\theta}^*\|_2^2 + D(1 + \|\boldsymbol{\theta} - \boldsymbol{\theta}^*\|_2^2)] - \alpha_t\phi(\|\boldsymbol{\theta} - \boldsymbol{\theta}^*\|_2) \\ &= \left[1 + \alpha_t^2 \frac{M}{a}(L^2 + D)\right] V(\boldsymbol{\theta}_t) + \alpha_t^2 MD - \alpha_t\phi(\|\boldsymbol{\theta} - \boldsymbol{\theta}^*\|_2). \end{aligned}$$

Now apply Theorem 6 with

$$z_t = V(\boldsymbol{\theta}_t), \eta_t = \alpha_t^2 \frac{M}{a}(L^2 + D), \gamma_t = \alpha_t^2 MD, \psi_t = \alpha_t\phi(\|\boldsymbol{\theta} - \boldsymbol{\theta}^*\|_2).$$

Then, if (3) holds so that $\sum_{t=0}^{\infty} \alpha_t^2 < \infty$, it follows that $\{V(\boldsymbol{\theta}_t)\}$ is bounded almost surely. In view of (19), this implies in turn that $\{\boldsymbol{\theta}_t\}$ is bounded almost surely. This is the first conclusion. In addition, there is a random variable ζ such that $V(\boldsymbol{\theta}_t)$ converges almost surely to ζ.

To prove the second conclusion, note that another consequence of Theorem 6 is that

$$\sum_{t=0}^{\infty} \alpha_t\phi(\|\boldsymbol{\theta} - \boldsymbol{\theta}^*\|_2) < \infty \tag{37}$$

almost surely. Let $\Omega_1 \subseteq \Omega$ denote the values of ω for which

$$\sup_t V(\boldsymbol{\theta}_t(\omega)) < \infty, V(\boldsymbol{\theta}_t(\omega)) \to \zeta(\omega), \text{ and } \sum_{t=0}^{\infty} \alpha_t\phi(\|\boldsymbol{\theta} - \boldsymbol{\theta}^*\|_2) < \infty.$$

Note that $P(\Omega_1) = 1$. It is now shown that $\zeta(\omega) = 0$ for all $\omega \in \Omega_1$. Suppose by way of contradiction that, for some $\omega \in \Omega_1$, we have that $\zeta(\omega) = 2\epsilon > 0$. Choose a T such that $V(\boldsymbol{\theta}_t(\omega)) \geq \epsilon$ for all $t \geq T$, and also define $V_M := \sup_t V(\boldsymbol{\theta}_t(\omega))$. Then we have that

$$\sqrt{\epsilon/b} \leq \|\boldsymbol{\theta}\|_2 \leq \sqrt{V_M/a}, \ \forall t \geq T.$$

Define

$$\delta := \inf_{\sqrt{\epsilon/b} \leq r \leq \sqrt{V_M/a}} \phi(r),$$

and observe that $r > 0$ because ϕ belongs to the class $\mathcal{B}$. Therefore

$$\sum_{t=T}^{\infty} \alpha_t\phi(\|\boldsymbol{\theta} - \boldsymbol{\theta}^*\|_2) \geq \sum_{t=T}^{\infty} \alpha_t r = \infty,$$

provided (4) holds.[5] But this contradicts (37). Hence no such $\omega \in \Omega_1$ can exist. In other words, $\zeta = 0$ almost surely, and $V(\boldsymbol{\theta}_t) \to 0$ almost surely. Finally, it follows from (19) that $\boldsymbol{\theta}_t \to \mathbf{0}$ almost surely as $t \to \infty$, which is the second conclusion. □

5 Proof of Theorem 3

Proof Following in Corless and Glielmo (1998), define the Lyapunov function candidate V as in (30). Then, as shown in Corless and Glielmo (1998), V satisfies (22) and (31). The latter is not of any concern to us. So we focus on proving (23).

Note that the solution function $\mathbf{s}(\cdot, \boldsymbol{\theta})$ satisfies

$$\mathbf{s}(t, \boldsymbol{\theta}) = \boldsymbol{\theta} + \int_0^t \mathbf{f}(\mathbf{s}(\tau, \boldsymbol{\theta}))\, d\tau. \tag{38}$$

Therefore

$$\nabla_{\boldsymbol{\theta}} \mathbf{s}(t, \boldsymbol{\theta}) = I + \int_0^\tau \nabla_{\boldsymbol{\theta}} \mathbf{f}(\mathbf{s}(\tau, \boldsymbol{\theta}))\, d\tau. \tag{39}$$

Next, the chain rule gives

$$\nabla_{\boldsymbol{\theta}} \mathbf{f}(\mathbf{s}(\tau, \boldsymbol{\theta})) = \nabla_{\boldsymbol{\phi}} \mathbf{f}(\boldsymbol{\phi})\big|_{\boldsymbol{\phi}=\mathbf{s}(\tau,\boldsymbol{\theta})} \nabla_{\boldsymbol{\theta}} \mathbf{s}(\tau, \boldsymbol{\theta}).$$

Now the global Lipschitz continuity of $\mathbf{f}$ implies that

$$\|\nabla_{\boldsymbol{\phi}} \mathbf{f}(\mathbf{s}(\tau, \boldsymbol{\phi}))\|_S \le L, \ \forall \boldsymbol{\phi}, \ \forall \tau.$$

Therefore (39) leads to (after dropping the subscript $\boldsymbol{\theta}$)

$$\|\nabla \mathbf{s}(t, \boldsymbol{\theta})\|_S \le 1 + \int_0^t L \|\nabla \mathbf{s}(\tau, \boldsymbol{\theta})\|_S\, d\tau.$$

Now Gronwall's inequality leads to the bound

$$\|\nabla \mathbf{s}(t, \boldsymbol{\theta})\|_S \le \exp(Lt), \ \forall t, \ \forall \boldsymbol{\theta}. \tag{40}$$

Next we proceed to find a bound on the second partial derivatives. It follows from (38) that

$$\frac{\partial s_i(t, \boldsymbol{\theta})}{\partial \theta_j} = \delta_{ij} + \int_0^t \frac{\partial f_i(\mathbf{s}(\tau, \boldsymbol{\theta}))}{\partial \theta_j}\, d\tau,$$

[5] Note that dropping a finite number of terms does not affect the validity of (4).

where δ_{ij} is the Kronecker delta. Next,

$$\frac{\partial^2 s_i(t,\boldsymbol{\theta})}{\partial\theta_j\partial\theta_k} = \int_0^t \frac{\partial^2 f_i(\mathbf{s}(\tau,\boldsymbol{\theta}))}{\partial\theta_j\partial\theta_k}\, d\tau. \tag{41}$$

We will use (41) later. Next, expand $V(\boldsymbol{\theta})$ as

$$V(\boldsymbol{\theta}) = \int_0^T e^{2\kappa\tau} \sum_{i=1}^d [s_i(\tau,\boldsymbol{\theta}) - \theta_i^*]^2\, d\tau.$$

Thus

$$\frac{\partial V(\boldsymbol{\theta})}{\partial\theta_j} = \int_0^T 2e^{2\kappa\tau} \sum_{i=1}^d [s_i(\tau,\boldsymbol{\theta}) - \theta_i^*] \frac{\partial s_i(\tau,\boldsymbol{\theta})}{\partial\theta_j}\, d\tau,$$

$$\begin{aligned}\frac{\partial^2 V(\boldsymbol{\theta})}{\partial\theta_j\partial\theta_k} &= \int_0^T 2e^{2\kappa\tau} \sum_{i=1}^d \frac{\partial s_i(\tau,\boldsymbol{\theta})}{\partial\theta_k} \frac{\partial s_i(\tau,\boldsymbol{\theta})}{\partial\theta_j}\, d\tau \\ &\quad + \int_0^T 2e^{2\kappa\tau} \sum_{i=1}^d [s_i(\tau,\boldsymbol{\theta}) - \theta_i^*] \frac{\partial^2 s_i(\tau,\boldsymbol{\theta})}{\partial\theta_j\partial\theta_k}\, d\tau \\ &= I_1 + I_2,\end{aligned}$$

where

$$I_1 = \int_0^T 2e^{2\kappa\tau} \sum_{i=1}^d \frac{\partial s_i(\tau,\boldsymbol{\theta})}{\partial\theta_k} \frac{\partial s_i(\tau,\boldsymbol{\theta})}{\partial\theta_j}\, d\tau, \tag{42}$$

$$I_2 = \int_0^T 2e^{2\kappa\tau} \sum_{i=1}^d [s_i(\tau,\boldsymbol{\theta}) - \theta_i^*] \frac{\partial^2 s_i(\tau,\boldsymbol{\theta})}{\partial\theta_j\partial\theta_k}\, d\tau. \tag{43}$$

We will prove the boundedness of each integral separately. Note that, as a consequence of (40), we have

$$\left|\frac{\partial s_i(\tau,\boldsymbol{\theta})}{\partial\theta_k}\right|, \left|\frac{\partial s_i(\tau,\boldsymbol{\theta})}{\partial\theta_j}\right| \le \|\nabla\mathbf{s}(\tau,\boldsymbol{\theta})\|_S \le \exp L\tau, \ \ \forall\tau, i, j, k.$$

So the first integral is bounded by

$$|I_1| \le \int_0^T 2de^{2\kappa\tau} e^{2L\tau}\, d\tau =: C_1 < \infty$$

for some constant C_1, whose precise value need not concern us. So we concentrate on showing that, under Assumption (F4), I_2 is also bounded globally.

Toward this end, we begin by observing that

$$\|\mathbf{s}(t,\boldsymbol{\theta})\|_2 \geq e^{-Lt}\|\boldsymbol{\theta}-\boldsymbol{\theta}^*\|_2,\ \forall t \geq 0.$$

The proof is elementary and is omitted. In particular

$$\|\mathbf{s}(t,\boldsymbol{\theta})\|_2 \geq e^{-LT}\|\boldsymbol{\theta}-\boldsymbol{\theta}^*\|_2,\ \forall t \in [0,T]. \tag{44}$$

Now we estimate the entity $\partial^2 f_i(\mathbf{s}(\tau,\boldsymbol{\theta}))/\partial\theta_j\partial\theta_k$ in (41). Note that

$$\frac{\partial f_i}{\partial\theta_j}(\mathbf{s}(\tau,\boldsymbol{\theta})) = \sum_{l=1}^{d} \left.\frac{\partial f_i(\boldsymbol{\phi})}{\partial\phi_l}\right|_{\boldsymbol{\phi}=\mathbf{s}(\tau,\boldsymbol{\theta})} \frac{\partial s_l(\tau,\boldsymbol{\theta})}{\partial\theta_j},$$

$$\begin{aligned}\frac{\partial^2 f_i(\mathbf{s}(\tau,\boldsymbol{\theta}))}{\partial\theta_j\partial\theta_k} &= \sum_{l=1}^{d} \left.\frac{\partial f_i(\boldsymbol{\phi})}{\partial\phi_l}\right|_{\boldsymbol{\phi}=\mathbf{s}(\tau,\boldsymbol{\theta})} \frac{\partial^2 s_l(\tau,\boldsymbol{\theta})}{\partial\theta_j\partial\theta_k} \\ &\quad + \sum_{l=1}^{d} \frac{\partial}{\partial\theta_k}\left[\left.\frac{\partial f_i(\boldsymbol{\phi})}{\partial\phi_l}\right|_{\boldsymbol{\phi}=\mathbf{s}(\tau,\boldsymbol{\theta})}\right] \frac{\partial s_l(\tau,\boldsymbol{\theta})}{\partial\theta_j}.\end{aligned} \tag{45}$$

The second term can be expanded as

$$\sum_{l=1}^{d}\left[\sum_{r=1}^{d} \left.\frac{\partial^2 f_i(\boldsymbol{\phi})}{\partial\phi_l\partial\phi_r}\right|_{\boldsymbol{\phi}=\mathbf{s}(\tau,\boldsymbol{\theta})} \frac{\partial s_r(\tau,\boldsymbol{\theta})}{\partial\theta_k}\right] \frac{\partial s_l(\tau,\boldsymbol{\theta})}{\partial\theta_j}.$$

Now Assumption (F4) and the bound (44) together imply that

$$\left|\left.\frac{\partial^2 f_i(\boldsymbol{\phi})}{\partial\phi_l\partial\phi_r}\right|_{\boldsymbol{\phi}=\mathbf{s}(\tau,\boldsymbol{\theta})}\right| \leq \|\nabla^2 f_i(\mathbf{s}(\tau,\boldsymbol{\theta}))\|_S \leq \frac{K}{\|\mathbf{s}(\tau,\boldsymbol{\theta})-\boldsymbol{\theta}^*\|_2} \leq \frac{Ke^{LT}}{\|\boldsymbol{\theta}-\boldsymbol{\theta}^*\|_2},\ \forall\tau\in[0,T].$$

Also, as shown in (40),

$$\left|\frac{\partial s_r(\tau,\boldsymbol{\theta})}{\partial\theta_k}\right|, \left|\frac{\partial s_l(\tau,\boldsymbol{\theta})}{\partial\theta_j}\right| \leq \|\nabla\mathbf{s}(\tau,\boldsymbol{\theta})\|_S \leq e^{L\tau} \leq e^{LT},\ \forall\tau\in[0,T].$$

Next, the global Lipschitz continuity of $\mathbf{f}$ implies that

$$\left|\frac{\partial f_i(\boldsymbol{\phi})}{\partial\phi_l}\right| \leq L.$$

Substituting all of these bounds into (41) gives

$$\left|\frac{\partial^2 s_i(t,\boldsymbol{\theta})}{\partial\theta_j\partial\theta_l}\right| \le \int_0^t L\sum_{l=1}^{d}\left|\frac{\partial^2 s_l(\tau,\boldsymbol{\theta})}{\partial\theta_j\partial\theta_k}\right| d\tau + \int_0^t \sum_{l=1}^{d}\sum_{r=1}^{d}\frac{Ke^{LT}e^{L\tau}e^{L\tau}}{\|\boldsymbol{\theta}-\boldsymbol{\theta}^*\|_2}\, d\tau$$
$$\le C_2 + \int_0^t L\sum_{l=1}^{d}\left|\frac{\partial^2 s_l(\tau,\boldsymbol{\theta})}{\partial\theta_j\partial\theta_k}\right| d\tau, \tag{46}$$

where

$$C_2 = \frac{d^2TKe^{3LT}}{\|\boldsymbol{\theta}-\boldsymbol{\theta}^*\|_2}$$

is inversely proportional to $\|\boldsymbol{\theta}-\boldsymbol{\theta}^*\|_2$. Now define

$$h_{jk}(t,\boldsymbol{\theta}) := \sum_{i=1}^{d}\left|\frac{\partial^2 s_i(t,\boldsymbol{\theta})}{\partial\theta_j\partial\theta_k}\right|.$$

Note that the right side of (46) does not depend on i. Therefore, (46) implies that

$$h_{jk}(t,\boldsymbol{\theta}) \le \sum_{i=1}^{d}\left[C_2 + \int_0^t L\left|\sum_{l=1}^{d}\frac{\partial^2 s_i(\tau,\boldsymbol{\theta})}{\partial\theta_j\partial\theta_k}\right|\right] d\tau$$
$$\le C_2 d + \int_0^t L d h_{jk}(\tau,\boldsymbol{\theta})\, d\tau.$$

So by Gronwall's inequality

$$h_{jk}(t,\boldsymbol{\theta}) \le C_2 d e^{LdT},\ \forall t\in[0,T].$$

Since h_{jk} is a sum, each individual component must also be smaller than h_{jk} in magnitude. Thus,

$$\left|\frac{\partial^2 s_i(t,\boldsymbol{\theta})}{\partial\theta_j\partial\theta_l}\right| \le C_2 d e^{LdT} \le \frac{C_3}{\|\boldsymbol{\theta}-\boldsymbol{\theta}^*\|_2}$$

for a suitable constant C_3. Therefore, we have established that the Hessian of each s_i decays as $\boldsymbol{\theta}$ gets farther away from $\boldsymbol{\theta}^*$. Now we return to I_2 as defined in (43), and observe that, as a consequence of Assumption (F3) of global exponential stability, we have

$$|s_i(t,\boldsymbol{\theta}) - \theta_i^*| \le \|\mathbf{s}(t,\boldsymbol{\theta}) - \boldsymbol{\theta}^*\|_2 \le \mu\|\boldsymbol{\theta}-\boldsymbol{\theta}^*\|_2,\ \forall t\ge 0.$$

Now in the definition of I_2, we get the bound

$$|s_i(t,\boldsymbol{\theta}) - \theta_i^*| \cdot \left|\frac{\partial^2 s_i(\tau,\boldsymbol{\theta})}{\partial\theta_j\partial\theta_k}\right| \le \mu\|\boldsymbol{\theta}-\boldsymbol{\theta}^*\|_2 \cdot \frac{C_3}{\|\boldsymbol{\theta}-\boldsymbol{\theta}^*\|_2} = \mu C_3.$$

Since the integrand in (43) is bounded and T is finite, it follows that I_2 is also bounded. This finally leads to the desired conclusion that $\|\nabla^2 V\|_S$ is globally bounded. □

Note that in the above proof, the finiteness of the constant T is crucial. Therefore the traditional infinite integral type of Lyapunov function defined in (29) is not directly amenable to such analysis. However, it is perhaps possible to replace the Lyapunov function candidate of (29) by another function of the form

$$V(\boldsymbol{\theta}) := \left[\int_0^\infty \|\mathbf{s}(t,\boldsymbol{\theta})\|^{2p}\, dt\right]^{1/p}.$$

However, Assumption (F4) is still required.

Note that Theorem 4 is a ready consequence of Theorems 2 and 3. Hence the proof is omitted.

6 Proof of Theorem 5

Proof Using Taylor's theorem, one can conclude that

$$\frac{J(\boldsymbol{\theta}_t - c_t\mathbf{e}_i) - J(\boldsymbol{\theta}_t + c_t\mathbf{e}_i)}{2c_t} = -[\nabla J(\boldsymbol{\theta})]_i + 2[\nabla^2 J(\boldsymbol{\theta} + \lambda\mathbf{e}_i)]_{ii}c_t,$$

for some $\lambda \in [-1, 1]$. Hence we can write $y_{t+1,i}$ of (5) as

$$y_{t+1,i} = -[\nabla J(\boldsymbol{\theta}_t)]_i + c_t r_{t,i} + \frac{1}{c_t}\xi_{t+1,i},$$

$$\mathbf{y}_{t+1} = -\nabla J(\boldsymbol{\theta}_t) + c_t\mathbf{r}_t + \frac{1}{c_t}\boldsymbol{\xi}_{t+1,i}, \tag{47}$$

where $\boldsymbol{\xi}_{t+1} = 2(\boldsymbol{\xi}^-_{t+1} - \boldsymbol{\xi}^+_{t+1})$, and

$$r_{t,i} = 2[\nabla^2 J(\boldsymbol{\theta} + \lambda\mathbf{e}_i)]_{ii}.$$

Though $\mathbf{r}_t$ is random (because $\boldsymbol{\theta}_t$ is random), (24) guarantees that

$$|r_{t,i}| \le M', \ \forall t.$$

(Here we use the fact that for any matrix A, we have that $A_{ii} \le \|A\|_S$ for all $i \in [d]$.) Consequently $\|\mathbf{r}_t\|_2 \le \sqrt{d}M' =: M_0$, for all t.

Now let analyze the recursion (2), using the hypotheses, in particular (27) and (28). This gives

$$\begin{aligned}\|\boldsymbol{\theta}_{t+1}\|_2^2 &= \|\boldsymbol{\theta}_t + \alpha_t \mathbf{y}_{t+1}\|_2^2 \\ &= \|\boldsymbol{\theta}_t - \alpha_t \nabla J(\boldsymbol{\theta}_t) + \alpha_t c_t \mathbf{r}_t + (\alpha_t/c_t)\boldsymbol{\xi}_{t+1}\|_2^2.\end{aligned}$$

If we expand the Euclidean norm squared on the right side, we get four square terms and six cross product terms. The conditional expectation of these terms can be bounded using the following facts, all of which hold almost surely and for all t:

$$E(\|\boldsymbol{\xi}_{t+1}^2\|_2^2|\mathcal{F}_t) \le D(1 + \|\boldsymbol{\theta}_t\|_2^2),$$

$$E(\boldsymbol{\xi}_{t+1}|\mathcal{F}_t) = \mathbf{0},$$

$$\|\mathbf{r}_t\|_2 \le M_0,$$

$$\|\nabla J(\boldsymbol{\theta}_t)\|_2 \le L\|\boldsymbol{\theta}_t\|_2,$$

$$\langle \boldsymbol{\theta}_t, \nabla J(\boldsymbol{\theta}_t)\rangle \ge h\|\boldsymbol{\theta}_t\|_2^2.$$

These substitutions lead to the following complicated expression which we will simplify in stages.

$$\begin{aligned}E(\|\boldsymbol{\theta}_{t+1}\|_2^2|\mathcal{F}_t) \le\ & \|\boldsymbol{\theta}_t\|_2^2[1 + \alpha_t^2 L^2 + (\alpha_t/c_t)^2 D] + \alpha_t^2 c_t^2 M_0^2 + (\alpha_t/c_t)^2 D \\ & + 2\|\boldsymbol{\theta}_t\|_2(\alpha_t c_t M_0 + \alpha_t^2 c_t L M_0) - 2\alpha_t\langle \boldsymbol{\theta}_t, \nabla J(\boldsymbol{\theta}_t)\rangle. \qquad (48)\end{aligned}$$

On the right side, there are constant terms, linear terms, and quadratic terms in $\|\boldsymbol{\theta}_t\|_2$. To make things uniform, we replace $\|\boldsymbol{\theta}_t\|_2$ by its upper bound[6] $1 + \|\boldsymbol{\theta}_t\|_2^2$, and $\alpha_t\langle \boldsymbol{\theta}_t, \nabla J(\boldsymbol{\theta}_t)\rangle$ by its lower bound $\alpha_t h\|\boldsymbol{\theta}_t\|_2^2$. This gives

$$E(\|\boldsymbol{\theta}_{t+1}\|_2^2|\mathcal{F}_t) \le \|\boldsymbol{\theta}_t\|_2^2(1 + p_{t,1}) + p_{t,2} - 2\alpha_t h\|\boldsymbol{\theta}_t\|_2^2, \qquad (49)$$

where

$$p_{t,1} = \alpha_t^2 L^2 + 2(\alpha_t c_t M_0 + \alpha_t^2 c_t L M_0) + (\alpha_t/c_t)^2 D,$$

$$p_{t,2} = 2(\alpha_t c_t M_0 + \alpha_t^2 c_t L M_0) + \alpha_t^2 c_t^2 M_0^2.$$

Define $p_t = \max\{p_{t,1}, p_{t,2}\}$. Then we can replace (49) by

$$E(\|\boldsymbol{\theta}_{t+1}\|_2^2|\mathcal{F}_t) \le \|\boldsymbol{\theta}_t\|_2^2(1 + p_t) + p_t - 2\alpha_t h\|\boldsymbol{\theta}_t\|_2^2. \qquad (50)$$

This is of the form (32) with

$$z_t = \|\boldsymbol{\theta}_t\|_2^2, \eta_t = \gamma_t = p_t, \psi_t = 2\alpha_t h\|\boldsymbol{\theta}_t\|_2^2.$$

[6] Actually $\|\boldsymbol{\theta}_t\|_2 \le (1 + \|\boldsymbol{\theta}_t\|_2^2)/2$. However, we drop the factor 2 to keep the formulas less cluttered.

So if we can establish that $\{p_t\}$ is summable, then it would follow from Theorem 6 that $\|\boldsymbol{\theta}_t\|_2^2$ is bounded almost surely. This is established next. Recall the two hypotheses that

$$\sum_{t=0}^{\infty} \alpha_t c_t < \infty, \sum_{t=0}^{\infty} (\alpha_t / c_t)^2 < \infty.$$

So, if $\{\alpha_t c_t\}$ is summable, then it is also square-summable, that is, $\{\alpha_t^2 c_t^2\}$ is summable. Moreover, since α_t is bounded by one, the sequence $\{\alpha_t^2 c_t\}$ is also summable. Finally, $\{\alpha_t^2\}$ is -summable because $c_t \leq \delta_0$, so that $(\alpha_t/c_t)^2 \geq \alpha_t^2/\delta_0^2$, and $\{(\alpha_t/c_t)^2\}$ is summable. These observations show that each of the terms that make up p_t, namely (excluding constants) α_t^2, $\alpha_t^2 c_t^2$, $\alpha_t c_t$, $\alpha_t^2 c_t$, $(\alpha_t/c_t)^2$, are all summable. Hence $\|\boldsymbol{\theta}_t\|_2^2$ is bounded almost surely. Moreover, there is a random variable ζ such that $\|\boldsymbol{\theta}_t\|_2^2$ converges almost surely to ζ. Next, suppose in addition that (4) also holds. Theorem 6 also leads to the conclusion that

$$\sum_{t=0}^{\infty} \alpha_t h \|\boldsymbol{\theta}_t\|_2^2 < \infty.$$

Now suppose that there is an $\omega \in \Omega$ such that $\zeta(\omega) > 0$. As in the proof of Theorem 4, this leads to a contradiction. Thus no such ω can exist, and $\zeta = 0$ almost surely. □

7 Examples and Applications

In this section, we give a few examples of the results presented thus far. Some of these might be termed as "toy" examples, while others represent applications of the theory to a few problems of interest in Reinforcement Learning.

Example 1 The first example illustrates how Theorem 1 goes beyond currently available theorems in Lyapunov stability theory. The current theorems, of which Vidyasagar (2002, Theorem 5.3.56) is typical, require $-\dot{V}(\theta)$ to be a function of Class $\mathcal{K}$. Now consider the one-dimensional ODE

$$\dot{\theta} = f(\theta),$$

where

$$\mathbf{f}(\theta) = \begin{cases} \theta, & \text{if } \theta \in [0, 1], \\ \theta^{-k}, & \text{if } \theta > 1 \\ -f(\theta), & \text{if } \theta < 0. \end{cases}$$

If we choose the Lyapunov function $V(\theta) = \theta^2$, then $-\dot{V}(\theta) = 2\theta^{-(k-1)}$. Hence, if $k > 1$, then $-\dot{V}(\theta)$ cannot be bounded below by a function of Class $\mathcal{K}$. If we choose

$$\mathbf{f}(\theta) = \begin{cases} \theta, & \text{if } \theta \in [0, 1], \\ e^{-(\theta-1)}, & \text{if } \theta > 1 \\ -f(\theta), & \text{if } \theta < 0, \end{cases}$$

then $-\dot{V}(\theta)$ will fail to be a function of Class $\mathcal{K}$ for *every* function of the form $V(\theta) = \theta^{2m}$, $m \geq 1$. However, Theorem 1 is applicable to all these situations and enables us to conclude that the equilibrium $\theta = 0$ is globally asymptotically stable.

Example 2 In Gladyshev (1965), Gladyshev demonstrates the convergence of the SA algorithm in the one-dimensional case, when the measurement y_{t+1} is of the form (1), with $f(0) = 0$, and

$$-\inf_{\epsilon \leq \theta \leq M} \theta f(\theta) > 0, \ \forall 0 < \epsilon < M < \infty. \tag{51}$$

Such a function $f(\cdot)$ is called a "passive" function in circuit theory. The objective of this example is to demonstrate that Gladyshev's result does *not* in general follow from those in Borkar and Meyn (2000), but *does* follow from Theorem 2.

Consider the one-dimensional ODE

$$\dot{\theta} = f(\theta),$$

where $f(0) = 0$, $\theta f(\theta) < 0$ for all $\theta \neq 0$, and in addition, $|f(\theta)| \to 0$ as $|\theta| \to \infty$. In this case, the scale-free function is defined in Borkar and Meyn (2000), namely

$$f_\infty(\theta) := \lim_{r \to \infty} \frac{f(r\theta)}{r} = 0, \ \forall \theta.$$

Hence, the ODE $\dot{\theta} = f_\infty(\theta)$ cannot be globally asymptotically state. On the other hand, if we use the Lyapunov function $V(\theta) = \theta^2$ (which is in effect what is done in Gladyshev, 1965), then

$$-\dot{V}(\theta) = -\theta f(\theta).$$

Hence, it follows from Theorem 2 that $\theta_t \to 0$ almost surely as $t \to \infty$, provided (3) and (4) hold.

In many problems in Reinforcement Learning, one wants to solve a linear equation of the form

$$\mathbf{v}^* = \mathbf{r} + \gamma A \mathbf{v}^*,$$

where $\mathbf{v}^*$ is called the "value" vector, $\mathbf{r}$ is called the "reward" vector, $\gamma \in (0, 1)$ is called the "discount factor," and A is a matrix whose spectral radius $\rho(A) \leq 1$. Often it is the case that the ℓ_2-induced norm $\|A\|_{2\to 2}$ greater than one.

Example 3 To apply SA to this problem, let us switch notation to be consistent with that in the paper and rewrite as

$$\boldsymbol{\theta} = \mathbf{r} + \gamma A\boldsymbol{\theta}.$$

If we define the function $\mathbf{f}$ via

$$\mathbf{f}(\boldsymbol{\theta}) = \mathbf{r} + \gamma A\boldsymbol{\theta} - \boldsymbol{\theta},$$

then the unique equilibrium of the associated ODE $\dot{\boldsymbol{\theta}} = \mathbf{f}(\boldsymbol{\theta})$ is indeed the desired solution $\mathbf{v}^*$. Note that, since $\gamma < 1$ and $\rho(A) \leq 1$, the eigenvalues of the matrix $\gamma A - I$ all have negative parts. Hence, $\mathbf{v}^*$ is a globally exponentially stable equilibrium. Then it follows from Vidyasagar (2002, Theorem 5.4.42) that, whenever Q is a symmetric, positive definite matrix, the so-called Lyapunov matrix equation

$$P(\gamma A - I) + (\gamma A - I)^\top P = -Q$$

has a unique positive definite solution for P. Thus, $V(\boldsymbol{\theta}) = \boldsymbol{\theta}^\top P\boldsymbol{\theta}$ satisfies the hypotheses of Theorem 2 (because the Hessian of $V(\cdot)$ is constant and thus bounded). Hence, we can conclude that the SA algorithm will converge to the desired solution $\mathbf{v}^*$, provided (3) and (4) hold. This conclusion holds even if $\|A\|_{2\to 2} > 1$.

8 Conclusions and Future Work

In this paper, we have presented some simple proofs for the convergence of the stochastic approximation (SA) algorithm, based on martingale methods and converse Lyapunov theory. The result presented here in the Lyapunov theory is new and may be of independent interest to researchers in nonlinear stability theory. The fact that the proof is based on the Lyapunov theory, and not the more complicated ODE method discussed in Derevitskii and Fradkov (1974), Ljung (1977), and Kushner and Clark (1978), opens the possibility that the same approach can be used to prove the convergence of the SA algorithm in more general settings, such as two-time scale SA (Lakshminarayanan & Bhatnagar, 2017) or projected gradient SA (Tsitsiklis & Van Roy, 1997). These lines of research are currently under investigation.

In Ljung (1978), a different class of stochastic algorithms is studied by Ljung. Specifically, the basic recursion is

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t + \alpha_t(\mathbf{f}(\boldsymbol{\theta}_t) + \boldsymbol{\xi}_{t+1} + \boldsymbol{\eta}_{t+1}), \tag{52}$$

where the function $\mathbf{f}(\cdot)$ is a *gradient vector field*, that is,

$$\mathbf{f}(\boldsymbol{\theta}) = -\nabla J(\boldsymbol{\theta})$$

for some function $J : \mathbb{R}^d \to \mathbb{R}$ with compact level sets. This means that, for every constant $c \in \mathbb{R}$, the set

$$\mathcal{S}(c) := \{\boldsymbol{\theta} \in \mathbb{R}^d : J(\boldsymbol{\theta}) \leq c\}$$

is compact. In addition, there are now *two* measurement noise sequences $\{\boldsymbol{\xi}_t\}$ and $\{\boldsymbol{\eta}_t\}$. The sequence $\{\boldsymbol{\eta}_t\}$ converges to zero almost surely as $t \to \infty$, while the sequence $\{\boldsymbol{\xi}_t\}$ *is not required to satisfy* the conditional zero-mean assumption (15). There are other minor differences, but these are the main differences between the set-up studied here and that studied in Ljung (1978). The convergence of the algorithm is established using what might be termed as "direct" methods. The two problem set-ups are not directly comparable. In Ljung (1978), the function $\mathbf{f}(\cdot)$ is restricted to be a gradient vector field, whereas there is no such restriction here. On the other hand, the conditional zero-mean assumption (15) is replaced by a more relaxed assumption. It would be highly desirable to explore whether the results of Ljung (1978) (or some broadly similar results) can be obtained via martingale methods.

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